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Foot and Mouth Disease

Syn: Aphthous fever, Fast Moving Disease, Endoepidemic disease

Foot and mouth disease is a highly contagious viral disease of cloven hoofed animals, including cattle, buffalo, pig, sheep and goat and wild animals. It is clinically characterized by high rise of temperature, lameness, profuse salivation, vesicles on the mouth, muzzle, teats and foot..

Etiology

- ☒ It is caused by Aphthovirus of the family Picornaviridae
- ☒ It is a single stranded RNA virus, the capsid encodes 4 structural polypeptides, namely VP1, VP2, VP3 and VP4
- ☒ There are seven major immunologically distinct serotypes O, A, C, Asia1, SAT1, SAT2 and SAT3 are recognized
- ☒ The serotype SAT1, SAT2 and SAT3 are restricted to African continent
- ☒ The serotype O, A have immunologically and serologically distinct subtypes contain more than 60 strains. There is no cross protection between serotype
- ☒ The **Serotype O is the most common serotype** worldwide. Other serotypes also cause serious outbreaks; however, serotype C is uncommon and has not been reported since 2004
- ☒ **In India serotypes, O, A, C and Asia 1** are known to occur. Among which A₂₂, A₅ & A₁₀ subtypes are more commonly reported
- ☒ **In india 80% of outbreak is caused by serotype O**
- ☒ The virus is inactivated at pH range 6.0 – 9.0. The 4% **sodium carbonate, 1-2% sodium hydroxide or formalin, 0.2% citric acid or acetic acid** is an effective disinfectant

Epidemiology

- ☒ Distribution : World wide
- ☒ Morbidity rate: 100%
- ☒ Mortality rate: 1-2%, sometimes may reach 100% in malignant form

Host affected

- ☒ Cattle, buffalo, pig, sheep and goat, deer, antelope, elephant and giraffe, among which cattle are highly susceptible
- ☒ Young calves are highly susceptible than adults
- ☒ Domestic pigs are amplifier host and is important in propagating the virus
- ☒ Sheep and goat act as maintenance host

Source of infection

- ☒ Foot and mouth disease virus (FMDV) can be found in all secretions and excretions from acutely infected animals, including expired air, saliva, milk, urine, faeces, semen and aborted foetus in sheep
- ☒ The amount of virus shed by each route can be influenced by the host species and viral strain. Pigs produce large amounts of aerosolized virus, and the presence of large herds of infected swine may increase the risk of airborne spread
- ☒ The carrier status occurs in **esophageal-pharyngeal fluid** of recovered animals 3.5 yrs in cattle, 9 months in sheep and 4 months in goat but **pigs do not be a carrier**

Transmission

- ☒ **The disease is mainly transmitted through inhalation of aerosol**, under favourable condition the virus can travel in air for long distance even upto 250 km
- ☒ Ingestion of contaminated feed and water
- ☒ Pigs acquire infection through ingestion of untreated swills or direct contact with an infected animal
- ☒ Through skin abrasions
- ☒ Sexual transmission occurs in SAT type viruses
- ☒ Mechanical transmission by fomites or vectors
- ☒ **People can act as mechanical vectors for FMDV**, by carrying the virus on clothing or skin. The virus might also be carried for short time in the nasal passages

Pathogenesis

After inhalation or ingestion of the virus, the Primary replication takes place in the mucosal and lymphatic tissue of the nasopharynx/pharynx/tonsil. Viremia follows primary multiplication with further replication in the lymph nodes, mammary glands and other organs as well as the epithelial cells of the mouth, muzzle, teats, interdigital space and coronary band. In these areas of stratified squamous epithelium, vesicle formation results from swelling and rupture of keratinocytes in the stratum spinosum.

Clinical manifestation

- ☒ Incubation period is variable 1-21 days, usually 3-6 days (In pigs, the incubation period is as short as 1-3 days)

Cattle

- ☒ Sudden onset of drop in milk yield
- ☒ High fever (40-41⁰C) accompanied by loss of appetite, depression, appearance of **painful stomatitis**
- ☒ **Profuse salivation**, the **saliva hanging in long, ropy strings**, a characteristic **smacking of the lips**, and the animal chews carefully
- ☒ Vesicles and bullae (1-2 cm in diameter) appear on the buccal mucosa, dental pad, tongue and udder
- ☒ The vesicles are thin walled contain a straw coloured fluid. Vesicles rupture within 24hrs, leaving a raw painful surface, which heals in about 1 week
- ☒ Vesicles appear on the feet, particularly in the clefts and on the coronet leads to **lameness**
- ☒ The lesions on the tongue often heal within a few days but those on the feet and nasal cavities are contaminated with bacteria, maggots interfere with wound healing
- ☒ **Vesicles may also develop on the teats, which results in severe mastitis**
- ☒ The virus does not cross the placenta but the abortion mostly during second trimester of pregnancy mainly due to fever
- ☒ Calves less 6 months old die suddenly due to **acute myocardial failure**
- ☒ The malignant form of infection in adults causes acute myocardial failure
- ☒ The common sequel of FMD is dyspnoea, **panting (heat intolerance)**, **over growth of hair** and poor semen quality in bulls

Sheep and goat

- ☒ The pronounced clinical sign is **sudden onset of acute lameness**
- ☒ The vesicles develop in the inter digital space of the feet, which rupture in about 2-3 days
- ☒ Oral lesions are rare or they may develop only on the tongue and upper palate but there is no salivation

Pigs

- ☒ Fever 107⁰ F
- ☒ Large vesicles and bullae develops on the snout and feet
- ☒ Lameness is the first sign, affected animal is lethargic, huddle among others
- ☒ Vesicles then develop on the coronary band and heel, and in the inter digital space
- ☒ The foot lesions are very painful; the pigs crawl rather than walk. The horns of the digits are sometimes sloughed

- ☒ Vesicles in the mouth are very less prominent
- ☒ Young pigs up to 14 weeks of age may die suddenly from heart failure; piglets less than 8 weeks of age are particularly susceptible

Necropsy findings

- ☒ Acute myocarditis occur in calves and malignant form of infection in adults
- ☒ Heart- Epicardial haemorrhage with or without pale area, ventricular walls appear streaked with patch of yellow tissue interspersed with normal myocardium giving the **typical tiger heart appearance**

Sample collection

- ☒ **Fresh vesicular fluids**, lesions of tongue, feet, udder or lips of about 1 gm, pieces of cardiac muscle, mammary gland and pancreas should be collected in **50% phosphate buffered glycerol saline** for virus isolation.
- ☒ The outside of the container should be disinfected with 4% sodium carbonate or 0.2% citric acid
- ☒ Paired sera sample

Diagnosis

- ☒ Based on clinical signs
- ☒ Viral isolation-Inoculation of virus or tissue homogenate in monolayers of bovine thyroid or pig kidney cells, Lamb kidney cells or BHK-21 cell line. The cytopathic effect (CPE) develops within 24-48 hours at 37° C
- ☒ The viral antigens are usually identified with ELISA and RT-PCR
- ☒ The persistent or subclinical infection can be detected in oesophageal/pharyngeal fluid by RT-PCR
- ☒ Virus infection associated antigen test (Enzyme Linked Immuno Electrotransfer Blot) is used to detect **non structural protein (NSP)** which are expressed during viral replication
- ☒ Complement fixation test (CFT) is the most commonly used to identify the type of virus
- ☒ The virus neutralization test and ELISAs are used to detect serotype specific antibodies in serum
- ☒ RT-LAMP- Reverse transcription loop mediated isothermal amplification
- ☒ Lateral flow Immunochromatographic test strip test

Differential diagnosis

- ☒ Vesicular stomatitis
- ☒ Vesicular exanthema of swine
- ☒ Blue tongue
- ☒ Swine vesicular disease

- ☒ Bovine viral diarrhoea
- ☒ Rinderpest
- ☒ Malignant catarrhal fever
- ☒ Pox infection of mammary gland
- ☒ Foot rot in sheep

Treatment

- ☒ Cleaning of oral lesions and foot lesions with mild antiseptic solution
- ☒ Antibiotics to prevent secondary bacterial infection
- ☒ In Kenya Ethano-veterinary practices are followed at the time of outbreak. Oral lesions are cleaned with Soda ash solution (97%) to remove necrotic tissue after that honey and finger millet flour was applied on the lesions to improve healing for 3 days

Prevention

- ☒ **Inactivated Oil adjuvented trivalent or polyvalent inactivated vaccines** are currently used
- ☒ Calves from unvaccinated dam should be vaccinated at **4th month of age** and revaccination should be done at 8th months of age
- ☒ Calves from vaccinated dam should be vaccinated at 6th month of age and revaccination should be done at 10th months of age
- ☒ Adults should be vaccinated once in 6 months regularly
- ☒ Sheep and pigs should be vaccinated at 6 months of age

Control

- ☒ Ring vaccination- vaccination of all susceptible population around 5-10 km radius from infected premises to prevent the further spread of infection
- ☒ Restriction of animal movement
- ☒ Quarantine of newly purchased animals
- ☒ Stamping out policy, slaughter of infected and in contact animals
- ☒ Proper disposal of dead animals
- ☒ Disinfection of premises with **4% Sodium carbonate or 1-2% sodium hydroxide or formalin** solution
- ☒ Animals in high risk area, the ruminants are vaccinated three times a year. With a medium risk, animals are vaccinated twice a year
- ☒ Avoid feeding of uncooked swill/garbage/meat scraps to the pigs
- ☒ Quarantine of all animals 16-24 km radius from outbreak area

Vesicular Stomatitis

Vesicular stomatitis is a viral disease of **horse, cattle and pig** is characterized by vesiculation, ulceration, erosion of oral and nasal mucosa, epithelial surface of tongue, coronary band and teats, along with crusting lesions of the muzzle, ventral abdomen and sheath.

Etiology

- ☒ It is caused by by Vesiculoviruses of the family Rhabdoviridae
- ☒ The virus contains two distinct serotypes **New Jersey and Indiana**. The Indiana has three subtypes
- ☒ The **New Jersey strain is most virulent and most common**. The virus was destroyed by boiling and common disinfectant

Epidemiology

- ☒ The disease is restricted to western hemisphere (America) however; it was previously described in France and in South Africa
- ☒ In United States sporadic outbreaks occur periodically in **late summer and autumn**. The incidence of disease sharply decline with onset of cold weather
- ☒ Horse, cattle, donkeys are more susceptible, but infection occur in pig, camelids, sheep and goat and human
- ☒ **Outbreak of disease is most common in horses and cattle**
- ☒ Calves are more resistant to infection than adults
- ☒ Saliva and vesicular fluid from infected animals act as source of infection

Transmission

The virus can be biologically transmitted by sand flies (*Lutzomyia* sp) and black flies (*Simulium* sp) and mechanically by *culicoides* sp

Clinical manifestation

- ☒ Incubation period- 3-15 days

Cattle

- ☒ Sudden appearance of mild fever
- ☒ Development of vesicles on the dorsum of tongue, dental pad, lips and buccal mucosa
- ☒ Ruptured vesicles results in irritation causes profuse roapy salivation and anorexia
- ☒ Ulcers and erosions of the oral mucosa and mucocutaneous junction of lips
- ☒ Decreased milk yield
- ☒ Lesions on the feet and udder occur rarely
- ☒ The extensive teat lesions may lead to development of mastitis

Horses

- ☒ Clinical signs are similar to cattle
- ☒ Fever, depression and inappetence
- ☒ Drooling of saliva
- ☒ Vesicles coalesce and rupture with detachment of epithelium and formation of shallow ulcer
- ☒ The lesions are limited to dorsum of the tongue
- ☒ Crusting lesions of muzzle, ventral abdomen, prepuce and udder
- ☒ Lesion may occur at coronary band and leads to lameness and deformity of hoof wall

Pigs

- ☒ Vesicles develop on or behind the snout or on the feet
- ☒ Lameness is frequent than in other animals

Sample collection

- ☒ Vesicular fluid and lesions

Diagnosis

- ☒ Based on clinical sign
- ☒ Isolation of virus in tissue culture
- ☒ Viral RNA is detected by RT-PCR and indirect sandwich ELISA
- ☒ The complement fixation (CF) test is also a good alternative. The virus neutralisation (VN) test may be used, but it is elaborate and time-consuming
- ☒ Detection of antibodies by liquid phase blocking ELISA, competitive ELISA, VNT, AGID and CIE

Differential diagnosis

- ☒ Foot and mouth disease
- ☒ Swine vesicular disease
- ☒ Vesicular exanthema

Control

- ☒ Animal movement from farm should be prohibited for 30 days until after all lesions have healed
- ☒ Hygienic and quarantine precautions

Rinderpest

Syn: Cattle Plague

It is an acute, highly contagious viral disease of wild and domesticated ruminants and clinically characterized by sudden onset of fever, oculonasal discharges, necrotic stomatitis, gastroenteritis and death.

Etiology

It is caused by Rinderpest virus, genus Morbillivirus belongs to family Paramyxoviridae. The rinderpest virus contain only one serotype with multiple strains vary with virulence, but all are immunologically identical. The infection with one strain protect against all other strains. The virus is antigenically related to other members of morbillivirus groups includes, PPR virus in sheep and goat, canine distemper in dog, measles in human, phocine distemper in seals and equine influenza recently described in Australia.

The rinderpest virus is quite fragile in the environment and rapidly inactivated at pH less than 5.6 or greater than 9.6., solar radiation and desiccation.

Epidemiology

- ☒ Historically, Rinderpest was a most devastating disease spread from Asia to Europe, Middle East and Africa
- ☒ Rinderpest is endemic in India and **eradicated on 2006**. The disease was eradicated from world and officially declared by OIE on **May 2011**.
- ☒ Non endemic area: Morbidity and mortality rate-100% and 90% respectively
- ☒ Endemic area - Case fatality rate - 30%

Host affected

- ☒ **All ruminants and pigs are susceptible**. Natural infection occurs commonly in cattle and buffalo
- ☒ Sheep and goats are generally less susceptible but may develop clinical disease
- ☒ Asiatic pigs are highly susceptible and European breeds are less susceptible
- ☒ Camels are apparently not infected and appear to have no role in rinderpest transmission and maintenance

Source of infection

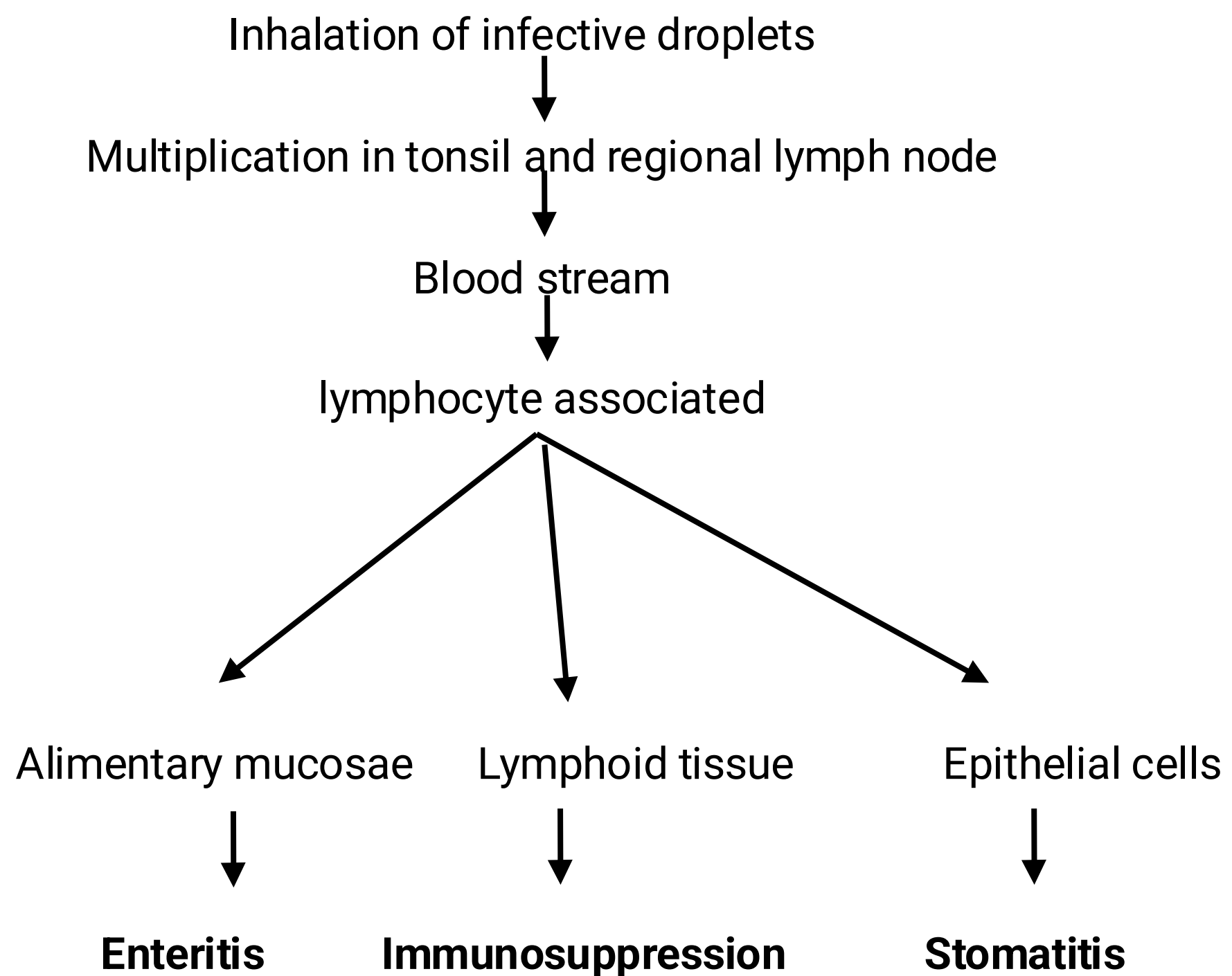
- ☒ Rinderpest virus is found in all expired air, nasal and ocular discharges, saliva, faeces, milk, semen, vaginal discharge and urine

Transmission

- ☒ **Close contact with an infected animals**
- ☒ Inhalation of virus in aerosol
- ☒ ingestion of contaminated feed and water
- ☒ Rinderpest virus is not transmitted by insect vectors

Immunity: Immunity after natural infection or vaccination is life long

Pathogenesis



Clinical signs

- ☒ It occur as peracute, acute or mild form
- ☒ Incubation period is 6-9 days

Cattle and buffalo

Peracute form

- ☒ Sudden onset of fever, inappetence, depression, congested visible mucous membranes, respiratory distress and death within two to three days, before oral erosions develops.

Acute form

- ☒ It occur in three phase
 - Prodromal phase
 - Mucosal phase
 - Convalescence phase

Prodromal phase (period of peak virus production)

- ☒ Fever (40.5-41.5° c) for several days without mucosal lesion
- ☒ Anorexia, fall in milk yield, lacrimation, harsh staring coat

Mucosal phase

- ☒ Characterized by inflammation of buccal, nasal and conjunctival mucosa
- ☒ Hyperemia of vulva and swelling of vagina
- ☒ Lacrimation is profuse and then purulent accompanied by blepharospasm
- ☒ Clear blood stained bubbly salivation is followed by purulent saliva and halitosis

- ☒ **Discrete grayish necrotic lesions**, appearing on **inside of the lower lip, adjacent gum**, on the **cheek mucosa at the commissures** and **lower surface of the tongue**
- ☒ Similar lesions are common on nasal, vulval and vagina mucosae
- ☒ Sloughing of necrotic material leaving raw, red area with sharp edges as these coalesce to form shallow ulcers
- ☒ **Severe diarrhoea**, sometimes dysentery with tenesmus, appear as lesions develop in abomasums and intestine
- ☒ **Abdominal pain**
- ☒ Severe dehydration
- ☒ Moist reddened skin lesions occur on perineum, scrotum, flanks, inner aspect of thigh and neck, later it covered with scab. But this is less common

Convalescence phase

- ☒ Mucosal lesions heals rapidly
- ☒ **Diarrhoea stops**
- ☒ Abortion in pregnant animal

Subacute form

- ☒ It occur in enzootic area with lower morbidity and mortality
- ☒ Inflammation of catarrhal only
- ☒ No dysentery
- ☒ Skin form- pustule develop on neck, withers, inside the thigh without systemic reaction
- ☒ Lymphocytolysis, activate the latent infection of *Anaplasma marginale*

Necropsy finding

- ☒ Zone of haemorrhage and erythema running transversely across the mucosa of colon produce a characteristic striped appearance is called as **zebra stripes**
- ☒ Histopathological examination of tonsil- intracytoplasmic and intranuclear inclusion bodies may be present

Sample collection

- ☒ **Buffy coat, lymph node biopsy, fresh spleen and gut lesions** should be collected aseptically and send it under refrigerated condition

Diagnosis

- ☒ Based on clinical symptoms and lesion
- ☒ Marked leukopenia
- ☒ Isolation and identification of organism in tissue culture
- ☒ Detection of antigen by AGID
- ☒ Antibody detection by virus neutralization test, complement fixation test, counter immunoelectrophoresis and ELISA

Differential diagnosis

- ☒ Foot and mouth disease
- ☒ Haemorrhagic septicaemia
- ☒ Malignant catarrhal fever
- ☒ Bovine viral diarrhoea
- ☒ Infectious bovine rhinotracheitis

Sheep and goat

- ☒ PPR
- ☒ Blue tongue
- ☒ Sheep and goat pox

Treatment

- ☒ Treatment is ineffective and should not be undertaken
- ☒ Fluid therapy may be given

Control

1. Rigid slaughter and quarantine measures
2. Endemic area- **annual vaccination of all cattle and domestic buffaloe >1yr of age with tissue culture rinderpest vaccine**
3. Ring vaccination of susceptible animal during outbreak
4. Continuous monitoring and surveillance of animals against rinderpest
5. Control of animal movement

Malignant Catarrhal Fever

Syn: Malignant head catarrh, Snotsiekte, Catarrhal fever, Gangrenous coryza

Malignant catarrhal fever (MCF) is an acute, generalized and usually fatal disease affecting many species of Artiodactyla. It is characterized by lymphoid cell accumulations in non lymphoid organs, vasculitis and **hyperplasia of T lymphocyte in lymphoid organs**. MCF usually appears sporadically and affects few animals in the population.

Etiology

- ☒ Alcelaphine herpes virus-1 (ALHV-1)- **Wildebeest associated malignant catarrhal fever virus** of the genus Maca virus
- ☒ Ovine herpes virus-2 (OvHV-2)- **Sheep associated malignant catarrhal fever virus** of the genus Maca virus

Epidemiology

- ☒ The Alcelaphine herpes virus-1 (AIHV-1) is maintained by inapparently infected wildebeest and transmits the disease to cattle in Africa and ruminant species (buffalo, deer and bison) in zoological garden
- ☒ **Ovine herpes virus-2 is prevalent in domestic sheep as a subclinical infection** and causes MCF in cattle in most region of the world

Source of infection

- ☒ **The virus is excreted only in natural host of wildebeest and sheep respectively.** These animals act as a source of infection for other animals
- ☒ The virus is not excreted from clinically affected cattle. So, infected cattle will not be source of infection for another cattle

Transmission

Wildebeest associated MCF Virus

- ☒ **Inhalation of aerosol or ingestion** of pasture contaminated by virus excreted in nasal and ocular discharge of young wildebeest
- ☒ Infected young wildebeest have persistent viraemia for upto 3 months of age

Sheep associated MCF Virus is spread by unknown means

- ☒ The epidemiological association of contact with lambing ewes suggest that perinatal lambs play a role in transmission similar to that of wildebeest calves. However, there is no evidence for transplacental transmission
- ☒ Infection can also occur when cattle and sheep housed in the same building

Clinical manifestation

- ☒ Incubation period is 3-8 weeks
- ☒ The symptoms occur as peracute, alimentary tract form, head and eye form and mild form

Peracute form

- ☒ Course of disease- 1-3 days
- ☒ There is usually a high fever, dyspnea and acute gastroenteritis

Head and eye form-most common

- ☒ Extreme dejection
- ☒ Anorexia
- ☒ Agalactia
- ☒ High fever (40-41.5° C)
- ☒ Rapid pulse rate (100-120/ min)
- ☒ Profuse mucopurulent nasal discharge
- ☒ Severe dyspnoea with stertor due to obstruction of nasal cavities with exudates
- ☒ Ocular discharge with oedema of the eyelids
- ☒ Blepharospasm

- ☒ Congestion of scleral vessels
- ☒ Superficial necrosis of anterior nasal and buccal mucosae
- ☒ **Discrete local area of necrosis appear on the gum, hard palate, gingivae, dorsum of the tongue, commissures of mouth and inside the lips**
- ☒ Cheek papillae is haemorrhagic especially at the tips
- ☒ Excessive ropy, bubbly salivation
- ☒ Necrosis of skin horn junction of feet especially at the back of the pastern
- ☒ **Nervous signs**- includes weakness in one leg, incoordination, demented appearance, muscle tremor, nystagmus, head pushing, paralysis and convulsions
- ☒ **Lymphadenopathy**
- ☒ **Opacity of the cornea, commence as a narrow gray ring at the corneoscleral junction** and hypopyon observed in some cases
- ☒ **In some cases gross hematuria with red coloration is most marked at the end of urination**
- ☒ Skin changes include local papule formation with clumping of hair into tufts over the loin and withers
- ☒ **Persistence of the fever is a characteristic of MCF**, even cases that persist for several weeks

Alimentary tract form

- ☒ The symptom resembles the head and eye form except marked diarrhoea and conjunctivitis

Mild form

- ☒ Transient fever and minor erosions appear on the oral and nasal mucosa. **The distinct clinical feature in chronic MCF is persistent bilateral ocular leukomata**

Necropsy finding

- ☒ Liver is swollen
- ☒ Haemorrhage in urinary bladder
- ☒ All lymph nodes are swollen and edematous and haemorrhagic
- ☒ Histopathology- **necrotizing vasculitis** with infiltration of lymphoblast like cells and macrophage in tunica media and tunica adventia is a pathognomic lesion of MCF

Sample collection

- ☒ Buffy coat, Lymph node, spleen and lung

Diagnosis

- ☒ Based on clinical symptoms
- ☒ Based on necropsy finding
- ☒ Isolation of virus is difficult
- ☒ Identification of viral nucleic acid by PCR

- ☒ Competitive inhibition ELISA used for detection of antibodies in serum
- ☒ Nested PCR is commonly used for diagnosis

Differential diagnosis

- ☒ Mucosal disease
- ☒ IBRT
- ☒ Rinderpest
- ☒ Sporadic bovine encephalomyelitis
- ☒ Theileriosis

Treatment

- ☒ There is no specific treatment; the anti-inflammatory drugs may ease the discomfort

Control

- ☒ Housing of sheep and cattle in separate shed
- ☒ Avoid mixed grazing of cattle with sheep
- ☒ No vaccine is currently available
- ☒ Avoid grazing of cattle with wildebeest

Infectious bovine rhinotracheitis (IBRT)

Syn: Red nose, Infectious pustular vulvo vaginitis, Infectious pustular balanoposthitis

Etiology

- ☒ Bovine herpes virus -1 (BHV-1) cause' s respiratory disease, conjunctivitis, abortion, encephalitis and other clinical form of the disease complex. Genetic analysis of various clinical isolates has found three subtypes
- ☒ BHV1.1-Respiratory subtype
- ☒ BHV 1.2- Genital subtype
- ☒ BHV1.3 -Encephalitic subtype, BHV1.3 is renamed as BHV1.5
- ☒ Rarely genital and respiratory form disease occurs together

Epidemiology

- ☒ **Occurrence**-The disease was recognized in Europe, Asia, North America, Africa and Newzealand
- ☒ Morbidity and mortality rate are higher in feedlot cattle than dairy cattle
- ☒ Morbidity and mortality rate in dairy cattle- 8% and 3% respectively
- ☒ Morbidity and mortality rate in feedlot cattle-20-30% and 1% respectively
- ☒ Case fatality rate in systemic form of infection in new born calves is almost 100%

Host affected

- ☒ All breeds and all age group of cattle are susceptible

Source of infection

- ☒ Nasal exudates, coughed up droplets, genital secretion, semen, foetal fluids and tissue of infected animals.

Transmission

- ☒ Respiratory disease- through inhalation of aerosol
- ☒ Genital disease- through venereal transmission

Immune mechanisms

- ☒ Immunity to the virus is complex and consists of relationship between local and systemic antibody and cell mediated immunity
- ☒ Colostral immunity in calves present upto 4-6 months of age

Pathogenesis

1. Respiratory disease

Inhalation of virus- nasal cavity ☒ upper respiratory tract ☒ Rhinitis, laryngitis and tracheitis

2. Encephalitis

Inhalation of virus ☒ nasal mucosae ☒trigeminal peripheral nerve ☒trigeminal ganglion ☒ non suppurative encephalitis

3. Abortion

Inhalation of virus ☒local mulitplication in tonsil ☒ systemic invasion ☒ placenta ☒infection in last trimester of pregnancy ☒ Abortion, mummification and still birth

Clinical manifestation

- ☒ Incubation period- 3-7 days

Respiratory disease

- ☒ **Rhinitis, tracheitis and conjunctitis (Red nose)**
- ☒ Sudden onset of anorexia
- ☒ Loud coughing
- ☒ Fever (upto 42° C)
- ☒ **Severe hyperaemia of nasal mucosa** with numerous cluster of grayish necrotic lesion in mucous membrane of nasal septum
- ☒ Serous nasal and ocular discharge
- ☒ Increased salivation
- ☒ Sometimes slight hyperexcitability may be noticed
- ☒ Fall in milk yield
- ☒ Increased respiratory rate, but shallow
- ☒ Respiratory distress is evident on exercise
- ☒ Short explosive cough is characteristic of some outbreak

Dairy cattle

Many animals in the herd become affected within a few days; the disease is usually mild characterized by inappetence, coughing, profuse bilateral serous nasal discharge, excessive salivation, moderate fever, and moderate drop in milk production, corneal edema, conjunctivitis, profuse ocular discharge and recovery within few days.

Feedlot cattle

Illness is often more prolonged, febrile period is longer, nasal discharge becomes more profuse and purulent and convalescent period is longer. Some recovered animals may have a persistent snoring respiration.

Ocular form of IBR

Conjunctivitis is a common finding in typical ‘ red nose’ ; one or both eyes may be affected. **IBR lesions are confined to the conjunctiva** and there is no lesion in cornea except diffuse edema. The conjunctiva is reddened and edematous with profuse serous ocular discharge.

Systemic disease in newborn calves

- ☒ In new born calves under 10 days of age, systemic form of disease is severe and highly fatal
- ☒ Sudden onset of anorexia, fever, excessive salivation and rhinitis, often accompanied by **unilateral or bilateral conjunctivitis are common**
- ☒ **Hyperaemia of oral mucous membrane, erosion of soft palate covered by tenacious mucus**
- ☒ **Acute pharyngitis covered by tenacious mucopurulent exudates**
- ☒ respiratory distress
- ☒ **Bronchopneumonia**, loud breath sound, crackles and wheezes associated with consolidation of lungs may be present
- ☒ Diarrhoea and dehydration occur in some calves

Abortion

- ☒ **Abortion in last trimester is common sequel** and occurs some weeks after clinical illness or parenteral vaccination of non immune pregnant cows with MLV vaccine of bovine tissue culture origin. Retention of placenta often follows
- ☒ The endometritis, poor conception rate and short oestrus can occur after insemination with infected bull semen.

Infectious pustular vulvovaginitis

- ☒ **Frequent urination and tail elevation**
- ☒ The vulva is swollen, papules, erosions and ulcers are present on the mucosal surface
- ☒ Mucosal ulcers may coalesce and sloughing of brown tissue may occur

- ☒ **Mucoid or mucopurulent exudates in vagina**
- ☒ Recovery usually occurs in 10-14 days

Balanoposthitis

- ☒ Papules, erosion and ulcer present on the glans penis and preputial mucosa

Necropsy finding

- ☒ Acute inflammation of the larynx, trachea and bronchi
- ☒ Profuse fibrino-purulent exudate in the upper respiratory tract in severe cases
- ☒ Chronic ulcerative gastro-enteritis in feedlot cattle
- ☒ Lung emphysema
- ☒ Secondary bronchopneumonia

Diagnosis

- ☒ Based on clinical signs
- ☒ Isolation of virus from nasal swab
- ☒ **PCR assay is a practical alternative for rapid detection of virus in clinical sample**
- ☒ Detection of antibody by ELISA, VNT and indirect immunofluorescence test
- ☒ IgM – ELISA test is useful for diagnosis of recent infection with BHV-1
- ☒ Micro ELISA is being used for control program of BHV-1 infection in Switzerland

Sample collection

- ☒ Live animal-Sterile nasal swab and semen, paired sera sample and aborted foetal contents
- ☒ Dead animal- lung

Differential diagnosis

- ☒ Pneumonic pasteurellosis
- ☒ Bovine viral diarrhoea
- ☒ Malignant catarrhal fever
- ☒ Calf diphtheria
- ☒ Viral pneumonia of calves
- ☒ Allergic rhinitis
- ☒ Systemic form of IBR in new born calves should be differentiated with pneumonia, septicaemia and toxemia

Treatment

- ☒ Broad spectrum antimicrobials are indicated if secondary bacterial tracheitis and pneumonia are present
- ☒ Isolation of infected animals and monitored frequently for evidence of secondary bacterial complication and then treated accordingly

Control

- ☒ Natural exposure

- ☒ Vaccination
- ☒ Eradication

Natural exposure

Cattle that have recovered from a natural infection with a virus are immune to further clinical disease. However natural exposure is risky because, not all animals in the herd will become infected and become immune.

Vaccination

Vaccines available for IBR are

- ☒ Modified live virus vaccine (MLV)
 - a. Parenteral vaccine – abortogenic in non immune cows
 - b. **Intranasal vaccine**- non abortogenic to pregnant animal, which is used during an outbreak
- ☒ Temperature sensitive BHV-1 MLV (virus multiplication restricted to the upper respiratory tract)- vaccine is efficacious and safe for use in pregnant animals
- ☒ **Inactivated vaccine**- vaccine do not cause abortion, mummification and safe for use in pregnant animals
- ☒ Subunit vaccine -contain major surface glycoprotein of BHV-1-stimulate protective immunity
- ☒ **Marker vaccine**-based on deletion mutants of one or more microbial antigen, which is used for differentiation of vaccinated from infected animals
- ☒ Combination or multivalent vaccine- for BHV-1, BRSV, PI-3 and BVDV

Eradication

- ☒ Annual serological testing of the national herd
- ☒ Restriction of trade with seropositive animals
- ☒ First priority of eradication was given to farms with breeding animals
- ☒ Stepwise elimination of seropositive animals

Bovine Viral Diarrhoea

Syn: Mucosal disease

Bovine viral diarrhoea is most common disease in young cattle, characterized by fever, inappetence and diarrhoea.

Etiology

Bovine viral diarrhoea virus (BVDV) is a SS-RNA virus, genus Pestivirus in the family Flaviviridae and is closely related to classical swine fever and Ovine Border disease viruses.

There are two biotypes designated as non-cytopathic (NCP) and cytopathic (CP) depending on their effect on tissue culture cells. **The non cytopathic type is most common and most important.** It cross the placenta, invade the fetus and establishes persistent infection in the fetus. The superinfection of persistent infected animal with cytopathic biotype virus causes mucosal disease. It occurs usually at 6-24 months or older.

Epidemiology

- ☒ The disease have been recorded in most of the countries
- ☒ Prevalence o infection is high, but the incidence of clinical mucosal disease is low
- ☒ Incidence of mucosal disease in herd is usually <5% upto 2 yrs of age
- ☒ Morbidity rate: 40%
- ☒ Mortality rate: 20 %

Source of infection

- ☒ **Persistently infected immunotolerant seronegative animals**
- ☒ The persistent infection develops when infection of foetus with a non-cytopathic strain occurs **before 120 days of gestation**
- ☒ These animals shed the virus in all secretion and excretions (nasal discharge, saliva, semen, faeces, urine, tears and milk).

Excretion of virus in the semen from both persistently and transiently infected bulls

Transmission

- ☒ Direct contact with an infected animal
- ☒ Nose to nose contact is effective method for transmission of virus from PI animal to susceptible animal
- ☒ Transplacental (in utero) transmission from dam to foetus
- ☒ Infection may be transmitted through natural service or artificial insemination
- ☒ Embryo transfer from persistently infected animals
- ☒ Inoculation of pregnant animal with live virus vaccine, their calves may develop disease
- ☒ Indirect transmission rarely occur through, farm workers, equipment and biting flies

Pathogenesis

It depends on multiple interactive factors. Host factors which influence the clinical outcome of BVDV infection include

- ☒ Whether the host is immunocompetent or immunotolerant to the virus
- ☒ Age of the animal
- ☒ Immune status
- ☒ Presence of stressors
- ☒ Transplacental infection and gestational age of the fetus
- ☒ Infection at 0-45 days of pregnancy causes early embryonic mortality

- ☒ Infection at **45-125 days of pregnancy results in persistent infection**, the foetus become **immunotolerant to the virus**
- ☒ Infection at 125-175 days of pregnancy results in numerous congenital defects (cerebellar hypoplasia, optic neuritis, ocular defects, retinal atrophy, cataract and microphthalmia). During this period organogenesis of nervous system and development of immune system takes place, which results in generation of inflammatory reaction to the virus. which is responsible for congenital defect in foetus
- ☒ **Infection at 180 days of gestation to term, the foetus become immuocompetent**, eliminates the virus. At birth foetus has antibody to the virus, but virus free

Clinical manifestation

Inapparent or subclinical infection

- ☒ The most frequent form of infection in cattle with high morbidity and low case fatality rate
- ☒ This occur in immunocompetent seronegative cattle
- ☒ Mild fever, leukopenia, inappetence and mild diarrhoea followed by rapid recovery

Peracute bovine viral diarrhoea

- ☒ Severe form of enteric infection in immunocompetent seronegative animals
- ☒ Fever, respiratory distress, severe depression, anorexia, profuse watery diarrhoea, dysentery, conjunctivitis and agalactia
- ☒ Abortion in late gestation
- ☒ Morbidity rate: 40%, Mortality rate : 25%

Acute mucosal disease

- ☒ It occur in persistently infected calves at 6-24 month age group
- ☒ Morbidity is low and case fatality rate is high (90%)
- ☒ Fever, tachycardia and polypnoea and lacrimation
- ☒ Affected animals are depressed, anorexic, drool saliva, wetting hair around the mouth
- ☒ Discrete shallow erosions on inside the lips, gum, dental pad, hard palate, commissures of the mouth and tongue, coronitis ad lameness
- ☒ Cooked up appearance of entire oral cavity
- ☒ Profuse watery diarrhoea, the faeces is foul smelling which contain mucus or blood
- ☒ Straining during defecation
- ☒ Dehydration
- ☒ Death occurs after 5-7days of illness

Thrombocytopenia and haemorrhagic disease

- ☒ Bloody diarrhoea, petechial and ecchymotic haemorrhage

- ☒ Bleeding at venupuncture site
- ☒ Epistaxis
- ☒ Platelet count < 25000

Chronic mucosal disease

- ☒ The animal become chronically ill
- ☒ There may be intermittent bouts of diarrhoea, inappetence, progressive emaciation, chronic bloat, rough hair coat, hoof deformities, chronic erosions on entire oral cavity and skin

Unthrifty PI animals

- ☒ Calves are smaller body size
- ☒ Curly hair coat
- ☒ Stunted growth
- ☒ High incidence of respiratory disease

Reproductive failure and neonatal disease

- ☒ Conception failure, increased embryonic mortality, fetal mummification, abortion, premature birth, congenital defect, birth of stunted weak calves

Sample collection

- ☒ Live animal-Whole blood, nasal swab, paired sera sample and semen
- ☒ The whole blood should be kept room temperature for 5 days for virus isolation before processing
- ☒ Dead animal-intestinal tissue, mesenteric lymph node, spleen, lymph node, lungs and liver

Diagnosis

Based on clinical signs- oral lesions and diarrhoea

Leukopenia

Identification of the viral agent from persistently viraemic healthy animals resulting from congenital infection can be readily identified by isolation of non-cytopathic virus in cell cultures from blood or serum

It is necessary to use an immune-labeling method to detect the growth of virus in the cultures

Persistence of virus should be confirmed by resampling after an interval of at least 3 weeks

These animals will usually have no or low levels of antibodies to BVDV

Indirect immunoperoxidase staining technique is used to detect BVDV antigen in semen

Identification of antigen in tissue sample-Immunohistochemical methods-immunofluorescence, immunoenzyme staining section in frozen tissue

Monoclonal antigen capture ELISA

Nucleic acid based detection methods

Nucleic acid hybridization probe and RT-PCR

Serum neutralization test and ELISA are used detect antibodies in serum

Differential diagnosis

- ☒ Clinical diagnosis is a complex problem. Signs of the different syndromes may appear alone or in combination
- ☒ Although many of the signs are similar to RP, the mortality in that disease is much higher
- ☒ The oral lesions are very important, but their absence does not exclude BVD
- ☒ This disease must be differentiated from conditions in cattle causing diarrhoea, erosions/ulcerations of the gastrointestinal tract, reproductive failure, teratology, skin disease, laminitis, poor growth and respiratory tract disease such as vesicular diseases, ingestion of caustic substances, mucosal disease complex: **Foot and mouth disease, Rinderpest, bluetongue, papular stomatitis and malignant catarrhal fever.**

Prevention

Vaccination

- ☒ Modified Live BVD vaccine - Not recommended for pregnant cattle or in stressed populations
- ☒ Killed BVD vaccine, which is safer for administration even to pregnant cattle. A booster dose after 3 to 4 weeks of initial vaccination
- ☒ Calves should be vaccinated at **4-6 months of age** and again when they are 8-12 months old
- ☒ Calves less than three months of age generally not vaccinated because of potential vaccination failure associated with the presence of colostral immunity
- ☒ Heifers and cows should be **revaccinated 30-60 days before breeding and booster dose one month before calving or in the last trimester.**

Control

- ☒ **Primary goal of a control strategy is removal of persistently infected animals** from a herd
- ☒ All cattle should be screened for the virus including calves born in the herd for nine months after initiation the programme
- ☒ Newly introduced animals should be tested for the disease before their introduction. Quarantine the newly purchased animals for 30 days
- ☒ Implementation of Strict managemental practices
- ☒ Cattle should not be allowed to mix with sheep and goats

- ☒ Periodical testing of bulls used for artificial insemination

Bovine Ephemeral Fever

Syn: Three day sickness

Ephemeral fever is an insect borne non contagious viral disease of cattle, characterized by high fever, salivation and lameness.

Etiology

- ☒ Ephemeral fever virus of genus Ephemerovirus in the family Rhabdoviridae
- ☒ It is a single stranded negative sense RNA virus. There are number of strains that vary antigenically.
- ☒ **The BEF virus is closely associated with the leukocyte-platelet fraction of blood**
- ☒ The virus is inactivated at pH <5.0 and >10.0

Epidemiology

- ☒ Ephemeral fever is enzootic in Africa, Asia, Middle East, East Indies and Australia
- ☒ Morbidity rate: 100% in highly susceptible population under favourable environmental condition
- ☒ Most cases are seen in the summer and early fall
- ☒ Outbreaks often follows period of rainfall
- ☒ In an enzootic area morbidity rate 5-10% and mortality rate 1%

Host affected

- ☒ Cattle and wild ruminants are highly susceptible
- ☒ All age group of cattle are susceptible and more common in animals less than 2yrs of age. Calves less than 6 months old is not affected by the natural disease.

Transmission

- ☒ The disease is transmitted by mosquitoes during blood sucking
- ☒ The disease can also be spread by intravenous inoculation of small amounts of blood.

Pathogenesis

Virus multiplication primarily occur in vascular system. There is generalized inflammation with vasculitis and thrombosis, serofibrinous inflammation in serous and synovial cavities leads to increased endothelial permeability.

Clinical manifestation

- ☒ Incubation period -2-4 days, sometimes as long as 10 days
- ☒ **Sudden onset of fever (40.5-41° C)**
- ☒ Anorexia
- ☒ Sudden fall in milk yield
- ☒ Serous nasal and ocular discharge

- ☒ Lacrimation
- ☒ **Shivering**
- ☒ Increased heart and respiratory rate
- ☒ **Muscle stiffness, weakness in one or more limbs and lameness occur on 2nd day**
- ☒ On third day of illness animal begins eating and ruminating and febrile reaction disappears, but lameness and weakness may persist for 2-3 more days
- ☒ **Hypocalcaemia**

Necropsy finding

- ☒ Serofibrinous polyserositis
- ☒ Pulmonary emphysema

Diagnosis

- ☒ Based on clinical signs
- ☒ Based on necropsy findings
- ☒ Leukocytosis
- ☒ Elevated level of plasma fibrinogen level for about 7 days
- ☒ Increased level of creatine kinase
- ☒ Hypocalcemia
- ☒ Serological test – blocking ELISA, SNT, CFT, FAT and AGID are used to identify antibodies

Differential diagnosis

- ☒ Botulism
- ☒ **Parturient paresis**
- ☒ Pneumonia
- ☒ Traumatic reticulitis

Treatment

- ☒ Non steroidal anti-inflammatory drugs phenylbutazone 8mg/ kg at 8 hours interval or ketoprofen 3mg/ kg b.wt once a day for three days
- ☒ Calcium borogluconate should be given for cow shows hypocalcemia
- ☒ Fluid therapy for dehydration
- ☒ Provide complete rest
- ☒ Recovered animal should not be stressed

Control

- ☒ Vaccination
- ☒ Live attenuated vaccine appear to be effective and should be used only in endemic area
- ☒ Formalin inactivated killed vaccine is not effective

Enzootic bovine leukosis

Syn: Bovine lymphosarcoma, Leukemia, Malignant lymphoma

Enzootic bovine leukosis (EBL) is a lymphoproliferative disease of cattle. It occurs in all breeds of both sexes.

Etiology

It is caused by bovine leukemia virus (BLV)-Exogenous C-type oncogenic retrovirus of retroviridae. The **virus infects a subpopulation of B lymphocytes**, and proviral DNA is integrated to a number of sites on the cellular DNA.

Transmission

- ☒ **Iatrogenic transmission occurs through inoculation of small amounts of infected blood** (e.g. infected needles, tattooing, dehorning and rectal palpation)
- ☒ Vertical transmission from the dam to the calf (3 – 20 % of calves may become infected) and by colostrum or milk (less than 2 %)
- ☒ Insect transmission is also a possibility; higher rates of infection were reported in the summer

Pathogenesis

Bovine leukosis is observed in two forms:

- a) Sporadic form – rare and occur in cattle below 3 yrs of age
- b) Enzootic form- most common and occur in cattle over 3yrs of age**

Infections are usually subclinical; only 30– 70% of infected cattle develop persistent lymphocytosis, and 0.1– 10% of the infected animals develop tumours. Signs will depend on the site of the tumours and may include digestive disturbances, inappetence, weight loss, weakness or general debility and sometimes neurological manifestation.

Clinical manifestation

- ☒ Weight loss
- ☒ Laboured breathing due to heart involvement
- ☒ Persistent diarrhoea following infiltration of the abomasum wall by neoplastic cells
- ☒ Marked enlargement of several superficial lymph nodes
- ☒ Edema of the brisket and the intermandibular region
- ☒ Paralysis of the hind legs due to tumour compression of the spinal cord
- ☒ Protrusion of the eye as a result of tumour invasion of the orbital cavity
- ☒ Debilitation or emaciation
- ☒ Pale mucosal surface
- ☒ Bloated animal

- ☒ Swelling of the neck when thymus is involved
- ☒ Cutaneous nodules in the terminal stage

Postmortem findings:

- ☒ Lymph node enlargement (clay-like consistency)
- ☒ Enlargement of spleen (splenomegaly)
- ☒ Thin watery blood
- ☒ Neoplastic lesions in the heart and intestines
- ☒ Enlarged haemolymph nodes

Diagnosis

- ☒ Based on clinical symptoms and lesions
- ☒ Virus can be detected in the culture supernatant following in-vitro culture of peripheral blood mononuclear cells (PBMC) from infected animals
- ☒ Proviral DNA can also be detected in PBMC or tumours of infected animals by PCR
- ☒ Detection of antibody by AGID in serum
- ☒ Detection of antibody by ELISA in serum or milk of infected animals

Differential diagnosis

- ☒ Lymphadenitis due to tuberculosis and actinobacillosis
- ☒ Hyperplastic haemolymph nodes
- ☒ **Congestive heart failure due to traumatic pericarditis**
- ☒ Enlarged spleen in septicaemic disease
- ☒ Compression of spinal cord by tumour is differentiated with dumb form of rabies

Treatment

- ☒ Currently is not available

Prevention and control

- ☒ Vaccine are not available for BLV
- ☒ Test and slaughter of seropositive animals in the herd
- ☒ Screening of antibodies in milk by ELISA
- ☒ Use single disposal gloves and needle to avoid transmission of disease through blood
- ☒ Cleaning and disinfection of dehorning and tattoo equipment and surgical instruments after use

Peste des petits ruminants (PPR)

Syn: Goat plague, Pneumoenteritis, KATA

It is an acute or subacute contagious viral disease of goats and sheep characterized by **fever, necrotic stomatitis, gastroenteritis, bronchopneumonia** and death

Etiology

- ☒ It is caused by peste des petites ruminant disease virus , genus morbillivirus belongs to family paramyxoviridae
- ☒ It is a negative sense single stranded enveloped RNA virus. The envelope contain two type of glycoprotein namely attachment protein and fusion protein. The attachment protein may be either a haemagglutinin and neurominidase protein or a protein without neuraminidase activity. The attachment proteins allow the virus to bind the cell surface receptor and the fusion protein causes the envelope to fuse with host cell membrane for replication.
- ☒ The virus is labile in the environment and susceptible to lipid solvents, non-ionic detergent and disinfectants

Epidemiology

- ☒ Distribution: Worldwide
- ☒ The disease is endemic in Indian subcontinent
- ☒ The case fatality rate is higher in goats (55-85%) than in sheep (<10%)
- ☒ Kids over 4 months and under 1 yr of age group are highly susceptible

Transmission

- ☒ **Close contact with infected animals**
- ☒ **Inhalation of fine infective droplets** from an infected animal through coughing and sneezing
- ☒ Ingestion of contaminated feed and water by using of common feed and water trough

Pathogenesis: similar to rinderpest in cattle

Clinical manifestation

- ☒ It occur as acute and subacute form
- ☒ Incubation period is 4-5 days

Acute form

- ☒ Fever (104-106° F) is accompanied by dullness, sneezing and serous discharge from eyes and nostrils, later it becomes mucopurulent
- ☒ Congested mucous membrane
- ☒ Some animals shows **profuse catarrhal conjunctivitis with matting of eyelids**
- ☒ **Discrete necrotic lesions develop in the mouth** and extend over the entire oral mucosa, forming **diphtheritic plaques (necrotic stomatitis)**
- ☒ **Diarrhoea** is profuse, mucoid or blood tinged

- ☒ **Bronchopneumonia** is characterized by coughing may develop at later stage of infection
- ☒ **Abortion** may occur in pregnant animal
- ☒ A common feature in later stages of the disease is the **formation of small nodular lesions in the skin on the outside of the lips around the muzzle (Orf like lesion)**

Subacute form

- ☒ **It is more common in sheep**, but they also occur in goats. The symptoms and lesions are less marked

Necropsy findings

- ☒ **Lungs were dark red or purple colour** mainly in the anterior and cardiac lobes and firm to touch
- ☒ Hemorrhagic ulceration is marked in the ileocaecal region, colon and rectum where they produce typical zebra stripes
- ☒ Intracytoplasmic eosinophilic inclusion bodies in intestinal crypt epithelium
- ☒ Intracytoplasmic and intranuclear inclusion bodies in large syncytial giant cells

Sample collection

- ☒ Live animal-Whole blood during febrile stage, oculonasal swab, diphtheritic lesion, faecal sample and paired sera sample
- ☒ Dead animal- Lung, intestine, mesenteric lymph node

Diagnosis

- ☒ Based on clinical signs and necropsy findings
- ☒ Isolation and identification of virus in cell culture
- ☒ Detection of antibody in serum by CIE, SNT and ELISA
- ☒ Detection of antigen by CIE and RT-PCR

Differential diagnosis

- ☒ Pneumonic pasteurellosis
- ☒ Contagious caprine pleuropneumonia
- ☒ Coccidiosis
- ☒ Contagious ecthyma
- ☒ Nairobi sheep disease

Treatment

- ☒ Antibiotics to control secondary bacterial infections
- ☒ Replacement fluid therapy for dehydration
- ☒ Nasal and mouth lesions are cleaned with mild antiseptics

Prevention and Control

- ☒ **Vaccination of kids and lambs over 4 months of age with live attenuated PPR vaccine** and annual revaccination every year

- ☒ **Ring vaccination** of kids and lambs over 4 months of age around 8 km radius from the infected premises to prevent further spread of infection, apart from this Prophylactic measures can be adopted in high risk population.
- ☒ The vaccine is safe, potent and acceptable for use even during pregnancy
- ☒ Segregation of recently purchased animals from unknown sources or sheep and goats that had been sent to shandy but returned unsold
- ☒ Restriction of shared grazing with nomadic or migratory flocks
- ☒ Cleaning and disinfection with 2% sodium hypochlorite

Sheep and goat pox

Sheep pox and goat pox are viral diseases of sheep and goats characterized by fever, generalized papules or nodules, vesicles (rarely) and internal lesions (particularly in the lungs).

Etiology

Sheep pox is caused by sheep pox virus and goat pox is caused by goat pox virus, these viruses are highly host specific, Kenya sheep and goat virus infect both sheep and goat equally all are member of genus **capripox virus**, belongs to family pox viridae.

Epidemiology

The disease is endemic in Africa northern part of the Equator, Middle East, Turkey, Iran, Iraq, Afghanistan, Pakistan, India, Nepal, parts of the Peoples in Republic of China and Bangladesh.

The Morbidity and mortality vary with the breed of the host and the strain of the virus. Mortality may be up to 50% in a fully susceptible flock, and as high as 100% in young animals. Imported breeds such as merinos highly susceptible than native breeds.

Mild infections are common in indigenous breeds; however, symptoms may be more severe in kids or lambs, stressed animals, animals that have concurrent infections, or animals from areas where pox has not occurred for some time.

Transmission

- ☒ **Direct contact with an infected animals** and contaminated equipment
- ☒ Inhalation of aerosols containing virus
- ☒ Stable flies (*Stomoxys calcitrans*) may transmit the viruses mechanically

Clinical manifestation of of sheep pox

- ☒ Incubation period is 12- 14 days
- ☒ The disease occur in two forms namely; malignant and benign form
- ☒ **Malignant form is common in young animals**

- ☒ The symptoms include high fever, marked depression; prostration and serous or mucopurulent discharge from eye and nose, affected lamb may die during this stage development of typical pox lesions
- ☒ Some animals shows **conjunctivitis, rhinitis, lymphadenopathy, anorexia** and depression, the initial rise in temperature is followed by development of macules (small circumscribed areas of hyperaemia), which are most obvious on unpigmented skin and then papules (hard swellings 0.5 and 1 cm in diameter), which may cover the entire body or **may be restricted to the groin, axilla and perineum**
- ☒ Papules on the mucous membranes of the eyes and nose ulcerate, so the discharge becomes mucopurulent, the mucosa of the mouth, anus and prepuce or vagina becomes necrotic
- ☒ **Breathing may become laboured and noisy** due to pressure on the upper respiratory tract from the **swollen retropharyngeal lymph nodes**, due to the developing lung lesions
- ☒ Secondary pneumonia is common and abortion is rare
- ☒ The benign form is common in adult, only skin lesion occur, particularly under the base of the tail and ear region. There is no systemic reaction
- ☒ Goat pox in sheep is more severe than sheep pox, the lesions occur on the lips, oral mucosa, teat and udder

Clinical manifestation of goat pox

- ☒ Symptoms are similar to sheep pox
- ☒ Young kids suffer from systemic disease, the pox lesions spread generally all over the skin, alimentary and respiratory mucosa
- ☒ Adult goat may have a systemic disease and extensive lesion, but symptoms are usually mild

Necropsy finding

- ☒ The lungs often contain discrete congested or edematous lesions or hard white nodules. Nodules in the lungs can be up to 5cm in diameter
- ☒ Pale foci are sometimes present on the surface of the kidney and liver

Sample collection

- ☒ Live animals- whole blood during peak temperature, scab 10% glycerol saline and paired sera sample
- ☒ Dead animal- skin lesion, lung lesion and lymph node

Diagnosis

- ☒ Based on clinical symptoms and lesion
- ☒ Virus isolation in tissue culture of ovine or caprine origin or lamb kidney cells

- ☒ Detection of viral antigen by PCR, AGID and FAT
- ☒ Detection of antigen by ELISA, the samples to be collected within first week of occurrence of clinical signs
- ☒ Detection of antibody by **virus neutralization test**, AGID, IFAT, Western blot analysis and ELISA

Differential diagnosis

- ☒ Contagious ecthyma
- ☒ Blue tongue

Treatment

Administration of antibiotics to control secondary infection and good nursing care are recommended. Infection results in good immunity.

Prevention

- ☒ Vaccination is the best method to prevent the disease in susceptible animals
- ☒ The **live attenuated sheep pox vaccine is administered above 3 months of age** in healthy sheep, and revaccination is done every year in endemic area

Control

- ☒ Elimination of infected and exposed animals by slaughter
- ☒ Proper disposal of cadaver and contaminated product by burning or deep burial
- ☒ Cleaning and disinfection of contaminated premises and equipment with **2% sodium hypochlorite** or 2% phenol
- ☒ If the disease spread extensively in endemic area mass vaccination with restriction of animal movement is advisable
- ☒ Quarantine the infected area is required to prevent the spread of the disease
- ☒ **Isolation of infected herds and sick animals for at least 45 days after recovery** is recommended

Rift valley fever

RVF is a vector-borne disease of sheep, cattle and goat is characterized by **high rate of abortion and neonatal mortality particularly in sheep**, goat and cattle considerably.

Etiology

It is caused by Rift valley fever virus, genus Phlebovirus of the family Bunyaviridae a negative-sense, single-stranded RNA virus, contain one serotype. The virus is inactivated at pH <6.8, highly susceptible to low concentration of formalin and strong solution of sodium or calcium hypochlorite.

Epidemiology

Rift valley fever was initially reported in rift valley in Kenya; now exist in sub-Saharan Africa, Egypt and Madagascar. Rift valley fever like disease is reported in sheep in India. The disease is usually epidemic following **heavy rains and flooding**.

- ☒ Mortality rate in New born lambs: 70-100%, Adult sheep: 10-30% and Cattle: 10%

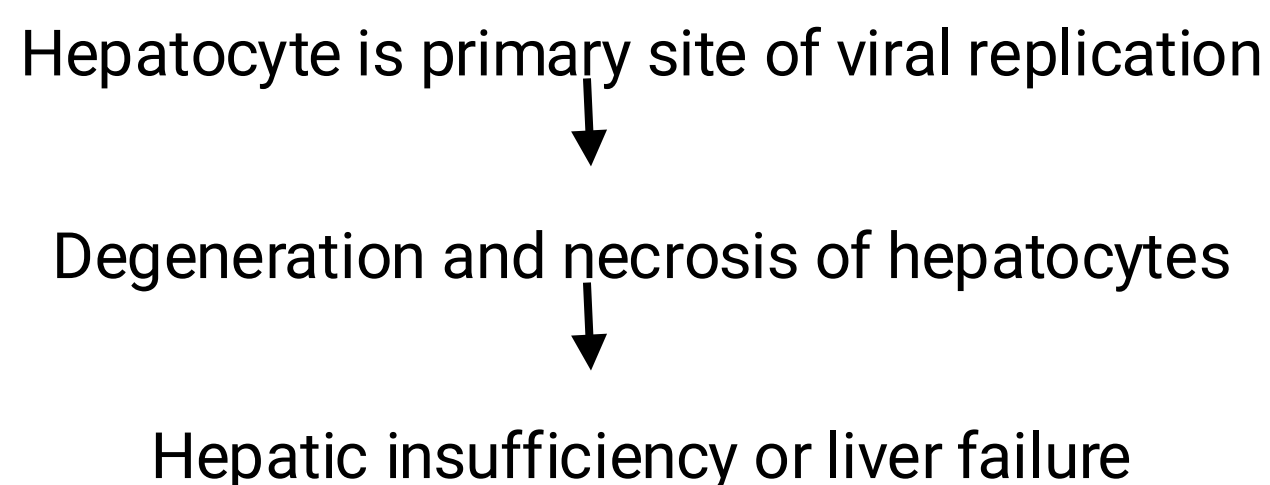
Host affected

- ☒ Sheep, cattle, goats, wild ruminants like antelope and wilde beest
- ☒ Humans are very susceptible (major zoonosis)

Transmission

- ☒ It is **transmitted by mosquitoes sand flies and culicoides**
- ☒ Direct contact with an infected animals and meat

Pathogenesis



Clinical manifestation

Neonates

- ☒ Incubation period: 12-36 hrs in lambs
- ☒ Sudden onset of high fever (biphasic fever temp 41°C)
- ☒ **Reluctant to move or feed, abdominal pain**
- ☒ Inco-ordination followed by collapse and death

Adult sheep and cattle

- ☒ Abortion- 85 % in cattle, and 100% in sheep
- ☒ Sudden death preceded by high fever, sick animal may regurgitate and develop **bad smelling bloody diarrhoea and icterus, which is common in cattle**
- ☒ Serosanguineous or **blood stained mucopurulent nasal discharge**
- ☒ Increased respiratory rate may also be seen

Necropsy findings

- ☒ Focal or generalised hepatic necrosis (white necrotic foci of about 1 mm in diameter)
- ☒ Brown-yellowish colour of liver in aborted foetuses

Sample collection

- ☒ Whole blood, paired sera sample and aborted foetus from live animals
- ☒ Liver, spleen, kidney, lymph node, heart blood and brain from dead animals

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Severe leukopenia
- ☒ Agar gel immunodiffusion test
- ☒ Viral antigen could be detected by RT-PCR
- ☒ Antibodies in serum by AGID,VNT and ELISA

Differential diagnosis

- ☒ Brucellosis
- ☒ Vibriosis
- ☒ Trichomoniasis
- ☒ Ovine enzootic abortion
- ☒ Blue tongue
- ☒ Enterotoxaemia
- ☒ Anthrax

Prevention and control

- ☒ Control of vector
- ☒ Movement of stock to high lying area
- ☒ Vaccination of lambs with live attenuated vaccine at 6 months of age
- ☒ Vaccination is normally not to be used for pregnant animals, because it may cause abortion, congenital defect or hydrops amnii in the ewe

Bluetongue

It is infectious non contagious arthropod borne viral disease of sheep, cattle, goat and other wild ruminants, clinically characterized by **fever, oculonasal discharge, hyperaemia and edema of lips and lameness.**

Etiology

It is caused by Blue tongue virus, genus Orbivirus belongs to family Reoviridae. The virus contains 25 serotypes and has considerable heterogeneity when ruminant or vector infected with more than one strain reassortment of gene segment may occur. Infection with one serotype may not be cross protective against others. The virus is inactivated at pH < 6.0 and susceptible to 3% sodium hydroxide solution and organic iodides.

- ☒ The prevalence of serotype in Asia is serotype 1,4,7,9,10,12,16,17,20,21 and 23
- ☒ The prevalence of serotype in India serotype 1, 2, 3, 4, 6, 9, 10, 12, 15, 16, 17, 18, 21 and 23

Epidemiology

- ☒ Blue tongue virus infects domestic livestock population in all tropical and subtropical countries and occurs in continents of Asia, Africa, North America, South America and Australia
- ☒ The distribution and intensity of infection is determined by climate, geography and altitude, which influence the occurrence and activity of culicoides vector
- ☒ Morbidity rate- 50-75% and Mortality rate upto 30%

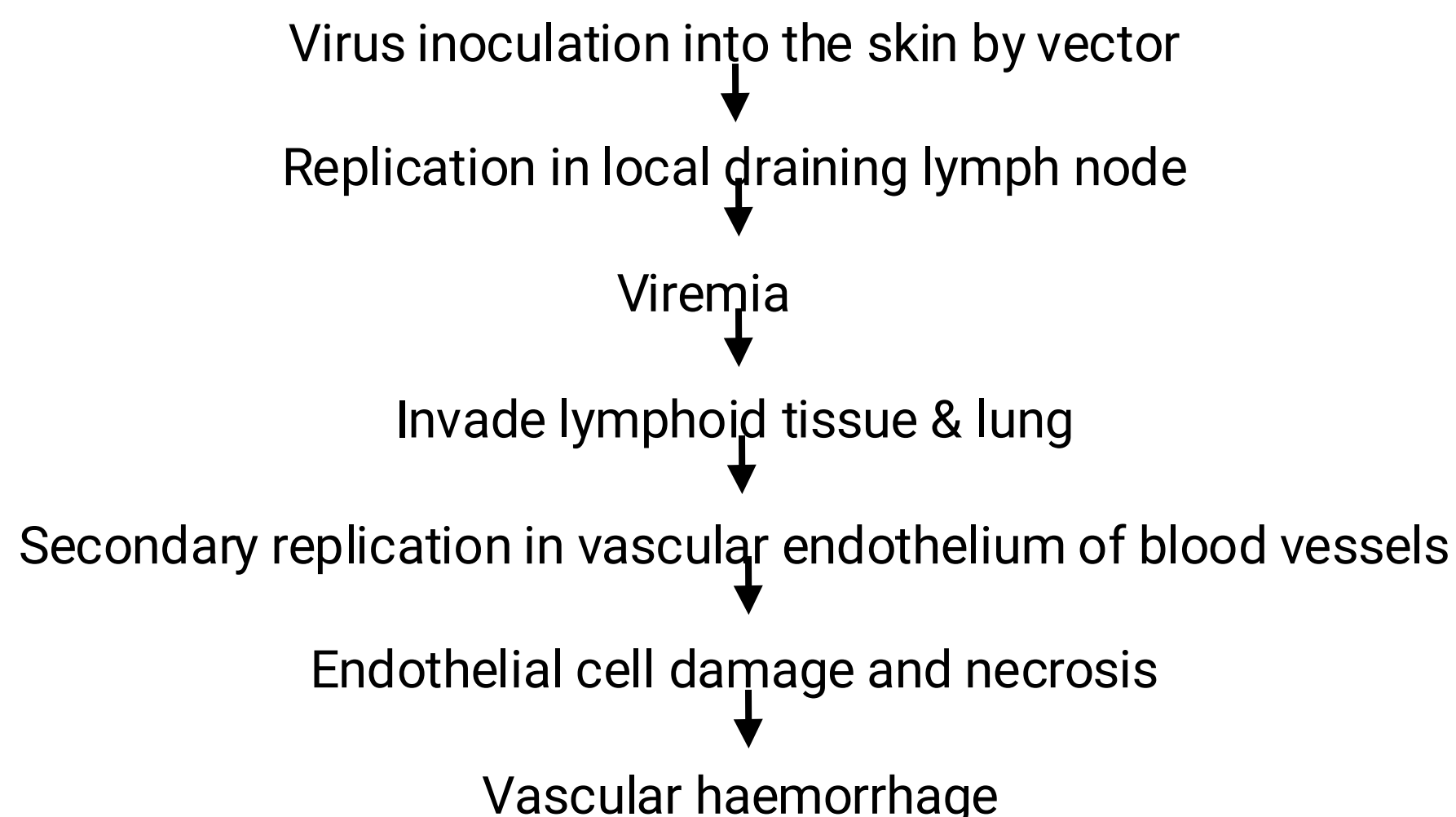
Host affected

- ☒ Bluetongue virus can infect many domesticated and wild ruminants including sheep, goats, cattle, water buffalo, African buffalo, bison and various cervids
- ☒ Cattle are major reservoir host for sheep
- ☒ All breeds and all age group of sheep are susceptible, but Merino and British breeds are more susceptible than others

Transmission

- ☒ It is mainly transmitted by biting midges in the genus Culicoides, which are biological vectors
- ☒ Mechanical transmission occur through sheep keds (*Melophagus ovinus*), cattle lice, ticks and other mosquitoes
- ☒ Bluetongue virus can also spread mechanically on surgical equipment and needles
- ☒ Venereal transmission through semen of infected bull

Pathogenesis



Clinical manifestation

- ☒ Incubation period is upto 10 days
- ☒ **Fever (40.5-41°C) persist for 5-6 days**
- ☒ Nasal discharge, salivation with reddening of buccal and nasal mucosa noticed after 48 hrs of rise of temperature

- ☒ Nasal discharge is mucopurulent and often blood stained
- ☒ **Swelling and edema of the lips, gums, dental pad and tongue**
- ☒ Facial swelling and drooping of ears
- ☒ Conjunctivitis and lacrimation
- ☒ Excoriation of buccal mucosa follows blood stained saliva and offensive odour in the mouth
- ☒ **Hyperaemia and ulceration are common at the commissures of the lips, buccal papillae, around the anus and vulva**
- ☒ Increased respiratory rate upto 100/min.
- ☒ Diarrhoea and dysentery is rarely noticed
- ☒ **Appearance of dark red to purple band in the skin just above the coronet** due to coronitis appear only in some animals
- ☒ **Lameness**
- ☒ **Wry neck** due to twisting of head and neck to one side due to direct action of virus on muscle tissue, which occur on 12th day of infection
- ☒ **Loss of wool**
- ☒ Long convalescence period
- ☒ **Cracking of hooves**

Cattle

- ☒ The symptoms are mostly inapparent
- ☒ Fever (104-106°F)
- ☒ Stiffness and laminitis in all four limbs
- ☒ Excessive salivation
- ☒ Edema of the lips, inappetence, nasal discharge and fetid breath
- ☒ Severe coronitis, sometime cause the sloughing of hooves
- ☒ Photodermatitis
- ☒ Congenital deformity in early pregnancy

Goats

- ☒ Mild to moderate fever
- ☒ Hyperaemia of mucosa and conjunctivitis

Necropsy finding

- ☒ **Distinct haemorrhage at the base of the pulmonary artery**
- ☒ **Focal necrosis of papillary muscles of left ventricle**
- ☒ Pulmonary edema, which may be accompanied by pleural and pericardial effusion is a common cause of death in fatal cases

Sample collection

- ☒ Whole blood and paired sera sample from live animals

- ☒ Spleen, lymph node and bone marrow from dead animals

Diagnosis

- ☒ Based on clinical symptoms and necropsy finding
- ☒ Isolation of virus in embryonated hens egg by intravenous route of inoculation at 15 days of embryonation
- ☒ Isolation and identification of virus in cell culture
- ☒ Reverse transcriptase-polymerase chain reaction (RT-PCR) tests are widely used to identify viral RNA in clinical samples, and can also identify the serotype
- ☒ AGID is used for detection of antibodies in serum
- ☒ Competitive ELISA is more sensitive and more specific than AGID for detection of antibodies in serum
- ☒ Marked leukopenia due to lymphopenia in severe cases
- ☒ Increased level of creatine kinase due to skeletal myopathy

Differential diagnosis

- ☒ Foot and mouth disease
- ☒ Epizootic haemorrhagic disease
- ☒ Contagious ecthyma and sheep pox

Treatment

- ☒ Cleaning of mouth lesions with mild antiseptic solution
- ☒ Fluid and electrolyte therapy
- ☒ Antibiotics to be administered to control secondary infection
- ☒ Protection of animals from hot sunlight

Prevention and control

- ☒ **Vaccination of Lambs and kids above 3 months of age**
- ☒ Live attenuated vaccines should not be used in pregnant ewes less than 8 wks of pregnancy, because risk of congenital deformity in lambs
- ☒ Currently **pentavalent inactivated blue tongue vaccine** (contain serotype 1, 2, 10, 16 & 23) is widely used and safe to use in pregnant animals
- ☒ Rams should be vaccinated before mating time
- ☒ Annual revaccination of animals one month before the expected occurrence of disease
- ☒ Reducing the risk of exposure by spraying of insecticide around the shed at night
- ☒ Avoid stabling or resting of animal in low lying damp pasture

Classical swine fever

Syn: Hog cholera, Swine fever

Classical swine fever (CSF) is a highly contagious, often fatal disease of pigs clinically characterized by high body temperature, lethargy, yellowish diarrhea, vomiting and purple skin discolouration of ears, lower abdomen and legs. CSF is potential to cause devastating epidemics particularly in countries that are free of the disease.

Etiology

It is caused by genus Pestivirus belongs to family Flaviviridae, it is a small enveloped RNA virus; contain only one serotype, although minor antigenic variability between the strain. The virus has antigenic relationship with bovine viral diarrhea virus. The virus is moderately fragile and does not survive in an environment. However, the virus can survive for prolonged periods in a moist, protein-rich environment such as pork tissue or body fluids particularly if kept cold or frozen, in frozen pork meat the virus can survive for several years. The virus is inactivated at $pH < 3.0$ or > 10 .

Epidemiology

- ☒ CSF has a worldwide distribution
- ☒ The disease is endemic in central and south America, Asia, Australia, Newland, Canada, South African countries, Europe, China and Japan, India, Myanmar, Nepal, Pakistan, Bangladesh and Philippines
- ☒ In India this disease was first reported in West Bengal in the year 1961
- ☒ Swine are the natural host of the virus
- ☒ All breeds and all age group of pigs are susceptible to this infection
- ☒ Wild pigs often remain as inapparent carrier of the virus
- ☒ Young pigs are more susceptible than the adult pigs

Risk factors

- ☒ Accidental introduction of CSFV into herds through illegally imported pork meat or pork products

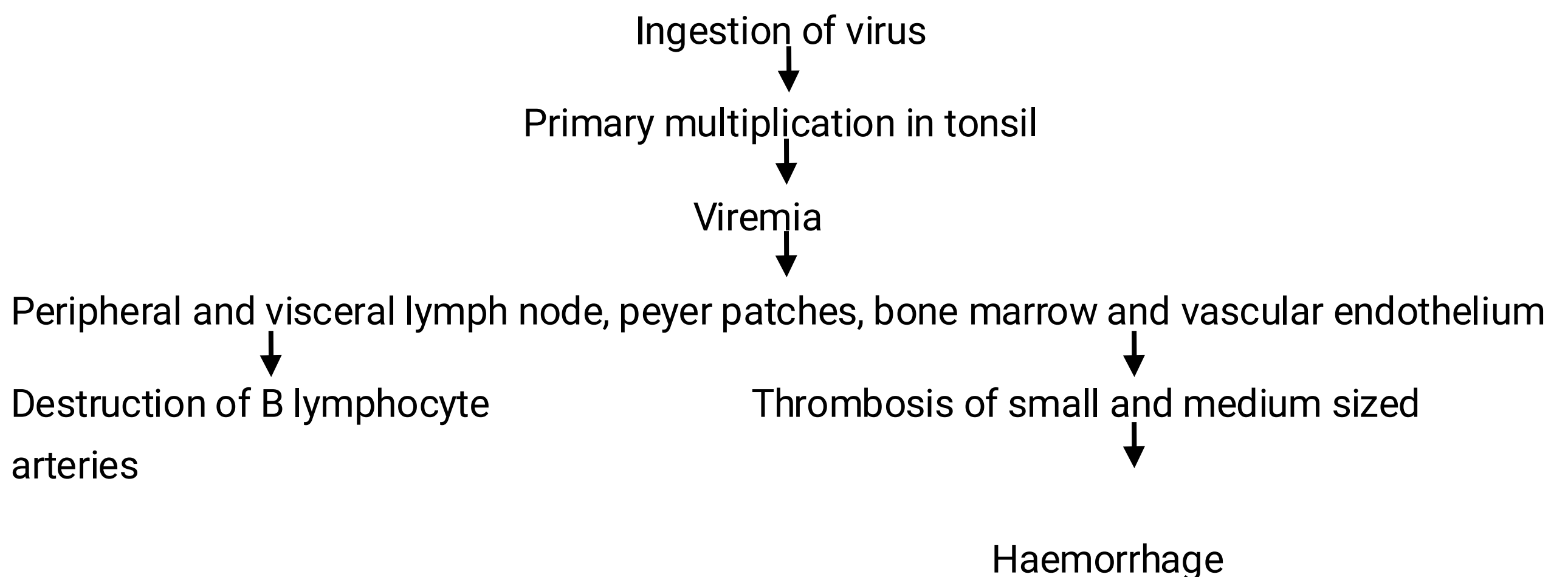
Source of infection

- ☒ Infected live animals can shed the virus in oronasal and ocular secretions, urine, faeces and semen before onset of clinical illness
- ☒ Uncooked pork products
- ☒ In Europe, CSF is endemic in wild boar population; infected wild boars are a source for CSF outbreaks among domestic pigs.
- ☒ The probable source for wild boar is contaminated garbage or spill over from the infected pigs
- ☒ Untreated swill feeding

Transmission

- ☒ **Ingestion of garbage containing uncooked pork products**
- ☒ Infection is transmitted readily by direct and indirect contact with an infected animals
- ☒ Inhalation is also a possible portal of entry

Pathogenesis



Note: vascular changes are more severe in the lymph node, spleen, kidneys and gastrointestinal tract

Clinical manifestation

- ☒ Incubation period is 3-7 days
- ☒ Clinical signs vary with the strain of CSFV, age and breed of the animals
- ☒ The disease may appear in three clinical forms

Peracute form

- ☒ **Most commonly noticed in young pigs**
- ☒ Disease terminates fatally within about 24 hours of developing disease
- ☒ No appreciable clinical manifestations except high rise of temperature and erythematous patches in the non-hairy parts of the skin

Acute form

- ☒ High fever (105-107° F)
- ☒ **Huddling**, weakness, drowsiness, anorexia and conjunctivitis
- ☒ Disinclination to move, when forced do move, **swaying movement of hindquarters often progress to posterior paresis**
- ☒ Constipation is followed by intermittent watery diarrhoea
- ☒ Some pigs may vomit yellow bile containing fluid
- ☒ **Skin is hyperaemic or may develop haemorrhages on the abdomen, inner thighs, ears or purple cyanotic discolouration on the snout, ears or tail**

- ☒ Pigs with acute classical swine fever often die within 1-3 weeks, and convulsions may occur in the terminal stages
- ☒ Convulsions may be accompanied by loud squealing may occur constantly or interval of several hours
- ☒ Tremor of body and limb muscle

Subacute and chronic form

- ☒ The symptoms are similar to acute form, but signs are less severe
- ☒ Poor reproductive performance may be the only sign in some breeding herds infected with less virulent strains
- ☒ Sows in these herds may abort or give birth to stillborn, mummified, malformed, weak or dead piglets

Necropsy findings

- ☒ Petechial or ecchymotic hemorrhages can often be seen on serosal and mucosal surfaces of the kidney (**Turkey egg appearance**), urinary bladder, epicardium, epiglottis, larynx, trachea, intestines, subcutaneous tissues and spleen
- ☒ Splenic infarcts (raised, dark, wedge-shaped lesions) are seen occasionally
- ☒ Necrotic foci or “ button” ulcers may be found in the intestinal mucosa (colon), epiglottis and larynx
- ☒ Straw colour fluid may be found in the peritoneal and thoracic cavities and the pericardial sac
- ☒ Non suppurative encephalitis
- ☒ In congenitally infected piglets, common lesions include cerebellar hypoplasia, thymic atrophy, ascites, and deformities of the head and legs. Edema and petechial hemorrhages may be seen in the skin and internal organ

Sample collection

- ☒ Live animals - Whole blood and swab from tonsil during fever
- ☒ Dead animal - tonsils, pharyngeal and mesenteric lymph nodes, spleen, kidneys, distal ileum or colon

Diagnosis

- ☒ Based on clinical signs and lesion
- ☒ Marked leukopenia
- ☒ Virus isolation can be carried out in porcine cell lines using homogenates of spleen and tonsil
- ☒ RT-PCR- Most commonly used for detection of viral nucleic acid
- ☒ Loop-mediated isothermal amplification (RT-LAMP) assays have also been published.
- ☒ CSFV antigen can be detected by FAT or ELISA

- ☒ Detection of antibodies in serum by **virus neutralization test**, fluorescent antibody virus neutralization test [FAVN] and neutralizing peroxidase– linked assay [NPLA] tests) and various ELISA test

Differential diagnosis

- ☒ Salmonellosis
- ☒ Erysipelas
- ☒ Pasteurellosis
- ☒ African swine fever
- ☒ Anticoagulant poisoning

Treatment

- ☒ There is no specific treatment, other than supportive care
- ☒ Hyperimmune serum may be given @ dose of 50-150 ml/ animal
- ☒ Simultaneous serum- vaccination method

Prevention and control

- ☒ Vaccination of pigs with live attenuated vaccine at the age of 1-2 months and booster at 6 months of age and followed by annual revaccination every year

Endemic area

- ☒ Purchase of animals from CSF free herd
- ☒ Quarantine of newly purchased animals for 4 months
- ☒ Testing of animals in quarantine before allowing to the rest of the herd

Epidemic area

- ☒ Eradication by slaughter of all- in contact and infected animal
- ☒ **Emergency vaccination** can be used for control of the disease in the event of severe outbreak with either Modified live vaccine (of either lapinized strain or cell culture adopted strain) or marker vaccine
- ☒ In wild boar outbreak, emergency vaccination using bait with modified live vaccine
- ☒ Slaughtered pigs are disposed of by burning
- ☒ Restriction of animal movement
- ☒ Disinfection of all vehicle used for transportation, , all pens and premises and utensils must be disinfected with 5% cresylic acid or sodium hydroxide solution
- ☒ Boiling of all contaminated clothing
- ☒ Feeding of boiled garbage
- ☒ Surveillance of animals
- ☒ Avoid contact of domesticated pigs with wild pigs

African swine fever

Syn: African pig disease, Wart hog disease

African swine fever (ASF) is highly contagious hemorrhagic disease of pigs that produce wide range of clinical signs and lesions that closely resembles to those of CSF.

Etiology

It is caused by African swine fever virus, a large enveloped DNA virus. It is currently classified as the only member of the family African swine fever like viruses (Asfiviridae). The virus is highly resistant to wide pH range, heat, dryness and survives in chilled carcasses for upto 6 months.

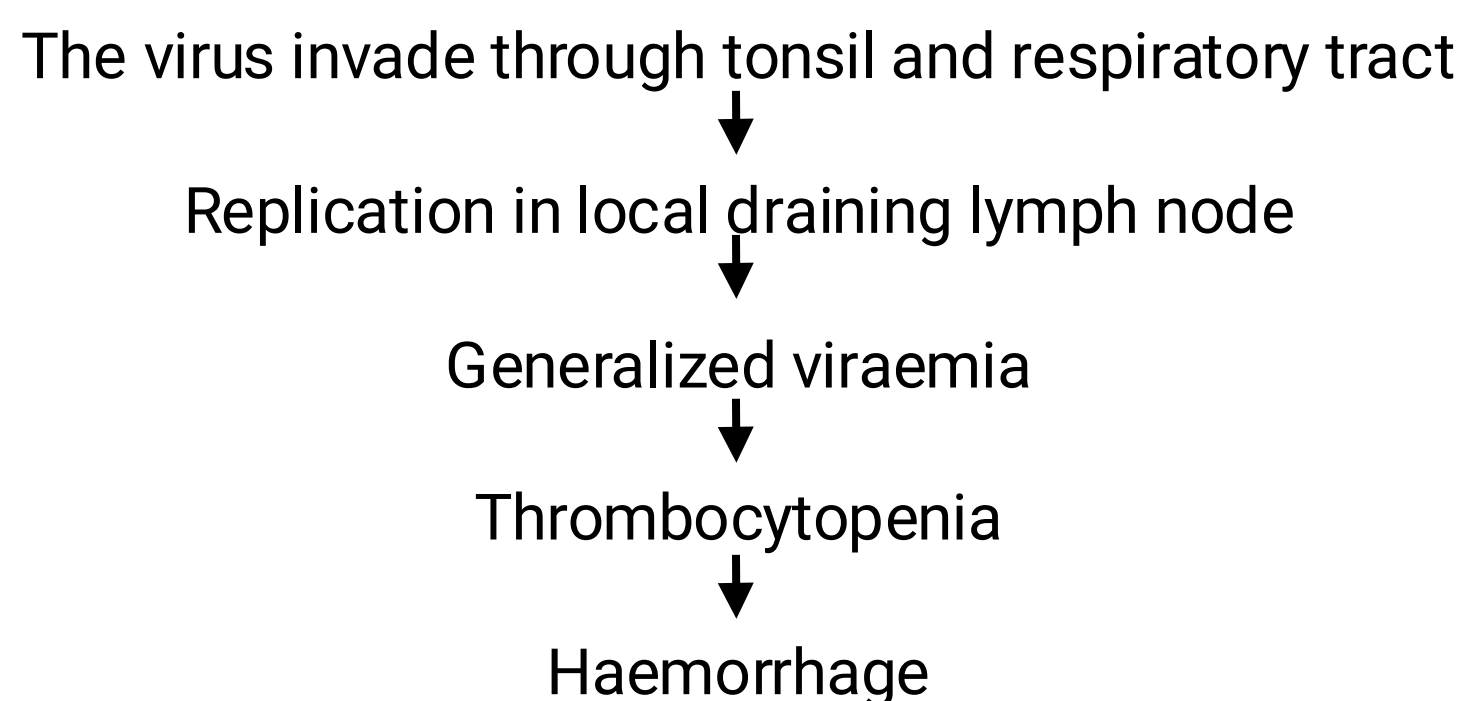
Epidemiology

- ☒ **Occurrence**-African swine fever is enzootic in African continent and can spread to neighbouring countries.
- ☒ Occurrence of clinical disease in South Africa has been cyclical with periods of 10-12 yrs and then absence of disease
- ☒ **Wart hogs, bush pigs and forest hogs act as reservoirs of the virus**
- ☒ Morbidity rate: 100% and case fatality rate > 90%

Transmission

- ☒ **Infection is primarily transmitted to domestic pigs through argasid tick**
- ☒ Ingestion of contaminated feed and water
- ☒ Feeding of uncooked garbage containing infected pig material

Pathogenesis



Clinical manifestation

- ☒ Incubation period is 5-15 days
- ☒ High fever (105°F) persist for 4 days without other signs, when fever subsides, marked cyanotic blotching of the skin is noticed
- ☒ Depression, anorexia, huddling together
- ☒ Disinclination to move
- ☒ Extreme weakness of hind quarters with difficulty in walking, dragging of hind legs

- ☒ Tachycardia
- ☒ Serous mucopurulent nasal discharge, dyspnea and cough
- ☒ Vomition, diarrhoea, sometimes dysentery may occur
- ☒ Bleeding from nose or rectum
- ☒ **Purplish discolouration of skin occur on snout, limbs, abdomen and ears**
- ☒ Death occur in one or two days after onset of obvious clinical signs

Necropsy finding

- ☒ Swollen and hemorrhagic gastrohepatic and renal lymph node
- ☒ Subcapsular petechiation of kidney
- ☒ Ecchymoses on the cardiac surface
- ☒ Hydrothorax
- ☒ Splenomegaly
- ☒ Gall bladder is edematous and haemorrhagic
- ☒ Marked karyorrhexis of lymphocyte in lymphoid organ

Sample collection

- ☒ Dead animals-spleen, lymph node, kidney, liver, colon and caecum

Diagnosis

- ☒ Based on clinical signs and lesion
- ☒ Blood picture: decreased WBC count, lymphopenia and neutrophillia
- ☒ Detection of antigen by PCR or ELISA
- ☒ Detection of antibodies in serum or tissue fluids by ELISA, CIE and indirect immunofluorescence test

Differential diagnosis

- ☒ Hog cholera
- ☒ Swine erysipelas
- ☒ Salmonellosis

Treatment: There is no treatment

Prevention and control

- ☒ Currently vaccines are not available for control of the disease
- ☒ Test and slaughter of all animals
- ☒ Disposal of dead carcass properly
- ☒ Control of movement
- ☒ Heat treatment of waste food
- ☒ Serological survey of all pig farms within a specific zone must be conducted

Swine vesicular disease

Swine vesicular disease (SVD) is a contagious disease of pigs, characterized by vesicles on the coronary bands, heels of the feet and occasionally on the lips, tongue, snout and teats. The disease may occur as subclinical, mild or severe, the latter usually only being seen when **pigs are housed on abrasive floors in damp conditions**.

The main importance of SVD is that it is clinically indistinguishable from foot and mouth disease (FMD), and any outbreaks of vesicular disease in pigs must be assumed to be FMD until investigated by laboratory tests and proven otherwise. However, subclinical infection has been the most frequent condition observed during recent years.

Etiology

The disease is caused by genus Enterovirus of family Picornaviridae. The virus contain only one serotype. The virus is antigenically related to human enterovirus B. The SVD virus is stable in an environment and pH range of 2.5– 12.0.

Epidemiology

The disease was first reported in Italy in 1966, and eradicated in most European countries in 1980. It has remained endemic in Italy.

Source of infection

- ☒ The virus excreted through nose and mouth occur upto 2 weeks, and faeces for 3 months

Transmission

- ☒ Direct or indirect contact with an infected animals
- ☒ Ingestion of contaminated pork and pork products
- ☒ Through skin abrasion

Clinical manifestation

- ☒ Incubation period is 2-7 days
- ☒ Fever (temp 41°C)
- ☒ Vesicles develop on coronary band particularly at the junction of heel rarely on the snout, dorsal surface of lips, tongue and teat
- ☒ Lameness and off feed for few days
- ☒ Lesion may be mild or inapparent when pigs kept on soft bedding
- ☒ The affected animal usually recovered 2-3 weeks, and show dark horizontal line on the hoof

Sample collection

- ☒ **Faecal sample**, vesicular fluid, affected skin and mucous membrane and whole blood

Diagnosis

- ☒ Isolation of virus in porcine kidney cell culture. Typical cytopathic effect is produced in 2 - 4 days of inoculation.
- ☒ Detection of SVD genome by ELISA or RT-PCR is same diagnostic value as virus isolation
- ☒ The ELISA (indirect sandwich ELISA) and RT-PCR may be suitable screening test for SVD

Prevention and control

- ☒ Currently no vaccines are available
- ☒ Prohibition of feeding of uncooked garbage to pigs
- ☒ Restriction of movement of pigs
- ☒ Serosurveillance of pig farm to detect subclinical infection
- ☒ Disinfection of premises with sodium hypochlorite

Pseudorabies

Syn: Aujeszky disease, Mad itch, infectious boar paralysis

Pseudorabies is an acute fatal disease of swine and other domestic and wild animals are affected incidentally.

Etiology

It is caused by porcine herpes virus-1, (Pseudorabies virus, a member of the family herpesviridae, subfamily alphaherpesvirinae. It is a DNA virus contains only one serotype. The virus is inactivated at 5.25% sodium hypochlorite.

Epidemiology

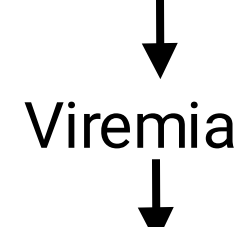
- ☒ The disease is worldwide in distribution
- ☒ All age group of pigs are affected
- ☒ Young animals are highly susceptible than adults
- ☒ Morbidity rate: 100% and Mortality rate: 50% in young piglets
- ☒ Economic loss may reach 100% in piglets less than 7 days old

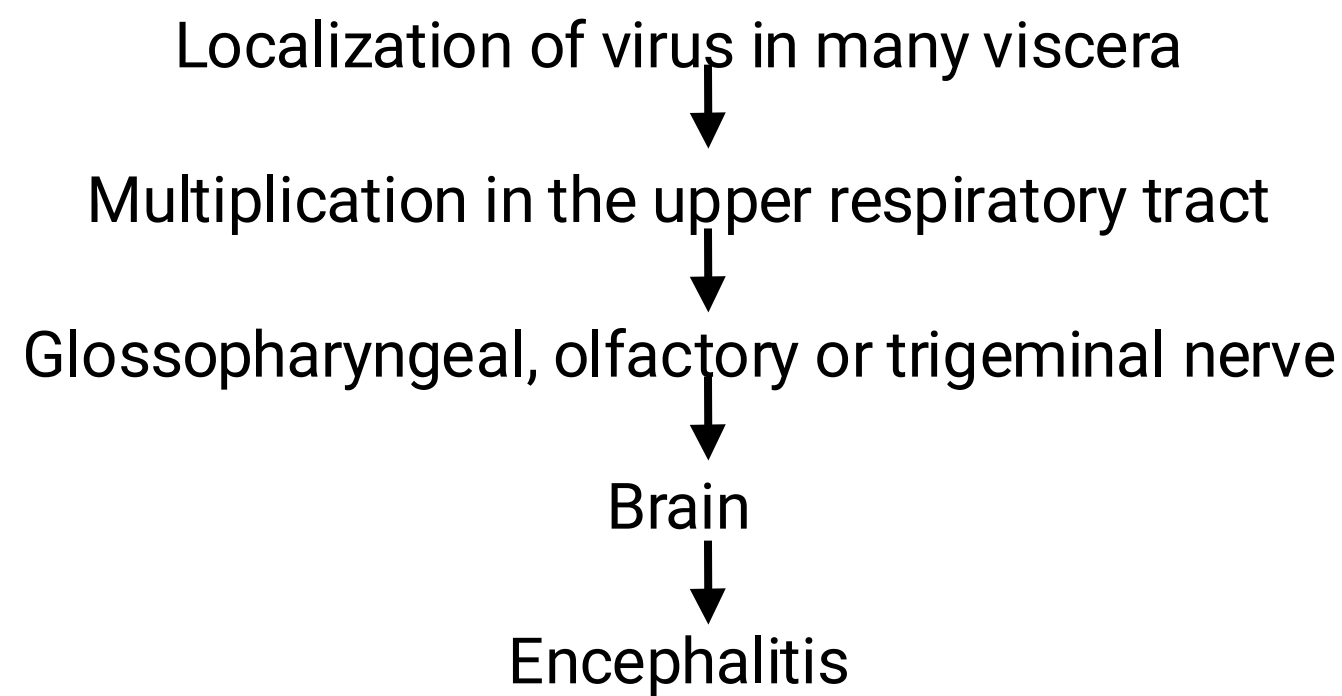
Transmission

- ☒ Nose to nose or oro-nasal contact
- ☒ Indirect transmission commonly occurs through inhalation of aerosol
- ☒ Rodents act as reservoir for virus

Pathogenesis

Virus enters the host through abraded skin or intact nasal mucosa





Clinical manifestation

The major systems affected in this disease are **respiratory, nervous and reproductive system**. Piglets less than one month old are more susceptible and died due to **encephalitis**

The symptoms include

- ☒ Fever (Temp. 107°C) occurs prior to the onset of nervous signs
- ☒ Inco-ordination of hind limbs followed by recumbency
- ☒ Muscle tremor
- ☒ Paddling movement
- ☒ Lateral deviation of head, frothing at the mouth
- ☒ Nystagmus
- ☒ Convulsion (epileptic form seizure)
- ☒ Snoring respiration
- ☒ Vomition and diarrhoea in some affected animals
- ☒ Death occur 12 hrs after onset of first signs

Grower and adult pig

- ☒ **Respiratory signs are common**
- ☒ Fever, sneezing, nasal discharge, coughing and severe dyspnoea
- ☒ Trembling, incoordination, paralysis and convulsion precede with death
- ☒ Infection in early pregnancy causes early embryonic mortality and abundant vaginal discharge
- ☒ Infection in late pregnancy may result in abortion, birth of mummified foetuses

Cattle

- ☒ **Local pruritis with violent licking**, chewing and rubbing of the part until great tissue destruction so, it is called as **mad itch**
- ☒ **Itching** is localized to any part of the body, but **most common on head, flank and feet**
- ☒ **Intense excitement, convulsions and constant bellowing**, salivation and respiratory distress and ataxia

- ☒ Manical behaviour, circling, opisthotonus and paralysis

Sheep

- ☒ Outbreak in sheep is associated with skin abrasions, acquired through shearing
- ☒ Fever
- ☒ **Nibbling of their fleece** and frenzied attempt to bite on the skin and rub it against the wall and bar of their pen

Goat

- ☒ Rapid death
- ☒ Unrest
- ☒ Frequently lying down and rising up
- ☒ Crying, profuse sweating
- ☒ Paralysis
- ☒ **No pruritis**

Dog and cat

- ☒ Clinical sign is similar to that of cattle
- ☒ Death is eminent within 2 days following manifestation of clinical signs

Necropsy finding

- ☒ Lungs show congestion, edema and some haemorrhage
- ☒ Splenomegaly
- ☒ Meningitis
- ☒ Necrotic lesion with **inclusion body formation in upper respiratory tract and lung is suggestive of pseudorabies in pig**
- ☒ In aborted foetus or very young piglets- **white spot on liver is pathognomic lesion**

Sample collection

- ☒ Live animals- oropharyngeal fluid, nasal and tonsil swab, paired sera sample
- ☒ Dead animal -lung, brain and tonsil from pig and spinal cord from cattle

Diagnosis

- ☒ Based on clinical signs and lesion
- ☒ Isolation and identification of the virus made by inoculation of tissue homogenate (tonsil and brain) into a susceptible cell line such as porcine kidney (PK-15) or SK6, or primary or secondary kidney cell
- ☒ The specificity of the cytopathic effect is verified by immunofluorescence, immunoperoxidase or neutralisation with specific antiserum
- ☒ Detection of viral antigen by PCR
- ☒ Detection of antibodies in serum by **VNT and ELISA**

Differential diagnosis

- ☒ Rabies

- ☒ Streptococcal meningitis
- ☒ Encephalopathy
- ☒ Salt poisoning
- ☒ Reproductive inefficiency associated with enterovirus (SMEDI) and parvovirus infections
- ☒ Polyencephalomalacia

Prevention and control

- ☒ The disease may be controlled by vaccination of piglets at 9-12 weeks of age with either modified live or inactivate vaccine
- ☒ Recombinant DNA derived, gene-deleted Marker vaccines(Lack of specific glycoprotein *gG*, *gE*, or *gC*) are available to differentiate vaccinated from infected animal
- ☒ Depopulation and repopulation of swine is practiced, when prevalence of infection above 50%
- ☒ The premises are cleaned, disinfected and left empty of pigs for 30 days
- ☒ Test and removal of pigs
- ☒ Due to low incidence of disease no vaccines are available in India

African horse sickness

African Horse Sickness (AHS) is an acute or sub acute, febrile, arthropod borne viral disease of equids, characterized by clinical signs and lesions associated with respiratory and circulatory impairment.

Etiology

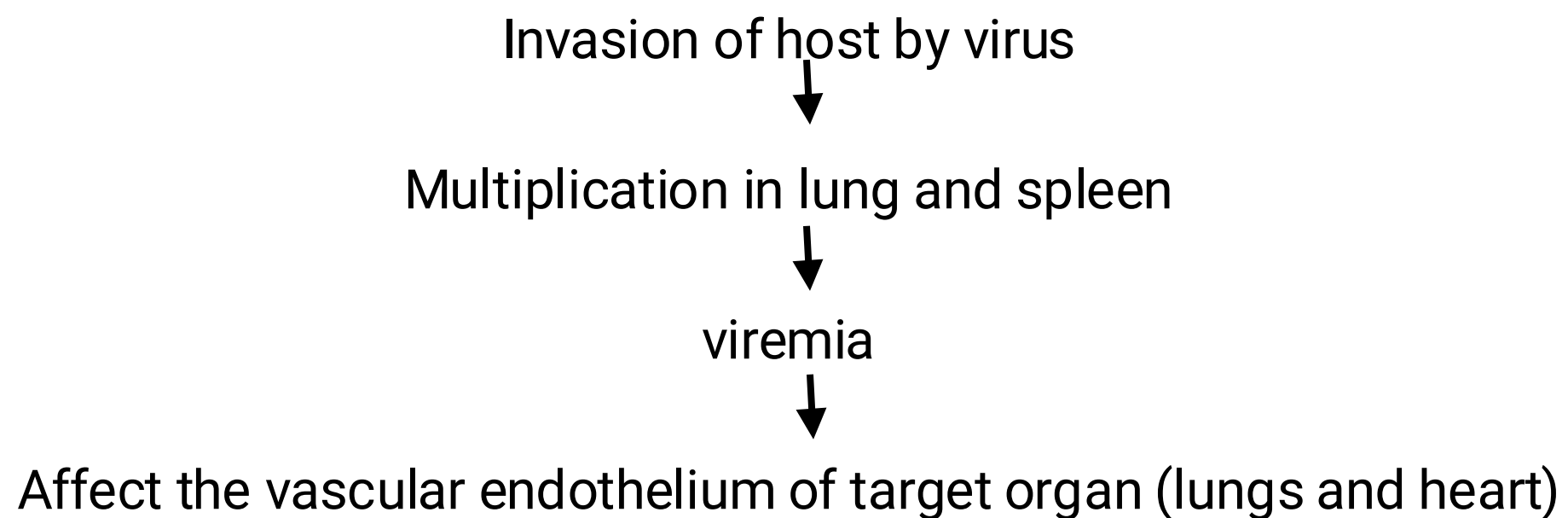
This disease is caused by orbivirus belonging to the family Reoviridae, which is a double stranded RNA virus. There are 9 immunologically distinct serotypes are recognized. On the other hand only a single antigenic type involved in the Asian outbreak in 1961. The virus is inactivated at pH< 6.0 or > 12 or by formalin and β - *Propiolactone*.

Epidemiology

- ☒ The disease is endemic in Africa, and has been recorded in Afghanistan, West Pakistan, Turkey and Iraq
- ☒ Mortality: In virulent form it is as high as 90% in horses and lower in mules. In milder outbreaks mortality is 25%
- ☒ It is a seasonal disease, common in late summer in warm, humid, low lying marshy districts
- ☒ Horses of all breeds are highly susceptible to natural infection
- ☒ Mules and zebras are less Affected and donkeys are generally resistant

- ☒ Transmitted by blood sucking mosquitoes (culicoides) midges and stomoxys

Pathogenesis



Clinical manifestation

- ☒ Incubation period is 5-7 days
- ☒ Four different clinical form of illness were described

Acute or pulmonary form (DUNKOP) – most common in epizootic area

- ☒ This form is more severe than cardiac form
- ☒ Fever followed by **labored breathing, severe paroxysms of coughing (spasmodic coughing)**
- ☒ **Yellowish profuse nasal discharge with froth**
- ☒ Animal stands with leg apart and head extended
- ☒ Profuse sweating, profound weakness and staggering gait progress to recumbency
- ☒ Total course of the disease is 4-5 days
- ☒ Animal die of anoxia

Sub acute form cardiac form (DIKKOP)- most common in enzootic area

- ☒ **Edema in the head, particularly temporal fossa, eyelids and lips**
- ☒ Edema develops in the chest, after one week of febrile period
- ☒ Oral mucosa is **bluish in color** and **petechiae may develop under the tongue**
- ☒ Auscultation of heart reveals **hydrocardidium, endocarditis and pulmonary edema**
- ☒ Restless, mild abdominal pain, paralysis of oesophagus
- ☒ Regurgitation of food and water through nostrils
- ☒ Animal stands leg apart and extension of head
- ☒ Animal die of congestive heart failure
- ☒ Recovery is prolonged

Mixed form

- ☒ Combination of both acute or pulmonary form and sub acute form or cardiac form.
The mixed form is not common in field outbreak.

Mild form or horse sickness fever

- ☒ Common in enzootic area
- ☒ Temperature rises to 40.5°C over a period of 1-3 days
- ☒ Poor appetite, conjunctivitis and moderate respiratory distress

Necropsy finding

- ☒ Pharynx, trachea and bronchi are filled with yellowish frothy serous fluid
- ☒ Severe hydrothorax, hydropericardium, pulmonary edema and moderate ascites
- ☒ Endocardial haemorrhage and myocardial degeneration

Sample collection

- ☒ Live animals: Blood from affected animal should be collected during early febrile phase
- ☒ Dead animals : Spleen, lung and lymph node

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Hematology and biochemical parameters- lymphopenia, neutropenia with left shift and thrombocytopenia, increased level of creatine kinase (CK), alkaline phosphatase (ALP), Lactate dehydrogenase (LDH), creatinine and bilirubin concentration
- ☒ Isolation of virus in baby hamster kidney-21 (BHK-21), monkey stable (MS) or African green monkey kidney (Vero), intravenously in embryonated eggs, and intracerebrally in newborn mice.
- ☒ Detection of viral antigen by ELISA and RT-PCR
- ☒ Detection of antibody in serum by AGID, IFAT, CFT VNT and ELISA

Treatment

- ☒ There is no specific treatment

Control

- ☒ Live attenuated cell culture polyvalent vaccine (contain 9 serotype) provide good protection, although annual revaccination is recommended
- ☒ Foals are vaccinated at 3-4 months of age
- ☒ Now, Inactivated and subunit vaccine is under development
- ☒ Restricting importation of horse from country known to have disease
- ☒ Quarantine of affected horse
- ☒ Stabling of horse indoors in insect proof stable
- ☒ Use fly repellent when risk is high
- ☒ Vector control

Equine infectious anemia

Syn: Swamp fever

It is a chronic persistent infectious viral disease of equines characterized by, intermittent fever, anaemia, emaciation, edema and generalized lymphoproliferative changes in the body.

Etiology

It is caused by Equine infectious anemia virus (EIAV) of genus Retrovirus, a member of subfamily Lentivirinae, family Retroviridae. The EIAV shares antigenic cross reactivity with human and feline immunodeficiency virus. The virus has one major group specific antigen p26. There is considerable antigenic drift in the surface glycoprotein (gp45 and gp90) responsible for emergence of novel antigenic strain is associated with the **relapsing febrile reaction is characteristic of the disease.**

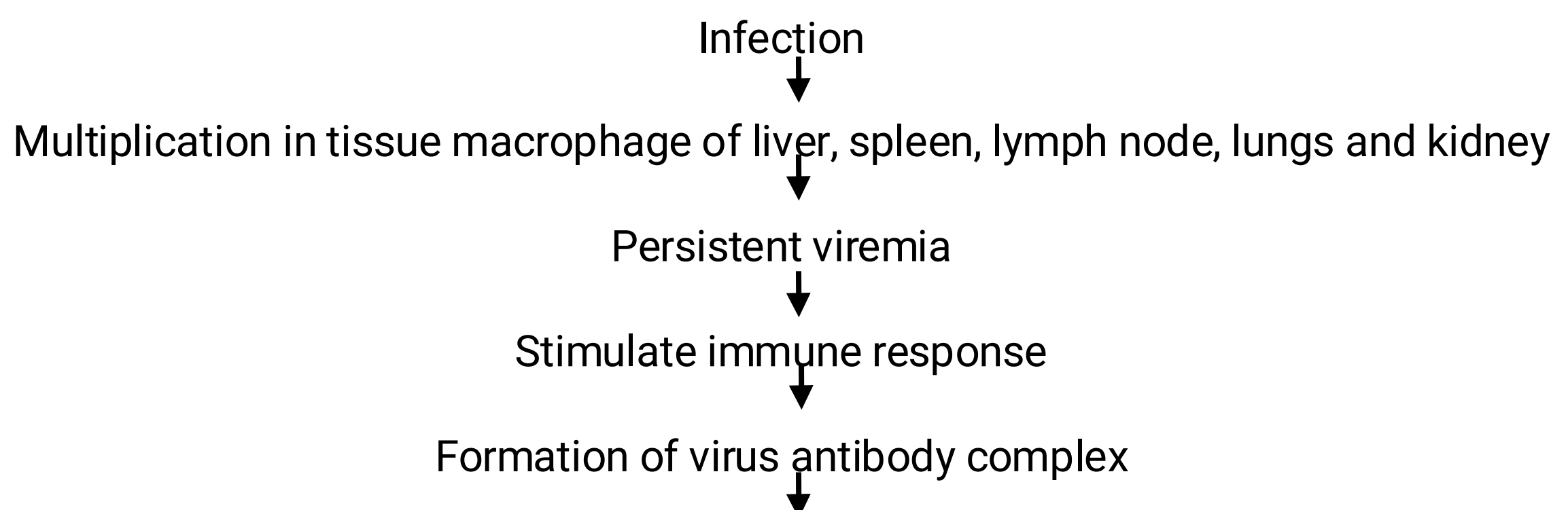
Epidemiology

- ☒ The disease is present in all continents except antartica
- ☒ The prevalence of infection depends on the density of equine population, proportion of carrier animal and vector population
- ☒ The outbreak of EIAV is rare in developed countries
- ☒ The morbidity rate varies with the strain of virus infected and inoculums delivered by insect
- ☒ **Host affected:** horse, donkey and mule
- ☒ **Source of infection:** persistently infected or clinically affected horse

Transmission

- ☒ **Transfer of contaminated blood or blood products by biting insects mechanically** occurs in summer season (under hot and humid condition)
- ☒ **Through contaminated surgical instruments or needles**
- ☒ Use of common needle when injecting group of horse
- ☒ Intrauterine infection can occur
- ☒ It can also transmitted through semen of infected stallion

Pathogenesis



Deposition of virus antibody complex on the surface of RBC and platelets



Phagocytosis



Intravascular hemolysis



Anemia and thrombocytopenia

Clinical manifestation

- ☒ Incubation period is 2-4 weeks
- ☒ It occurs as acute, subacute or chronic form

Acute form

- ☒ Anorexia, depression and profound weakness and ataxia
- ☒ **Intermittent fever**
- ☒ **Jaundice, edema of the ventral abdomen, prepuce and legs**
- ☒ Petechial haemorrhage under the tongue and conjunctiva
- ☒ Increase in heart rate and intensity of heart sound
- ☒ Respiratory signs are not marked, but may be thin serosanguineous nasal discharge
- ☒ Considerable **enlargement of spleen can be detected per rectum**
- ☒ Pregnant mares may abort
- ☒ Many animals show temporary recovery 3 days-3 weeks of illness
- ☒ Others progressively weak, recumbent and die after 10-14 days of illness
- ☒ Relapse occurs in temporarily recovered animals
- ☒ Relapse continues to occur at intervals of 2 weeks and ceases about a year

Chronic form

- ☒ Appetite is good, may show allotriphagia
- ☒ Affected animals are recovered completely but relapse occurs in the later years

Necropsy finding

- ☒ Considerable enlargement of spleen, liver and local lymph node
- ☒ Reddened bone marrow
- ☒ Glomerulonephritis

Sample collection

- ☒ Live animals- buffy coat and paired sera sample
- ☒ Dead animal- spleen, liver, bone marrow and perihepatic lymph node

Diagnosis

- ☒ Based on clinical signs and lesions

- ☒ Haematology and biochemical changes include anemia, thrombocytopenia, presence of sideroleukocyte, Hyper gamma globulinemia, increased bilirubin level and decrease in serum iron concentration
- ☒ AGID (**Coggin' s test**), competitive ELISA used to detect antibodies against group specific p26 core antigen of EIAV after 45 days of infection
- ☒ Western blot test useful to detect antibodies against surface glycoprotein gp45 and gp90 (relapse occur with this antigen) after 60 days of infection
- ☒ Experimental transmission of the disease to susceptible horse

Differential diagnosis

- ☒ Purpura haemorrhagica
- ☒ Babesiosis
- ☒ Equine viral arteritis
- ☒ Equine granulocytic ehrlichiosis
- ☒ Leptospirosis
- ☒ Autoimmune haemolytic anemia
- ☒ Idiopathic thrombocytopenia
- ☒ Parasitism

Chronic disease

- ☒ Internal abscessation
- ☒ Chronic hepatitis, neoplasia
- ☒ Streptococcal infection- cough and respiratory distress are common

Treatment

- ☒ No specific treatment available
- ☒ Supportive therapy include blood transfusion, haematinic drugs may used to facilitate clinical recovery

Prevention and control

- ☒ Killed whole virus vaccine is available and safe to use
- ☒ Subunit vaccine enhance the disease occurrence
- ☒ Lifelong quarantine of infected animals
- ☒ Quarantine and testing of new stock
- ☒ Compulsory testing of imported horse
- ☒ Restriction of introduction of infected horse into the clean area
- ☒ Maintenance of strict hygiene during blood collection and vaccination
- ☒ Draining of marshy area and control of biting insects
- ☒ Disinfection of tattoo equipment and surgical instruments by immersion of less corrosive phenolic compound for 10 min.

- ☒ Cleaning and disinfection of premises with sodium hypochlorite and iodine compound

Equine influenza

Syn: Equine distemper, Typhoid fever, pink eye

Equine influenza is an acute highly viral disease of horses characterized by generalized septicemia, respiratory problem accompanied by severe persistent dry cough.

Etiology

Equine influenza is caused by **influenza A/ equine-1 (H7N7) or influenza A/equine-2 (H3N8)** virus is an Orthomyxovirus. There is no cross protection between subtypes. The entire outbreak in the past two decades has been caused by H3N8. The virus can travel upto 30 kms in an environment.

Antigenic drift in the major surface protein **hemagglutinin** contributes to continuing susceptibility of horse to infection and reduced vaccine efficacy.

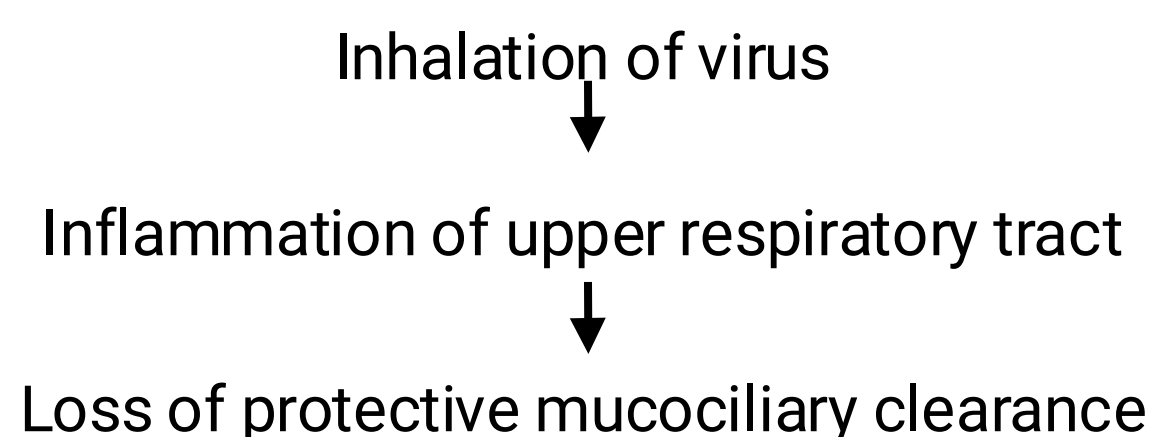
Epidemiology

- ☒ The disease is worldwide in distribution except Australia and Newzealand. In contrast, the H3N8 subtype isolated from horses in China was more closely related to avian strains
- ☒ All age group of horse including newborn foals are susceptible. However, most cases of the disease occur in 2 years old or younger horse
- ☒ Outbreak can occur at any time of the year

Transmission,

- ☒ Direct contact with an infected animals
- ☒ Inhalation of aerosol discharged from an infected material
- ☒ Fomite transfer on veterinary clothing and equipment

Pathogenesis



Clinical manifestation

- ☒ Incubation period- 2-3 days
- ☒ Fever (101-106°F)

- ☒ **Dry hacking cough** in the beginning and moist cough later
- ☒ During early stage of disease, nasal discharge is not prominent sign, only mild watery discharge occurs
- ☒ **Pain on palpation of submaxillary lymph node** without marked swelling
- ☒ Abnormal lung sound is characterized by crackles, **wheezes** and increased intensity of normal breath sound
- ☒ The secondary bacterial complication causes severe bronchitis, pneumonia and edema of the legs. The animal shows depression, anorexia and reluctance to move

Sample collection

Live animals

- ☒ Nasopharyngeal swab should be collected by inserting a sterile cotton swab approximately 30 cm into the nostril in the adult horse during rise of temperature
- ☒ Paired sera sample

Diagnosis

- ☒ Based on clinical signs
- ☒ Isolation of virus in embryonated hens egg or cell culture
- ☒ Detection of viral antigen by **haemagglutination test (HA)**, Fluorescence antibody technique (FAT), ELISA and polymerase chain reaction (PCR)
- ☒ Detection of antibody titre in serum by **haemagglutination inhibition test (HI)** and single radial hemolysis test (SRH)

Treatment

- ☒ Antibiotics should be used to prevent secondary bacterial complications
- ☒ Anti-inflammatory drugs should be used judiciously
- ☒ Provide rest and dust free environment

Prevention

- ☒ Several inactivated vaccines are commercially available
- ☒ However, immunity is usually short-lived and booster injections are required in accordance with manufacturer's instructions
- ☒ Inactivated subunit vaccine containing immunogenic protein in an immune stimulating complex (ISCOM) induce a more sustained and greater immune response than conventional vaccine
- ☒ Older horse should receive frequent booster before racing and show season
- ☒ Mares should be revaccinated before being shipped to breeding farm
- ☒ Foals should be vaccinated at 6 months of age and booster dose given at 21 days later and revaccination should be done every 6 months interval

Control

- ☒ Avoid movement of horse between barn and paddock

- ☒ Isolation of infected animal for 21 days
- ☒ Cleaning, disinfection of infected premises
- ☒ Vaccination of clinically normal horse during the face of the outbreak may enhance an immunity

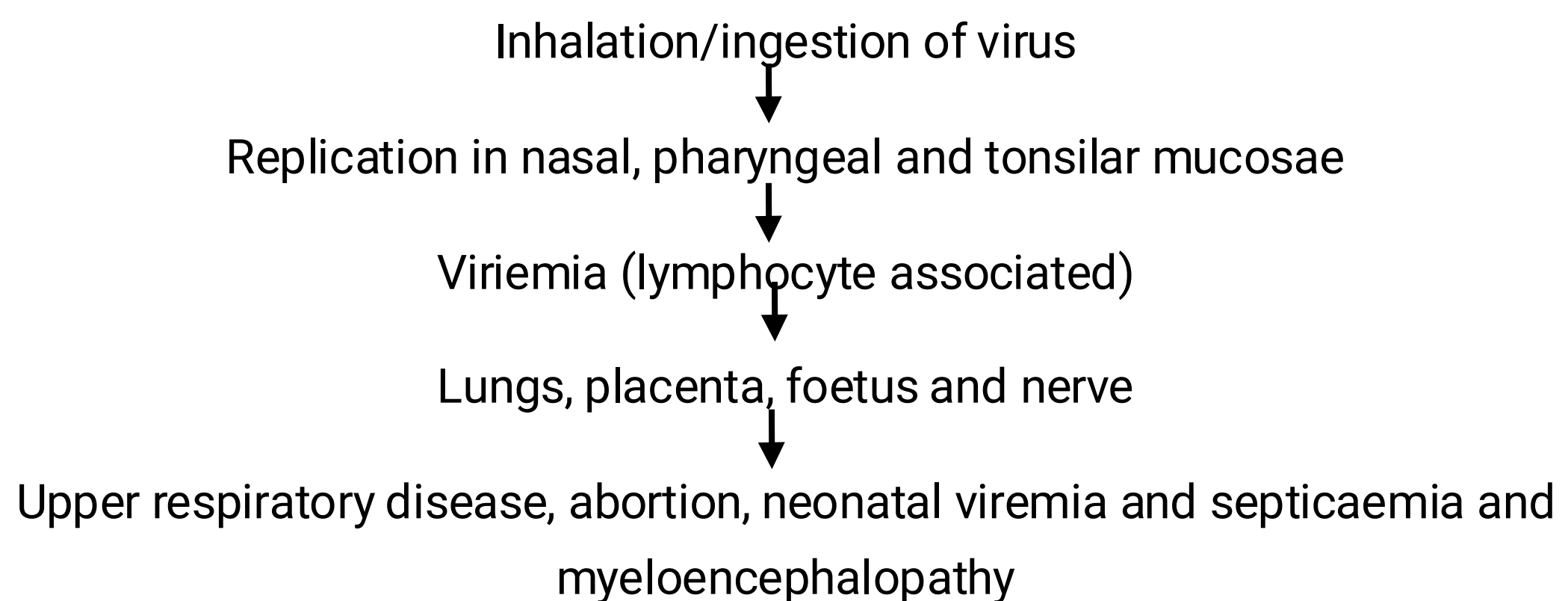
Equine viral rhinopneumonitis

- ☒ It is a contagious viral disease of horse caused by equine herpes virus (EHV-1 and EHV-4)
- ☒ EHV-4 is a major cause of upper respiratory disease than EHV-1
- ☒ EHV-1 causes abortion and myeloencephalopathy sporadically
- ☒ Morbidity rate: 1-90% and mortality rate-0.5-40 %

Transmission

- ☒ Inhalation of infected droplets
- ☒ Ingestion of material contaminated by nasal discharge and aborted foetus

Pathogenesis



Clinical manifestation

Clinical signs indistinguishable from other respiratory disease

1. Respiratory disease

- ☒ Incubation period -2-20 days
- ☒ Fever (temp. 39 to 40.5°C) conjunctivitis, coughing and mild inflammation of upper respiratory tract, slight enlargement of the lymph node of throat
- ☒ Inapparent infection is common
- ☒ Edema of the limbs and diarrhoea occur rarely
- ☒ Length of illness 2-5 days
- ☒ Nasal discharge and cough may persist for 1-3 weeks
- ☒ EHV-2 or equine cytomegalovirus causes purulent nasal discharge, fever and lymphadenopathy

2. Abortion

- ☒ Abortion occur particularly at **8-10 months of pregnancy**
- ☒ No retention of placenta

3. Neonatal viremia and septicaemia

- ☒ In utero EHV-1 infection causes abortion or birth of infected foals. Some of which normal at birth and become weak and die within 3-7 days

4. Myeloencephalopathy

- ☒ Clinical signs are due to involvement of spinal white matter
- ☒ **Ataxia, paresis, stumbling, toe dragging** and pivoting circumduction is more severe in the hind limbs
- ☒ Signs are usually symmetrical
- ☒ **Hypotonia of the tail and anus**
- ☒ **Faecal and urinary incontinence, dribbling of urine may soil the perineum and legs**
- ☒ Hemiplegia and paraplegia manifesting as a recumbency and inability to rise
- ☒ Less commonly lingual or pharyngeal paralysis, head tilt, nystagmus and strabismus

Necropsy finding

- ☒ Straw yellow colour fluid in the pericardial and peritoneal cavities
- ☒ Focal hepatic necrosis

Sample collection

- ☒ Nasal swab, whole blood (buffy coat) and aborted foetus

Diagnosis

- ☒ Based on clinical signs and lesion
- ☒ Hematology-leukopenia
- ☒ Isolation of virus in cell culture
- ☒ Detection of antigen by PCR
- ☒ Detection of antibodies in serum by ELISA

Differential diagnosis

- ☒ Leptospirosis
- ☒ Rabies
- ☒ Equine polyneuritis
- ☒ Trauma
- ☒ Acute spinal cord compression
- ☒ Equine degenerative myelopathy
- ☒ Neonatal septicaemia

Treatment

- ☒ There is no specific treatment

- ☒ Antibiotics should be used to prevent secondary bacterial complications

Prevention

- ☒ Modified live or inactivated vaccine available for EHV-1
- ☒ The combined inactivated vaccine available for EHV-1 and EHV-4
- ☒ The mares are vaccinated with inactivated vaccine during 5th, 7th and 9th month of gestation. Additional vaccination at breeding and one month before foaling
- ☒ Modified live vaccine to foals at 3-4 months of age, non pregnant mares, two doses administered at 3 months apart and followed by revaccination every 9 months

Control

- ☒ Enhancing immunity of individual by vaccination
- ☒ Minimising risk of introducing EHV-1 infection to the farm or stable
- ☒ Hygienic maintenance of farm to prevent spread of the disease
- ☒ Isolation of infected horse from stable immediately

Canine parvoviral enteritis

- ☒ It is a highly contagious viral disease of young dogs characterized by acute haemorrhagic gastroenteritis.

Etiology

- ☒ It is caused by canine parvovirus -2 (CPV-2) belongs to family Parvoviridae
- ☒ It is a single stranded DNA virus
- ☒ The virus is stable in the environment. They are resistant to lipid solvents and wide pH range (pH 3.0 to 9.0). The virus is inactivated by formalin, beta-propiolactone, sodium hypochlorite and oxidizing agents.

Epidemiology

- ☒ Distribution: Worldwide
- ☒ **Rottweiler, Doberman, Bull terrier, Spaniels and German shepherd** have been described to increased risk of disease
- ☒ **Young unvaccinated or incompletely vaccinated dogs at 6 wks to 6 month age group are most susceptible**
- ☒ The severity of infection increases, when dog have concurrent infection with intestinal parasitism, enteric pathogens (eg. *Clostridium* spp, *campylobacter* sp, *Giardia* spp and Coronavirus infection)

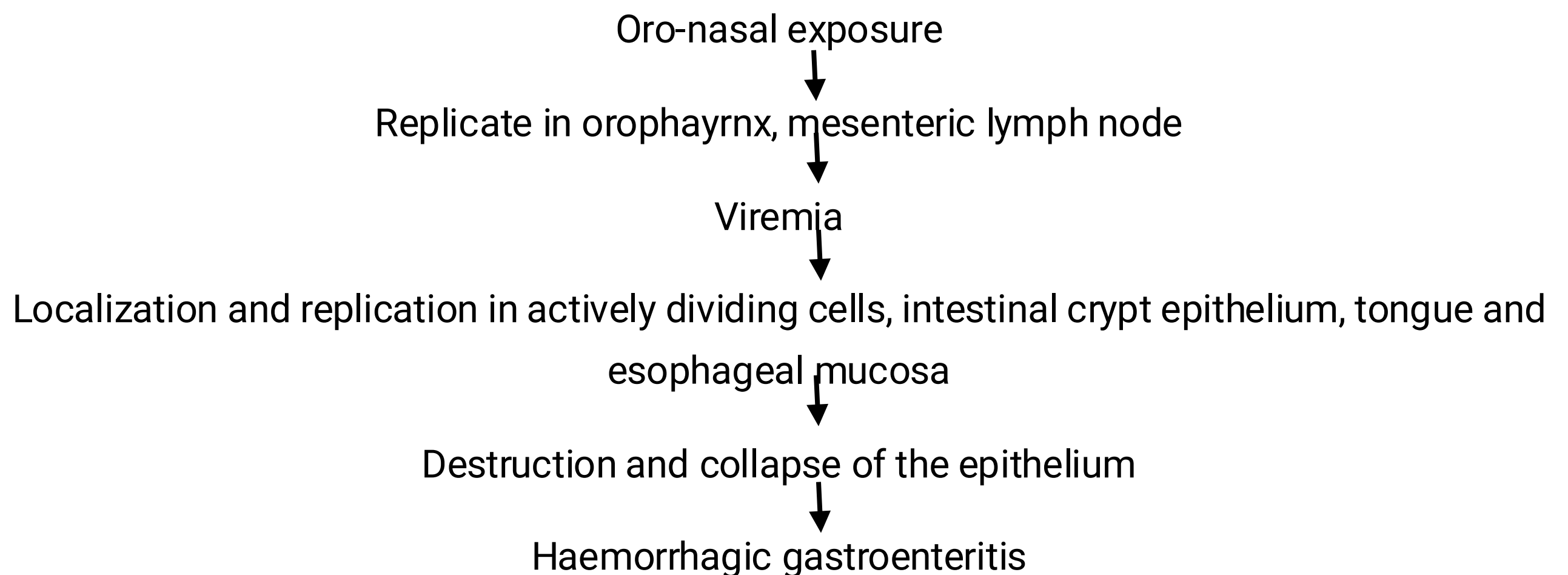
Source of infection

- ☒ Infected animal shed virus in **faeces for upto 10 days** after recovery

Transmission

- ☒ Direct contact with an infected animals
- ☒ Indirectly through contaminated fomites

Pathogenesis



Clinical manifestation

- ☒ Incubation period is 3-7 days
- ☒ Clinical signs manifested as
 - a) Parvoviral enteritis
 - b) Parvoviral myocarditis

a) Canine parvoviral enteritis

- ☒ Lethargy, anorexia, fever
- ☒ **Vomition is often severe**
- ☒ Rapid onset of dehydration
- ☒ **Haemorrhagic small bowel diarrhoea**
- ☒ Elevated rectal temperature (40°C-41 °C)
- ☒ Abdominal pain
- ☒ Severely affected animal may present collapsed , prolonged capillary refill time, poor pulse quality
- ☒ Tachycardia
- ☒ Hypothermia
- ☒ Neurological signs may occur as a result of haemorrhage in CNS or disseminated intravascular coagulation or hypoglycemia
- ☒ Immunity after natural infection is persist for 20 months

b) Canine Parvovirus myocarditis

- ☒ The myocarditis can develop from infection in utero or in pups younger than 6 weeks of age

- ☒ All pups in a litter are usually affected
- ☒ Pups with parvoviral myocarditis is often die, they showed symptoms of dyspnoea, crying and retching before death
- ☒ Cardiac dysfunction can be preceded by enteric form
- ☒ Sudden onset of congestive heart failure occur in apparently normal pups at 6 weeks to 6 months of age
- ☒ Death is due to cardiogenic shock

Necropsy finding

- ☒ Discoloured intestinal walls
- ☒ Haemorrhagic intestinal contents
- ☒ Congestion of abdominal and thoracic lymph node

Sample collection: faeces and paired sera sample

Diagnosis

- ☒ Based on clinical signs and lesion
- ☒ Clinical pathology: Leukopenia, lymphopenia and neutropenia
- ☒ Serum biochemistry: **Prerenal azotemia, hypoalbuminemia, hypokalemia, hyponatremia, hypochloremia and hypoglycemia**
- ☒ Detection of antigen in faeces by ELISA and PCR, HA&HI test
- ☒ Antibodies in serum by HI test

Treatment

- ☒ **Restoration of body fluids and electrolytes** by administration of isotonic balanced electrolyte solution
- ☒ Oral electrolyte solution is indicated in the absence of vomition
- ☒ Use antimicrobial agents to prevent secondary infections (mostly for gram negative bacteria)-Gentamicin- 6-8 mg/kg b.wt given after fully hydration, alternatively third generation penicillin or cephalosporin can be used
- ☒ Antiemetic drugs- metoclopramide @0.2-0.4 mg/ kg b.wt and ondansetron @ 0.1- 0.15 mg/ kg.b.wt
- ☒ Motility modifiers- Loperamide HCL
- ☒ Gastric protectants- Ranitidine 2-4 mg/kg b.wt
- ☒ **Whole blood or plasma transfusion @10-20 ml/ kg b.wt**
- ☒ Plasma volume expanders @10-20 ml/ kg b.wt
- ☒ **Colloids @ 20 ml/kg b.wt should be used 1/3rd of total fluid requirement to maintain plasma oncotic pressure**
- ☒ Transfusion of specific hyperimmune serum

Prevention and control

- ☒ Live attenuated vaccine administered to pups at 6 -8 weeks of age, two booster at 3-4 weeks interval after primary dose then annual revaccination every year
- ☒ Strict hygienic measures should be adopted in kennel
- ☒ Cleaning and disinfection of kennel with 1:30 bleach solution
- ☒ Segregation of infected animal from the normal healthy animals

Canine parvovirus-1 infection

- ☒ It is a minute virus of canine. The Physical and chemical properties virus is similar to CPV-2
- ☒ The disease is restricted to **pups less than 3 weeks of age**

Clinical manifestation

- ☒ Infrequently mild diarrhoea, enteritis, pneumonitis, myocarditis and lymphadenitis in pups between 5- 21days of age
- ☒ Affected pups usually have diarrhoea, vomiting and crying constantly
- ☒ Some puppies have respiratory signs without enteric signs
- ☒ Transplacental infection causes bitch failure to conceive or abortion

Necropsy finding

- ☒ Thymic edema and atrophy and enlarged lymph node

Treatment and prevention

- ☒ Mortality may be reduced by ensuring environmental temperature of new born pups kept warm, adequate nutrition and hydration
- ☒ Currently no vaccine is available

Infectious canine hepatitis

Syn : Rubarth' s disease, contagious hepatitis

Infectious canine hepatitis is a contagious viral disease of dogs, characterized by biphasic fever, congestion of mucous membrane, severe depression, leukopenia and **prolonged bleeding time (thrombocytopenia).**

Etiology

- ☒ It is caused by adenovirus- 1 (CAV-1) which is antigenically related to CAV-2
- ☒ It is a linear non enveloped double stranded DNA viruses

Epidemiology

- ☒ Distribution: Worldwide

- ☒ Dogs and foxes are mostly susceptible
- ☒ All age group of dogs are susceptible
- ☒ Dogs less than one year old is highly susceptible to acute form of the disease

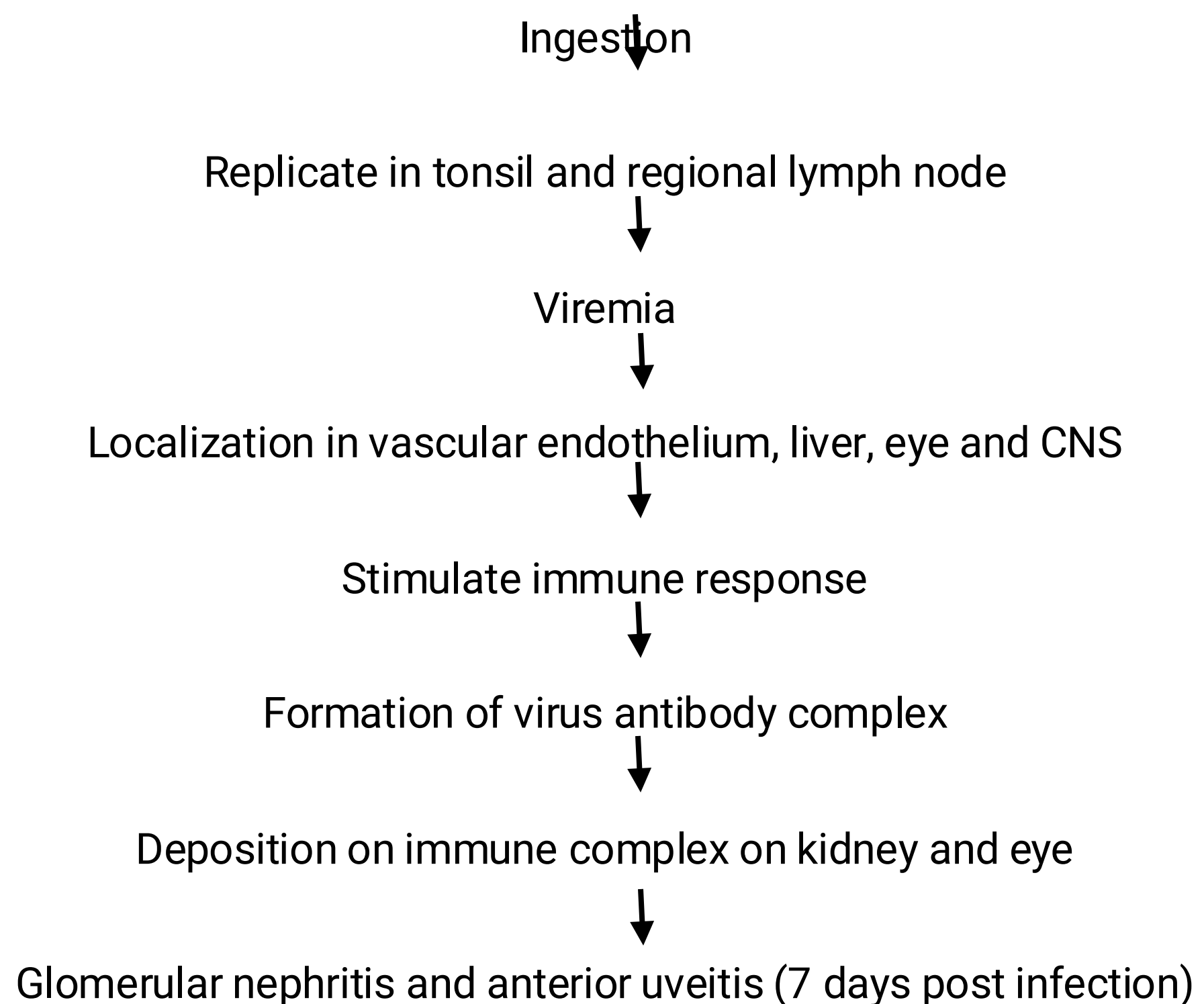
Source of infection

- ☒ The virus is excreted through **faeces, urine and saliva of infected animals.**
- ☒ The clinically recovered dogs shed the virus in their **urine for greater than 6 months**

Transmission

- ☒ Ingestion of food and water contaminated by urine, faeces and saliva of the infected animal

Pathogenesis



Anterior uveitis or corneal edema is self limiting unless additional complications **and** cleared 21 days of post infection

Clinical manifestation

- ☒ Incubation period for this disease is 6-9 day
- ☒ **Biphasic fever**

- ☒ Increased rectal temperature (103-106°F), accelerated pulse and respiratory rate
- ☒ Anorexia, **vomition, abdominal pain and diarrhoea** with or without evidence of haemorrhage
- ☒ Enlargement of tonsillar lymph node
- ☒ Coughing
- ☒ **Cervical lymphadenomegaly**
- ☒ Subcutaneous edema of the head, neck and dependent portion of trunk
- ☒ **Petechial and ecchymotic haemorrhage on buccal mucosa, haemorrhage around the deciduous teeth and epistaxis**
- ☒ Icterus is uncommon in acute form of ICH
- ☒ Abdominal distension is due to accumulation of serosanguineous fluid or haemorrhage
- ☒ Tucked up appearance of abdomen on palpation of xiphoid region
- ☒ Transient corneal opacity and edema is due to iridocyclitis. The condition is called as **blue eye**
- ☒ CNS signs include depression, disorientation, seizures and terminal coma
- ☒ Death in dog can result from damage to brain (hepatoencephalopathy), lungs and other parenchymal organs or from the development of DIC

Complication

- ☒ Dogs are more prone to **bacterial pyelonephritis** as a result of renal damage after ICH infection
- ☒ Disseminated intravascular coagulopathy
- ☒ **Decreased synthesis of clotting factors in liver is leading to bleeding defect**

Necropsy finding

- ☒ The abdominal cavity contain bright red colour fluid
- ☒ Liver is enlarged, dark and mottled appearance, fibrinous deposit on the surface of the liver
- ☒ Gall bladder is enlarged and edematous
- ☒ Enlarged spleen
- ☒ Greyish necrotic foci on the surface of kidney cortex

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Haematological findings- leukopenia, lymphopenia, neutropenia and **thrombocytopenia**
- ☒ Proteinuria (TP >50 mg/ dl), **increased concentration of protein in CSF fluid** and aqueous humor
- ☒ **Increased level of AST, ALT, Bilirubin and ammonia level in serum**

- ☒ **Hypergammaglobinemia and Hypoglycemia**
- ☒ Antigen detection by PCR and immunohistochemical staining of the tissue
- ☒ Serological tests – AGID, CFT and VNT are used to detect antibodies in serum

Treatment

Symptomatic and supportive treatment can be tried

Fluid therapy with polyionic isotonic ringers' solution

For animal with hypoglycemia, **hypertonic glucose infusion** should be continued at the rate of **0.5-0.9g/ kg/hr** for efficient utilization

Therapy is directed to reduce the ammonia concentration in blood, which is achieved by decreasing the protein catabolism ammonia and reduce the ammonia resorption in renal tubules

- a) Decreased feeding of protein diet
- b) Cleansing and evacuation of bowel content with acidifying ammonia
- c) Non absorbable antibacterial drug (Neomycin) is given to reduce the ammonia producing bacteria in the bowel
- d) Renal reabsorption is reduced by use of urinary acidifier (Ascorbic acid) and oral or parenteral administration of potassium

Blood or plasma transfusion can be tried, when an animal suffering from severe anemia

Prevention

- ☒ Vaccination of pups at 8-10 weeks and 10- 12 weeks of age with modified live vaccine and revaccination should be done every 3 years
- ☒ Combined vaccine (DHLPPi vaccine) at 6-8 weeks of age and booster dose 3-4weeks later after that annual revaccination every year
- ☒ Cleaning and disinfection of shed with 1-3 % sodium hypochlorite

Canine Distemper

Syn: Carre's disease, Hard-pad disease, Canine influenza

Canine distemper is a highly contagious, systemic, viral disease of dogs and other carnivores, clinically characterized by diphasic fever, leukopenia, GI and respiratory catarrh, and frequently causes pneumonic and neurologic complications.

Etiology

- ☒ It is caused by genus **Morbilivirus**, family Paramyxoviridae, a single stranded enveloped RNA virus

- ☒ This virus is antigenically related to rinderpest, peste des petits ruminants and measles viruses
- ☒ There is only one serotype causing this disease in canines, which contains three strains, namely A, B and C. The strains A and B were more virulent than C strain. All these strains vary in their pathogenicity and tissue tropism.
- ☒ It is relatively fragile in an environment
- ☒ The virus is sensitive to lipid solvents, such as ether, phenols and quaternary ammonium compounds

Epidemiology

- ☒ World wide in distribution
- ☒ **Canine distemper virus causes higher morbidity and mortality** than any other virus that infects dogs
- ☒ Antigenic drift and strain diversity is increasingly documented in association with outbreaks in wild species, domestic dogs, and exotic animals held in zoos and parks

Host affected

- ☒ The disease is seen in dog, fox, wolf, raccoon, ferret, mink, skunk, wolverine, marten, badger, otter binturong, palm civet, red panda, bear and large felines
- ☒ **The disease is more common in dogs at 3-6 months of age.**
- ☒ The incidence of disease is gradually decline as age increases

Source of infection

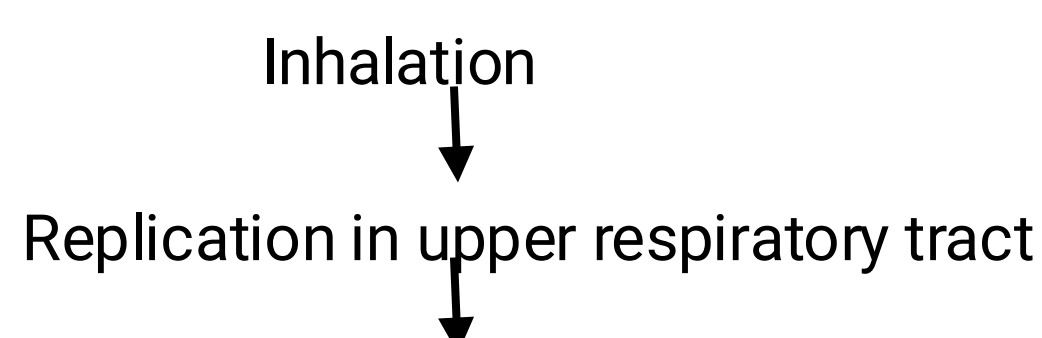
- ☒ Some infected dogs may shed virus from secretion for several months

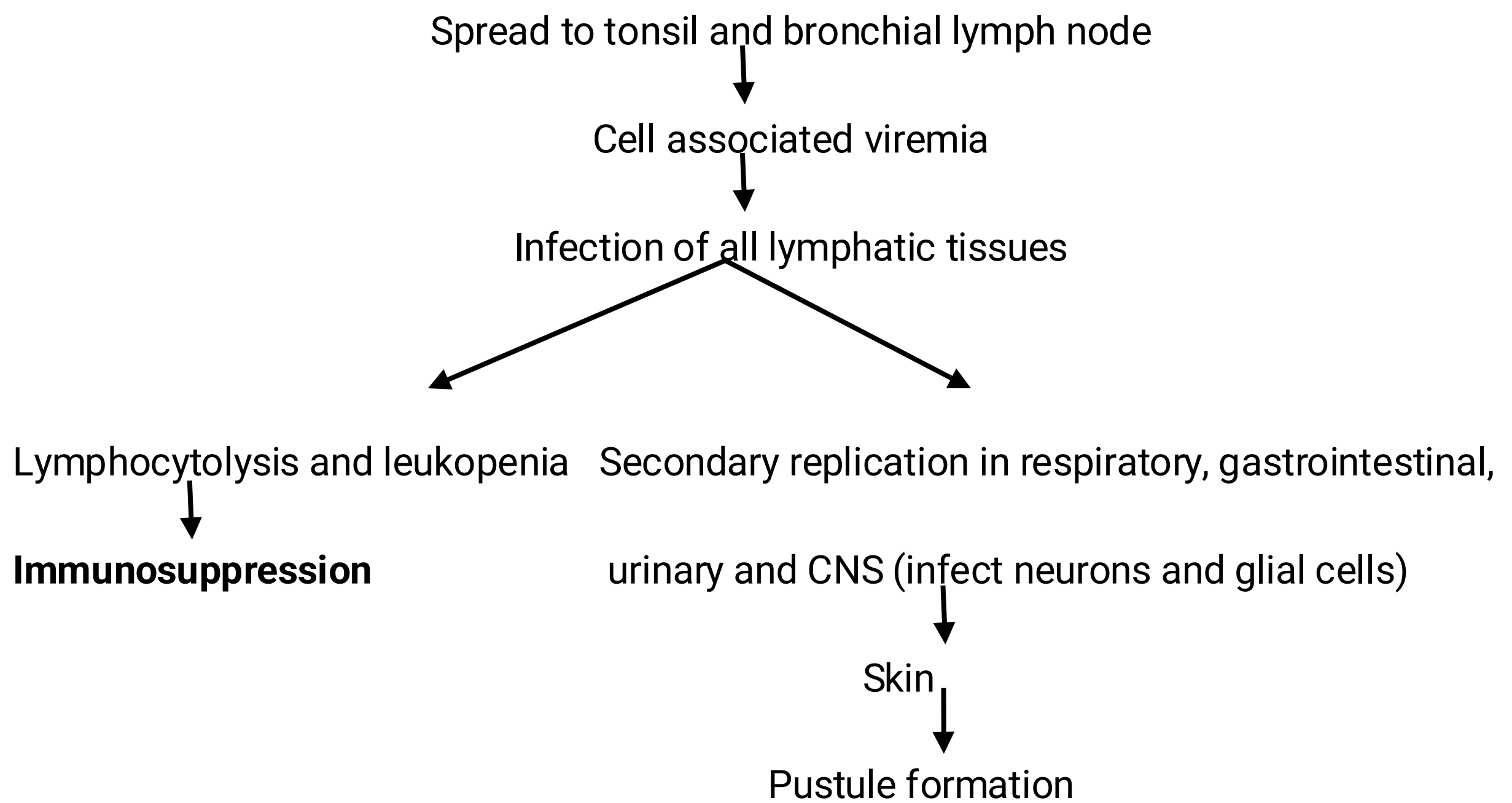
Transmission

- ☒ Inhalation of aerosol droplets secreted from an infected animal
- ☒ Direct contact with an infected animal

Pathogenesis

Virus initially replicates in the lymphatic tissue (tonsil and bronchial lymph node) of the respiratory tract. A cell-associated viremia results in infection of all lymphatic tissues, which is followed by infection of respiratory, GI, and urogenital epithelium, as well as the CNS and optic nerves. Disease follows virus replication in these tissues. The degree of viremia and extent of viral spread to various tissues is moderated by the level of specific humoral immunity in the host during the viremic period.





Clinical manifestation

- ☒ Incubation period is one week but may extend to 4 weeks or more
- ☒ The severity and duration of illness depends on virulence of virus, age and immune status of infected animal
- ☒ **Fever is biphasic** (initial elevation of temperature may not be noticed). During the second period of pyrexia, **oculonasal discharge, pharyngitis and tonsillar enlargement** are noticed
- ☒ The second rise of temperature is accompanied by rhinitis, **conjunctivitis**, gastroenteritis and **bronchopneumonia**
- ☒ Coughing, vomition and diarrhoea are consequence of secondary infections
- ☒ A **skin rash or pustules** may be present on the abdomen
- ☒ Some affected dog have hyperkeratosis of nose and foot pads so, it is called as **hard pad disease**
- ☒ Neurological signs include **paresis, myoclonus and seizures**
- ☒ Convulsions may begin as petit mal 'chewing gum fits' where the animal may salivates and makes chewing movements with the jaws before convulsions, after that convulsions become more frequent and severe resulting in grand mal epileptiform of seizures
- ☒ The course of the systemic disease may be as short as 10 days, but the onset of neurologic signs may be delayed for several weeks or months as a result of **chronic progressive demyelination within the CNS**
- ☒ **Chronic distemper encephalitis (old dog encephalitis, (ODE)**, a condition often marked by ataxia, compulsive movements such as head pressing or continual

pacing, and incoordinated hypermetria, may be **seen in fully vaccinated adult dogs** without a history suggestive of systemic canine distemper infection.

- ☒ The disease is caused by an inflammatory reaction associated with persistent canine distemper virus infection in the CNS, but mechanisms that trigger this syndrome are unknown

Clinical signs of canine distemper manifested as five different forms

Pulmonary form

- ☒ A coryza like syndrome is characterized by oculonasal discharge, pharyngitis and bronchitis are seen
- ☒ **Bronchopneumonia is the common feature**
- ☒ The pulmonary form is more prevalent than digestive form
- ☒ The pneumonic condition may persist for a long period.

Gastrointestinal form

- ☒ This form is characterized by loss of appetite, vomition and abdominal pain
- ☒ The faeces is semisolid or loose and foul smelling
- ☒ The diarrheic faeces may contain blood
- ☒ The hemorrhagic enteritis is commonly seen in young pup

Ocular form

- ☒ This is manifested as **swollen eyelids, conjunctivitis and purulent discharge from the eyes**
- ☒ Ulceration of the cornea and transient keratitis may also be observed

Nervous form

- ☒ The affection of the nerve cells lead to neurological disorders
- ☒ These are characterized by restlessness, **excitement, chewing movements, excessive salivation and convulsion**
- ☒ The nervous manifestations are noted when the animal can resist the primary infection and suffer from secondary complication
- ☒ Lymphopenia is the distinct feature of distemper but this may not occur in dogs with delayed encephalitis
- ☒ A condition known as “ old dog encephalitis” characterized by mental disorders, motor deterioration and death as clinical features
- ☒ The nervous form may be characterized by **jerky movements of group of muscles**
- ☒ The **muscular spasms may be observed in the lips, cheeks, jaws, head and neck or limb muscles**

Cutaneous form

- ☒ There is appearance of rash, vesicles and pustules on the body of the affected dog especially on the ventral aspect of the abdomen and on the inner parts of the thigh

- ☒ In some cases the skin of the foot pads and nose may become hard due to hyperkeratitis and the condition is ascribed as hard pad disease

Clinical pathology

- ☒ Lymphopenia
- ☒ Viral inclusion bodies in circulating leukocytes during very early course of the disease
- ☒ Thoracic radiographs may reveal an interstitial pattern typical viral pneumonia.

Necropsy finding

- ☒ Thymic atrophy
- ☒ Histopathology: interstitial pneumonia, intracytoplasmic and intranuclear inclusion bodies in respiratory, urinary and GI epithelium
- ☒ Non inflammatory demyelination, nonsuppurative leptomeningitis, and intranuclear inclusion bodies predominately within glial cells

Sample collection

- ☒ Smears from conjunctival, tracheal, vaginal, or other epithelium, buffy coat, urine sediments and bone marrow aspirates

Diagnosis

- ☒ Based on clinical signs and necropsy findings
- ☒ Immunofluorescent assay or reverse transcriptase (RT- PCR) are commonly used for diagnosis of canine distemper virus
- ☒ Commercially available quantitative RT-PCR and two-step RT-PCR is commonly used to distinguish vaccine strains from emerging wild-type strains
- ☒ Viral antigen immunofluorescent assay (IFA) or fluorescent in situ hybridization for viral DNA can be performed on biopsies from the footpads or from the haired skin of the dorsal neck.
- ☒ ELISA can be performed to detect antibodies in serum and CSF. The **antibody titre is relatively higher in CSF compared to blood**. It is useful to differentiate vaccinated from infected animal

Differential diagnosis

- ☒ Leptospirosis
- ☒ Infectious canine hepatitis
- ☒ Rocky mountain spotted fever
- ☒ Organophosphorus and lead poisoning

Treatment

- ☒ Treatments are symptomatic and supportive, aimed at limiting secondary bacterial invasion, supporting fluid balance, and controlling neurologic manifestations.

- ☒ Broad-spectrum antibiotics, balanced electrolyte solutions, parenteral nutrition, antipyretics, analgesics, and anticonvulsants are used, and good nursing care is essential.
- ☒ No single treatment is specific or uniformly successful.
- ☒ Experimental in vitro work with antiviral agents shows promise, but these agents have not yet been widely used
- ☒ Some dogs with chronic progressive or vaccine-induced forms of neurologic disease may respond to immunosuppressive therapy with anti-inflammatory drugs or greater dosages of glucocorticoids

Antibiotics

- ☒ Ampicillin, Amoxicillin 20 mg/kg b.wt TID for 7 days
- ☒ Doxycycline 5-10 mg/kg b.wt BID for 7 days
- ☒ Chloramphenicol 40-50 mg/kg b.wt TID for 7 days
- ☒ Cephapirin 10-30mg/kg b.wt I/M or I/V TID for 3-5 days

Anticonvulsive agents

- ☒ Phenobarbital 10-20mg/Kg IV once then followed by 2-8 mg/Kg b.wt orally
- ☒ Seizures are best managed with parenteral diazepam 0.5-2.0 mg/Kg rectally or slow IV for status epilepticus

Anti-inflammatory agents

- ☒ For CNS edema- Dexamethasone 1-2 mg/Kg b.wt
- ☒ For optic neuritis- 0.1 mg/Kg b.wt for 3-5 days

Other supportive therapy

- ☒ Parenteral antiemetics
- ☒ Lactated ringer's solution IV
- ☒ B-vitamins
- ☒ Ascorbic acid/Vitamin A 20000 IU for 5 days

Prevention:

- ☒ Pups are vaccinated with MLV vaccine at **6 wk old and at 3- to 4-wk intervals until 16 wk old and then annual revaccination every year**
- ☒ **Onderstepoort and Rockborn strain is commonly used as vaccine strain**
- ☒ The rockborn strain causes greater chance of virus induced illness than onderstepoort
- ☒ MLV vaccines should not be used in late-pregnant or early-lactation bitches. MLV vaccines can produce postvaccinal illness in some immunosuppressed dogs
- ☒ A recombinant canarypox vector vaccine expressing distemper virus proteins is licensed for use in ferrets

- ☒ Disinfection of a CDV contaminated environment can be accomplished with commonly used disinfectants because the enveloped virus destroyed outside the host.
- ☒ Measurement of **serum neutralizing antibody** titer is more accurate for withstanding the infection after vaccination
- ☒ Serum neutralizing antibody titer equal to or greater than 100 is protective when the dog received maternal antibody
- ☒ **Serum neutralizing antibody titer 20** is considered protective when measuring vaccination response (when dog not received maternal antibody through milk)

Rabies

Syn: Hydrophobia, Lyssa, Mad dog disease

It is an acute progressive viral encephalomyelitis that primarily affects carnivores and bats, although it can affect any mammals. The disease is fatal once the clinical signs are appear.

Etiology

- ☒ It is caused by Lyssa virus of family Rhabdoviridae
- ☒ It appears as filamentous, bell or bullet shaped and approximately 60-175 nm in size.
The virion is rounded at one end and flat or truncated at the other
- ☒ It is readily inactivated by sunlight, drying in air 40-70% alcohol, quaternary ammonium compounds, tincture-iodine, carbolic acid and any lipid solvents
- ☒ The virus is very resistant to autolysis and putrefaction. The virus will persist for months in infected nervous tissue in 50% glycerol. It can be preserved at – 70°C or by lyophilization

Epidemiology

- ☒ At the present time rabies occurs in most parts of the world except in Japan, UK, New Zealand, Antarctica, Australia, Hawaii islands and Switzerland. In India the incidence is very high

Host affected

- ☒ Fox, wolf, coyotes and jackal are extremely susceptible
- ☒ Guinea pig, Hamster, Bat, Mongoose, mice, rabbit, skunk and cattle are highly susceptible
- ☒ Dog, sheep, goat, horse and human are moderately susceptible
- ☒ Poultry and opossum are resistant
- ☒ Reservoir animal for rabies vary throughout the world

Transmission

- ☒ **Introduction of virus laden saliva into tissue usually by the bite of rabid animal**
- ☒ Saliva from salivary gland or brain less likely to cause infection by entering the body through fresh wounds or intact mucous membrane
- ☒ Hematogenous spread does not occur
- ☒ Aerosol transmission is not dangerous in most circumstances, however, aerosol transmission has occurred under very specialized conditions in which high concentration of virus in suspended particles

Pathogenesis

Following introduction of virus into tissue, the virus travels via peripheral nerves to the spinal cord and ascends to the brain, replication in brain causes neuronal damage. The virus spread centrifugally within nerve cell processes and is released at axonal terminal, where it infects the salivary glands. The virus sheds in saliva intermittently.

Clinical manifestation

- ☒ The incubation period in natural outbreak of dog rabies is 3-8 weeks, but it may be as short as 10 days to as long as one year
- ☒ The clinical signs of rabies is rarely definitive
- ☒ The most reliable signs regardless of species, are **acute behavioural changes and unexplained progressive paralysis**
- ☒ Behavioural changes may include sudden onset of anorexia, signs of apprehension or nervousness, irritability and hyper excitability, altered phonation and changes in temperament
- ☒ **Uncharacteristic aggressiveness- normally docile animal may suddenly become vicious**
- ☒ The clinical course may be divided into three general phases
- ☒ Prodromal, acute excitative, and paralytic end stage

Prodromal phase

- ☒ Affected animals are often confused and disoriented
- ☒ Wild animal may lose their natural fear of humans

Furious form

- ☒ The animal becomes irritable, and with slightest provocation aggressively use its teeth, claws, horns or hooves, alertness and anxiety
- ☒ Noise may invite attack
- ☒ Attacking other animals
- ☒ Eating inanimate object
- ☒ Papillary dilation

Paralytic form

- ☒ It manifest as ataxia, paralysis of throat and masseter muscles often with profuse salivation and inability to swallow
- ☒ Drooping of jaw
- ☒ Paralysis progress rapidly to all parts of the body, coma and death in few hours

Cattle

- ☒ **Abnormal bellowing**, which may continue or intermittently until before death
- ☒ Drooling of saliva
- ☒ The eye and ear follows sound and movement
- ☒ **Yawning movements are more accurately described as voiceless attempt to bellow**
- ☒ In paralytic form, knuckling of the hind fetlock, sagging and **swaying of the hindquarters while walking, deviation or flaccidity of tail to one side**
- ☒ Decreased sensation usually accompanied this weakness
- ☒ Tenesmus with paralysis of anus, resulting in the suckling in and blowing out of air

Sheep

- ☒ The clinical symptoms are similar to cattle, some animals show sexual excitement, attacking human, wool pulling, sudden falling after violent exertion, muscle tremor and salivation
- ☒ Excessive bleating does not occur

Goat

- ☒ Goats are aggressive and continuously bleating

Horse

- ☒ Evidence of distress, extreme agitation, these signs may accompanied by rolling
- ☒ Horse may bite or strike viciously
- ☒ Animal frequently have self implicated wound
- ☒ Muzzle tremors and pharyngeal paresis are common
- ☒ Sudden onset of lameness in one limb followed by recumbency, high stepping gait, blindness, paddling, convulsions and terminally paralysis

Dog

- ☒ Dog becomes irritable, restless, nervous, deprived appetite, aggressive, and often dangerous as it loses all fear of humans and bites at anything that gains its attention
- ☒ There is usually exaggerated response to light and sound
- ☒ **Dog exhibit characteristic change in its barking and may howl in an unusual tone due to the paralysis of laryngeal muscle**
- ☒ Salivation and frothing at the mouth becomes progressively more profound

- ☒ Terminally, there are often convulsive seizures, coma and respiratory arrest, with death occurring 2 to 14 days after the onset of clinical signs

Cat

- ☒ In cats the clinical signs are similar to those of dogs but last for 2-4 days before death occurs

Pig

- ☒ Tendency to attack, twitching of the nose, rapid chewing movements, excessive salivation, walk backward and terminally paralysis

Sample collection

- ☒ Live animals-Saliva, corneal/conjunctiva smear
- ☒ Dead animals- Hippocampus major, cerebellum, cerebral cortex and placed in 50% glycerol saline to preserve the virus
- ☒ No refrigeration is required

Diagnosis

- ☒ Based on clinical signs
- ☒ Demonstration of Negri bodies in impression smear collected from hippocampus major in dog and cerebellum in cattle by Sellers staining
- ☒ Negri bodies are intracytoplasmic, magenta red or cherry red spherical or oval bodies with characteristic basophilic inner granules in the cytoplasm of neurons
- ☒ Animal inoculation: For isolation of virus, the brain or other tissue specimens are prepared as 10% suspension
- ☒ Suckling mice or hamsters of 3-6 weeks age are generally used for isolation and identification.
- ☒ Route of inoculation is intracerebral. Infected mice almost always develop clinical symptoms within 17 days of inoculation
- ☒ **Direct fluorescent antibody test (FAT)** on acetone fixed brain tissue smears. It is gold standard test for diagnosis of rabies
- ☒ **Rapid fluorescent focus inhibition test (RFFIT)** rarely used to confirm diagnosis due to late seroconversion

Differential diagnosis

- ☒ Nervous ketosis
- ☒ Polioencephalomalacia
- ☒ Lead poisoning
- ☒ Lactation tetany
- ☒ Vitamin A deficiency
- ☒ Pseudorabies
- ☒ Enterotoxaemia

Treatment

- ☒ Supportive care for rabies infected animal is not recommended

Prevention and control

- ☒ For control of rabies in dog population, vaccination of minimum 70% of dog should be necessary

Vaccine types

1. Parenteral modified live vaccines
 - ☒ Early vaccines were developed for virus grown in nervous tissue and later avian embryos
 - ☒ Low egg passage vaccine (50 serial passage of virus in chicken embryo)
 - ☒ High egg passage vaccine (180 serial passage of virus in chicken embryo-safe for dog)
2. Oral modified live vaccine
 - ☒ Oral bait vaccine is used for vaccination of feral or wild animals
 - ☒ SAG2 strain is avirulent strain
 - ☒ Protective neutralizing antibody titre have been demonstrated for upto 2 years after vaccination
3. Parenteral inactivated cell culture vaccine
4. Oral or parenteral recombinant vaccine
5. Nucleic acid vaccine
6. Anti-rabies virus specific agents

Pre exposure prophylaxis

Dog

- ☒ **Inactivated cell culture vaccine administered to puppies and kitten at the age of 3 months** and repeated one year later. Subsequent vaccinations are repeated every 1-3 yrs depending on the product and local health regulations
- ☒ Vector recombinant rabies glycoprotein vaccine licensed for use in cat, which can be administered at the age of 8 weeks
- ☒ Multiple dose primary canine rabies vaccination schedule with annual boosters may be considered for dogs in rabies endemic countries

Management of suspected rabies cases-Exposure of Pets

- ☒ Any unvaccinated dog, cat or ferret exposed to rabies is euthanized immediately. If owner unwilling to do this the animal should be placed in strict isolation (i.e no human and animal contact) for 6 months and vaccinated against rabies 1 month before release
- ☒ If an exposed animal is currently vaccinated, it should be revaccinated immediately and closely observed for 45 days

Control

- ☒ Notification of suspected cases
- ☒ Euthanasia of dog with clinical signs of rabies/ dog bitten by a suspected rabid animal
- ☒ Reduction of contact rate between susceptible dogs
- ☒ Dog movement control
- ☒ Mass vaccination by campaign
- ☒ Continuing vaccination of young dogs
- ☒ Stray dog control
- ☒ Dog registration
- ☒ Control of rabies in wildlife population relies on population reduction of wildlife
- ☒ Use of oral bait vaccine for control of rabies in wild animals

Zoonotic risk

Pre exposure prophylaxis

- ☒ It is recommended for people in high risk groups such as veterinary staff, animal control officer, rabies diagnostic laboratory worker and some travellers working in canine rabies enzootic area
- ☒ Pre exposure vaccine is administered on days 0, 7 and 21 or 28
- ☒ However, in the event of subsequent rabies virus exposure and must be supplemented by a limited post exposure regimen (2 doses of vaccine I/M on days 0 and 3)

Post exposure prophylaxis

- ☒ Immediate washing and flushing wound with soap water, or water alone for 15 minutes
- ☒ Disinfection of wound with 70% ethanol, iodine (tincture or aqueous solution), or other substances with virucidal activity
- ☒ Bleeding at any wound site indicates potentially severe exposure and must be infiltrated with either human or equine rabies immunoglobulin
- ☒ Administration of antibiotics and tetanus prophylaxis
- ☒ If human unvaccinated previously infiltrate human rabies immunoglobulin 20IU at the site of bite on day 0-7 and also administer inactivated cell culture vaccine on day 0, 3, 7 and 14 and 28 I/M on upper deltoid region.

Categories of contact and recommended post exposure prophylaxis

Category	Type of contact with rabid animal	Recommended post exposure prophylaxis for people

Category I	Touching or feeding animals, licks on intact skin	None
Category II	Nibbling of uncovered skin, minor scratches or abrasions without bleeding	Immediate vaccination and local treatment of the wound
Category III	Single or multiple transdermal bites or scratches, licks on broken skin; contamination of mucous membrane with saliva from licks, contacts with bats.	Immediate vaccination and administration of rabies immunoglobulin along with local treatment of the wound

Post exposure recommendations for rabies in human

Potential source of infection	Situation	Animal disposition	Post exposure prophylaxis for people
Dog, cat, ferret	All bites, healthy, owned	Confine the animal and observe for at least 10 days, especially unprovoked attack	None or consider
	All bites, healthy, stray dog	If animal available, euthanize immediately and submit head for examination	Yes; stop if laboratory result negative or continue if animal unavailable
Wild carnivore	All bite	If animal was captured, euthanize immediately, submit head for examination	Yes; stop if laboratory result negative or continue if animal unavailable
Small rodents	All bite	Usually not examined	None, but consult public health officials

Newcastle disease

Syn: Ranikhet disease, Avian Pneumoencephalitis, Avian distemper, Doyle's disease

Newcastle disease is a highly contagious acute viral disease of domestic poultry and other birds, clinically characterized by acute respiratory disease (distress, coughing, sneezing), nervous signs (wing paralysis, inco-ordination, torticollis), bile stained greenish diarrhoea and death.

Etiology

- ☒ It is caused by **avian paramyxovirus serotype 1** (APMV-1), genus: Avula virus, family Paramyxo viridae is an RNA virus. There are 11 serotypes are recognized, but serotype 1 only causes disease
- ☒ The NDV encodes six protein (Nucleocapsid protein, phosphoprotein, Matrix protein, RNA dependent RNA polymerase and Fusion protein (contain Haemagglutinin and Neurominidase)
- ☒ APMV-2&3 causes respiratory disease in turkey
- ☒ Based on virulence and tissue tropism Newcastle disease virus classified as velogenic (virulent), Mesogenic (moderately virulent), lentogenic (low virulent)
- ☒ Velogenic and mesogenic strain mostly causes disease, the low virulent lentogenic strain is widely used as live vaccine
- ☒ The virus is comparatively stable in an environment and survives for long periods at ambient temperature, especially in faeces and can persist in houses (in faeces, dust etc) for up to 12 months
- ☒ The virus is inactivated at temperatures 56⁰C for 3 hours or 60⁰ C for 30 min, acid pH, formalin and phenol

Epidemiology

- ☒ Virulent NDV strains are endemic in poultry in most parts of Asia, Africa, and some countries of North and South America, USA and Canada
- ☒ The disease was first observed in 1926, in Java, Indonesia and in Newcastle-upon-Tyne, England
- ☒ The name Newcastle disease was coined by Doyle. In India in July 1927, the virus was recognized in Ranikhet by Edwards. Hence it named as Newcastle disease or Ranikhet disease

Host affected

- ☒ Domestic chicken is the principal host. In addition to chickens, turkeys, pheasants, guinea fowl, ducks, geese, and pigeons, psittacine birds and wide range of captive and free ranging semi domestic and free-living birds including migratory waterfowl
- ☒ NDV has been reported in 241 species of birds
- ☒ Mortality is highly variable upto 100%

Source of infection

- ☒ Infected chickens, other domestic and wild birds may be sources of NDV

- ☒ The virus is shed in all secretions and excretions (faeces, nasal discharge and dust particle) for at least 4 weeks
- ☒ Infected birds shed virus in exhaled air, respiratory discharges, and faeces
- ☒ Some psittacine species may become persistently infected with velogenic virus and excrete it intermittently for more than 1 year without showing clinical signs

Transmission

- ☒ Chickens are readily infected by **inhalation of aerosols**
- ☒ **Ingestion** of contaminated water or food with faeces of infected birds
- ☒ Infection can also spread through inanimate objects, contaminated equipments or by movement of people, by non-avian species like flying insects
- ☒ Virus may also be disseminated by frozen chickens, uncooked kitchen waste, foodstuffs, bedding, manure and transport containers

Pathogenesis

Following inhalation or ingestion of virus, the initial replication occurs in epithelium of respiratory and intestinal tract is followed by haematogenous spread to the spleen and bone marrow. Secondary viremia results in infection of other organs including lungs, intestine and CNS. The extent of viral spread within the body related to virulence of strain, which is determined by the amino acid sequence of fusion protein.

Clinical manifestation

- ☒ Incubation period is usually 5 days
- ☒ Severity of infection depends on the virulence of the infecting virus, age and immune status of host. Based on tissue tropism and virulence of virus, the clinical signs are classified into following forms

Viscerotropic velogenic (Doyle's form or Asiatic form)

- ☒ Very acute form with **hemorrhagic lesions** of the digestive tract
- ☒ Respiratory signs includes **gasping, coughing, sneezing, and rales** predominate in infections, swelling of head and neck, cyanosis of comb and wattle and greenish watery diarrhoea
- ☒ Birds survive the acute phase may develop neurological signs

Neurotropic velogenic (Beach's form)

- ☒ Infection with neurotropic strain results in respiratory disease followed by nervous signs such as paralysis of wing, leg, torticollis and muscle spasm with high mortality
- ☒ **Nervous signs with diarrhoea are typical in pigeons**, and nervous signs are frequently seen in cormorants and exotic bird species also
- ☒ Poorly vaccinated birds may develop torticollis, ataxia, or body and head tremors 10– 14 days after infection and may recover with supportive care

Mesogenic (Beaudette's form)

- ☒ Acute respiratory and some times lethal nervous infections of young chicks
- ☒ There is a loss of appetite, listlessness, abnormal thirst, huddling, weakness, and somnolence
- ☒ G.I symptom includes **crop dilatation, presence of foamy mucus and fibrinous exudates in the pharynx**, a similar discharge also noticed from **the beak**, and yellow-green diarrhoea
- ☒ In layers, there is a sudden decrease in egg production together with depigmentation and/or loss of shell and reduction in the albumen quality of eggs

Lentogenic (Hitchner's form)

- ☒ Causes mild or inapparent respiratory infections of chicks

Necropsy findings

- ☒ **Non-suppurative encephalomyelitis in CNS**
- ☒ Virulent velogenic strains cause predominantly **hemorrhagic lesions**, particularly in **trachea, oesophagus/proventriculus caecal tonsil, proventriculus-pin point haemorrhage at the tip of papillae** and in the posterior half of the duodenum, the jejunum, and ileum
- ☒ In addition to this **egg peritonitis, edematous, hemorrhagic or degenerated ovary** is also seen

Sample collection

- ☒ Live birds: **Cloacal swabs, droppings, oro-nasal swabs and paired sera sample**
- ☒ Dead birds: **Brain, bile, caecal tonsils**, liver, spleen, lung, kidney and heart tissue or pooled organ samples

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Isolation of virus in an allantoic route of inoculation of embryonated egg at 9-11 days of embryonation
- ☒ Demonstration of **viral antigen on impression smears by IFT** and other serological like **HA and ELISA**
- ☒ Demonstration of viral antigen by RT-PCR are highly valuable
- ☒ The **haemagglutination inhibition test** is used most widely used to diagnose antibodies against NDV. It is useful to assess the vaccinal immune status of the birds to be tested and on prevailing disease conditions
- ☒ The virulence of virus is confirmed by **intracerebral pathogenicity index (ICPI) or intravenous pathogenicity index (IVPI)** by intracerebral or intravenous injection of suspected sample in a day old SPF chicks

- ☒ The pathogenicity index <60 hrs- Velogenic, 60-90 hrs- Mesogenic and >90 hrs for lentogenic strain
- ☒ Nucleotide sequence analysis to detect genetic differences for comparison of isolates from different outbreaks and to identify the source of those infections.

Differential diagnosis

- ☒ Avian influenza
- ☒ Infectious bronchitis
- ☒ Infectious laryngotracheitis

Prevention and control

Vaccination

- ☒ Vaccines are available for chickens, turkeys, and pigeons and are used to induce an antibody response
- ☒ Unfortunately, **ND vaccines do not provide sterile immunity**
- ☒ There are three types of vaccine used for ND
 - Live lentogenic- F strain and Lasota**
 - Live mesogenic - Komarov, Roakin, Mukhteswar, R₂B**
 - Inactivated vaccines
- ☒ Live lentogenic vaccines are usually derived from field viruses that have been shown to have low pathogenicity for poultry but produce an adequate immune response
- ☒ Healthy chicks are vaccinated with live vaccine F strain as early as day 1– 4 of life through nares or conjunctival sac
- ☒ Live lentogenic vaccines, chiefly **B1 and LaSota strains**, are widely used and typically administered to poultry by **mass application in drinking water or by spray**. Mucosal immunity induced in birds by vaccination with live vaccines applied by these routes. These birds shed the low amount of vNDV, when infected with vNDV compared with the immune response induced by an inactivated vaccine. This type immunization is also called as priming.
- ☒ Mass vaccination methods are less labor intensive but if not applied properly may lead to <85% of the flock being immunized, (For herd immunity greater than 85% of birds in a flock should be immunized)
- ☒ Mesogenic strains are used only in areas, where ND is epidemic and in village chickens
- ☒ These vaccines are suitable only for secondary vaccination because of their high virulence
- ☒ Normally mesogenic vaccine, such as Komarov, Roakin, Mukhteswar and R₂B are used as secondary vaccines after a primary vaccination with a lentogenic vaccine

- ☒ Inactivated vaccines are usually given after an **initial priming vaccination with a live vaccine**
- ☒ **Vaccine is usually administered through water, aerosol spray, eye or nasal drops, or by injection**
- ☒ Oil-adjuvanted inactivated vaccines are also used after live vaccine in breeders and layers and may be used alone in situations where use of live virus may be contraindicated (eg, in pigeons)
- ☒ The frequency of revaccination to protect chickens throughout life largely depends on the risk of exposure and virulence of the field virus challenge
- ☒ Fowlpox or turkey herpesvirus– vectored NDV vaccines are commercially available for chickens and have the advantage of being able to be administered in ovo at the hatchery

Vaccination schedule for layer against ND

Ranikhet F strain	5-7days	Intra ocular or intra nasal
Lasota strain	5-6 weeks	Intra ocular or drinking water
Ranikhet (RDVK) Killed	10-12 weeks	Subcutaneous or wing web
Ranikhet (RDVK) Killed	18-19 weeks	Subcutaneous (before commencement of lay)
Lasota vaccination is repeated during laying period at every 6 weeks interval		

- ☒ Broilers are vaccinated with Ranikhet disease F strain at 7, and 21 days

Control

- ☒ Implementation of strict biosecurity measures
- ☒ National Level- Quarantine, slaughter and vaccination (Ring vaccination)
- ☒ Farm level- Strict hygienic practices and medical measures
- ☒ Import restrictions on chickens and eggs
- ☒ **Quarantine of psittacine birds** in the same air space as non-immune chickens.

Infectious bronchitis

Syn : Gasping disease, Blind layer

Infectious bronchitis (IB) is an acute, highly contagious viral disease of commercial chicken flocks throughout the world. It is usually characterized by respiratory signs (coughing, sneezing, tracheal rales with accumulation of mucus in the bronchi) decreased egg production and poor egg quality in breeders and layers, interstitial nephritis in association with myopathy and proventriculitis, particularly in chicks.

Etiology

- ☒ Infectious bronchitis virus (IBV), genus Coronavirus is a member of the family Coronaviridae. It is a positive stranded linear ssRNA virus
- ☒ The severity of disease are influenced by the strain of the virus, age and immune status of host and concurrent infection with *Mycoplasma gallisepticum*, *Mycoplasma synoviae*, *Escherichia coli*, and/or *Avibacterium paragallinarum* can exacerbate disease, diet of the chickens and cold stress

Epidemiology

- ☒ This disease is worldwide in distribution
- ☒ In recent years, a novel IBV genotype, the **QX strain**, has become increasingly common in Asia and Europe
- ☒ Chicken is the only bird that is usually infected by IBV, although some other birds may be subclinically infected
- ☒ All age group of birds are susceptible, but the disease is most severe in baby chicks
- ☒ Morbidity is commonly close to 100% and mortality rate is 5%

Source of infection

- ☒ Infected chickens shed the IBV in **respiratory discharges and faeces**
- ☒ Naturally infected chickens and those vaccinated with live IBV may shed virus intermittently for up to 20 wk after infection

Transmission

- ☒ Inhalation of aerosol
- ☒ Ingestion of contaminated feed and water, and contact with contaminated equipment and clothing

Clinical manifestation

- ☒ The incubation period of IB is 18-36 hour depending on the route of inoculation
- ☒ The characteristic respiratory signs of IB in chicks are **gasping, coughing, sneezing and have a tracheal rales and nasal discharges for 10-14 days**
- ☒ Wet eyes may be observed and occasional swollen sinuses
- ☒ The chicks appear depressed, huddled under heat source. The feed consumption and weight gain are significantly reduced
- ☒ In chickens over 6 weeks of age and in adult birds the signs are similar to those in chicks, but nasal discharge does not occur as frequently and the disease may go unnoticed unless the flock is examined carefully by handling the birds or listening to them at night when the birds are normally quiet
- ☒ Infection with nephropathogenic strains can cause initial respiratory signs, then later depression, ruffled feathers, **wet droppings, increased water intake**, and death

- ☒ Nephropathogenic strains can induce **interstitial nephritis** with high mortality (up to 60%) in young chicks
- ☒ In laying flocks there is drop in egg production upto 70% and **misshapen eggs with thin, soft, rough or pale shell with watery albumin** are seen in addition to respiratory signs
- ☒ The **egg albumin may be thin and watery without definite demarcation** between the thick and thin albumen
- ☒ In most cases, egg production and egg quality return to normal, but this may take up to 8 wks
- ☒ Infection of young chicks may cause permanent damage to the oviduct, resulting in layers or breeders that never reach normal levels of production. These birds are called as **blind layers**.

Necropsy findings

- ☒ Infected chickens have serous, catarrhal or caseous exudates in the trachea, nasal passages and sinuses
- ☒ Airsacs may appear cloudy or contain yellow caseous exudates
- ☒ A caseous plug may be found in the lower trachea or bronchi of the chicks
- ☒ When birds infected at very young age may have cystic oviducts, whereas those infected while in lay have an oviduct of reduced weight and length and regressed ovaries
- ☒ In affected layer birds fluid yolk material may be found in the abdominal cavity
- ☒ Infection with nephropathogenic strains results in swollen, pale kidneys and distension of the tubules and ureters with urate crystals

Sample collection

- ☒ Paired sera sample, tracheal swab, trachea, caecal tonsils and kidney tissue

Diagnosis

- ☒ Based on clinical history and lesions
- ☒ Isolation of virus by inoculation of homogenates of tracheal, cecal tonsil, and/or kidney tissue into 9- to 11-day-old SPF chicken embryos. The growth of IBV indicated by embryo stunting and curling, and deposition of urates in the mesonephrons with variable mortality
- ☒ Alternatively, isolation of virus by tracheal organ cultures, the growth of virus indicated by cessation of ciliary motility
- ☒ The MAbs have been best used after propagation in chicken embryos, to detect viral antigen in the chorioallantoic membranes by immunofluorescence or immunoperoxidase staining, or in the allantoic fluid by ELISA

- ☒ Rapid detection of viral antigen by RT-PCR, and detection of specific serotype and strain by restriction fragment length polymorphism analysis, analyzed by nucleotide sequencing
- ☒ The S1 region of the spike glycoprotein gene determines the serotype of virus
- ☒ Typing of viruses can help to distinguish vaccine strain from field strains and may help to diagnose outbreaks caused by serotypes distinct from those of the vaccines used in the flock
- ☒ Antibodies in serum is diagnosed by HI test

Differential diagnosis

- ☒ New castle disease
- ☒ Infectious laryngotracheitis

Management procedures

- ☒ No medication alters the course of IBV infection
- ☒ Antimicrobial therapy may be used to reduce caused by secondary bacterial infection
- ☒ In cold weather, increasing the ambient temperature may reduce mortalities
- ☒ **Reduce protein concentration in feed and provide electrolytes in drinking water,** when outbreaks caused by nephropathogenic strains

Vaccination

- ☒ Both live and inactivated virus vaccines used for IB immunization
- ☒ Live vaccines are used in broilers and for initial vaccination of breeders and layers
- ☒ **Inactivated oil emulsion vaccines are used primarily at point of lay in breeders and layers**
- ☒ **Live vaccine administered individually by eye drop, intra tracheal or intranasal route**
- ☒ Mass application methods include administration of vaccine as coarse spray, aerosol and drinking water
- ☒ Inactivated vaccines require injection of individual birds. These vaccines are usually given after priming with live virus and are administered a few weeks before production commences
- ☒ The most commonly used live vaccines in the USA contain derivatives of the **Massachusetts**, Connecticut, and Arkansas strains
- ☒ In Australia- VicS and Armidale strains
- ☒ In Europe- 4/91 strain and those derived from QX-like viruses

Time of vaccination

- ☒ Two weeks (14 days) of age is frequently used as the time for initial immunization
- ☒ Booster doses administered at 28 days and 16-18 weeks of age.

Control measures

- ☒ Strict isolation and repopulation with only day old chicks, following the cleaning and disinfection of the poultry house

Infectious laryngotracheitis

Syn: Haemorrhagic tracheitis

It is a viral respiratory tract infection of chicken that may result in severe production losses due to mortality or decreased egg production. The diseases occur as epizootic and mild enzootic form.

The epizootic form of infection is characterized by severe dyspnoea, gasping and expectoration of bloody mucus and high mortality.

The mild enzootic form of infection is characterized by nasal and ocular discharge, mucoid tracheitis, sinusitis, conjunctivitis, general unthriftiness and low mortality.

Etiology

- ☒ Gallid herpesvirus 1 is belonging to the sub family Alphaherpesvirinae, family Herpesviridae
- ☒ It is a double stranded DNA virus
- ☒ Latency is a common characteristic feature of this virus
- ☒ The virus is sensitive to detergents and lipid solvents

Epidemiology

- ☒ Chicken is the primary natural host for this virus.
- ☒ Birds of all age groups are affected but this disease is mostly common in adult birds
- ☒ Mortality may reach 50%

Transmission

- ☒ Natural portals of entry for infectious laryngotracheitis virus (ILTV) are through the upper respiratory and ocular routes
- ☒ Ingestion is also a mode of infection
- ☒ Transmission occurs more readily from acutely infected birds than through contact with clinically recovered carrier birds shed the virus intermittently

Clinical manifestation

- ☒ Incubation period is 6-12 days
- ☒ Nasal discharge, moist rales followed by **gasping and coughing, rattling noise and extension of neck during inspiration** are seen
- ☒ Anorexia, depression, reduced egg production

- ☒ **Mouth and beak may be blood stained from the tracheal exudates**
- ☒ Bird infected with low virulent strain causes mild respiratory signs and reduced egg production, unthriftiness, watery eyes, conjunctivitis, swelling of infra orbital sinuses and persistent nasal discharge
- ☒ Recovered bird act as carrier and become source of infection for susceptible birds

Necropsy finding

- ☒ Blood, mucus and yellow caseous exudates in trachea
- ☒ Microscopically desquamative, necrotizing tracheitis

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Isolation of virus in chicken embryo by chorioallantoic route of inoculation
- ☒ Cell cultures of choice are Chicken embryo liver cells (CEL), and Chicken kidney cells. A maximum of two passages in CEL and CK cell cultures required
- ☒ Histopathology-Intra nuclear inclusion bodies in tracheal epithelium
- ☒ Detection of viral antigens by PCR

Differential diagnosis

- ☒ Diphtheritic form of fowl pox
- ☒ New castle disease
- ☒ Infectious bronchitis
- ☒ Fowl adenovirus
- ☒ Aspergillosis

Prevention and control

- ☒ Rearing of birds under dust free environment
- ☒ Mild expectorant can be used
- ☒ Vaccination should be practiced in endemic area
- ☒ Vaccination of adult birds with modified live attenuated vaccine shorten the course of disease during an outbreak
- ☒ Layer flocks generally vaccinated twice before the onset of egg production. The modified live attenuated vaccine is typically administered by eye drop at 7 weeks of age and again at 15 weeks of age by eye drops, spray, drinking water
- ☒ Recombinant fowl pox and herpes virus of turkey viral-vectored vaccines expressing genes of ILT virus are commercially available for in ovo, subcutaneous and wing web inoculation at least 8 weeks of age or 4 weeks prior to onset of egg production

Avian influenza

(Syn: Fowl plague)

Avian influenza (AI) is an infection of domestic poultry as well as pet, zoo, and wild birds caused by influenza A virus of the H5 or H7 subtypes. These viruses are divided into high pathogenicity avian influenza (**HPAI**) viruses and low pathogenicity avian influenza (**LPAI**) viruses based on **intravenous pathogenicity index (IVPI)**. High pathogenicity avian influenza (HPAI) viruses have an intravenous pathogenicity index (IVPI) in six-week-old chickens were greater than 1.2 or, as an alternative, cause at least 75% mortality in four-to eight-week-old chickens infected intravenously. Low pathogenicity avian influenza (LPAI) viruses are all influenza A viruses of H5 and H7 subtypes that are not have high pathogenicity.

Etiology

- ☒ It is caused by Type A Influenza virus, genus Orthomyxoviruses of family Orthomyxoviridae. It is a single stranded , segmented RNA virus
- ☒ AI viruses are further divided into **16 hemagglutinin (H1-16)** and **9 neuraminidase (N1-9)** subtypes based on hemagglutinin inhibition and neuraminidase inhibition tests respectively
- ☒ Most AI viruses (H1-16 subtypes) are of low pathogenicity, but some of the H5 and H7 AI viruses are highly pathogenic for chickens, turkeys, and related gallinaceous domestic poultry
- ☒ **New subtype of influenza A virus emerge periodically as a result of point mutation (antigenic drift) and genetic reassortment**
- ☒ The virus is labile in the environment and are sensitive to heat, pH Changes, lipid solvents and oxidizing agents

Epidemiology

- ☒ Avian Influenza (AI) viruses have been circulating worldwide for centuries with four known outbreaks recorded in the last century. The present wave of highly pathogenic avian influenza (HPAI) emerged in Hong Kong in 1997
- ☒ In India first outbreak of avian influenza reported on 18.02.2006 in states of Maharashtra, Gurajarat and followed by second outbreak in Madhya Pradesh in March-April-2006, subsequent outbreaks have been recorded in Manipur, West Bengal, Tripura, Assam, Sikkim, Odisha, Meghalaya, Karnataka, Bihar, Chhattisgarh, Kerala and Chandigarh and Kerala since 2007-2015
- ☒ LPAI viruses are distributed worldwide and are frequently recovered from apparently normal ducks, shorebirds and migrating waterfowl. HPAI viruses arise from mutation of some H5 and H7 low pathogenic viruses and cause devastating epidemics

- ☒ In most wild birds, AI viral infections are subclinical except for the recent H5N1

Host affected

- ☒ Chicken, ducks, turkeys, guinea-fowl, goose, quail , pet, zoo, and wild birds

Source of infection

- ☒ The **ducks and migratory water fowl are source of infection**; these birds excrete the virus through **faeces**
- ☒ Faeces and nasal discharge of infected chicken

Transmission

- ☒ **Ingestion of food and water contaminated with the faeces and respiratory secretion** of infected birds
- ☒ Through contaminated equipments and clothing
- ☒ Air borne transmission may be important over limited distance

Pathogenesis

Spread of influenza virus in tissue is depends on type of protease in host tissue and structure of haemagglutinin molecule. In the majority of influenza A virus subtype haemagglutinin is cleaved by host protease such as trypsin like enzymes found only in the epithelial cells of the respiratory and digestive tracts. This is facilitating the development of generalized infection.

Clinical manifestation

Highly pathogenic avian influenza (HPAI)

- ☒ Sudden onset of unusual mortality
- ☒ In acute cases, respiratory distress, sinusitis, lacrimation, **cyanosis and edema of the head, comb, wattle, and snood (turkey); edema and red discoloration of the shanks and feet** due to subcutaneous ecchymotic hemorrhages; petechial hemorrhages on visceral organs and in muscles, and blood-tinged oral and nasal discharges
- ☒ In severely affected birds, **greenish diarrhoea is common**. Birds that survive the peracute infection may develop CNS involvement evident as **torticollis opisthotonos, incoordination, paralysis, and drooping wings**

Low pathogenic avian influenza (LPAI)

- ☒ LPAI viruses typically produce respiratory signs such as sneezing, coughing, ocular and nasal discharge and swollen infraorbital sinuses
- ☒ **Sinusitis is common in domestic ducks, quail, and turkeys**
- ☒ Decreased egg production and fertility in layers
- ☒ A few layer and breeder chickens may have acute renal failure and visceral urate deposition (visceral gout)

- ☒ Lesions in the respiratory tract typically include congestion and inflammation of the trachea and lungs
- ☒ The morbidity and mortality is usually low unless accompanied by secondary bacterial or viral infections or aggravated by environmental stressors

Zoonotic risk

- ☒ Human get infection through direct contact with live or dead infected poultry, but a few cases have resulted from consumption of uncooked poultry products, defeathering of infected wild swans
- ☒ Respiratory infection has been the most frequent presentation of human H5N1 cases

Sample collection

- ☒ Oropharyngeal, tracheal and cloacal swabs and faeces (approximately 1g) preserved in viral transport medium or tissue culture medium for virus isolation

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Isolation of virus from clinical samples by allantoic route of inoculation at 9-11 days of embryonated egg
- ☒ **Haemagglutination test** for detection of virus in clinical sample
- ☒ Intravenous pathogenicity test should be done in chicken at 4-8 weeks of age to assess pathogenicity of the virus
- ☒ Genome sequencing can be used to determine the amino acid composition at the cleavage site of haemagglutinin molecule to assess pathogenicity
- ☒ Detection of antibody in serum by **haemagglutination inhibition test**
- ☒ Agar gel immunodiffusion test (AGID) and competitive ELISA can also be used for antibody detection
- ☒ Reverse transcriptase- polymerase chain reaction (RT-PCR) and real time polymerase chain reaction techniques are commonly used for rapid detection of virus in clinical sample

Differential diagnosis

- ☒ New castle disease
- ☒ Septicaemic pasteurellosis
- ☒ Water deprivation syndrome

Prevention and control

- ☒ Currently, only inactivated whole AI virus, recombinant fowlpox-AI-H5, and recombinant herpesvirus-turkey-AI-H5 (rHVT-AI-H5) vaccines are licensed in the USA.
- ☒ The use of any licensed AI vaccine requires approval of the state veterinarian. In addition, use of H5 and H7 AI vaccines in the USA requires USDA approval.

- ☒ Treating of birds infected with LPAI with broad-spectrum antibiotics to control secondary pathogens and increasing house temperatures may reduce morbidity and mortality
- ☒ Treatment with antiviral compounds is not approved or recommended.
- ☒ Suspected outbreaks should be reported to appropriate regulatory authorities

Biosecurity measures

- ☒ Restriction of entry of vehicles inside the farm
- ☒ Restriction of movement of people inside the farm and other birds in the farm
- ☒ The personnel working inside the farm should wear protective clothing all the time inside the farm, including face-masks and gloves, gumboots
- ☒ The infected farm premises area should be invariably **disinfected by spraying disinfectants like 2% Sodium hypochlorite or 4% Formalin to reduce the virus load**
- ☒ Treating all the equipment with 2% sodium-hypochlorite solution for 48 hrs
- ☒ Use 2% NaOH or KMnO_4 solution at the entrance of the premises
- ☒ White-washing of concrete areas with lime
- ☒ Cages and other large metal structures may be decontaminated by heat treatment (flame gun)
- ☒ Water-reservoirs must also be emptied, washed and disinfected
- ☒ Feed tanks (silos) need to be emptied, washed with a hot water-pressure pump and subsequently fumigated
- ☒ Wash hands and feet of farm workers and the visiting officials with soap and disinfectant with approved detergent or rectified spirit
- ☒ Use virucidal agent Vircon-S®, D-125, Trilocid concentrate for cleaning of floor

Fowl pox

Syn: Avipox, avian diphtheria, variola gallinarum

Fowl pox is a slowly spreading viral disease of chicken, characterized by development of discrete nodular proliferative skin lesions on the non-feathered part of the body (cutaneous form) or fibrino-necrotic and proliferative lesions in the mucous membrane of upper respiratory, mouth and oesophagus (Diphtheritic form).

Etiology

Avipox virus of family poxviridae, a single linear double stranded DNA with hairpin loop at each end. The virus is inactivated by 1% caustic potash and resistant to chloroform and ether. The virus can survive in dried scab for months or even years.

Epidemiology

- ☒ **Mortality rate:** Cutaneous form- low and Diphtheritic form- high
- ☒ Avipox virus infection of canaries and finches can significantly cause high rate of mortality than those of chicken

Transmission

- ☒ Mechanical transmission of virus to injured or lacerated skin
- ☒ Inhalation or ingestion of virus

Clinical signs

- ☒ Incubation period: 4-10 days in chicken, turkey and pigeon, and about 4 days in canaries

Cutaneous form (Dry pox)

- ☒ Characterized by appearance of **nodular lesions on the comb, wattle, eyelids and non feathered area of the body**
- ☒ Cutaneous eye lesions will interfere with birds ability to reach food and water leads to starvation and death

Diphtheritic form (Wet pox)

- ☒ Cankers or **diphtheritic yellowish lesion** occur on the mucous membrane of the **mouth, oesophagus or trachea** with accompanying coryza like mild or severe respiratory signs similar to ILT
- ☒ Lesions in the corner of the mouth, on the tongue, throat and upper part of the throat interfere with eating, drinking and breathing
- ☒ Emaciation, poor weight gain
- ☒ Unthriftiness and reduced egg production

Necropsy finding

- ☒ Characteristic cutaneous form of lesion- epithelial hyperplasia (scab in head, feet and legs)
- ☒ Diphtheritic form- white opaque nodule or yellowish patches develop on the mucous membrane of the mouth, oesophagus, tongue and upper trachea
- ☒ Histopathology-intracytoplasmic eosinophilic inclusion bodies (borrel bodies) on the affected tissue

Sample collection

- ☒ Live birds-Scab in 10% glycerol saline and paired sample
- ☒ Dead birds- trachea and oesophagus

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Isolation of virus in embryonated egg by chorioallantoic route of inoculation (10-12 days)
- ☒ Isolation of virus in cell culture

- ☒ Molecular diagnosis of viral antigen by PCR and Restriction fragment length polymorphism (RFLP) is used to identify the strain of virus
- ☒ Antibody detection by AGID, VNT and ELISA

Differential diagnosis

- ☒ Infectious laryngotracheitis

Prevention and control

- ☒ Fowl pox vaccine is commonly applied to chicken by the wing web method at 6-8 weeks and also to be administered as early as 4 week old. The pullet birds are vaccinated 2 months before egg production is expected to start. Vaccination should not be done when birds in laying
- ☒ Live virus vaccines used for immunization of birds against pox. Vaccines of fowl pox and pigeon pox virus origin is routinely used for vaccination of chicken and turkeys
- ☒ Turkeys are vaccinated at 2-3 months old; breeders should be revaccinated before egg production. Site: midway of thigh stick or wing web method
- ☒ Fowl pox vaccine should not to be used in pigeons
- ☒ Recombinant fowl pox vaccine is also available for vaccination

Avian encephalomyelitis

Syn: Epidemic tremor

Avian encephalomyelitis is an acute viral disease of very young chickens (1-21 days) characterized by **ataxia, paralysis, tremors in head and neck**.

Etiology

It is caused by genus Enterovirus in the family Picornaviridae, a single stranded RNA virus. The virus is susceptible formaldehyde fumigation and inactivation with β -Propiolactone. There are two distinct pathotypes of virus namely **field strain (enterotropic)** and shed in faeces, **embryo adopted strain is highly neurotropic** and causes severe neurological signs.

Host affected

- ☒ Chicken, turkeys, Japanese quail, pheasants and pigeon
- ☒ Morbidity rate: 40-60% and Mortality rate: 25% and may exceed 50%

Source of infection

- ☒ The very young chicks excrete the virus through faeces for more than two weeks

Transmission

- ☒ **Vertical transmission**

- ☒ Horizontal transmission (Ingestion is the portal of entry of virus)

Clinical manifestation

- ☒ Dull expression of eyes
- ☒ Progressive ataxia from incoordination of muscle
- ☒ Sit on their hocks, lateral recumbency, fine tremor of head and neck
- ☒ Chicks with marked ataxia and prostration are frequently trampled by their penmates
- ☒ Reduced egg production and hatchability
- ☒ Opacity of lens, either unilateral or bilateral
- ☒ Nervous signs may be seen at or soon after hatching, but most commonly at 1 week of age

Pigeon

- ☒ Paralysis of the wings, opisthotonus, torticollis and head tremor
- ☒ Diarrhoea

Necropsy findings

- ☒ Whitish area in the muscularis of ventriculus (due to mass of infiltration of lymphocytes)
- ☒ CNS- disseminated non-suppurative encephalomyelitis, perivascular infiltrate in all portion of brain and spinal cord are seen

Sample collection: Brain, pancreas and duodenum

Diagnosis

- ☒ Isolation of virus by inoculation of tissue homogenate in yolk sac route 5-7 days of embryonated egg- muscular dystrophy is evident in embryo
- ☒ Nested PCR is used to detect viral antigen in clinical sample
- ☒ VNT and ELISA are used for detection of antibodies in serum

Control

- ☒ Live and inactivated vaccines are available
- ☒ The live vaccine most commonly used, this given in drinking water
- ☒ Vaccination by wing web inoculation of AEV practiced, but this some risk of causing clinical disease
- ☒ Generally vaccination is done after 8 weeks of age and at least 4 weeks before egg production

Infectious bursal disease

Syn: Gumboro disease, Avian bursitis, Infectious nephritis.

Infectious bursal disease is an acute highly contagious viral disease of young chicken that affect lymphoid tissue primarily bursa of fabricius characterized by high mortality in chicks and prolonged **immunosuppression**.

Etiology

- ☒ It is caused by an infectious bursal disease virus genus Avibirnavirus a member of the family Birnaviridae. It is a non enveloped double stranded segmented RNA virus. It is very stable in an environment and difficult to eradicate from premises. The severity of infection is depends on age, breed of chicken and virulence of virus.
- ☒ The virus contains two segments A and B. The segment A encodes viral structural proteins VP2, VP3, VP4 and VP5. The VP2 elicits neutralizing antibody response. The segment B encodes VP1-involved in replication and transcription of virus and thus playing some role in the virulence of the virus. The strains of IBDV are Variant, classical and very virulent strain.

Epidemiology

- ☒ It was first recognized as a specific disease entity by cosgrove in 1962 and was referred to as avian nephrosis because of extreme kidney damage succumb to infection
- ☒ The first outbreak occurred in the area of Gumboro, Delaware, so it is called as Gumboro disease
- ☒ In India this disease was first reported by Mohanty *etal* (1971) in Uttar Pradesh.
- ☒ Chicken are the only natural hosts for the virus in which natural infection occurs
- ☒ All breeds are susceptible but, white leghorns breeds are more susceptible than other breeds
- ☒ Variant IBDV induce little mortality with marked bursal lesions
- ☒ Classical IBDV- induce-10-15 % mortality
- ☒ Very virulent infectious bursal disease virus (vvIBDV) induce approximately 50-100% mortality

Source of infection: The virus shed in faeces of infected bird

Transmission

- ☒ Ingestion of contaminated feed and water with faeces of infected birds
- ☒ Fomite transfer from infected shed to non infected shed

Clinical manifestation

- ☒ It occur as subclinical and clinical infection
- ☒ **Subclinical infection occur in less than 3 weeks of age**
- ☒ Clinical infection occur at 3-6 weeks of age, but severe infection occurred in white leghorn chicken upto 18 weeks old
- ☒ Early subclinical infection is most important than clinical infection

- ☒ **Subclinical infection causes severe long lasting immunosuppression** due to destruction of immature lymphocytes in bursa of fabricius, thymus, spleen and atrophy of bursa
- ☒ The **humoral (B cell) immune response is severely affected** and cell mediated (T cell) immune response also affected to lesser extent, then bird become immunocompromised and highly susceptible to other infection

Clinical infection

- ☒ Incubation period is 2-3 days after exposure to virus
- ☒ Chickens exhibit severe prostration, incoordination, **whitish or watery diarrhoea, soiled vent feathers, vent picking and inflammation of the cloaca**, anorexia, depression, ruffled feathers, trembling and death
- ☒ Affected birds became dehydrated and subnormal temperature in terminal stages
- ☒ Recovery occur within one week
- ☒ Morbidity 100%, mortality 20-30%

Necropsy finding

- ☒ Bursa is highly enlarged twice or 2-4 times of its normal size after 3 days of infection. The affected bursa is swollen, edematous yellowish in colour and occasionally have haemorrhage
- ☒ Haemorrhage at the junction of proventriculus and gizzard
- ☒ Congestion and haemorrhage of the pectoral, thigh and leg muscle
- ☒ Recovered bird has small atrophied cloacal bursa due to destruction and lack of bursal follicle after 12th day of infection

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Isolation by virus in embryonated chicken egg by inoculation into chorioallantoic route
- ☒ Isolation of virus in cell culture (cells from cloacal bursa) or chicken embryo fibroblast
- ☒ The ELISA is most commonly used for evaluation of antibodies in serum. ELISA does not differentiate between antibodies to serotype1 and serotype 2
- ☒ **Virus neutralization test** is used to identify antibodies to different serotype of IBDV
- ☒ Quantitative agar gel precipitation test used for detection of antibodies in serum

Differential diagnosis

- ☒ Coccidiosis
- ☒ Nephropathogenic strain of IBV
- ☒ Water deprivation syndrome

Treatment

- ☒ No therapeutic or supportive treatment has been found to change the course of the disease

Prevention and Control

- ☒ Rigorous disinfection of contaminated farms after depopulation
- ☒ Live vaccine of chicken embryo or cell culture origin (low pathogenic virus) can be given through eye drops or drinking water route at 1-21 days of age
- ☒ Breeder flocks should be vaccinated one or more times during growing period, first with live vaccine and again before egg production with **oil adjuvanted inactivated vaccine**. Which **induce more uniform and persistent level of antibody** than live vaccine
- ☒ The immune status of breeder flocks should be monitored by periodical testing of flock with VNT and ELISA
- ☒ Revaccination is followed to maintain adequate immunity to progeny, when antibody level is fall down
- ☒ IBD subunit vaccine has been attempted mainly from baculovirus. A baculovirus expressed recombinant VP2 protein has been used commercially in broiler breeds, and immunity transferred to their progeny.

Marek's Disease

Syn: Range paralysis, Skin leukosis, Neural lymphomatosis, Gray eye

Marek's disease is a lymphoproliferative disorder of chicken is characterized by **oncogenic transformation of T lymphocyte** that infiltrate into lymphoid and other visceral organs.

Etiology

- ☒ It is caused by Alphaherpesvirus, belongs to family Herpes viridae, which contain 3 serotypes
- ☒ Serotype 1 – oncogenic, serotype 2 & 3 is non oncogenic in nature
- ☒ The serotype 3 is called as turkey herpes virus (HVT)
- ☒ Serotype 1 contain variable pathotypes namely mild (m), virulent (v), very virulent (vv) and very virulent plus (vv+)
- ☒ The virus is stable in an environment and inactivated by a variety of common chemical disinfectants within ten minute treatment period
- ☒ All the herpes viruses are sensitive to ether. The infectivity of dried MDV infected feather is destroyed by chlorine, organic iodine, quaternary ammonium compound, cresylic acid, synthetic phenol and sodium hydroxide

Epidemiology

- ☒ Distribution: Worldwide
- ☒ Chicken, turkey and quail are commonly affected
- ☒ **The disease most commonly occur between 12- 30 weeks of age**
- ☒ White leghorn breeds are highly susceptible than other breeds
- ☒ Transportation, vaccination, handling and beak trimming are stress factors that increase susceptibility to diseases
- ☒ The concurrent infection with **Infectious bursal disease (IBD)**, **Chicken infectious anemia (CIA)** and presence of **high level of aflatoxin in feed** increases the susceptibility to disease

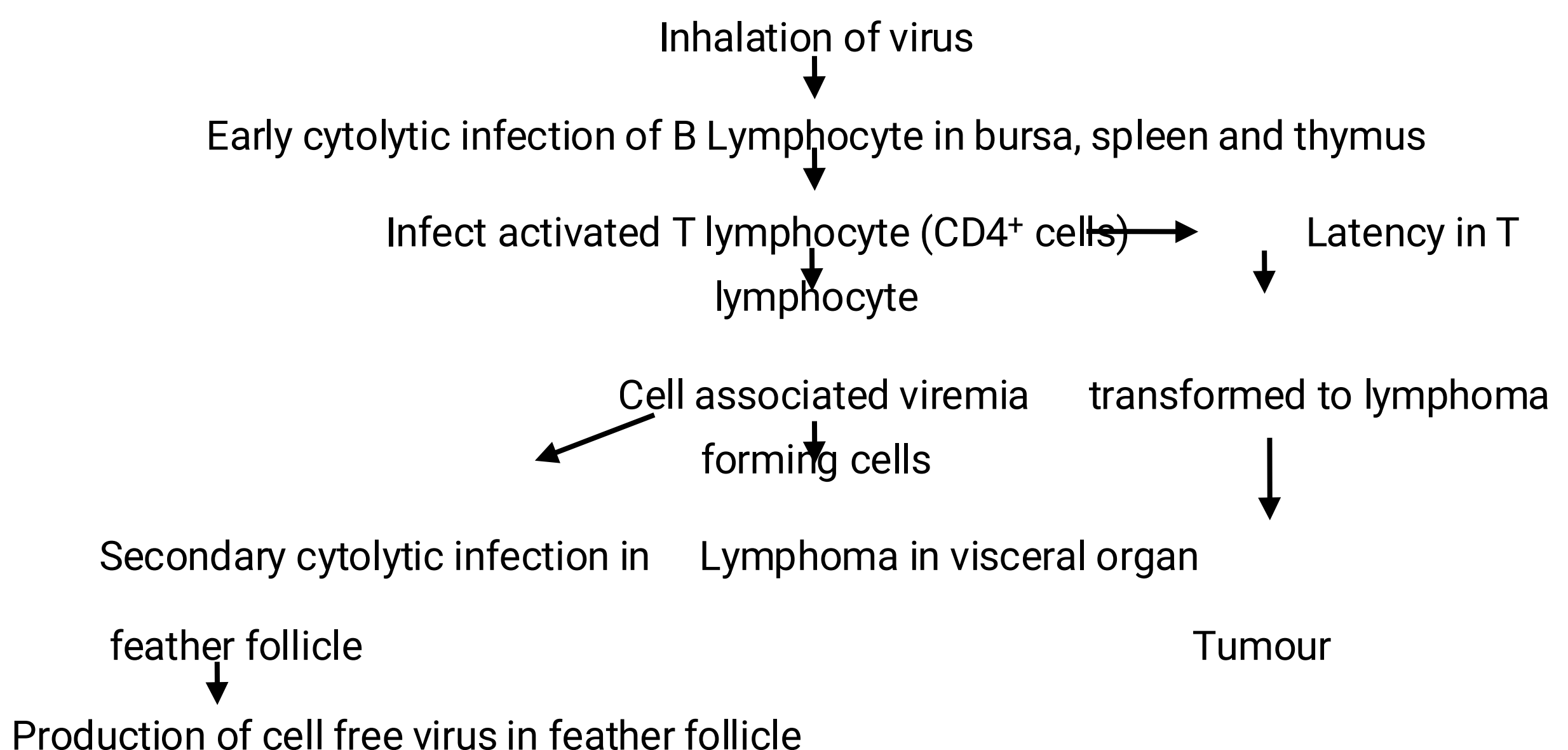
Source of infection

- ☒ Feather follicle, dander, litter, the virus is infective for several months
- ☒ Infected birds become carrier for life and remain source of infection by excretion of virus through feather follicle

Transmission

- ☒ Inhalation of virus from feather follicle, dander and dust in poultry house environment
- ☒ Darkling beetle can transmit the virus mechanically

Pathogenesis



Note: **Latently infected T lymphocyte transformed into lymphoma forming cells**

MATSA: Marek' s Disease Tumour Associated Surface Antigen

Clinical manifestation

- ☒ It occur as classical form and acute form

Classical form

- ☒ Partial or complete paralysis of leg and wings results in **stretching of one leg forward and another backward**
- ☒ Torticollis occur when nerve controlling muscle is affected
- ☒ Involvement of vagus nerve results in paralysis and **dilatation of crop**. These bird shows symptom of gasping and respiratory distress

Acute form

- ☒ Usually lymphoma formation in visceral organ
- ☒ Generalized signs are depression, weight loss, anorexia and diarrhoea
- ☒ Mortality is 10-30 % and in major outbreaks can go upto 70%

Acute cytolytic disease

- ☒ Infection caused by very virulent plus strain
- ☒ Severe atrophy of lymphoid organs
- ☒ Acute cytolytic form also called as early mortality syndrome
- ☒ Mortality is high usually occur between 10-14 days of age

Transient paralysis

- ☒ Uncommon manifestation of MDV infection occur between 5-18 weeks of age
- ☒ Sudden development of varying degree of ataxia, paresis or paralysis of the legs, wings and neck

Immunosuppression

- ☒ Loss of lymphocyte as a consequence of virus replication
- ☒ Virus induced changes in regulation of immune responses
- ☒ Tumour cell induced immunosuppression

Clinical manifestation of Marek's disease in different forms

Ocular form

- ☒ Blindness in birds due to mononuclear cell infiltration in the iris causing “ **grey eye**” or “ **pearl eye**”

Skin or cutaneous form

- ☒ Distinct white nodules on the skin and in extreme cases looks like brownish nodules

Muscular form

- ☒ Superficial and deep muscles like pectoral muscles affected
- ☒ Whitish grey and there are tiny white streaks to nodular tumours in the muscles.

Necropsy finding

- ☒ Enlargement of vagus, brachial or sciatic nerve, loss of striations and glistening nerve
- ☒ Nodular lymphoid tumour in liver, spleen, gonads, heart, lung, kidney and proventriculus and feather follicle of skin

- ☒ Acute cytolytic form- extensive atrophic changes in the thymus and bursa of fabricious may result in complete disappearance of the thymus and bursa

Sample collection

- ☒ Feather follicle, buffy coat and visceral organ (tumor growth)

Diagnosis

- ☒ Based on clinical symptoms and necropsy finding
- ☒ Histopathological examination- pleomorphic infiltration of lymphocytes and presence of MD cell in liver and spleen
- ☒ The infection is usually detected by inoculating live buffy coat cells on to monolayer cultures of chicken kidney cells or duck embryo fibroblasts, in which characteristic viral plaques develop within a few days
- ☒ Detection of antigen by AGID and PCR
- ☒ Detection of antibody by AGID

Differential diagnosis

- ☒ Lymphoid leukosis

Treatment: There is no specific treatment

Prevention and control

- ☒ **Vaccination of day old chicks with either monovalent (HVT) or bivalent (HVT+SB-1) live vaccine with correct dose**
- ☒ CVI988/Rispen strain used to protect against very virulent plus (vv+) strain of serotype 1
- ☒ In ovo- vaccination of chicks at 18 days of embryonation
- ☒ Proper handling, thawing and reconstitution to ensure adequate dose to be administered
- ☒ Recombinant vaccine is available for fowl pox and Marek' s disease virus
- ☒ Recently, marek' s disease virus lacking meq oncogene I confer protection against challenge with vv+ strain of MDV
- ☒ Adequate cleaning and disinfection of poultry house before introduction of chicks
- ☒ Rearing of birds in all – in all-out system
- ☒ Selection of genetically resistance strain
- ☒ Avoid feeding of aflatoxin level in feed above permissible level (>0.02 ppm)

Avian leukosis

Syn: Avian sarcoma, Big liver disease, visceral lymphoid leukosis, lymphomatosis

The avian leukosis sarcoma viruses cause erythroid, lymphoid and myeloid leukosis, a variety of other tumours, such as fibrosarcomas, hemangiomas, nephroblastoma and bone disorder osteopetrosis

- ☒ **Lymphoid leukosis- common in layers**
- ☒ Myeloid leucosis and erythroid leukosis- common in meat type birds (Broilers)

Etiology

- ☒ It is caused by an Alpharetrovirus of family Retroviridae
- ☒ The viral structural protein p27 is major group specific antigen encoded by gag gene. Which is common to all viruses of leukosis or Sarcoma group
- ☒ ALSV in chickens are classified into six subgroups (A, B, C, D, E & F) on the basis of differences in their viral envelope glycoprotein gp 85
- ☒ Most of the viruses causing avian leukosis are exogenous. Endogenous ALGVs, which mostly belong to subgroup E and occur commonly in chickens and other avian species are of little or no pathogenicity

Host affected

- ☒ Chicken are natural host for the virus, the virus also isolated from pheasants, partridge and quail
- ☒ Adult bird greater than 16 weeks of week old are commonly affected

Lymphoid leukosis

Lymphoid leukosis is a common neoplasm caused by **ALV subgroup A and B**. ALV-J has weak ability to cause lymphoid leukosis in layers.

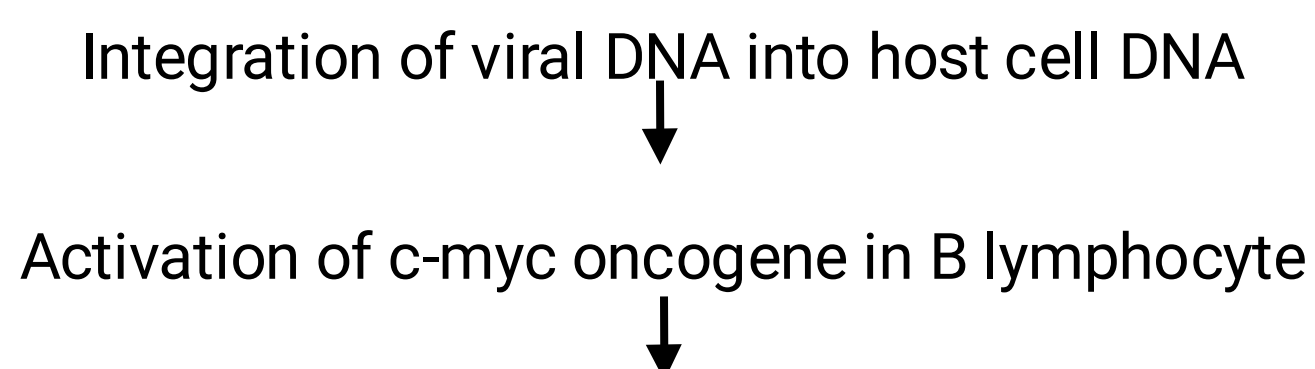
Host affected

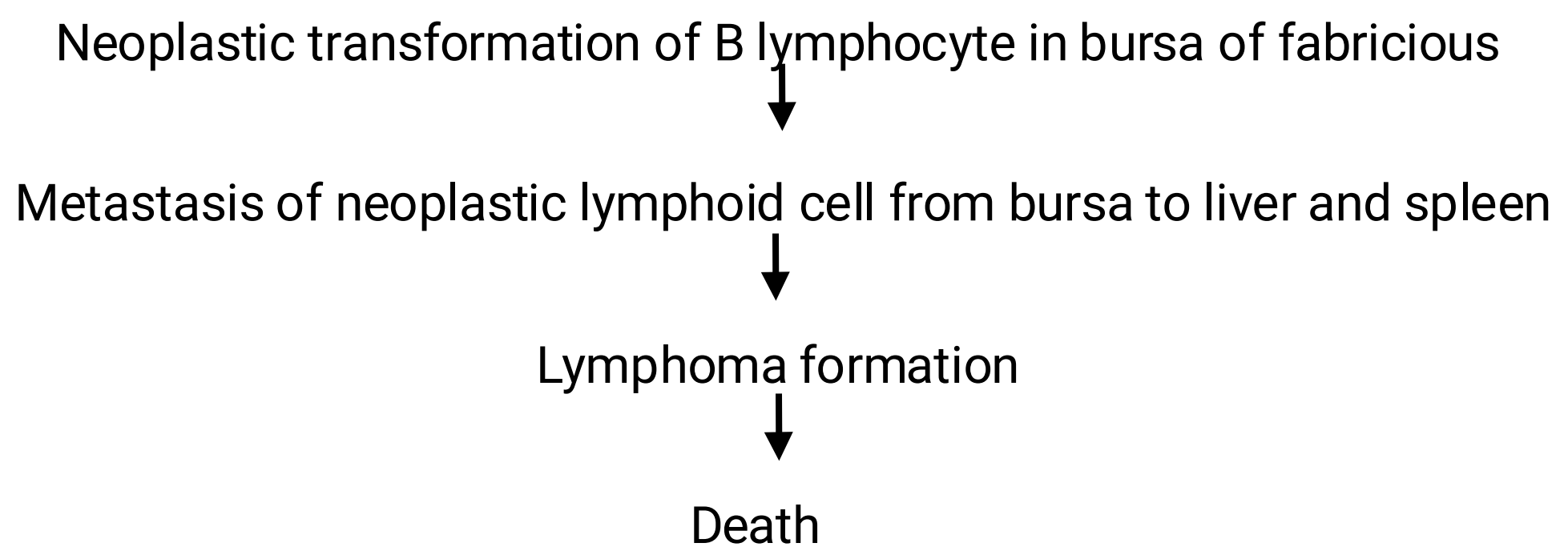
- ☒ Chicken are natural host for the virus, the virus also isolated from pheasants, partridge and quail.
- ☒ **Adult bird greater than 16 weeks of week old are commonly affected**

Transmission

- ☒ **Vertical transmission- most common (from hen to progeny through the egg)**
- ☒ Horizontal transmission through faeces of infected chicks to susceptible host
- ☒ Congenitally infected chicks develop immunological tolerance to the virus and fail to develop neutralizing antibodies and permanently shed the shed organism.

Pathogenesis





Note: ALV infection in the absence of neoplastic disease can adversely affect egg production traits in layers, including age at first egg production, number, size, fertility and hatchability.

Broilers- suppress the growth rate

Clinical manifestation

- ☒ The incubation period is variable and may be as long as months or years
- ☒ Unthriftiness, inappetence, weakness, diarrhoea, dehydration and emaciation
- ☒ **Pale comb**
- ☒ **Enlarged abdomen is due to tumour growth in liver**
- ☒ Infected chicken is depressed before death
- ☒ Reduced egg production occur in subclinical infection

Necropsy finding

- ☒ Diffuse enlargement of liver and nodular tumour is common in liver, spleen and bursa and occasionally found in kidney, gonads and mesentery
- ☒ Histopathological examination- uniform infiltration large lymphoblast cells in liver, spleen and bursa

Sample collection: Buffy coat, egg white, paired sera sample, liver, spleen and bursa of Fabricius

Diagnosis

- ☒ Isolation of virus in cell culture
- ☒ Histopathological examination
- ☒ Cytological examination of May-Grunwald-Giemsa stained impression smear of fresh tumour
- ☒ Detection of antigen by RT-PCR
- ☒ Detection of antibodies by ELISA

Control

- ☒ Eradication of avian leukosis virus from primary breeding flocks
- ☒ Breeder flock is evaluated by **shedding of virus in albumin of egg by ELISA**

- ☒ Eggs from shedder are discarded
- ☒ Selection of genetically resistance line for breeding
- ☒ There is no commercial vaccine

Erythroid leukosis (Erythroblastosis)

- ☒ It is caused by ALV- J subgroup
- ☒ Meat type bird (broilers) are commonly affected
- ☒ Adult chickens are mostly affected
- ☒ Anemia with immature RBCs in the blood
- ☒ Disease originate from bone marrow
- ☒ In Malignant form- the cells remain within the blood vessels throughout the course of the disease
- ☒ Erythrostasis in sinusoids of liver, spleen and bone marrow imparting cherry red colour
- ☒ Liver and spleen is moderately enlarged

Myeloid leukosis (Myeloblastosis and Myelocytomatosis)

- ☒ It is caused by ALV- J subgroup
- ☒ It involves extravascular and intravascular proliferation of myeloid (granulocytic series)
- ☒ Myeloblastosis – liver and spleen are diffusely enlarged, the liver frequently as a granular, morocco leather appearance
- ☒ Myelocytomatosis- tumour is nodular or diffuse have a creamy white colour. The tumour growth occur in liver, spleen, kidney, flat bones of rib, skull, sternum and pelvis

Osteopetrosis

- ☒ It is a sporadic disease that occurs primarily in males
- ☒ The shafts of the long bones are thickened, often resulting in a stilted gait
- ☒ Affected birds may be anemic and also have lesions as that of lymphoid leukosis.

Inclusion body hepatitis/hydropericardium syndrome

Syn: Hepatitis hydropericardium

- ☒ It is an acute disease of young chickens associated with anemia, hemorrhagic disorders and hydropericardium
- ☒ It is caused by **avian adenovirus** the virus contain 12 serotype among which **serotype 4 and 8 causes disease**
- ☒ **Broilers at 3-7 weeks are severely affected and mortality rate is high in broiler**

- ☒ Sometimes layer and breeder pullets also affected. The disease was first reported in Pakistan in the year 1987 and first reported in India in the year 1994
- ☒ The IBH is rare in turkey, characterized by sudden increase in mortality
- ☒ If concurrent infection with IBD and CIA will result in a more severe disease
- ☒ Source of infection-The virus is excreted in the faeces for about 3 weeks, with peak excretion between 4 and 7th day after infection
- ☒ **Transmission** -Vertical and horizontal transmission

Clinical manifestation

- ☒ Sudden onset of mortality is usually seen in chicken less than 6 weeks of age and as young as 4 days old
- ☒ Lethargy, huddling with ruffled feathers and crouching position
- ☒ Yellow mucoid dropping
- ☒ Depression of overall feed conversion ratio and weight gain
- ☒ Duration of infection is 9-14 days

Necropsy findings

- ☒ Straw coloured transudate in pericardial sac
- ☒ Generalized congestion
- ☒ Enlarged pale and friable liver

Diagnosis

- ☒ Based on clinical signs and lesions
- ☒ Detection of AAV by PCR
- ☒ Detection of antibodies in serum by double immunodiffusion test (DID)
- ☒ Histopathology- basophilic intranuclear inclusion bodies in liver and myocardium

Sample collection: Faeces, pharynx, kidney and affected organ

Treatment: There is no specific treatment

Prevention and control

- ☒ Live attenuated and inactivated vaccines are available
- ☒ Breeder flock should be vaccinated at 10-14 weeks of age with live attenuated vaccine or inactivated vaccine
- ☒ Live attenuated vaccine should be given in drinking water
- ☒ Vaccination of chicks less than 10 days of age, if chicks lack maternal antibodies

Fasciolasis

Syn: Liver fluke, liver rot, Pipe stem liver

Fasciolosis is one of the widespread causes of mortality in sheep and cattle characterized by weight loss, **anaemia and hypoproteinaemia**. It is caused by *Fasciola hepatica* and *Fasciola gigantica*. The *Fasciola hepatica* found in temperate areas and in cooler areas of high altitude in the tropics and subtropics. The *F. gigantica* which predominates in tropical areas. **Fasciolosis rarely fatal in cattle, but often fatal in sheep.**

Host affected

Definitive host

- ☒ Most mammals, sheep and cattle are the most commonly affected than other ruminant animals.
- ☒ Unusual host: Horse & Man

Intermediate hosts (Amphibious snail)

- ☒ *Fasciola hepatica*: ***Lymnea truncatula*** is most common amphibious snail is found throughout the world
- ☒ *Fasciola gigantica*: *L.auricularia*, *L.accuminata* and *L.rufescens* (India & Pakistan)

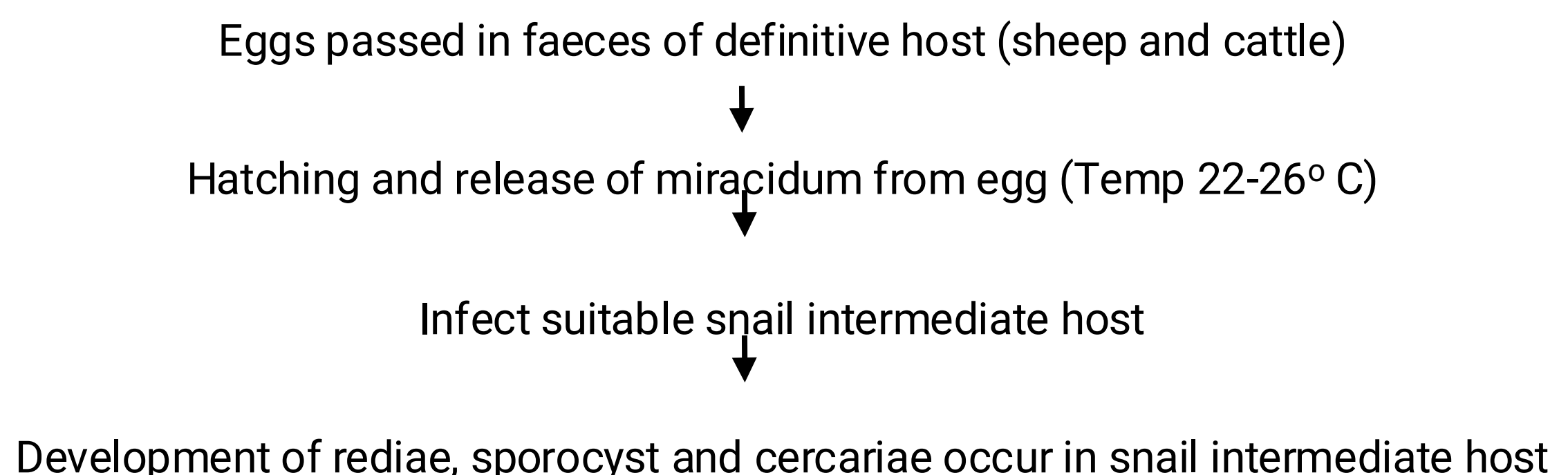
Location of parasite in host

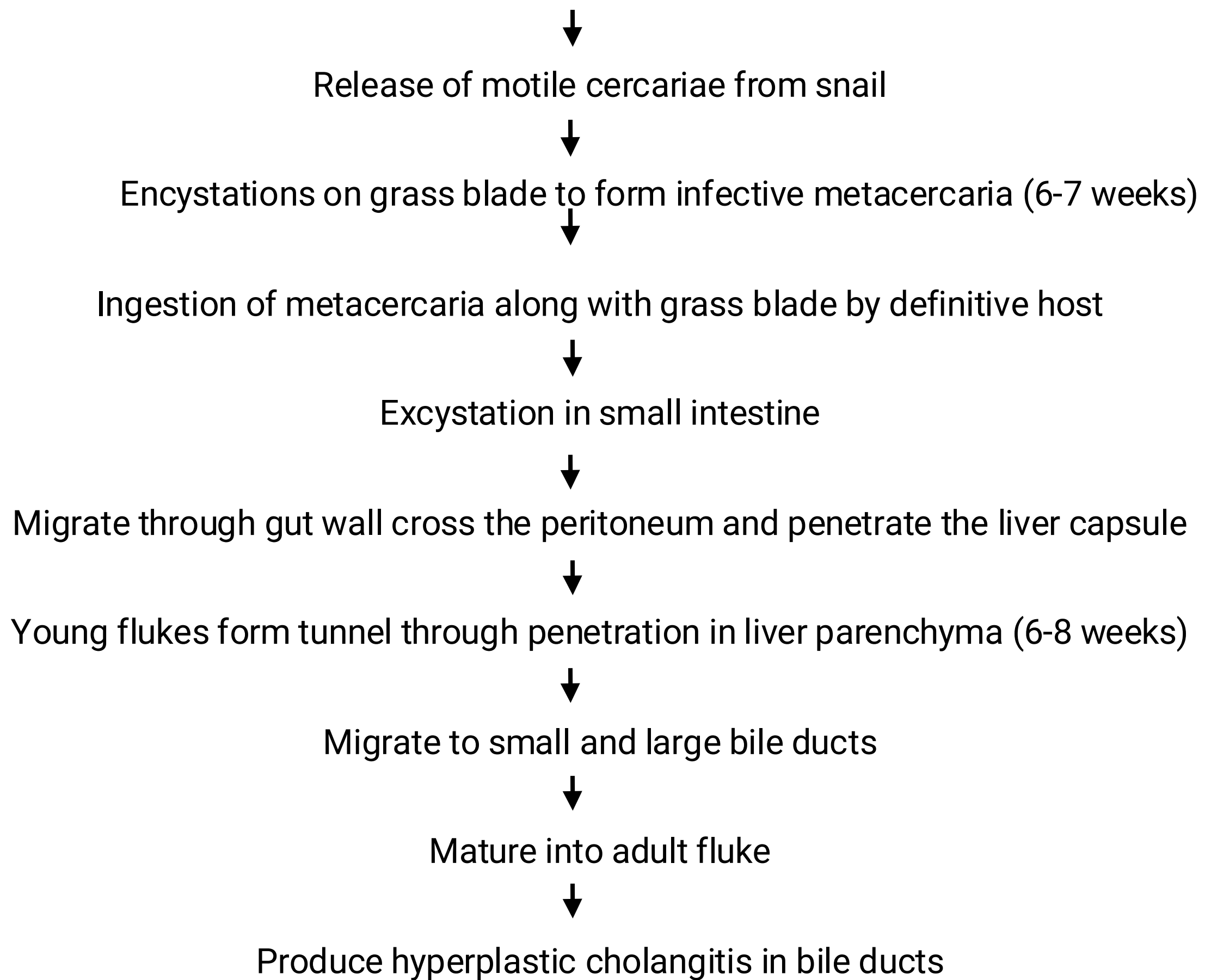
- ☒ Adult fluke- Bile ducts
- ☒ Immature fluke-liver parenchyma
- ☒ Occasionally aberrant encapsulated fluke found in other organ such as lungs

Epidemiology

- ☒ Snail habitats: ***L.truncatula*** found in wet mud to free water, banks of ditches or streams and edge of small ponds. Temporary habitat in hoof marks of animals (clay soil)
- ☒ Temperature: A mean day/night temperature >10° C is necessary both for snail to breed and development of *F. hepatica* with moist environmental condition
- ☒ **Summer infection of snails**, in which metacercariae appear on pasture from August to October
- ☒ **Winter infection of snails** in which metacercariae appear on the pasture in May to June

Life cycle





The minimum period for completion of entire life cycle is 17-18 weeks

Prepatent period: 10-12 weeks

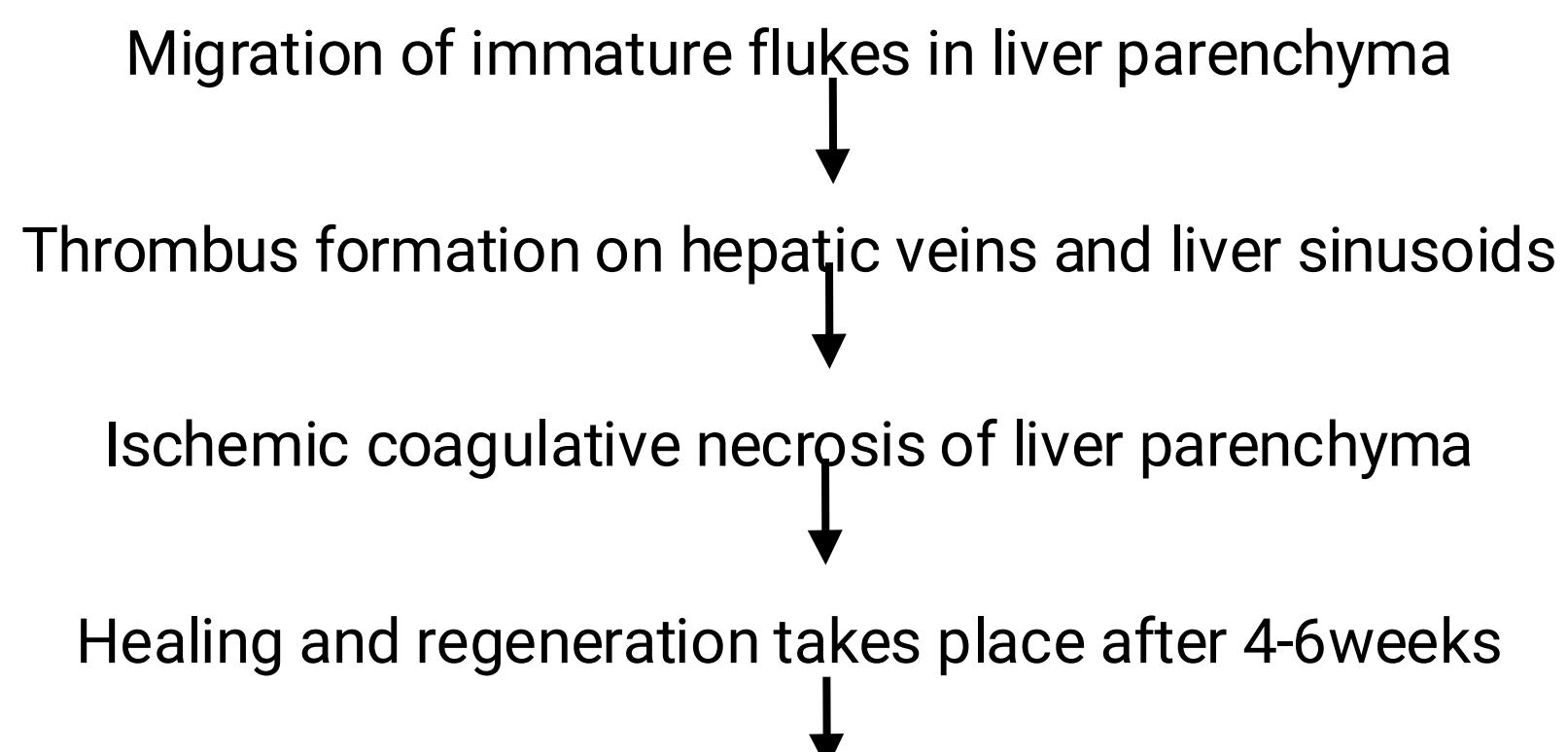
Pathogenesis

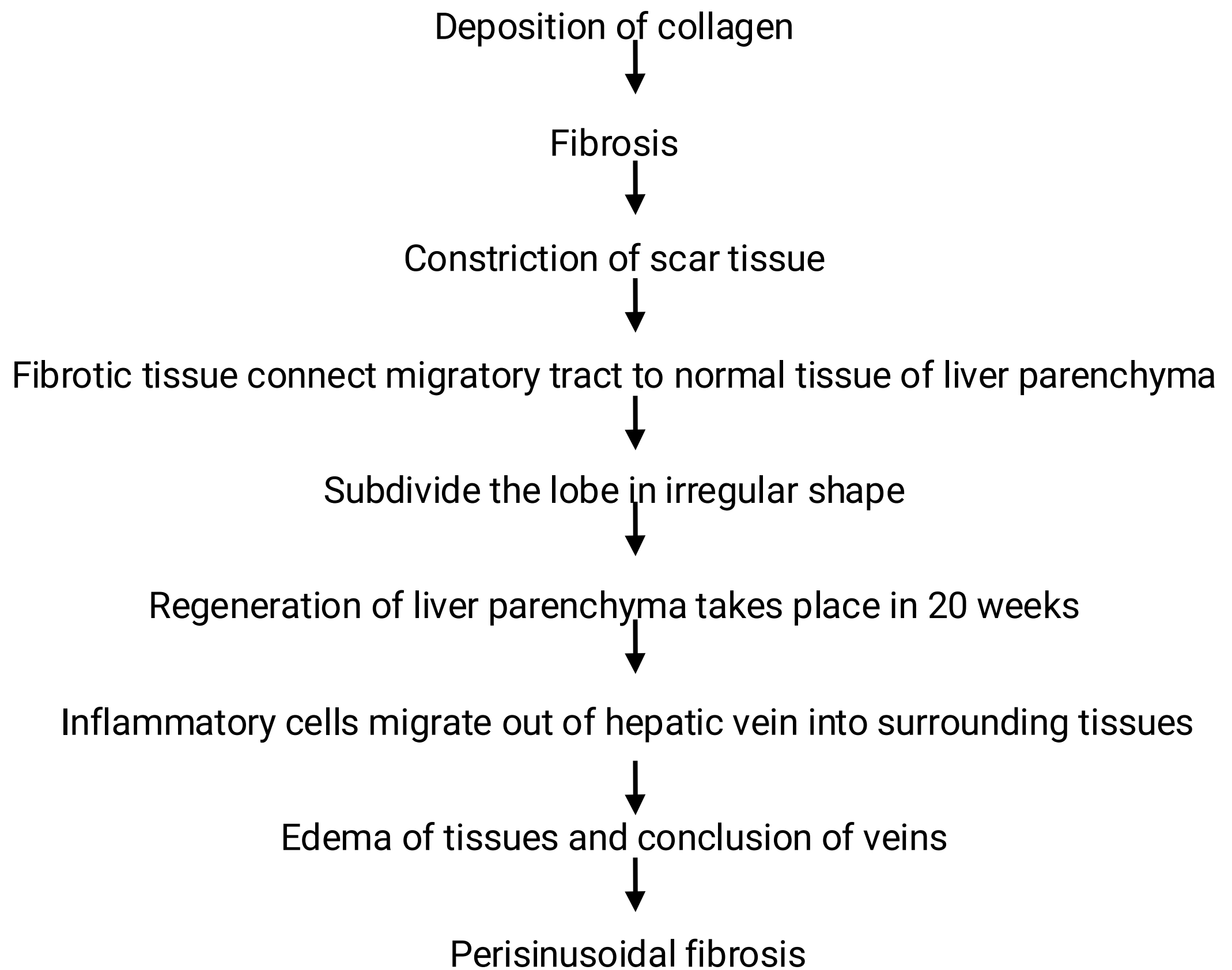
Acute

It occurs in two phases

- ☒ First phase- Migration of immature fluke in liver parenchyma results in liver damage and haemorrhage
- ☒ Second phase- Presence of adult fluke in bile duct causes damage to biliary mucosa by cuticular spine leads to cholangitis

Chronic





Clinical manifestation

It occur as acute, subacute and chronic form

Acute ovine fasciolosis

- ☒ 'The acute disease occurs 2-6 weeks after the ingestion of large numbers of metacercariae, usually over 2000
- ☒ Outbreak of acute fasciolosis is generally presented as **sudden death** during autumn and early winter
- ☒ On examination of remaining flock one may find some sheep are **weak and pale mucous membrane**, dyspnoeic, some instances have **palpable enlarged liver associated with abdominal pain and ascites** and weight loss due to liver damage
- ☒ Outbreak of disease is complicated by concurrent infection with ***clostridium novyi*** resulting in **black disease**

Subacute ovine fasciolosis

- ☒ This form of the disease occurring 6-10 weeks after ingestion of approximately 500-1500 metacercariae
- ☒ It is presented as a rapid and severe haemorrhagic anaemia with hypoalbuminaemia, and if untreated, can result in a high mortality rate
- ☒ Marked pale mucous membrane, enlarged and palpable liver
- ☒ Submandibular and facial edema

- ☒ Ascities

Chronic fasciolasis

- ☒ It is seen mainly in late winter/early spring season this is most common form of the disease
- ☒ It occurs 4-5 months after the ingestion of moderate numbers of metacercariae 200-500. Chronic infection may be asymptomatic or lead to intermittent abdominal pain, cholelithiasis, cholangitis, obstructive jaundice, or pancreatitis
- ☒ The main pathogenic effect is **anemia and hypoalbuminemia** and more than 0.5 ml blood/fluke can be lost in bile duct each day
- ☒ Additional loss of plasma proteins occurs by leakage through the hyperplastic biliary mucosa
- ☒ Pharyngeal fascioliasis has been reported in human after consumption of infected raw liver in the Middle East

Bovine fasciolasis

- ☒ Clinical signs same as that of sheep
- ☒ Acute and subacute fasciolasis may occasionally occur especially in young calves
- ☒ Chronic form is most common in cattle

Necropsy finding

Acute fasciolasis

- ☒ Severe haemorrhage in liver parenchyma due to rupture of blood vessels by migrating liver flukes
- ☒ Subcapsular haemorrhage are common
- ☒ Blood-stained fluid is often present in the abdominal cavity and natural orifices. These symptoms and lesions are differentiated with anthrax

Chronic fasciolasis

- ☒ Hepatic fibrosis and hyperplastic cholangitis
- ☒ Liver has irregular outline, pale and firm and ventral lobe become atrophied
- ☒ Heavy infection can cause sclerosing cholangitis and biliary cirrhosis. Ectopic lesions may occur in the intestinal wall, lungs, or other organs

Diagnosis

- ☒ Microscopic examination of stool or duodenal or biliary material for eggs (oval brown colour operculated egg. Which is distinguished from rumen flukes)
- ☒ Antibody assays is useful in early stage of infection before eggs are produced
- ☒ Computerized tomography (CT) frequently shows hypo dense lesions in the liver during the acute stage of infection
- ☒ Ultrasonography, CT, MRI, ERCP, or cholangiography can detect biliary tract abnormalities in chronic disease

- ☒ Increased plasma concentration of γ -glutamyl transpeptidase, which is due to bile duct damage, and also helpful during the late maturation period
- ☒ Copro antigen ELISA is used for detection of *Fasciola hepatica* in cattle and sheep

Treatment

- ☒ Several drugs are available to treat infected ruminants, including **triclabendazole**, clorsulon (cattle and sheep only), albendazole, netobimin, closantel, rafoxanide, and oxclozanide

Older drug (mostly effective against mature flukes)

Carbon tetrachloride (oldest – 50 years)

- Recommended for adult fluke of sheep only (not for Cattle)

Hexachlorethane – 90% Effective against mature flukes

- Cattle – 220 to 400 mg/kg in divided doses (3-4)
- Sheep – 20 to 30 grams / animal

Hexachlorophene - 90% Effective against mature flukes

- Cattle – 10 to 20 mg / kg
- Sheep – 15 to 20 mg / kg

Hetol – against adult flukes

- Cattle – 125mg / kg
- Sheep – 150 mg / kg

Bithionol

- Cattle – 30 – 35 mg / kg (66 to 68%)
- Sheep - 40 mg / kg (90 to 100% -chronic fascioliasis)

Recent drugs (Active against mature & immature flukes)

Diamphenethide – 100 to 150 mg / kg (90 to 100 % for Adult)

Oxclozanide

- Cattle – 10 to 15 mg / kg (mature -100%)
- Sheep – mature - 15 to 20 mg / kg & Immature - 45 mg /kg

Rafoxanide

- Sheep & Cattle – 7.5 mg / kg
- Mature - more than 90 % & Immature – 50 to 90%

Nitroxynil

- Sheep & Cattle
- Mature (100%)- 10 mg / kg & Immature (90%) – 15 mg / kg

Benzimidazole (Albendazole & Oxfendazole)

- Albendazole – Sheep - 7.5 mg /kg & Cattle – 15 mg / kg

Latest drug (Active against mature and highly effective against immature flukes)

Triclabendazole – 10 mg/kg body

- Highly effective against immature flukes
- Sheep – single dose remove all stages (over one week old)
- Cattle – two doses remove over 4 weeks aged flukes
- Movement of stock from the infected pasture

Closantel - 7.5 mg/ kg

- ☒ Rafoxanide , Nitroxynil and **Oxyclozanide** most commonly used for subacute and chronic fasciolasis
- ☒ Anthelmintic resistance by *F. hepatica* to various compounds, including albendazole, clorsulon, and triclabendazole, has been demonstrated
- ☒ The timing of treatment is also important so that the pharmacokinetics of the drug used will result in the optimal removal of flukes. Each flukicide has varying efficacy against different ages of fluke
- ☒ In areas where heavy infections are expected, **sheep may require treatment in September or October, January or February, and again in April or May** to reduce both the chances of acute or chronic infections and the output of fluke eggs for development of future disease
- ☒ Cattle-antihelmintics should be given every December and May to prevent pasture contamination on spring and winter season

Immunization

Natural immunity

- ☒ Cattle - Reinfection of *F. hepatica* induce moderate to high immunity
- ☒ Sheep - fails to develop protective immunity

Active immunity

- ☒ Cattle – Three doses of irradiated metacercariae of *F. Hepatica*
- ☒ Age - 6 – 9 months
- ☒ Sheep – Not effective
- ☒ Sheep infected with *Taenia hydatigena* give immunity to *F. hepatica* but reverse not possible

Control

1. Reduction of pasture contamination

- ☒ Reduction of pasture contamination with parasitic egg at most suitable time for development i.e April to August with suitable antihelmintic drugs
- ☒ Late April to early May- treat the adult sheep with drug effective for immature and mature flukes
- ☒ In June- 4-6 weeks after April/May treatment again treat the adult sheep with drug effective for both immature and mature flukes
- ☒ October- treat entire flock with drug effective for immature stage

- ☒ January- treat entire flock with drug effective for mature stage

2. Snail control

- ☒ Survey of snail habitat to implement control measures
 - ☒ Drainage of water stagnation area to reduce snail population
 - ☒ Fencing of snail habitat and treat with molluscicide (Copper sulphate 1 in 100000 solution for water reservoir and N-Trityl Morpholine 0.45 kg/680 litre/hectare)
 - ☒ **Copper sulphate is most commonly used molluscicide**
 - ☒ Application of copper sulphate on pasture 10-35 kg/hectare. It should be applied along with sand for easy application. Avoid grazing of sheep on treated pasture for 6 weeks after application because copper sulphate is toxic to sheep
 - ☒ Rearing of ducks, geese and frog in infected water reservoir
3. Computerized prediction systems based on rainfall, evapotranspiration, number of wet days per month, and/or prevalence is used to predict the timing and severity of disease

Amphistomiasis

- ☒ Amphistomes are called as Stomach flukes or Conical flukes

Name of trematodes	Host	Location
<i>Paramphistomum gotoi</i>	Cattle	Fore stomach
<i>Paramphisomum cervi</i>	Cattle	Fore stomach
<i>Cotylophoron cotylophorum</i>	Sheep, Goat & Cattle	Rumen & Reticulum
<i>Calicophoron calicphorum</i>	Sheep & Cattle	Rumen & Reticulum
<i>Ceylonocotyle streptocoelium</i>	Cattle, Sheep & Antelope	Rumen & Reticulum
<i>Ceylonocotyle scoliocoelium</i>	Cattle, Sheep & Antelope	Rumen & Reticulum
<i>Gigantocotyle explanatum</i>	Buffalo * & Cattle	Bile duct, Gall bladder & Duodenum
<i>Gastrothylax crumenifer</i>	Sheep, Cattle & Buffalo	Rumen & Reticulum

<i>Fischoederius elongatus</i>	Cattle & other Bovidae	Rumen
<i>Fischoederius cobboldi</i>	Zebu cattle	Rumen
<i>Carmyerius spatiosus</i>	Cattle & Antelope	Rumen
<i>Carmyerius gregarius</i>	Cattle & Buffalo	Rumen
<i>Gastrodiscus aegyptiacus</i>	Equine & Pig	Large & Small intestine
<i>Gastrodiscus secundus</i>	Horse & Elephant	Colon
<i>Gastrodiscoides hominis</i>	Man & Pig	Colon
<i>Pseudodiscus collinsi</i>	Horse	Colon

Snail of veterinary importance in India

Name of the Snail	Name of trematode
<i>Indoplanorbis exustus</i>	<i>Gigantocotyle explanatum</i> , <i>Cotylophoron cotylophorum</i> and <i>Gastrodiscus secundus</i>
<i>Gyraulus convexiculus</i>	<i>Gastrothylax crumenifer</i>
<i>Lymnaea luteola</i>	<i>Fischoederius elongatus</i> and <i>Fischoederius cobboldi</i>

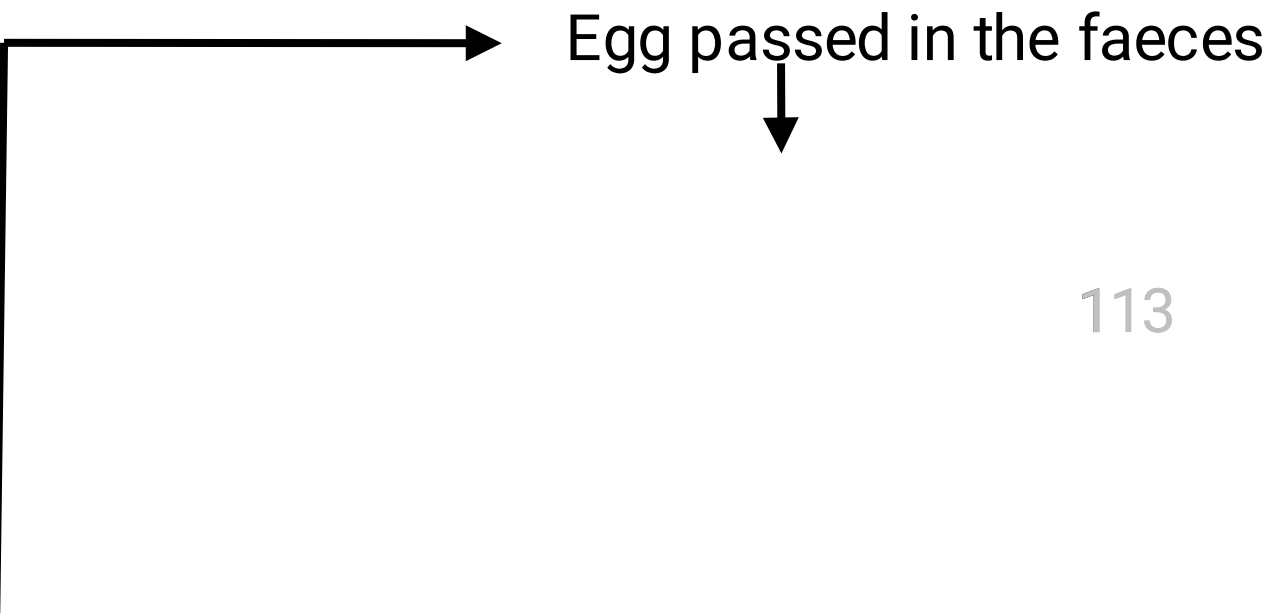
Morphology

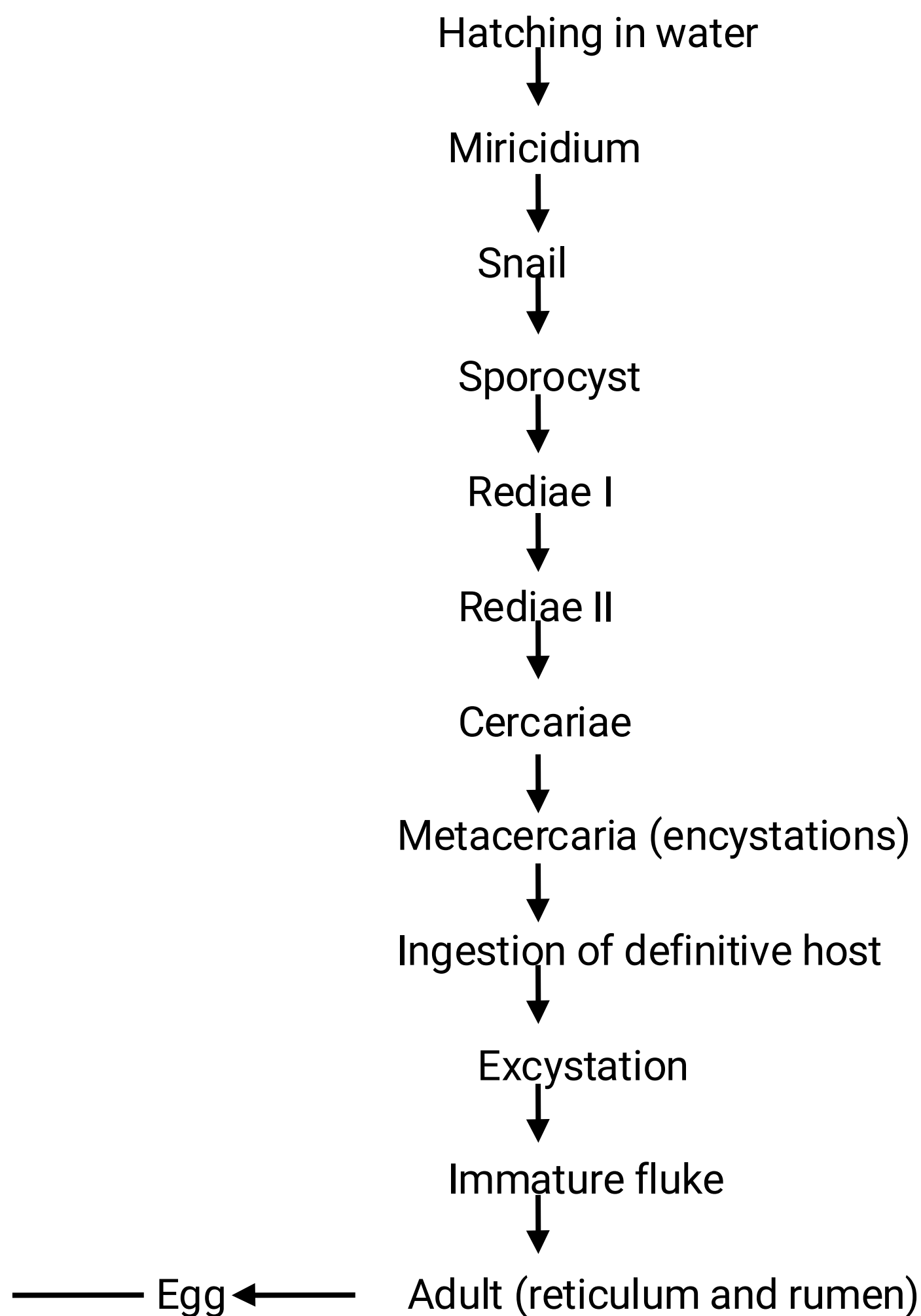
- ☒ Small, conical in shape & pink in colour
- ☒ Eggs are thin shelled, oval shape, distinct operculum with a small knob at the posterior end

Epidemiology

- ☒ Animals are reared in extensive systems of management are more susceptible
- ☒ Animals are grazing near pond area/ other natural resources, especially during drier months are more susceptible
- ☒ The palatable grazing, concentration of cattle, snail and metacercariae over small area leading to heavy infections
- ☒ Incidence of disease is high during summer months with high rain fall
- ☒ Young animals are highly susceptible than adult
- ☒ Preinfection – give resistant

Life cycle





Pathogenesis

Immature flukes (Pitto)

- ☒ Immature flukes embedded in mucosa of duodenum and upper ileum, plug feeder of mucosa (drawing piece of mucosa into the sucker) causes haemorrhages and necrosis
- ☒ In heavy infections it causes **frank haemorrhagic dueodenitis**

Clinical manifestation

- ☒ Severe enteritis with enormous number of migrating flukes in the dueodenum
- ☒ Profuse persistent fetid watery diarrhoea, increased thirst and drink more water
- ☒ Marked weakness, depression, dehydration and anorexia
- ☒ Submaxillary edema with pallor of mucosa
- ☒ Anemia and hypoproteinaemia
- ☒ Dry rough hair coat

Necropsy findings

- ☒ Subcutaneous oedema
- ☒ Accumulation of fluid in the body cavities

- ☒ Presence of large no. of flesh colour worms in the intestine (dueodenum)
- ☒ *Gigantocotyle explanatum*- superficial haemorrhage in the bile duct and gall bladder at the attachment
- ☒ In heavy infection- liver is pale and shows some degree of fibrosis

Diagnosis

- ☒ Based on grazing history and clinical signs
- ☒ Faecal examination
- ☒ Presence of immature worms in fluid faeces
- ☒ At post mortem-presence of large number of brownish pink parasites on intestinal mucosa

Treatment

- ☒ **Oxyclozanide – 18.7 mg / kg for two days (Immature & Mature - 100 % effective)**
- ☒ Bithionol sulphoxide – 40 mg/ kg (activity against Immature – 100%)
- ☒ Resorantol – 65 mg / kg (Immature – 65% &Mature – 100%)
- ☒ Niclosamide - 90 mg / kg (Immature – 99.9% &Mature – 8%)
- ☒ Niclofolan - 6 mg / Kg (Immature – 96% & Mature – 43%)
- ☒ Hexachlorophene – 20 mg/ kg - single dose
- ☒ Rafoxanide @7.5-10 mg/kg
- ☒ **Resorantol and Oxyclozanide** are the drug of choice for both immature and mature Amphistomes

Control

- ☒ Avoid grazing near pond area
- ☒ Prevent access of natural water resources by fencing
- ☒ Provide wholesome water supply at convenient places
- ☒ Drainage of water pools & swampy area
- ☒ Snail control
- ☒ Prophylactic deworming

Control of Snail

- ☒ Physical – Improving drainages
- ☒ Chemical - Applying molluscicides
- ☒ Copper sulphate - 1 in 1,00,000 solution for water reservoir
- ☒ Copper sulphate powder - 10 – 35 kg / hectare, applied along with sand for easy application and Sheep are not allowed to graze until rainfall occurs
- ☒ N-tritylmorpholine – 0.45 kg/ 680 litres / hectare applied during spring or mid summer season
- ☒ Biological- rearing of ducks, geese and frog in infected water sources prey the snail

Schistosomiasis

- ☒ Schistosomiasis is a common parasitic infection in cattle and rarely in other domestic animals in Africa and Asia
- ☒ Most of the infections in endemic areas are subclinical
- ☒ High prevalence rates of subclinical infections cause significant losses in terms growth and productivity

Schistome species of Veterinary Importance

Schistosome sp.	Host	Location
<i>S. bovis</i>	Cattle, sheep and goat	Portal and mesenteric veins
<i>S. spindale</i>	Ruminants, pig and dog	Mesenteric veins
<i>S. indicum</i>	Ruminants, equines and camels	Portal and mesenteric veins
<i>S. haematobium</i>	Pig, monkeys and man	Veins of mesenteric, bladder, urethra and ureter
<i>S. nasalis</i>	Buffalo , cattle, goat, sheep and horse	Nasal vein
<i>S. incognitum</i>	Pig and dog	Mesenteric vein
<i>S. mattheei</i> ,	Domestic and wild ruminants, equines and man	Portal and mesenteric vein, veins of urogenital tract and stomach
<i>S. japonicum</i>	Man and animals	Portal and mesenteric veins
<i>S. mansoni</i>	Man	Mesenteric vein
<i>S. mekongi</i>	Dog, man	Liver, spleen, gastrointestinal tract and oesophagus

Prevalence of schistome species in India: ***S.spindale*, *S. nasalis*, *S.indicum* and *S.incognitum***

Intermediate host

Name of the Snail	Name of trematode
<i>Indoplanorbis exustus</i>	<i>S. nasalis</i> , <i>S. indicum</i> and <i>S. spindale</i>
<i>Lymnaea luteola</i>	<i>S. nasalis</i> and <i>S. incognitum</i>
<i>Lymnaea acuminata</i>	<i>S. spindale</i>

Morphology

- ☒ Schistosomes are called as blood flukes
- ☒ Unisexual – dimorphic trematode
- ☒ Female - long (12 – 28 mm) & slender
- ☒ Male - Comparatively small in length (9-22 mm) & having gutter-like groove (Gynaecophoric canal) in which female lodges at the time of copulation
- ☒ Cercariae - *Furcocercous cercariae*, tail is forked
- ☒ Egg - 100 – 500 µm long and spindle shape without operculum. Some species having lateral spines

Developmental stages

- ☒ Redia and metacercariae stages are absent
- ☒ Two generation of sporocysts (I and II)
- ☒ Immature schistosomes is called as **Schistosomula**

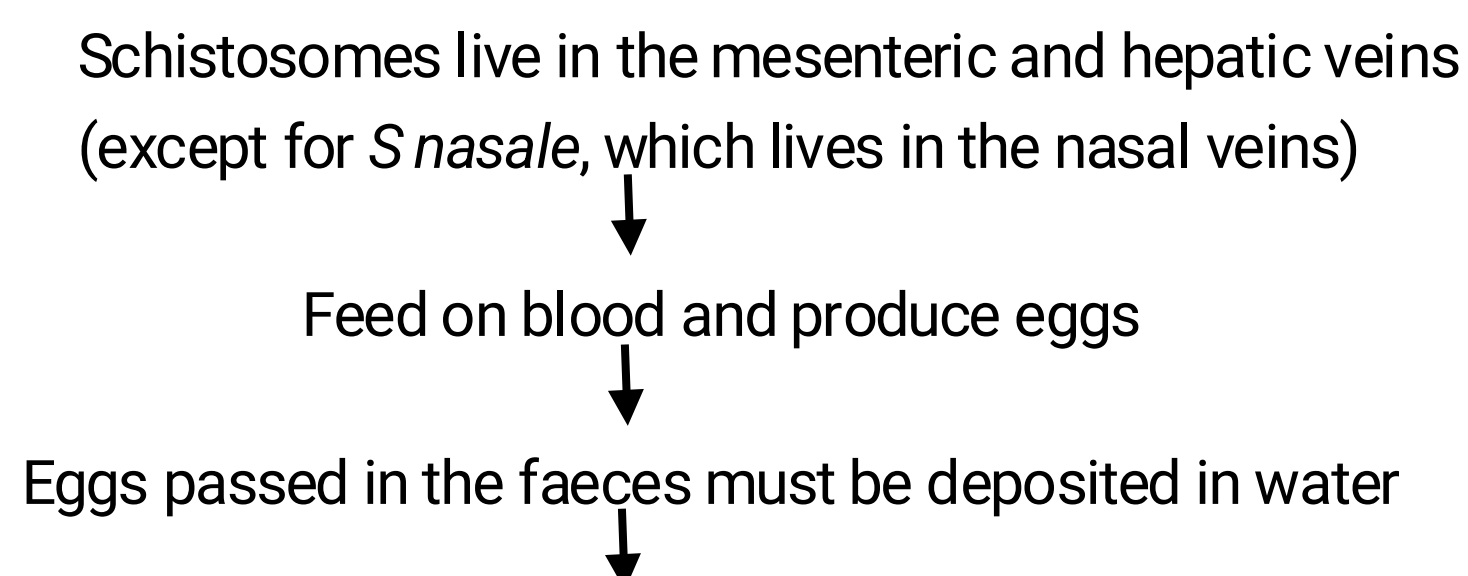
Epidemiology

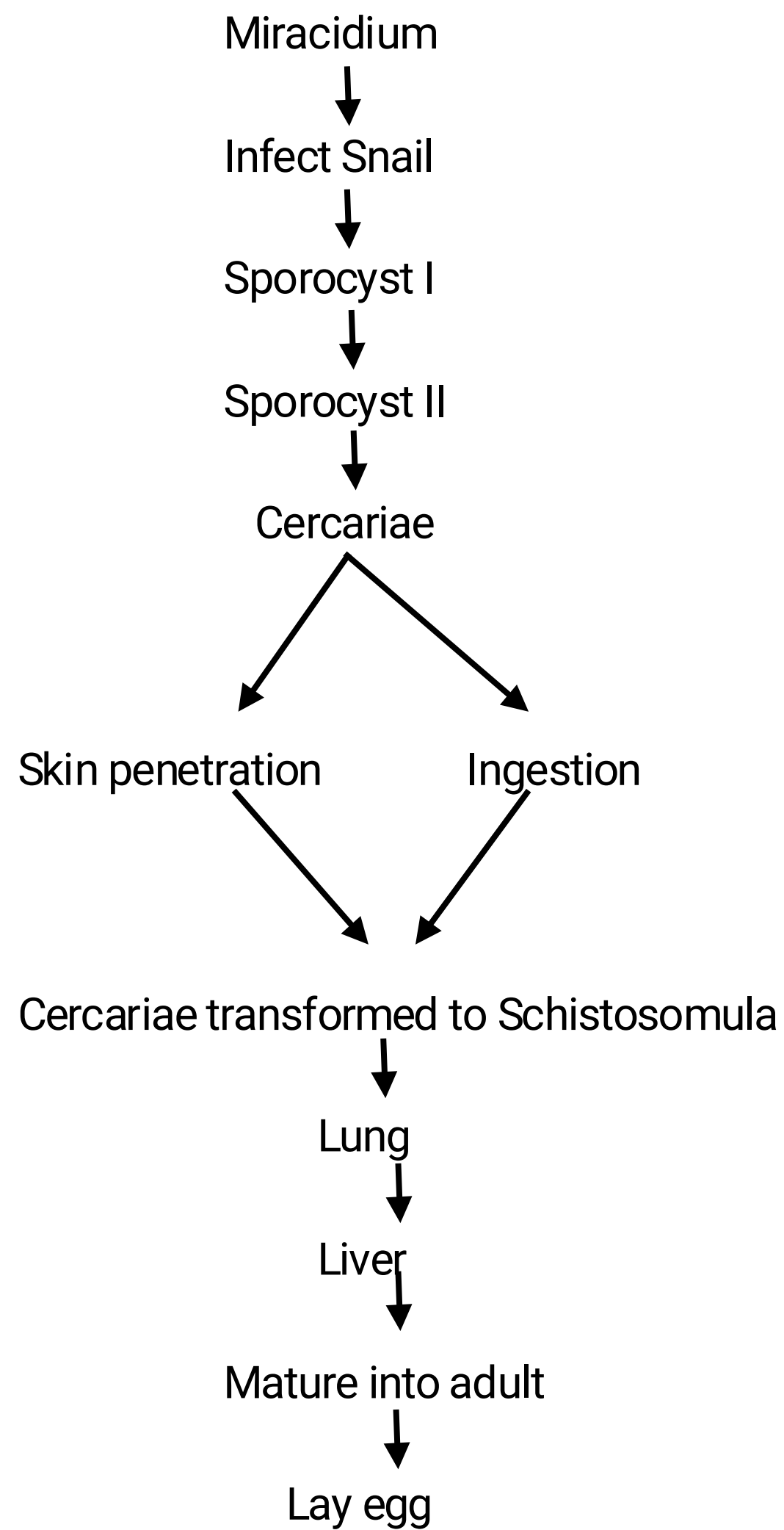
- ☒ **Cattle is highly susceptible than buffalo, sheep and goat**
- ☒ Older bovines (>3 years) are highly susceptible
- ☒ Infection also occurs in 5- 6 months old calves
- ☒ Both male and female animals are equally affected

Transmission

- ☒ Horses and cattle may get infection when standing in shallow infected water source of dam or river
- ☒ Buffalo get infection while wallowing in the infected water sources
- ☒ Slow moving water favour the entry of cercariae into body of animal
- ☒ Consumption of water contaminated with cercariae
- ☒ Highest rainfall (Season – Sepember to October)

Life cycle



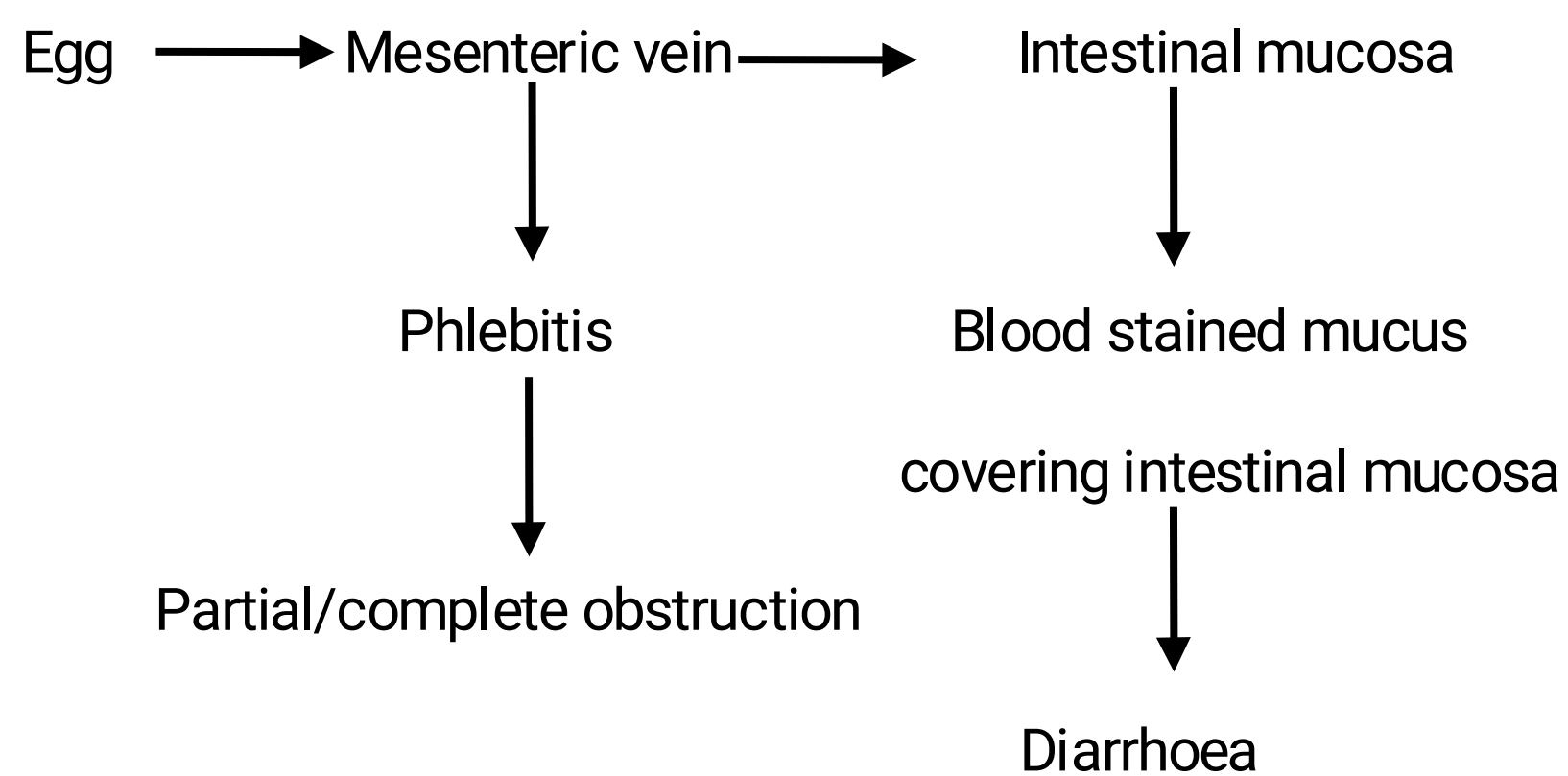


Pathogenesis

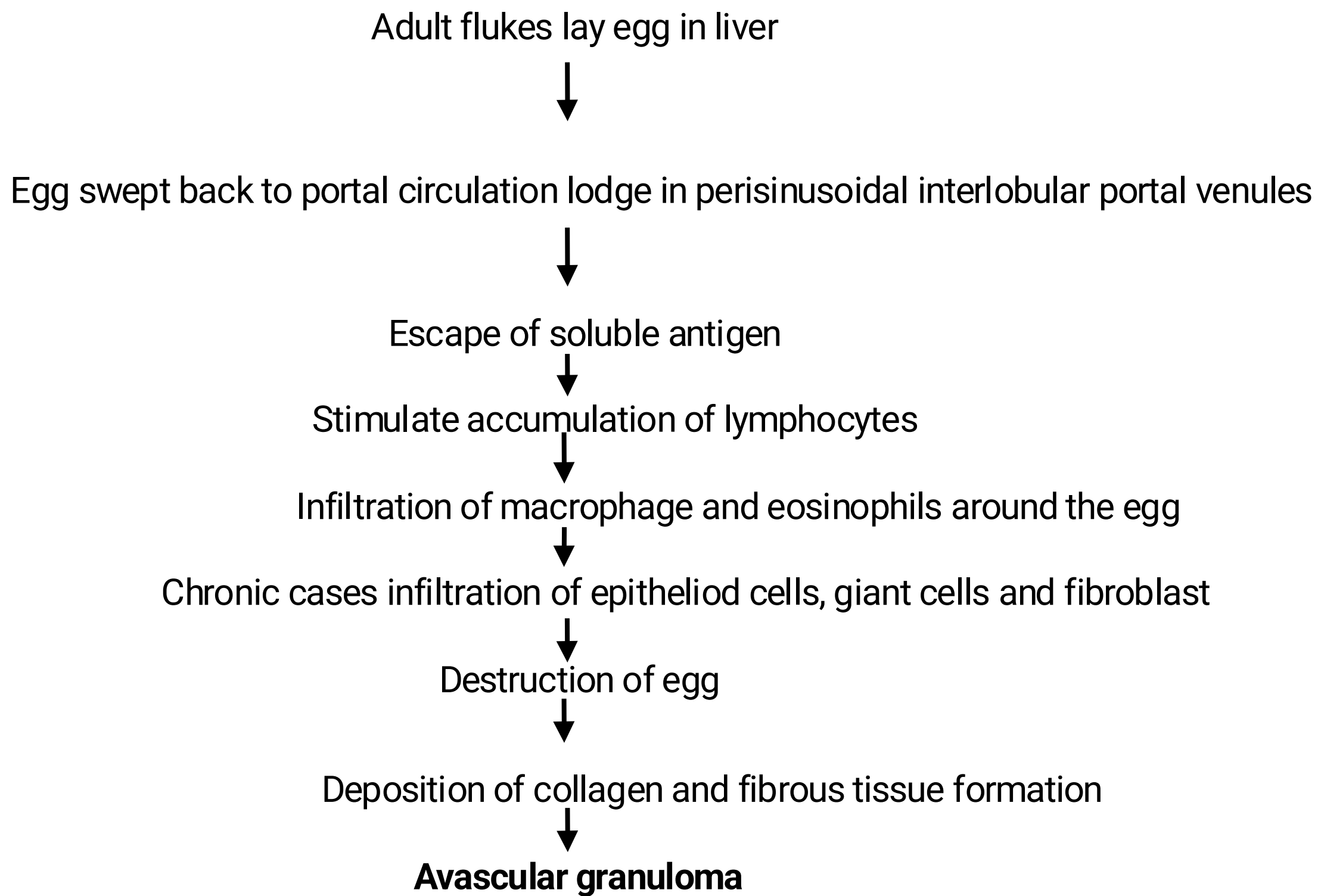
- ☒ Acute intestinal syndrome
- ☒ Chronic hepatic syndrome

Acute intestinal syndrome

Heavily infected animal passes large no. of egg through intestinal mucosa



Chronic Hepatic syndrome (Clay pipe stem fibrosis)



Nasal schistosomiasis/ pseudotubercle/actinobody formation/snoring disease

Clinical signs

Cattle

- ☒ Rhinitis , mucopurulent discharge, signs of coryza, sneezing and dyspnoea and snoring respiration
- ☒ Adult parasite causes dilation and thrombosis of vein
- ☒ Nasal mucosa studded with small granulomas and small abscess containing eggs
- ☒ In chronic cases – proliferation of nasal epithelium and fibrous tissue formation may occur
- ☒ Space occupying lesion in nasal septum results in breathing difficulties leads to snoring respiration

Buffalo

- ☒ Pin sized eruption and congestion of nasal mucosa is called as **Actinobodies**
- ☒ Diarrhoea, sometimes blood stained and contains mucus
- ☒ Anorexia, thrust and emaciation

Clinical signs in other *schistosome* sp

- ☒ Migration of large no. of worms through lung causes temporary cough
- ☒ Profuse watery diarrhoea or dysentery
- ☒ Dehydration and anorexia
- ☒ Anemia and hypoalbuminemia

- ☒ Decrease in production and loss of weight

Immunity

- ☒ **Natural immunity** – occur in cattle after infection

- ☒ **Artificial immunity**

Administration of irradiated Schistosomula of *S. mattheei* and *S. bovis* S/C or I/M to cattle or sheep induce protection greater than 60% against schistosoma infection

Clinical pathology

- ☒ Anemia, Eosinophilia, hypoalbuminemia, hypergammaglobulinemia
- ☒ Dyshaemopoiesis and expansion of plasma volume

Diagnosis

- ☒ Based on clinical signs
- ☒ Detection of *schistosome* sp egg in faecal sample examination
- ☒ Detection of *schistosome* sp egg in nasal washing
- ☒ Measuring cell mediated immune response

Differential diagnosis

- ☒ Rhinosporidiosis
- ☒ Tumour

Treatment

- ☒ Tartar emetic - 2 mg / kg
- ☒ Sodium antimony tartrate – 1.5 mg / kg twice a day for 2 days
- ☒ For cattle praziquantel @ 25mg/kg is highly effective but, it require two treatment at 3-5 weeks apart
- ☒ Trichlorophon - 30 – 40 mg/Kg three doses
- ☒ Niridazole: Sheep – 100 mg/Kg for 3 days, Pig: 25-30 mg/kg
- ☒ Praziquantel: Sheep and goat – 60 mg/kg

Control

- ☒ Management practices and snail control similar to that of Amphistomiasis and Fasciolasis
- ☒ Biological control *Schistosomes* larvae
- ☒ Larval stages of *Echinostoma* spp. is used as predatory on schistosome larvae within the snail intermediate host
- ☒ Microsporidial protozoa – *Nosema eurytremae* can cause extensive damage to intramolluscan stage of schistosomes
- ☒ Avoid contact with infected water sources

Cercarial dermatitis/Swimmers itch / Clay-diggers itch / Hunters itch / Rice-Paddy itch

- ☒ It is caused by skin penetration of cercariae of non human schistosomes (Avian schistosomes *Trichobilharzia* sp. and *Austrobilharzia* sp) and animal Schistosomes (*Heterobilharzia* spp.) in man. Initial exposure to cercariae causes a mild erythema and edema, but repeated exposure causes severe pruritis and papular or pustular eruptions
- ☒ **Treatment** - Application of benzyl benzoate over the body and antihistaminic therapy
- ☒ **Prevention** - Wearing protective clothing while working in the infected water sources

Cysticercosis

Taeniasis is an intestinal infection of adult tapeworm (*Taenia solium* and *Taenia saginata*) following ingestion of contaminated pork and beef.

Cysticercosis is an infection of pig and cattle with larvae of *T. Solium* and *T. Saginata* respectively, which develops after ingestion of ova excreted in human faeces. Both adult and larval stage of *Taenia* sp is most important. The intermediate stage of tapeworm is found in muscles of cattle and economic problem to pork and beef industry and public health issues.

The definitive and intermediate host of *Taenia* sp

Adult tapeworm	Definitive host	Metacestode stage	Intermediate host	Predilection site of larvae	Type of Cysticercosis
<i>Taenia saginata</i>	Man	<i>Cysticercus bovis</i>	Cattle	Muscle	Bovine cysticercosis or measly beef
<i>Taenia solium</i>	Man	<i>Cysticercus cellulosae</i>	Pig , man	Muscle	Porcine cysticercosis or measly pork
<i>Taenia ovis</i>	Dog	<i>Cysticercus ovis</i>	Sheep	Muscle	Ovine cysticercosis

<i>Taenia hydatigena</i>	Dog	<i>Cysticercus tenuicollis</i>	Sheep, cattle, pig	Peritoneum	Abdominal cysticercosis
<i>Taenia pisiformis</i>	Dog	<i>Cysticercus pisiformis</i>	Rabbit	Peritoneum	Abdominal cysticercosis
<i>Taenia taeniformis</i>	Cat	<i>Cysticercus fasciolaris</i>	Mouse, Rat	Liver	Other cysticercosis
<i>T. krabbei</i>	Dog	<i>Cysticercus tarandi</i>	Reindeer	Muscle	Other cysticercosis

Taenia saginata

Morphology

- ☒ The length of adult tape worm is 5-15 cm. The scolex has neither rostellum and nor hooks. Which is found in small intestine
- ☒ **Metacestode stage-*Cysticercus bovis*** is greyish white about 1.0 cm in diameter and found in skeletal, cardiac muscle, masseter, intercostals muscle, heart and tongue

Epidemiology

- ☒ World wide distribution
- ☒ Abnormal eating habits of cattle due to certain mineral deficiencies (pica) may result in cravings that increase the exposure through the ingestion of faeces
- ☒ Sewage water is an important route of dissemination of infection
- ☒ Birds that consume infected sewage water and disseminate the infection

Transmission in cattle

- ☒ Passive ingestion of contaminated pasture
- ☒ Neonatal calves may be infected when they are handled by infected person (with motile proglottids)

Transmission in human

- ☒ Ingestion of raw or under cooked beef (Measely beef)
- ☒ Changing of food habits – fast foods
- ☒ If cattle slaughtered 9 months after infection (without reinfection) - infection rate in human beings is very less

Cysticercosis storm

- ☒ In developed countries like Australia and Britain – human sewage is utilized as fertilizer for pasture land in the form of sludge (sedimented or bacterial digested faeces)

- ☒ If cattle are exposed to contaminated pasture land immediately after application of sludge. The high incidence of infection of *T. Saginata* was noticed

Life Cycle

An infected human may pass millions of eggs daily, either free in the faeces or as intact segments each containing about 2,50,000 eggs, and these can survive on pasture for several months. After ingestion by a susceptible bovine the onchosphere travels via the blood to striated muscle. It is first grossly visible about two weeks later as a pale, semi-transparent spot about 10mm in diameter, but is **not infective to man until about 12 weeks**, later it has reached its full size of 1.0cm, which is enclosed by thin fibrous capsule. **The longevity of the cysts ranges from weeks to years in muscles.** When they die they are usually replaced by a caseous, crumbly mass which may become calcified. Both living and dead cysts are frequently present in same carcass. Man becomes infected by ingesting raw or inadequately cooked meat. Development to patency takes place in 2-3 months.

Clinical manifestation

Cattle

- ☒ Under natural conditions clinical symptoms are not evident
- ☒ Experimental infection of calves with *T.saginata* develops myocarditis and heart failure

Human

- ☒ Diarrhoea and hunger pain, but the infection is usually asymptomatic

Diagnosis

- ☒ Identification of cyst in muscles during meat inspection

Treatment

- ☒ Praziquantel has shown efficacy in experimental infection

Prevention

Immunoprophylaxis against *T. saginata*

- ☒ **Homologous vaccine** - In-vitro culturing of oncosphere of *T. saginata* releases secretory and excretory antigen
- ☒ **Active immunization in calves**- Secretory and excretory antigen administered in calves before infection (after infection there is no useful)
- ☒ **Passive immunization**--Secretory and excretory antigen administered to periparturient cows either through intramuscularly or intra mammary route
- ☒ Calves are immunized while consuming colostrum / milk from immunized dam

Heterologus vaccine

- ☒ In-vitro culturing of oncospheres of *T. taeniaformis* releases secretory and excretory antigen, which capable of inducing high level protective immunity against heterologus parasite – *T. Saginata*

- ☒ Intramuscular injection of eggs of *T. hydatigena* in cattle induces partial immunity against heterologous parasite.

Control

- ☒ Maintenance of high standard of human sanitation
- ☒ Compulsory meat inspection
- ☒ Thorough cooking of meat
- ☒ Regulations usually require that infected carcasses are frozen at -10° C for at least 10 days which is sufficient to kill the cysticerci
- ☒ In agricultural practice the use of human sludge as a fertilizer should be confined to cultivated fields or to those on which cattle will not be grazed for at least two years

Taenia solium

Morphology

- ☒ The adult tape worm has scolex, which contain rostellum and two concentric rows of hooks
- ☒ Egg - spherical in shape
- ☒ **Metacestode stage - *Cysticercus cellulosae*** found in pig and man
- ☒ Location - Skeletal and cardiac muscles

Epidemiology

- ☒ Almost similar to *T. saginata*, differences are
 - Infection rate – 10 proglottids X 40,000 = 4,00,000 eggs per day
 - Gravid proglottids do not leave spontaneously from the host and voided passively through faeces

Man acts as an IH

- ☒ Autoinfection - by reverse peristalsis is the most important source of infection
- ☒ Accidental ingestion of *T. solium* eggs along with food and water
- ☒ Nail biting habits – especially with dirty hands
- ☒ If man acts as an IH, the life cycle is arrested in metacestode stage itself

Definitive host

- ☒ *Cysticercus* reaches small intestine and become changed in to adult worm. Adult worm lay the eggs (prepatent period – 100 days) – not much pathogenic changes

Intermediate Host

- ☒ Infected by passive ingestion of embryonated egg or proglottids in faeces

Life cycle

Humans develop intestinal infection with adult worms after ingestion of contaminated pork or may develop cysticercosis after ingestion of *T. solium* eggs (making human intermediate hosts)

- ☒ Humans ingest raw or undercooked pork containing cysticerci (larvae)
- ☒ After ingestion, cysts evaginate, attach to the small intestine by their scolex, and mature into adult worms in about 2 months
- ☒ Adult tapeworms produce proglottids, which become gravid; they detach from the tapeworm and migrate to the anus
- ☒ Detached proglottids, eggs, or both are passed from the definitive host (human) in feces
- ☒ Pigs or humans become infected by ingestion of embryonated eggs or gravid proglottids in fecal contaminated food. Autoinfection may occur in humans if proglottids pass from the intestine to the stomach via reverse peristalsis
- ☒ After eggs are ingested, they hatch in the intestine and release oncospheres, which penetrate the intestinal wall
- ☒ Oncospheres travel through the bloodstream to striated muscles, brain, liver, and other organs, where they develop into cysticerci causes **cysticercosis**
- ☒ In man *C. cellulosae* primarily found on subcutaneous tissues and secondarily on brain and ocular cavity

Clinical manifestation

Pig

- ☒ Clinical signs are inapparent

Human

- ☒ Adult worms may cause mild gastrointestinal symptoms or passage of a motile segment in the stool
- ☒ Cysticercosis is usually asymptomatic unless larvae invade the CNS, resulting in **neurocysticercosis**. Infection in the brain (cerebral cysticercosis) may result in severe symptoms due to mass effect and inflammation induced by degeneration of cysticerci and release of antigens, which can cause seizures and various other neurologic signs like increased intracranial pressure, hydrocephalus, altered mental status or aseptic meningitis.
- ☒ Cysticerci may also infect the spinal cord, muscles, subcutaneous tissues and eyes
- ☒ Substantial secondary immunity develops after larval infection

Diagnosis

- ☒ Neurocysticercosis may be recognized on brain imaging studies (computerized axial tomography (CAT) and MRI techniques). The cysticerci mostly found in ventricle of the brain

- ☒ Fewer than half of patients with neurocysticercosis have adult *T. solium* in their intestines and thus eggs or proglottids in their stool
- ☒ Finding of antibody to cysticerci in the cerebrospinal fluid

Treatment

Pig

- ☒ Albendazole @ 50 mg/kg – effective against mature metacestode
- ☒ Praziquantel @ 50 mg/kg – highly effective against both mature and immature metacestodes.

Human

- ☒ Symptomatic treatment of neurocysticercosis with corticosteroids, anticonvulsive agents, and in some situations, albendazole or praziquantel also can be used
- ☒ Adult worms can be eliminated with praziquantel @10mg/kg b.wt and Niclosomide 2g (total dose)
- ☒ Surgical removal of cyst

Prevention and control

- ☒ Treating of infected persons (especially animal handlers)
- ☒ Public education and hygiene
- ☒ Proper meat inspection – made multiple incisions of suspected carcasses
- ☒ Thorough cooking of meat
- ☒ Freezing of carcasses
 - – 5° C for 15 days
 - – 10° C for 9 days
 - – 15 to – 30° C for 6 days

Taenia hydatigena

- ☒ This is the largest tapeworm found in dogs and wild canids
- ☒ The oncosphere is infective to sheep, cattle and pigs are carried in the blood to the liver in which they migrate for about four weeks before they emerge on the surface of this organ and attach to the peritoneum. Within four weeks each oncosphere develops into ***cysticercus tenuicollis*** (metacestode stage)
- ☒ Infection is prevalent particularly in sheep but, infection generally only detected at meat inspection infrequently; however, large numbers of developing cysticerci migrate contemporaneously in the liver of the sheep or pig producing '**hepatitis cysticercosa**'. This condition resembles acute fascioliasis, which is often fatal.

Occasionally developing cysticerci are killed, when sheep are previously exposed to infection.

- ☒ **Pathology:** The subcapsular phase of liver is studded with greenish nodules around 1cm dia
- ☒ **Diagnosis:** faecal examination for onchosphere or gravid segments in faeces
- ☒ **Treatment:** Praziquantel, mebendazole, fenbendazole, nitroscanate, and dichlorophen

Coenurosis

Syn: Gid, Sturdy)

Etiology

Coenurosis	Adult tapeworm	Definitive host	Metacestode stage	Intermediate host	Location of metacestode stage
Coenurosis sheep	in <i>Taenia multiceps</i>	Dog and fox	<i>Coenurus cerebralis</i>	Sheep, Cattle	Brain and spinal cord
Coenurosis goat	in <i>Taenia gaigeri</i>	Dog and fox	<i>Coenurus cerebralis</i>	Goat	Brain, spinal cord, intermuscular and subcutaneous connective tissue
Coenurosis rabbit	in <i>Taenia seriallis</i>	Dog and fox	<i>Coenurus serialis</i>	Rabbit	Intermuscular and subcutaneous connective tissue

*Adult worms are found in small intestine of definitive host

Metacestode stage: *Coenurus cerebralis* found in intermediate host

Coenurus cerebralis: It is a large fluid filled cyst about 5 cm or more in diameter, the cyst wall containing several hundred scolices in clusters

Taenia multiceps

Host affected: Older lambs and adult sheep are at high risk

Transmission

- ☒ The animal acquire the infection while grazing on contaminated pasture with *T. multiceps* egg

Life cycle and pathogenesis

After ingestion, some oncospheres hatch, then reach the brain, developing by endogenous budding into a metacestode (larval) stage known as *Coenurus cerebralis*. Initial invasion and development of the oncospheres may be responsible for **acute suppurative meningoencephalitis**. The fully developed coenurus may be 5–6 cm in diameter (it takes 8 months for maturation in CNS). The cyst produces space occupying lesions and exerts pressure on tissues of brain and spinal cord. In some cases it causes softening of skull bone and finally leads to perforation and causes increased intracranial pressure, which results in ataxia, hypermetria, blindness, head deviation, stumbling, and paralysis. This clinical condition is colloquially known as **gid, sturdy, or staggers**

Clinical manifestation

The clinical sign varies with the location of cyst on brain and spinal cord includes circling, visual defects, ataxia and hyperaesthesia.

Location of cyst

- a) Parietal region – left side on surface of cerebral hemisphere

Animals held its head in affected side (left) and moves in circle in left direction and right eye may be blind

- b) Parietal region – right side on surface of cerebral hemisphere

Animals held its head in affected side (right) and move in circle in right direction and left eye may be blind.

- c) Anterior part of the brain

The animal's head lowered and held against chest, walk in straight line until to hit on hard objects and remain motionless for some time

- d) Ventricle of the brain-opposite to above direction

- e) Cerebellum-animal become hyperaesthetic, walk in jerky or staggering gait

- f) Spinal cord- Progressive paresis of one or both hind limbs

Diagnosis

- ☒ Live animals: Through antemortem cyst is difficult to detect
- ☒ In sheep, palpation of the skull caudal to the horn buds may reveal refraction
- ☒ Ophthalmoscopic examination of the eye prior to development of clinical signs
- ☒ Dead animals: post mortem examination

Treatment

- ☒ Generally useless, high doses of albendazole may be tried
- ☒ Surgical removal of cyst, if it present in the surface of brain

Control

- ☒ Avoid feeding of infected brain and spinal cord to dogs
- ☒ Regular deworming of dog is necessary

Echinococcosis

Etiology

- ☒ *Echinococcus granulosus* (*E. granulosus granulosus* and *E. granulosus equinus*) and *Echinococcus multilocularis*
- ☒ It is a smallest cestode of domestic animals (6mm), consist of 3 or 4 segments. The gravid segment is half the length of rest of segment.

Strain	<i>E.granulosus granulosus</i>	<i>E.granulosus equines</i>
	<i>E.multilocularis</i>	
Distribution	Worldwide	Europe
Definitive host	Dog and other carnivores	Dog and other carnivores
Intermediate host	Domestic and wild ruminants, pig and man	Horses and donkey

Location

- ☒ Adult worm-small intestine
- ☒ Immature/Metacestode/ hydatid stage: **Lung and liver**
- ☒ Sheep-70% of cyst found in lung and 25%in liver
- ☒ Horse and cattle: 90% of cyst found in liver

Epidemiology

It is based on pastoral and sylvatic cycle

- ☒ **Pastoral cycle:** The dogs are infected by feeding of ruminant offal containing hydatid cyst, which mature into adult and excrete the gravid segment through faeces. The man get infection by accidental ingestion of vegetable and other foodstuff contaminated with dog faeces
- ☒ **Sylvatic cycle:** Wild canids and ruminants are involved in transmission

Life cycle

Proglottids/egg — > Intermediate host (6 - 12 months) — > Definitive host (40 - 50 days)

The prepatent period in the definite host is around 40-50 days after which only one gravid segment is shed by the tapeworm per week. The oncospheres are capable of prolonged survival outside the host, being viable on the ground for about two years. After ingestion by the intermediate host, the onchosphere penetrates the gut wall and travels in the blood to the liver or in the lymph to the lungs. These are the two commonest sites for larval development, but occasionally onchospheres escape into the general systemic circulation and develop in other organs and tissues, where they develop into large, thick-walled, unilocular hydatid cysts that bud protoscolices endogenously about 6-12 months. Hydatid cysts have been rarely reported in the CNS of domestic animals and are rare in people, in which they produce symptoms similar to those of brain tumor.

Hydatid cyst

Types of cyst

- ☒ **Sterile cyst:** not infective for dog. In horse – 27% of cysts are sterile, in sheep - 51% of cyst are sterile and in cattle & pig most of cysts are sterile
- ☒ **Fertile cyst:** infective for dog
- ☒ **Size:** 5 – 10 cm in size (maximum of 50cm)
- ☒ **Cyst wall-**Outer layer is thick and made up of concentrically arranged laminated membrane. The inner layer: germinal layer and granular in nature
- ☒ **Cyst fluid** - pale yellow in colour and contains 17 to 200 mg of protein per ml
- ☒ **Brood capsule** – protoscolices attached on the inner granular layer or detached and float free in the hydatid fluid is called as “ Hydatid sand”
- ☒ **Daughter cyst** develops within the mother cyst, If mother cyst rupture release protoscolices & brood capsules and develops as external daughter cyst

Clinical manifestation

- ☒ **Dog** – clinical signs are not evident
- ☒ In domestic animals hydatid cyst in the liver or lungs is usually tolerated without any clinical signs, and the majority of infections are only revealed at the abattoir.
- ☒ **Human:** respiratory signs are common. Presence of several hydatid cysts in liver causes abdominal distension. If cyst rupture causes death by anaphylaxis

Necropsy finding

- ☒ Presence of fluid filled cyst in liver, lung and other organs

Diagnosis

- ☒ Presence of hvdatid cyst is clinical entity rarely suspected in domestic animals
- ☒ Diagnosis of infection in dogs with adult tape worm is difficult, because the segments are small and are only shed sparsely

- ☒ In some countries control regimes have involved the administration of purgative anthelmintics such as arecoline hydrochloride, so that the whole tapeworm is expelled in mucus and can be searched for in the faeces
- ☒ Man: Intradermal casoni' s test

Treatment

- ☒ Echinococcus tapeworms are more difficult to remove compared to *Taenia* sp.
- ☒ Praziquantel is highly effective @ 5.0-12.5 mg/kg b.wt. After treatment it is advisable to confine dogs for 48 hours to facilitate the collection and disposal of infected faeces
- ☒ **Man:** surgical excision of cyst
- ☒ Aspiration of cyst fluid and irrigation of cyst cavity with 2.5 % - 10% formalin (not commonly used) or own serum. Care should be taken to avoid spillage of cyst fluid, while aspiration
- ☒ High dose of mebendazole, albendazole and praziquantel may be effective

Control

- ☒ Regular anthelmintic treatment for adult tape worm
- ☒ Avoid feeding of cyst containing offals
- ☒ Proper disposal of sheep carcass in the farm
- ☒ Denying dog access to abattoir

E. multilocularis

- ☒ *DH:* Wild canids, domestic dog and cat
- ☒ *Location:* Small intestine
- ☒ *IH:* Rodents, Large mammals and man
- ☒ *Metacestode stage:* Hydatid cyst in liver is mostly multilocule
- ☒ *Pathogenesis*

Egg — > IMH — > small intestine — > Release of oncosphere — > Development of adult metacestode stage — > Liver and lung — > systemic circulation

Clinical signs

Clinical signs based on severity of infection and location of cyst

- ☒ Heavy infection causes impairment of affected organs functions
- ☒ Rupture of cyst leads to anaphylatic shock
- ☒ Alveolar cyst, a diffuse growth with many compartment containing gelatinous matrix into which scolices budding off

Diagnosis

- ☒ Animals: rare only by post mortem examination
- ☒ Man: Serology : CFT and CIE

Treatment

Man

Aspiration of cyst fluid and irrigation of cyst cavity with 2.5 % - 10% formalin (not commonly used) or own serum. Care should be taken to avoid spillage of cyst fluid while aspiration

Surgical removal of cyst

Higher doses of mebendazole, albendazole and praziquantel

Control: Similar to *T. ovis*

Clinical manifestation of other dog tape worm infection

Dipylidium caninum

- ☒ This is the commonest tapeworm of the domestic dog and cat.
- ☒ **Intermediate hosts:** Fleas (*Ctenocephalides canis*, *C felis* and *Pulex irritans*) and lice (*Trichodectes canis*)
- ☒ **Site:** Adult worm in Small intestine of definitive host; **Cysticercoid** in fleas and lice
- ☒ **Clinical manifestation**
 - The gravid segment actively crawl from anus
 - Excessive grooming of the perineum (rubbing of anus along the floor), but impacted anal glands are most common cause for this behaviour
- ☒ **Diagnosis :** crawling of gravid segment from anus and presence of egg pocket in faecal examination
- ☒ **Treatment:** Administration of Nitroscanate and praziquantel accompanied by use of insecticide to control fleas

Gastrintestinal Nematodiasis

a. Toxocariasis

Ascaridoids is one of the largest nematodes in domestic animals. Presence of adult worm in the intestine may cause unthriftiness and occasionally intestinal obstruction.

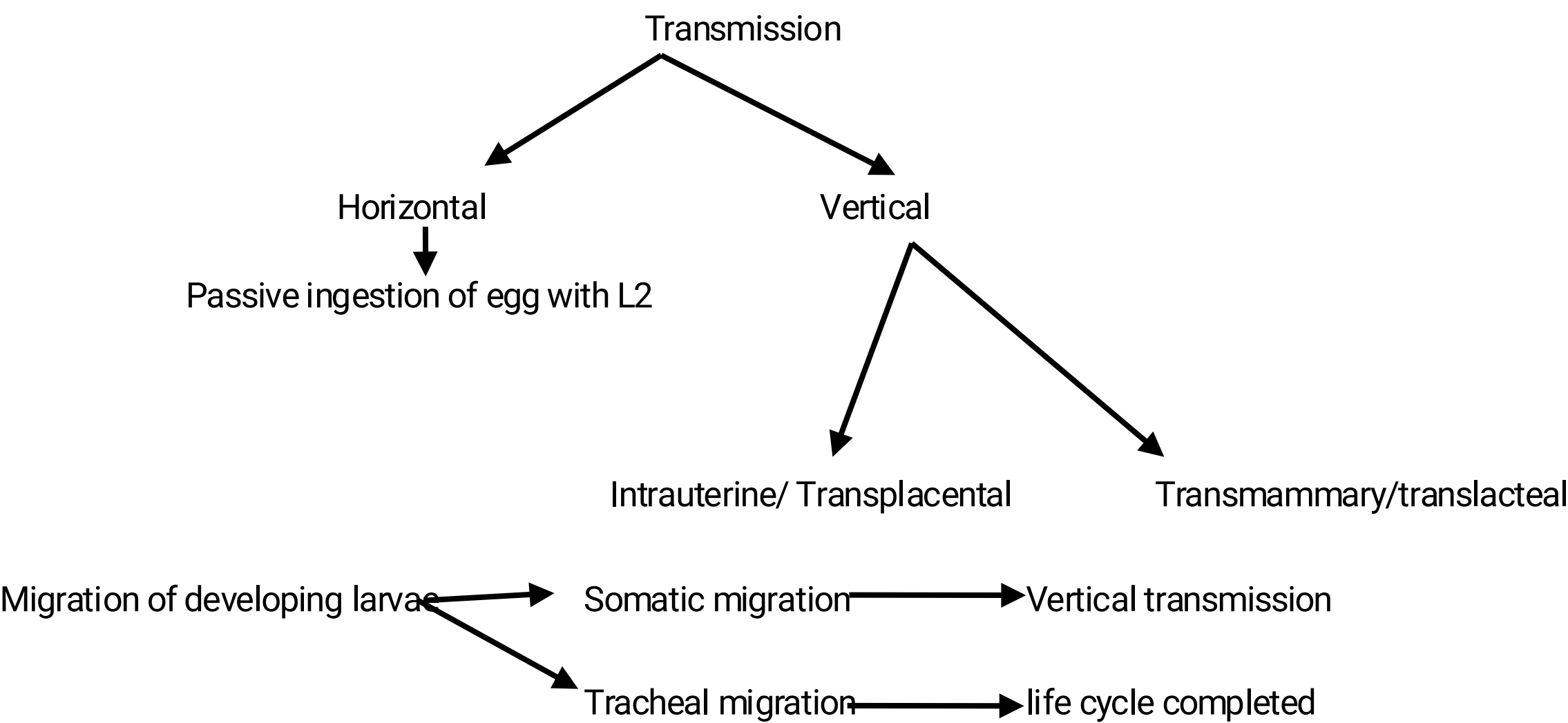
<i>Toxocara</i> sp	<i>T. vitulorum</i>	<i>T. Canis</i>	<i>T. Cati</i>
Host	Cattle, Zebu and	Dog and fox	Cat and wild

	Indian buffalo		felids
Location	Small intestine	Small intestine	Small intestine

Morphology

- ☒ Large white worm
- ☒ The egg is dark brown and subglobular, with a thick pitted shell

Transmission and life cycle: Direct and complex in nature



***Toxocara vitulorum* in bovines**

- ☒ It is the largest intestinal parasite of cattle

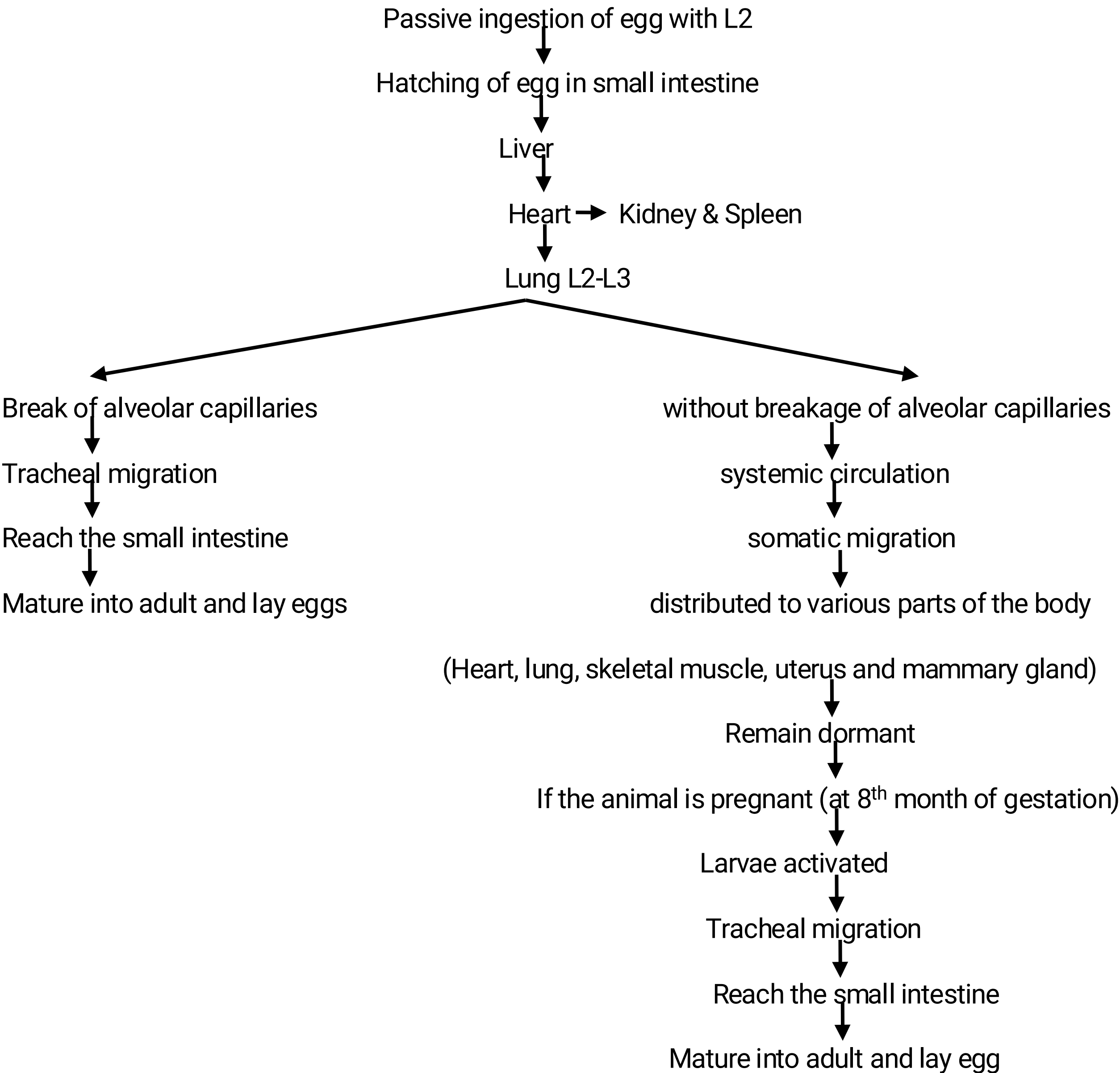
Host affected

- ☒ Cattle, buffalo, sheep and goat
- ☒ Buffalo calves are highly susceptible due to prenatal and neonatal infection
- ☒ Larvae present in dams milk up to 30 days after parturition
- ☒ If no tissue migration - prepatent period: 3 - 4 weeks
- ☒ Pregnant cows body tissues acts as a reservoir of larvae for its offspring (features is the reservoir of larvae in the tissues of cows with subsequent milk borne transmission, ensuring that calves are exposed to infection from the first day of life).

Male calf	Age less than 4 month old	Horizontal transmission	No vertical transmission
	Age greater than 4 month old	Infection - not completion of life cycle	No vertical transmission
	Age less than 4 month old	Horizontal transmission	No vertical transmission

	Age greater than 4 month old	No horizontal transmission	Vertical transmission
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Life cycle



Pathogenesis

- ☒ Light infection - less than 70 larvae / calf - infection unnoticed
- ☒ Heavy infection - 70 -500 larvae / calf - clinical disease
- ☒ Lung: Verminous pneumonia – haemorrhagic & necrotic foci

Clinical signs

- ☒ Intermittent diarrhoea, steatorrhoea and mud coloured evil smelling faeces, pot-belly appearance and poor thriving

- ☒ Colic signs resembling intestinal obstruction (may be piling of adult worm in intestine)
- ☒ Calf hood mortality in buffalo calves

Diagnosis

- ☒ Based on history and clinical signs
- ☒ Identification of characteristic eggs

Treatment

- ☒ Benzimidazole compounds-fenbendazole and oxfendazole can be used
- ☒ Morantel citrate - 10 mg / kg -mature and immature worm
- ☒ Levamisole - 7.5 mg / kg - mature and immature worm
- ☒ Piperazine - 250 mg / kg b.wt

Control

- ☒ The prevalence of infection can be drastically reduced by treating calves at 3rd and 6th weeks of age to prevent the developing worms reaching patency.

Prophylactic Schedule: (Buffalo calf)

- ☒ First dose - 15th days of age
- ☒ Second dose - 30th days of age
- ☒ Third dose - 3rd month of age
- ☒ Fourth dose - 6th month of age
- ☒ Fifth dose - one year of age

Toxocara canis

- ☒ Host: Dog and fox
- ☒ Location: Small intestine
- ☒ Morphology: similar to *Toxocara vitulorum*

Epidemiology

- ☒ **Puppies less than six months are highly susceptible**
- ☒ Paratenic host: Rodents and birds ingest egg with L₂, the L₂ larvae settled in tissues of paratenic host.
- ☒ Definitive host pickup infection while ingestion of paratenic host infected with L₂ larvae
- ☒ Recent evident that bitches may be reinfected during late pregnancy or lactation leading directly to transmammary infection of the suckling pups
- ☒ Somatic tissues of bitch constant reservoir of infection for puppies as well as adults and unsusceptible to most anthelmintics

Prepatent period

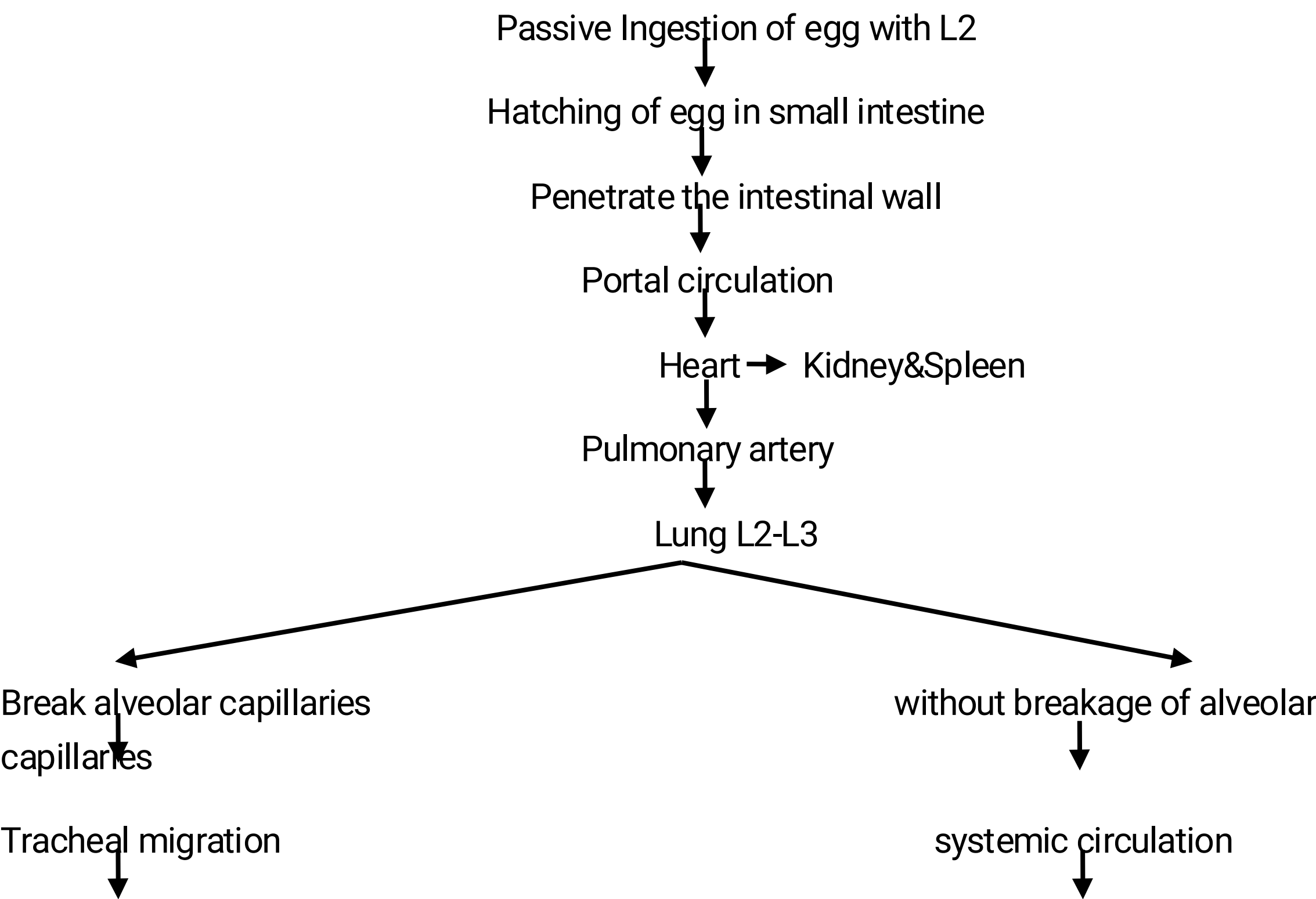
- ☒ Ingestion of eggs with L₂ without somatic migration: 3- 4 weeks
- ☒ Ingestion of paratenic host with L₂: 4 – 5 weeks
- ☒ In puppies due to prenatal infection: 3 weeks

Transmission and life cycle: Direct and complex in nature

Transmission

Passive ingestion of egg with L2	Male	>3 months	Somatic migration	Development arrested
	Female	>3 months		Transplacental transmission (Pregnant bitches)
				Transmammary/ Translacteal transmission (Whelped bitches)
	Male &female	< 3 months	Tracheal migration (L2-L3)	No vertical transmission

Life cycle



Reach the small intestine
↓
Mature into adult and lay eggs

somatic migration in bitches
↓
L2 settled at heart, lung, kidney, brain, G.I tract
↓
mammary gland
↓
Remain dormant
↓
If the animal is pregnant (at 42 days of gestation)
↓
L2 larvae cross the placenta
↓
Tracheal migration L2-L3
↓
Reach the small intestine
↓
Mature into adult and lay egg (L3-L5)

Weakened immunity

- ☒ Occurrence of eggs in faeces of bitches after whelping as a result of weakening of immunity and contaminated environment
- ☒ Lactogenic hormone (prolactin), which permits mobilization of larvae from somatic tissues to small intestine through lung, larvae undergo maturation and become adult
- ☒ Habit of whelped bitches to eat faeces of puppies, which may contain immature worms. The immature worms enter small intestine of bitches and become adult
- ☒ Post parturient infection are eliminated within a few weeks of the termination of lactation
- ☒ Not all larvae are mobilized during pregnancy and some may remain dormant in tissues and eliminated through subsequent pregnancies
- ☒ The duration of larvae in bitches may be long and animals infected for up to 385 days and are being capable of transmitting infection to puppies
- ☒ Factors which induce mobilization and migration are unclear but probably there is a hormonal basis involved

Pathogenesis

- ☒ In moderate infections, the larval migratory phase is accomplished without any apparent damage to the tissues
- ☒ In heavy infection the pulmonary phase of larval migration is associated with pneumonia. This is sometimes accompanied by pulmonary oedema. The adult worms cause mucoid enteritis; there may be partial or complete occlusion of the gut and in rare cases, perforation with peritonitis or in some instances blockage of the bile duct.

- ☒ Aberrant site
 - Bile duct-inflammatory changes and occlusion
 - CNS-Nervous disorder

Clinical manifestation

- ☒ In mild to moderate infections, there are no clinical signs during the pulmonary phase of larval migration
- ☒ In heavy infection migratory larvae causes pneumonic signs such as coughing, increased respiratory rate and frothy nasal discharge
- ☒ Heavy infection in puppies leads to death of whole litters
- ☒ Most fatalities from *T. canis* infection occur during pulmonary phase and pups which have been heavily infected transplacentally and may die within a few days of birth
- ☒ The adults in the intestine may cause pot-belly, with failure to thrive, and occasional diarrhoea. Entire worms are sometimes vomited or passed in the faeces
- ☒ Aberrant site in CNS - convulsion (yet to be proved)
- ☒ Death due to intestinal obstruction

Diagnosis

- ☒ Based on history and clinical signs
- ☒ Demonstration of eggs on direct smear method/Centrifugal sedimentation/Flotation method
- ☒ Excretion of adult worm in the faeces

Treatment

- ☒ In dogs, compounds approved for treatment of roundworm infections include fenbendazole, milbemycin, moxidectin, piperazine, and pyrantel
- ☒ Fenbendazole 100 mg/kg b.wt single dose or 50mg/ Kg b.wt daily for three days
- ☒ Piperazine - mature worm - 100 mg/ kg and immature worm - 200 mg /kg
- ☒ Pyrantel pamoate – 5-10 mg/kg b.wt
- ☒ Milbemycin oxime 0.5 mg/Kg b.wt
- ☒ Diethyl carbamazine (DEC) - 50 mg/ kg (contraindicated for dogs with patent infection of *D. immitis*)
- ☒ Dichlorvos – 12 – 15 mg/ kg body weight
- ☒ Trichlorophan - 75 mg/kg
- ☒ Nitroscanate - 50 mg/kg
- ☒ Mebendazole - 10 mg/ kg (twice a day for 2 days)

Control

- ☒ Good hygienic maintenance in kennels
- ☒ The main control measures relies on reducing environmental contamination

- ☒ Environmentally resistant larvated eggs on the ground and somatic larvae in the bitch are the main reservoirs of infection
- ☒ Perinatal transmission of infection can be greatly reduced by treating bitches with **daily doses of fenbendazole (25 mg/kg, PO) from day 40 of gestation to day 2 after whelping**
- ☒ Ivermectin (0.3 mg/kg, S/C) on days 0, 30, and 60 of gestation, and 10 days after whelping,
- ☒ Ivermectin (0.5 mg/kg) on days 38, 41, 44, and 47 of gestation, or ivermectin (1 mg/kg) on days 20 and 42 of gestation; these uses of ivermectin are extra-label
- ☒ To minimize egg output, pups should be treated as early as possible; ideally, treatment should be given 2 wk after birth and repeated at 2-wk intervals upto 2 month of age, and then monthly upto 6 month of age
- ☒ Nursing bitches should be treated at the same times as puppies. In cats, perinatal transmission can be greatly reduced by treating queens with a single dose of emodepside/praziquantel spot-on in the last week of pregnancy. Kittens should be treated with an appropriate anthelmintic at 3, 5, 7, and 9 wk of age, and then monthly to 6 month of age.
- ☒ Nursing queens should be treated at the same time as kittens. In other animals, the appropriate frequency of preventive treatment for roundworms should be based on a risk assessment of the animal's environment
- ☒ Prenatal and neonatal infections of young puppies are controlled by treating of pregnant bitches and her litter with suitable prophylactic schedule as follows

Puppies		Bitches	
Time schedule	Purpose	Time schedule	Purpose
-	-	Pregnant bitches 3 weeks before whelping to 2 days after whelping - daily high doses of fenbendazole	To eliminate transplacental and transmammary infection
2 nd weeks of age (1 dose)	To eliminate prenatal	Same time when pups are treated	To eliminate transmammary

	infection		infection
4 th week of age (II dose)	To eliminate prenatal infection	Same time	To eliminate transmammary infection
6 th week of age (III dose)	Neonatal infection through milk or from environment	-	-
Monthly once upto 6months	Routine prophylactic	Once in every 3-6 months	Routine prophylactic

- ☒ Kennels should be made up of impervious surfaces and floor should be thoroughly cleaned
- ☒ Control the entry of rodents in the kennels

Toxocara cati

- ☒ Host: cat and wild felids
- ☒ Location: small intestine
- ☒ Morphology: Male - 6 cm and female - 10 cm long, Egg: Similar to *T. canis* (Colourless thick pitted albuminous layer)

Life cycle: Direct

- ☒ Prenatal infection does not occur
- ☒ Paratenic host involves in the life cycle are earth worm, rodents, cockroaches, chicken, sheep and other animals fed with infective eggs
- ☒ Transmammary infection only occurs
- ☒ Larvae occur in the milk throughout lactation when queens infected with eggs and transported to the mammary gland after an extended period in other tissues before lactation commences

Pathogenesis and clinical signs

- ☒ Pot belly appearance, diarrhoea and poor hair coat

Diagnosis and treatment: Similar to *T. canis*

Control

- ☒ Removal of kitten from infected dams
- ☒ Similar prophylactic schedule as that of *T. Canis*

Toxocara leonina

- ☒ Host: Dog and cat
- ☒ Location: Small Intestine
- ☒ Paratenic host: Mice
- ☒ Morphology: Male – 7 cm and female – 10 cm, Egg: Slightly oval with smooth thick shell

Life cycle: Direct – but not complex

Egg (L₂) – > Passive ingestion – > Small intestine – > L₂ – > Adult -> eggs

Pathogenesis and clinical signs

- ☒ Pot bellied, intermittent diarrhoea and possibly anemia

Diagnosis and treatment: Similar to *T. canis*

Control

- ☒ Hygienic maintenance
- ☒ Prevent the exposure
- ☒ Prevent capturing of paratenic host

b.Strongyloidiosis

Etiology

- ☒ Family Strongyloididae comprises of free living, parasitic and both forms
- ☒ Parasitic form- parthenogenetic in nature

Strongyloides sp infection in various domestic animals

Species	Host affected	Location
<i>Strongyloides papillosus</i>	Sheep, goat, cattle, rabbit and wild ruminants	Small intestine
<i>S. westeri</i>	Horses, pig and zebra	Small intestine
<i>S. stercoralis</i>	Dog , cat and man	Small intestine
<i>S. cati</i>	Cat	Small intestine
<i>S. ransomi</i>	Pig	Small intestine
<i>S. avium</i>	Poultry	Small intestine

Morphology

- ☒ Egg: Blunt ends, thin shelled and contain fully developed embryos when passed in the faeces of host

Host affected

- ☒ Piglets, foals, calves, lambs - neonatal infection (through colostrum) after birth - mobilization of arrested larvae in tissues of ventral abdominal wall of the dam, which are subsequently mobilized and excreted through milk
- ☒ Dam is act as reservoir of larva for very young animal, clinical strongyloidosis in foals and pigs as a result of successive progeny from same dam causes heavy infection
- ☒ *S.ransomi* causes prenatal infection in pig

Environment

- ☒ Larvae are not ensheathed and susceptible to adverse environmental condition
- ☒ Warm and moist condition favours accumulation of infective larvae

Pathogenesis

S. papillosus

- ☒ Catarrhal enteritis
- ☒ Skin penetration by infective larvae causes erythematous reaction, which favours introduction of foot rot organism into skin around feet

S. stercoralis

- ☒ In puppies during summer months causes catarrhal enteritis

S. ransomi

- ☒ It infect mainly suckling piglets
- ☒ Piglets get infection while suckling, because the infective larva is adhere on udder and teat
- ☒ Piglets can also infected through skin penetration of larva from soil and litter
- ☒ Protein losing gastroenteropathy
- ☒ Erythematous skin lesion may be seen, pulmonary disorder is not frequent in natural outbreaks. verminous pneumonia in all animals, petechial/ ecchymotic haemorrhagic in lungs

Clinical manifestation

- ☒ Skin lesion
- ☒ Severe diarrhoea which may be blood stained, dehydration, anorexia, loss of weight and finally death (especially piglets mortality - 50 %)
- ☒ Foals - acute diarrhoea

Immunity

- ☒ Young animals infected with few infected larvae develops marked immunity
- ☒ Foals - satisfactory immunity occur 15 to 23 weeks of after birth

Diagnosis

- ☒ Based on clinical signs
- ☒ Demonstration of eggs and larvae in faeces

Treatment

- ☒ Sheep-Thiabendazole @ 75 mg/ kg orally
- ☒ Pigs- Thiabendazole @ 50 mg/ kg through feed and Levamisole @ 5 - 10mg / kg
- ☒ For Dog
 - Diethylcarbomazine @ 100 mg / kg
 - Dithiazanine @ 5 mg /kg (10 daily doses)
 - Pyvinium pamoate @ 20 mg /kg (5 days)
 - Thiabendazole @ 50 -75 mg/ kg – highly effective
- ☒ For foals
 - Cambendazole @ 20 mg /kg
 - Fenbendazole @ 50 mg /kg

Prophylaxis

- ☒ For prenatal and transcolostral infection in anticipated cases, treat before manifestation of clinical signs
- ☒ For pigs
 - Mebendazole @ 72 – 104 mg / kg for daily 12 – 14 days prior to farrowing
 - Levamisole @ 140 mg /kg - total dose
 - Ivermectin -Single dose - 4 – 16 days prior to farrowing

c.Ancylostomiosis

It is one of the widespread causes of morbidity and mortality in pups primarily due to their blood sucking activities in small intestine.

Etiology

- ☒ *Ancylostoma caninum* – principal cause of canine hookworm disease in most tropical and subtropical areas of the world
- ☒ *A. tubaeforme* – cats
- ☒ *A. braziliense* – dogs and cats
- ☒ *A. ceylanicum*-dogs, cats and people
- ☒ *Uncinaria stenocephala*-dogs and cats
- ☒ These worms have hook posture at anterior end, which is responsible for blood sucking activities in intestine
- ☒ The infective larvae of canine hookworms, particularly those of *A.braziliense*, may penetrate and wander under the skin of people and cause **cutaneous larva migrans**.

Epidemiology

- ☒ Worldwide in the tropics and warm temperate areas
- ☒ In endemic areas the disease is most common in dogs under one year old
- ☒ In older animals, the gradual development of age resistance makes clinical disease less likely, particularly in dogs reared in endemic areas whose age resistance is reinforced by acquired immunity
- ☒ Predisposing factor- failure to removal of soiled bedding in kennels, damp, porous and cracked floors, can lead to a massive build-up of infection

Transmission

- ☒ Ingestion of infective L₃ larvae from the environment
- ☒ Skin (percutaneous) penetration of infective L₃ larvae of *A.caninum*, *A.braziliense*, *A.tubaeforme*, or *A. ceylanicum*
- ☒ Transmission of larvae of *A.caninum* through colostrums or milk of infected bitches
- ☒ Suckling of new born pups on soiled mammary gland

Life cycle

Infection by either ingestion or skin penetration in young pups is followed by migration of the larvae through the blood to the lungs, where they moult to L₄ in bronchi and trachea, which is coughed up and swallowed then mature in the small intestine. However, animals >3 months old, *A. caninum* larvae, after migration through the lungs, development are arrested in the somatic tissues (skeletal muscle). These arrested larvae are activated during pregnancy, then accumulate in the mammary glands. Arrested development may also occur in the mucosa of the small intestine; activation may occur after removal of adult worms from the intestine.

The reactivated L₃ larvae passed in milk of bitch for 2 weeks after whelping. This trans mammary infection is often responsible for **severe anemia in litters of young pups** in their second or third week of life. **Infection of the bitch** on a single occasion has been shown to produce **transmammary infections in at least three consecutive litters**.

Pathogenesis

The acute or chronic haemorrhagic anemia starts about 8th day of infection in pups by plug feeding activities of immature adults on mucosa containing arterioles. Each worm removes about 0.1 ml of blood within 24 hrs.

Clinical manifestation

- ☒ Anemia results directly from the blood sucking and the bleeding ulcerations at the site of bite
- ☒ Initially acute normocytic, normochromic anemia followed by **hypochromic, microcytic anemia in young puppies** is the characteristic feature
- ☒ Surviving puppies develop some immunity and show less severe clinical signs

- ☒ Debilitated and malnourished animals may continue to be unthrifty and suffer from chronic anemia
- ☒ **Diarrhoea with dark, tarry faeces accompanies severe infections. Anemia, anorexia, emaciation, and weakness develop in chronic disease**
- ☒ Skin infection with *U. stenocephala* causes dermatitis in the inter digital space

Necropsy findings

- ☒ Liver-fatty infiltration is seen
- ☒ Lung consolidation

Diagnosis

- ☒ Based on clinical signs
- ☒ Presence of thin-shelled, oval eggs is readily seen in fresh faeces of infected animals either by centrifugation sedimentation/Floatation technique
- ☒ Haematological analysis to confirm anaemia

Treatment

- ☒ In dogs, fenbendazole, moxidectin, and pyrantel are approved for treatment of *A. caninum* and *U. stenocephala* infections
- ☒ When anemia is severe, chemotherapy may have to be supported by blood transfusion or supplemental iron, followed by a high-protein diet until the hemoglobin level become normal
- ☒ When neonatal pups die due to hookworm infection, subsequent litters from the bitch should be treated weekly for *A. caninum* for upto 12 wk beginning at 2 wk of age.
- ☒ In addition, fenbendazole (**25 mg/kg, PO**) given daily to **pregnant bitches three weeks before whelping and two days after whelping** greatly reduces transmammary transmission to the pups
- ☒ Treatment of the bitch with ivermectin (0.5 mg/kg) on two occasions (4– 9 days before whelping and 10 days later), or with moxidectin/imidacloprid spot-on on day 56 of pregnancy, has the same effect (extra-label use).

Control

- ☒ Weaned pups and adult dogs are treated every three month
- ☒ Bitches should be free of hookworms before breeding and kept out of contaminated areas during pregnancy
- ☒ Sanitary quarters should be provided for whelping and nursing bitches
- ☒ Concrete runways that can be washed at least twice a week in warm water
- ☒ Sunlit clay or sandy runways can be decontaminated with sodium borate (1 kg/2 m²).

d. Bunostomiosis

Bunostomum trigonocephalum

- ☒ Host: Sheep, goat and cattle (rarely)
- ☒ Location: ileum and jejunum
- ☒ Life cycle: Similar to *Ancylostoma*, prepatent period: 30 – 56 days
- ☒ Epidemiology: Similar to *Ancylostoma*

Occurrence is more common in warmer climate than cold climate

Mixed infection of *Strongyle* sp and *Bunostomum* sp are common

Clinical manifestation

- ☒ Anaemia, oedema on intermandibular region – “ Bottle jaw”
- ☒ Diarrhoea is not infrequent, faeces is dark in colour due to altered blood pigments
- ☒ Prostration and finally death

Diagnosis

- ☒ Based on clinical signs
- ☒ Faecal examination

Treatment

- ☒ Fenbendazole @ 5.0 mg /kg b.wt
- ☒ Albendazole @ 7. 5 mg /kg b.wt
- ☒ Thiabendazole @ 75 mg /kg b.wt

Control

- ☒ Avoid wet floor shelter – larvae highly susceptible to dryness

Bunostomum phlebotomum

- ☒ *Host*: Young calves, sheep is rarely affected
- ☒ *Location*: Duodenum

Clinical manifestation

- ☒ Anaemia, diarrhoea and submandibular oedema
- ☒ In stabled cattle causes itching and stamping of legs
- ☒ Treatment and control similar to *B. trigonacephalum*

e. Strongylosis

Strongylosis is most frequently a common problem in young horses reared on permanent horse pastures, although cases of severe disease may occur in adult animals kept in suburban paddocks and subjected to overcrowding and poor management.

Etiology

Species	Host	Location	Morphology

<i>Strongylus vulgaris</i>	Horse and donkey	Caecum and colon	Buccal capsule have two ear shaped teeth
<i>Strogylus equinus</i>			Three conical teeth
<i>Strongylus edentatus</i>			No teeth

It is a robust dark red worm easily seen on intestinal mucosa. Eggs are passed in faeces and development of L₃ takes place in an environment.

Host affected

- ☒ Infection is more severe in foals, adults are resistant to reinfection

Source of infection

- ☒ Infective larvae which have developed during grazing season survived on pasture over winter
- ☒ Eggs passed on current grazing season by nursing horse and shared grazing on same area

Life cycle

Infection is by ingestion of the L₃, which penetrate the intestinal mucosa and moult to L₄ in the submucosa. These then enter small arteries and migrate on the endothelium to their predilection site in the **cranial mesenteric artery and its main branches**. After a period of certain development the larvae moult to L₅ and return to the intestinal wall (the arterial lumina). Nodules are formed around the larvae mainly in the wall of the caecum and colon, when due to their size, they can travel no further within the arteries and subsequent rupture of these nodules releases the young adult parasites into the lumen of the intestine. The prepatent period is 6-7 months.

Pathogenesis

The larvae migrate to cranial messentric artery and its main branches cause thrombus formation provoked by larval damage to the endothelium together with marked inflammation and thickening of arterial wall. True aneurysm with dilatation of arterial wall was encountered in repeated infection.

Clinical manifestation

- ☒ It usually seen upto 2-3 yrs of age
- ☒ Unthriftiness, poor performance, weight loss, anaemia and diarrhoea
- ☒ Poor appetite
- ☒ **Intermittent colic**

Diagnosis

- ☒ Based on EPG - 1,000

☒ Fecal culture - Identification of L₃ larvae

☒ Based on clinical signs

Aneurysm in cranial mesenteric artery – palpated through rectal examination (Large pulsating bodies - 6 – 7 cm)

Hyper motility of intestine heard on abdominal auscultation

Vertebral percussion -pain evinced on lumbar and sacral region during percussion

Differential diagnosis

Anemia is differentiated with babesiosis, equine infectious anaemia, ascariasis and nutritional deficiency in grasses.

Diarrhoeais is differentiated with salmonellosis

Treatment

☒ Treatment may targeted against immature and adult stage

☒ Fenbendazole @ 30mg/kg single dose or Fenbendazole @ 7.5 mg / kg daily for 5 days effective against mucosal larvae and kill over 90% of hypobiotic larvae

☒ Ivermectin @0.2 mg/kg is effective against mucosal stage of larvae and repeated at 8-10 weeks interval

☒ Mebendazole@5-10mg/kg, Oxibendazole@10mg/kg is effective against adult worms

☒ Tetrahydropyrimidines-Pyrantel@ 19mg/kg, Pyrantel [pamoate@6.6mg/kg](#)

Prophylactic Schedule

☒ High stocking density & favorable environmental condition - deworm once in six weeks

☒ High stocking density & unfavorable environmental condition (winter)- deworm once in two weeks)

☒ Low stocking density - deworm once in three months

Treatment for colic

☒ If animal have verminous aneurysm give 6 % iron dextran & 5% dextrin through I/V for antithrombin activity

Prophylaxis and control

☒ Routine anthelmintics - all grazing animals over two months of age treated with broad spectrum anthelmintics (change drugs once in 6 months to prevent anthelmintic resistance)

☒ Pasture and Grazing management - not over grazed or over stocked

Alternative grazing: Horse + Sheep; Horse + Cattle (cattle will assist reduction of larvae burden in the pasture because, strongyles do not infect ruminants)

Rotational grazing: nourishing mares and their foals, don't graze the same area for successive year

Special attention for foals - fed only wholesome fodder

Proper disposal of manure

Heaping of manure will generate heat and destroy eggs

Newly purchased animal should be treated with anthelmintics and quarantine for 48-72 hrs before introduced into herd

f. Oesophagostomiosis

(Nodular worm of ruminants and pigs, Pimply gut)

It is a major cause of enteritis in ruminants and pigs in tropical and subtropical areas and usually associated with nodule formation in the large intestine.

Etiology

Species	Host	Location
<i>Oesophagostomum columbianum</i>	Sheep and goat	Caecum and colon
<i>O. Venulosum</i>	Sheep and goat	Caecum and colon
<i>O. radiatum</i>	Cattle and buffalo	Caecum and colon
<i>O. dentatum</i>	Pig	Caecum and colon
<i>O. quarispinulatum</i>	Pig	Caecum and colon

Morphology

- ☒ Adult worm Length: 1 – 2 cm
- ☒ Egg: Thin shelled embryonated eggs (Blastomere 8 – 16 cell stage)

Life cycle

The infection occurs by ingestion of infective L₃ larvae. The L₃ larvae enter the mucosa of any part of small or large intestine. The L₃ moult to L₄ emerges on mucosal surface and migrates to colon and develops into adult stage. The prepatent period is about 45 days. **On reinfection with most species, the larvae may remain arrested as L₄ in nodules for up to one year.** However infection with *Oe. Venulosum* nodules are absent.

Pathogenesis

In the intestine *Oe. Columbianum* L₃ migrates deep into the mucosa of caecum, provoking an inflammatory response with the **formation of nodules**, which are visible to the naked eye (**pimply gut**). On reinfection, this response is more marked, increase in size of nodules (upto 2.0 cm dia) containing eosinophilic greenish pus and L₄. When larva emerges, there may be an ulceration of mucosa leads to diarrhoea one week after primary infection. In heavy it causes ulcerative colitis.

Clinical manifestation

Acute

- ☒ Young lambs are highly susceptible
- ☒ Severe dark green diarrhoea, the faeces may contain excess mucus as well as streak of blood
- ☒ The sheep become weak, lose weight despite a good appetite, and show intermittent diarrhea and constipation
- ☒ Nodule formation usually is more pronounced in sheep than in cattle. Affected sheep walk with a stilted gait and often have a humped back
- ☒ Stenosis and intussusception may develop in severe cases

Chronic

- ☒ It occur primarily in sheep
- ☒ Inappetence, emaciation with intermittent diarrhoea
- ☒ Anaemia is the main clinical signs

Pig

- ☒ Pregnant sows show inappetence, become very thin following farrowing, reduced milk production, which indirectly affect the growth performance of litter

Immunity

- ☒ Prolonged survival of L₄ on nodules in the gut wall and the lack of an effective immunity, control is difficult
- ☒ In cattle good immunity developed against *O.radiatum* partly due to age and previous exposure

Diagnosis

- ☒ Based on clinical signs
- ☒ Post mortem examination-presence of nodules in the colon (pimply gut)
- ☒ Acute case-eggs are not usually present
- ☒ Chronic cases- eggs are identified in faeces
- ☒ The L₃ identified in faecal culture

Treatment

- ☒ Benzimidazole compound at recommended dose

- ☒ Levamisole and morantel citrate at recommended dose

Control

- ☒ Prevent heavy exposure in susceptible hosts
- ☒ Reduce pasture contamination
- ☒ Minimize the effects of worm burden
- ☒ Encourage the development of development of immunity in the animals

g.Ostertagiasis

It is a major cause of parasitic gastritis in ruminants in temperate areas of the world.

Species	Host	Location
Major pathogens		
<i>Ostertagia ostertagia</i>	Cattle	Abomasum
<i>Ostertagia(Teladorsagia) circumcinata</i>	Sheep and goat	Abomasum
<i>Ostertagia trifurcate</i>	Sheep and goat	Abomasum
Minor pathogens		
<i>Ostertagia lyrata and O.chochida</i>	Cattle	
<i>Ostertagia leptospicularis</i>	Cattle, sheep and goat	

The adult worms are **slender reddish-brown** up to 1.0cm long, occurring on the surface of the **abomasal mucosa and are only visible on close inspection**. The larval stages occur in the gastric glands and can only be seen microscopically following processing of the gastric mucosa.

Distribution: Worldwide, Ostertagia is especially important **in temperatue climate** and in **subtropical regions with winter rainfall**.

Ostertagia ostertagia

Ostertagia ostertagi is the most common cause of parasitic gastritis in cattle. The disease often simply known as ostertagiasis. It is characterized by weight loss, diarrhoea

and typically affects young cattle during first grazing season, although sporadic individual cases in adults are reported.

Life cycle: Direct

The eggs are passed in the faeces and develop into infective L₃ larva within the faecal pat under optimal condition. The L₃ Larvae remain in the faecal pat until moistened by rain. The L₃ larvae migrate from faeces onto herbage. After ingestion, the L₃ exsheaths in the rumen and further development takes place in the lumen of abomasal gland. Two parasitic moults occur before the L₅ emerge from gland around 18 days after infection. For completion of entire life cycle takes three weeks. But under certain circumstances many of the ingested L₄ become arrested in development at the early fourth larval stage (EL₄) for periods of up to six months.

Pathogenesis

The developing parasites cause a **reduction in the functional gastric gland mass** responsible for the **production of the highly acidic proteolytic gastric juice**; in particular, the **parietal cells**, which produce **hydrochloric acid**, are replaced by rapidly dividing, undifferentiated, non-acid-secreting cells. Initially, these cellular changes occur in the parasitized gland, but as it becomes distended by the growing worm which increases from 1.3-8.0mm in length, these changes spread to the surrounding non parasitized glands, results in thickened hyperplastic gastric mucosa. This results in leakage of pepsinogen into the circulation leading to elevated level of plasma pepsinogen and the loss of plasma proteins into the gut lumen eventually leading to **hypoalbuminemia**.

Clinical manifestation

It occur in two form

Type I

- ☒ It is usually seen in calves grazed intensively during their first grazing season. Infection usually occurs in mid July onwards (July-October month)
- ☒ Persistent profuse greenish watery diarrhoea
- ☒ loss of weight
- ☒ Morbidity is high and mortality low

Type II

- ☒ The disease occur in yearling and in late winter or spring following first grazing season results from maturation of larva ingested during the previous autumn (arrested development at L₄ stage)
- ☒ Intermittent profuse watery diarrhoea, anorexia, thirst and loss of weight
- ☒ Hind quarters highly soiled with faeces
- ☒ Hypoalbuminemia
- ☒ Submandibular edema

- ☒ Morbidity is low and mortality rate is very high

Treatment

- ☒ Benzimidazole (Albendazole, Fenbendazole or Oxfendazole), Probenzimidazole (febental, netobiomin, and thiophnate), Levamisole or ivermectin and milbemectins are effective against immature and mature adult worms.

Diagnosis

- ☒ Base on clinical signs
- ☒ Seasonal occurrence
- ☒ Grazing history
- ☒ Faecal egg count
- ☒ Increased level of plasma pepsinogen (Tyrosine) > 3.0 IU/L (Normal 1.0 IU/L)
- ☒ Post mortem examination-presence of slender small red colour worms on abomasal mucosal surface

Control

- ☒ Limiting exposure to infection by grazing on new pasture
- ☒ Prophylactic antihelmintic treatment
- ☒ Rotational grazing of young and adult stock- It involves continuous rotation of each paddock, in which susceptible young calves graze ahead of immune adults

Ovine and caprine ostertagiosis: Pathogenesis, clinical signs and treatment similar to cattle

h.Trichostrongylosis

Etiology

Species	Host affected	Location
<i>Trichostrongylus Colubriformis</i>	Sheep, goat, cattle, camel and antelope	Abomasum, anterior part of small intestine
<i>Trichostrongylus axei</i>	Sheep, goat, cattle, deer and antelope	Abomasum
	Pig, horse, donkey and man	Stomach
<i>Trichostrongylus tenius</i>	Domestic and wild ducks, geese, fowl, G. fowl and turkey	Caecum and small intestine

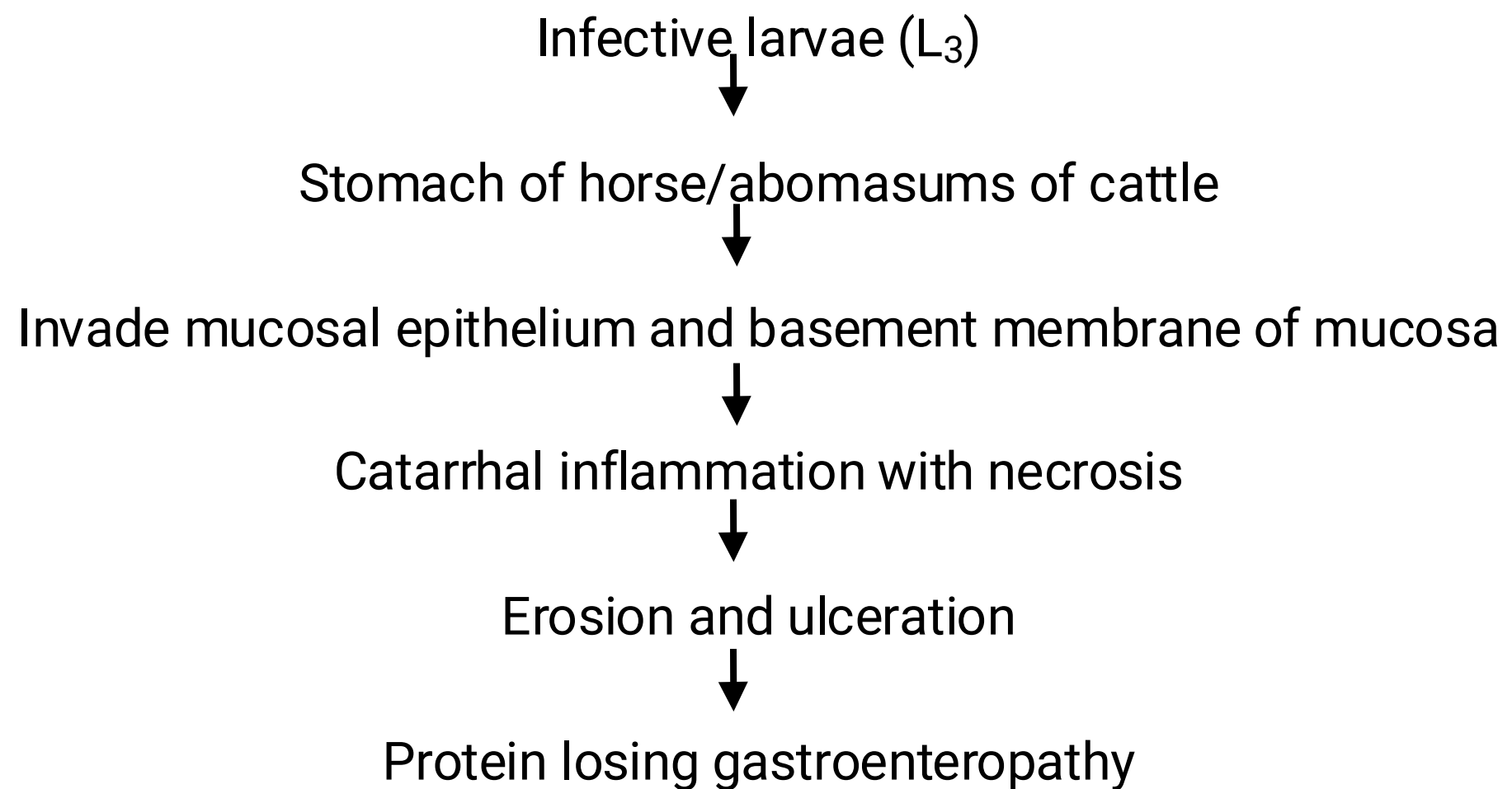
- ☒ It is a small slender pale reddish brown worm
- ☒ Egg: oval, thin shelled segmented yolk (32 cell stage when laid)

Life cycle

- ☒ Infection of definitive host: Through passive ingestion
- ☒ Exsheathment (abomasum or small intestine)

Pathogenesis

- ☒ *T. Axei*



T. colubriformis: Similar to *T. axei*

- ☒ It occurs in anterior part of small intestine
- ☒ In mucosa - formation of tunnels between epithelium and basement membrane
- ☒ Distortion and displacement of mucosal epithelium
- ☒ Loss of plasma protein into intestine results in hypoalbuminaemia

In general Trichostrongylus spp

- ☒ Intestinal parasitism - increased gastric secretion of cholecystokinin act on appetite centre in brain, leads to decreased appetite
- ☒ Loss of serum protein is due to anorexia, reduced feed intake and reduced food conversion ratio
- ☒ Reduced absorption of calcium and phosphorus from intestine leads to retarded bone growth and osteoporosis

Clinical manifestation: Gastrointestinal disturbances – diarrhoea

- ☒ Acute cases

Sudden death, such animal neither emaciated nor anaemic
Leg weakness and unable to stand

- ☒ Chronic cases

variable in appetite, emaciation, rough and dry hair coat
alternatively constipation and diarrhoea, faeces is dark in colour, so the worm is called as “ **Black scour worms**”
anemia is not well pronounced (mild one)

In avian species

- ☒ Haemorrhagic typhilitis with diarrhoea, loss of appetite, emaciation and anemia

Immunity

- ☒ Infection induces immunity to reinfection (intake of infective larvae or presence of adult worm)
- ☒ Cross immunity is observed between *Trichostrongylus* spp, not at generic level

Diagnosis

- ☒ Demonstration of eggs in faeces
- ☒ Faecal culture

Treatment: Same as that of ostertagia

Control

- ☒ Reduce pasture contamination
- ☒ Minimize the antihelmintic resistance by rotational use of antihelmintics
- ☒ Antihelmintic treatment of weaned lambs before entered into the grazing area
- ☒ Reduce the periparturient rise (PPR) of egg count in pregnant ewes by treating with antihelmintic at the end of gestation and one month after lambing
- ☒ Alternate or mixed grazing of animals with different host species and different age groups, immune and nonimmune animals.

i. Haemonchosis

(Stomach worm or Wire worm of Ruminants)

Etiology

- ☒ *H. contortus*-sheep, goat and cattle
- ☒ *H. placei*-cattle
- ☒ It is a blood-sucking abomasal nematode may be responsible for extensive losses in sheep and cattle, especially in tropical areas

Morphology

- ☒ Adult worm- Larger in size (1-3 cm) and visible under naked eye, male even reddish in colour and female uneven reddish in colour due to twisting of ovaries around reddish intestine. The buccal capsule has slender tooth or lancet.

Epidemiology

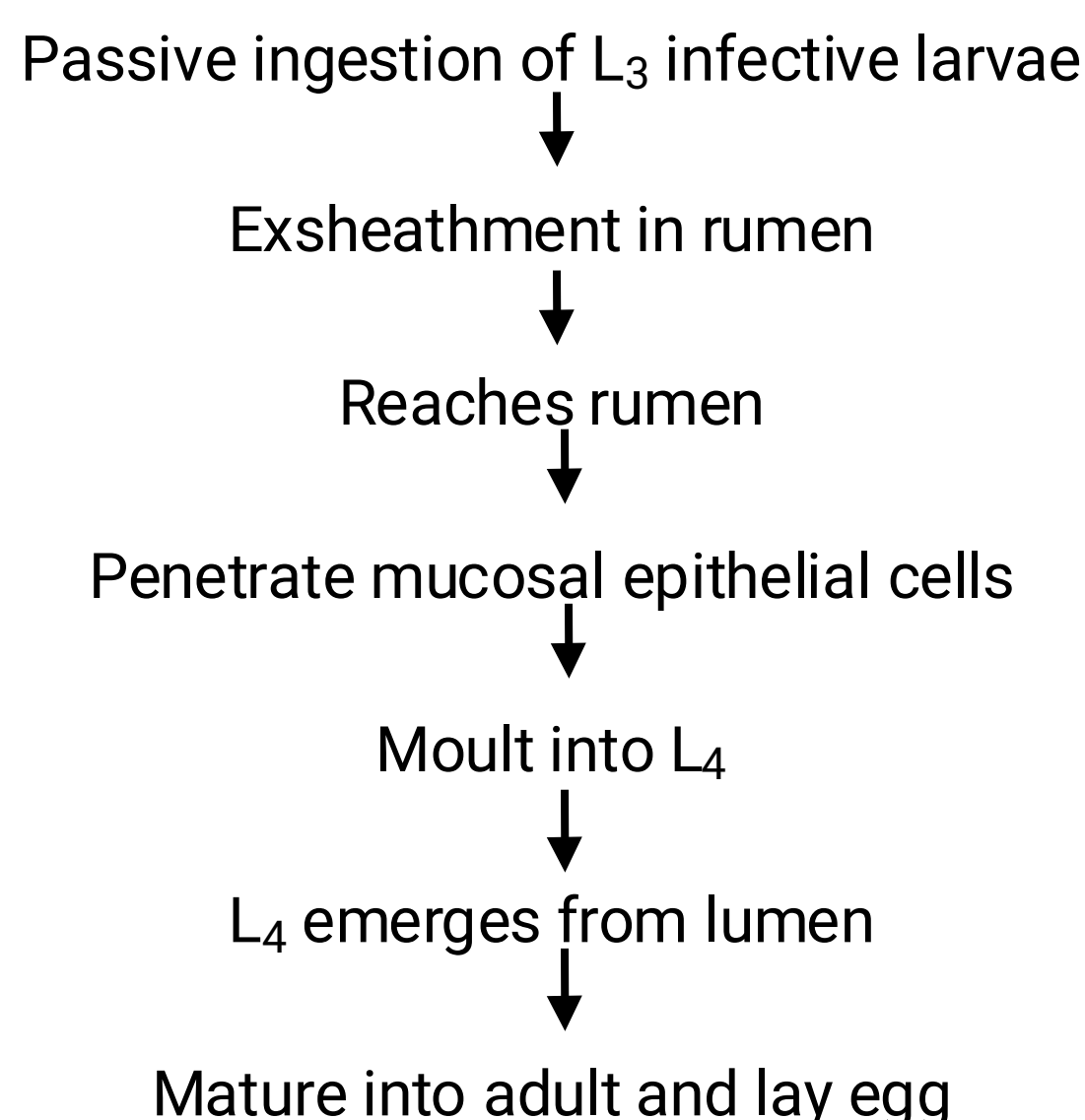
- ☒ The frequency and severity of outbreak is largely depends on rainfall in that particular area
- ☒ Survivability of larva is associated with ability to undergo hypobiosis

- ☒ Resumption of development occurs just before the onset of seasonal rains
- ☒ The sudden occurrence of acute clinical haemonchosis is depends on high faecal worm egg output (2000 and 20000epg)and massive pasture contamination with L3 larvae
- ☒ Sheep in endemic areas develop an acquired immunity, so that continuous contamination of the pasture occur
- ☒ Epidemiology *H. placei* is similar to *H. contortus*.
- ☒ In addition to that cattle fecal pad acts as a reservoir of infective larvae of *H. placei*
- ☒ In which the larvae survive for several months and ultimately released when fecal pad is sufficiently moistened by rain

Host affected

- ☒ *H. contortus*: Sheep
 - Age: Weaner age group (3 to 6 months) and lambed ewes are highly susceptible
 - Breed: Native breeds are highly resistant (Presence of anti-larval IgA antibodies) to infection than exotic and crossbred sheep
 - Hemoglobin type: Sheep having homozygous Hb type AA resistant to infection than sheep having heterozygous Hb type AB
- ☒ *H. placei*: Cattle less than 2 years old are highly susceptible.

Life cycle



Prepatent period

- ☒ *H. contortus*: 15 days

- ☒ *H. placei*: 30 days

Pathogenesis

Acute haemonchosis

- ☒ Acute haemorrhagic anemia due to the blood sucking habits of the worms
- ☒ Each worm suck 0.05 ml of blood/day
- ☒ Sheep, Goat and cattle infected with 5000 *H. Contortus* may lose about 250 ml blood daily
- ☒ Anemia become apparent after two weeks of infection and is characterized by progressive and dramatic fall in PCV. During subsequent weeks haematocrit value stabilizes at low level, but at expense of two to three fold compensatory expansion of erythropoiesis
- ☒ However continual loss of iron and protein in gastrointestinal tract and increasing inappetence, the bone marrow eventually become exhausted. The haematocrit value again falls before death
- ☒ When ewes are affected, the consequent agalactia may result in the death of the suckling lambs

Hyperacute haemonchosis

- ☒ Heavy infection of sheep with 30000 worm, the apparently healthy sheep suddenly die due to severe acute gastroenteritis

Chronic haemonchosis

- ☒ It is important as acute haemonchosis.
- ☒ It develops during a prolonged period of dry season, when reinfection are negligible, but pasture become deficient in nutrients
- ☒ Continuous loss of blood results from persistence of small worm burden (100-1000 infective larvae) in intestine

Clinical manifestation

Hyperacute case

- ☒ Sheep die suddenly with haemorrhagic gastritis

Acute haemonchosis

- ☒ **Anemia, submandibular edema, ascites**, lethargy, dark coloured faeces, loss of wool, diarrhoea is not generally a feature

Chronic haemonchosis

- ☒ Progressive weight loss, weakness, neither severe anemia, nor gross edema being present

Post mortem finding

- ☒ Hydrothorax, hydropericardium and ascites
- ☒ Watery blood

- ☒ Abdominal fat – liquefied and become gelatinous in nature
- ☒ Liver – light brown in colour is due to fatty changes
- ☒ Abomasum- **reddish-brown fluid ingesta** and **presence of worm on the mucosal folds** and bite marks noticed on the abomasal mucosa

Diagnosis

- ☒ Based on clinical signs
- ☒ Identification of eggs in faeces
- ☒ Fecal culture
- ☒ Autopsy of animal in the suspected flocks

Treatment

- ☒ When an acute outbreak has occurred the sheep should be treated with one of the benzimidazoles, levamisole, avermectin, milbemycin or salicylanilide and allowed to graze in fresh pasture
- ☒ Chronic haemonchosis is dealt with a similar fashion. If possible the new pasture should have a good nutritional value; alternatively some supplementary feeding may be given

Control

Self cure phenomenon

- ☒ In an endemic area, the faecal egg count of sheep drops to zero level after a period of heavy rainfall is due to expulsion of major part of adult worm burden. This is called as self cure phenomena. The expulsion of adult worm as consequence of immediate hypersensitivity reaction to the antigens derived from developing larvae
- ☒ Loss or expulsion of worm burden in the gastro-intestinal tract of the suitably infected and sensitized sheep can be induced by challenge dose of infective larvae
- ☒ Self-cure and protection against infection are not necessarily interrelated - Immunity is not an absolute
- ☒ Latent period of self cure induction is 6-7 weeks

Host factor which induce Self cure phenomena

- ☒ *Species*: Sheep – it is well pronounced
- ☒ *Breed*: Native breed of sheep having good response than exotic and crossbred sheep (due to presence of anti-larval IgA antibodies in the native breed)
- ☒ *Age*: Sheep greater than 6 months old having good response
- ☒ *Hemoglobin type*: Sheep with homozygous Hb type AA having good response than homozygous Hb type BB
- ☒ *Stage of lactation*: Relaxation of immunity – poor response
- ☒ *Level of lymphocytes in peripheral blood*: Sheep with low level of lymphocytes having low immune response

Immunity

- ☒ Cross immunity was observed between species of *Haemonchus*
- ☒ Cross immunity occur between genus of *Haemonchus* and *Trichostrongylus*, but it is one way
- ☒ *Haemonchus* induce immunity against *Trichostrongylus* and not vice versa
- ☒ Exsheathed larvae induce good immunity than ensheathed larvae.
- ☒ Irradiated larvae induce good immunity
- ☒ Size of the challenge dose is inversely related to the self-cure

Verminous bronchitis

Lung worms of domestic animals includes

Lung worm	Host	Location
<i>Dictyocaulus viviparous</i>	Cattle	Bronchi
<i>Dictyocaulus filarial</i>	Sheep and goat	Bronchi
<i>Protostrongylus rufescens</i>		Bronchi
<i>Mullerius capillaries</i>		Lung
<i>Metastrongylus apri</i> (<i>M.elongatous</i>)	Domestic and wild Pigs	Bronchi and bronchioles
<i>M.salmi</i>		
<i>M.pudenodotectus</i>		
<i>Dictyocaulus arnfieldi</i>	Donkey and Horse	Bronchi
<i>Oslerus osleri</i>	Dog	Trachea
<i>Aelurostrongylus abstrusus</i>	Cat	Lung
<i>Capillaria aerophila</i>	Cat	Trachea and bronchi

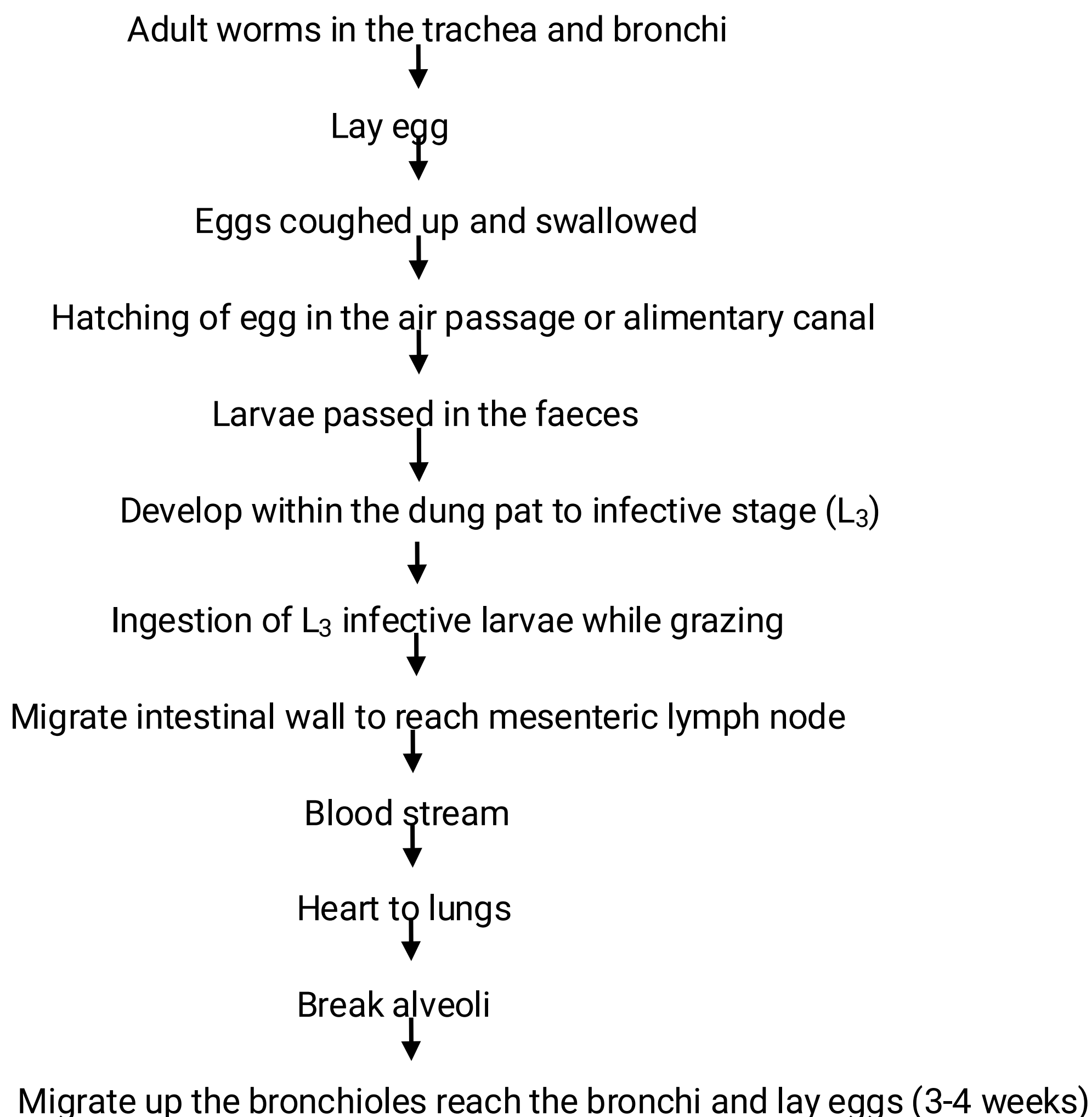
Verminous bronchitis in cattle (*Dictyocaulus viviparous*)

Epidemiology

- ☒ Bovine parasitic bronchitis is a sporadic and unpredictable disease, because immunity develops more quickly than other nematodes
- ☒ Disease is common in temperate countries due to high rainfall, prevent desiccations of the larvae

- ☒ Disease is more common in young calves
- ☒ Wind borne field-to-field transmission of larvae of lung worm by *Pilobolus sporangiae*, which is common in cattle faeces
- ☒ The fungus *Pilobolus* may accumulate larvae of *D. viviparous* on the upper face of the spornangium, which when it explodes may propel the larvae as far as 3 meters
- ☒ little migration of larvae from the faecal pads on to herbage, except during heavy rainfall

Life cycle and transmission



Pathogenesis & clinical signs- husk or hoose

It depends on location of larva within the respiratory tract, number of larva ingested and animal's immune status.

Prepatent phase

- ☒ It is associated with blockage of many respiratory bronchioles with an eosinophilic exudates in response to developing larva, this results in obstruction of the airways and collapse of alveoli distal to the block

- ☒ Clinically associated with tachypnea, coughing and animal may die due to development of severe interstitial emphysema

Patent phase

- ☒ Lasting from days 25 to 55 is associated with adult parasite in the bronchi and trachea
- ☒ The severe damage to the epithelium of bronchi causes, marked exudation and blockage of air passages
- ☒ Development of chronic nonsuppurative eosinophilic granulomatous pneumonia in response to eggs and larvae aspirated into the bronchioles and alveoli, leading to consolidation of lung lobules
- ☒ Death is due to interstitial emphysema, pulmonary edema and secondary bacterial infections

Late prepatent phase

- ☒ Epithelialization of the alveolar epithelium and hyaline membrane formation occur
- ☒ Animals show dyspnea and coughing, with rapid loss of condition
- ☒ Harsh respiratory sounds with bronchi and emphysematous crackling can be heard
- ☒ Complication include pulmonary oedema and emphysema

Postpatent phase

- ☒ It is a recovery stage
- ☒ Respiratory rate decreases, coughing is less frequent and weight gain is resumed
- ☒ Larvae which reach the lungs are destroyed by the immune response with the formation of lymphoreticular granulomas or nodule results in bronchiolar obstruction

Diagnosis

- ☒ Based on the clinical signs and post mortem examination
- ☒ Presence of first stage larva in the faeces. larva usually not found in faeces during prepatent and post patent phase
- ☒ Bronchoscopy and radiography may helpful to find the larva and lesions in the bronchi
- ☒ Recovery of larva by Baermanns technique

Differential diagnosis

- ☒ Epizootic bronchitis
- ☒ Viral pneumonia
- ☒ Pasteurella infection

Treatment

- ☒ Ivermectin, doramectin, eprinomectin and moxidectin, fenbendazole, oxfendazole and albendazoles are effective against all stage of *D.viviparous*

- ☒ Ivermectin 200 µg/Kg
- ☒ Fenbendazole – 5.0 mg/kg
- ☒ Albendazole - 7.5 mg /kg
- ☒ Oxfendazole - 4.5 mg/kg
- ☒ Levamisole @ 7.5 mg/Kg effective against immature worm, second dose repeated 2 weeks later
- ☒ Diethylcarbamazine – 22 mg/kg intramuscularly for three days or a single dose of 50 mg/kg is primarily effective against the immature parasites

Control

- ☒ Grazing management should be improved
- ☒ Provide clean pasture for young calves
- ☒ Animals continuously exposed to infection are at little risk provided the rate of acquisition of the infection is sufficient to stimulate a satisfactory immunity and not enough to cause clinical illness
- ☒ Prophylactic anthelmintic treatment reduces pasture contamination
- ☒ Vaccination: Two doses of 1000 irradiated larvae given at 4 weeks interval
- ☒ Age of immunization – two months or older and exposure of infection is avoided until two weeks after second dose

Verminous bronchitis in sheep and goat

Etiology

Dictyocaulus filaria, Mullerius capillaris and protostongylus rufescens

Transmission

- ☒ Infective *Dictyocalus filaria* larvae found on grass and ingested during grazing
- ☒ *M. capillaris* and *P. rufescens* are transmitted when molluscan intermediate host are accidentally ingested by grazing animals
- ☒ Lambs 4-6 months of the age are most severely affected, but sheep of all age groups are susceptible

Life cycle: similar to *D.viviparous* in cattle

Pathogenesis

Presence of live larvae in bronchi, producing catarrhal parasitic bronchitis. The inflammatory process spreads to the surrounding tissues and exudate frequently passes back into the bronchioles and alveoli cause atelectasis and catarrh or pneumonia. Secondary bacterial infection leads to extensive areas of pneumonia

Clinical manifestation

- ☒ It is usually chronic in nature

- ☒ The animal may coughed up and tenacious mucus exudes from the nostrils, but cough is not always present
- ☒ Dyspnea is obvious signs
- ☒ Rapid respiration
- ☒ Abnormal lung sounds on auscultation
- ☒ Temperature is not elevated unless pneumonia develops

Necropsy finding

- ☒ Localized and cone shaped area of pneumonia, accompanied by atelectasis and compensatory emphysema

Diagnosis

- ☒ Presence of first stage larvae in fresh faeces
- ☒ Eggs may found in the sputum or nasal discharge
- ☒ Bronchoscopy and Radiography-to find lesions in bronchi and lungs

Treatment

- ☒ Ivermectin, doramectin, eprinomectin and moxidectin, fenbendazole, oxfendazole and albendazoles are effective against all stage of *D.filaria*
- ☒ Ivermectin 200 µg/Kg
- ☒ Fenbendazole – 5.0 mg/kg
- ☒ Albendazole - 7.5 mg /kg
- ☒ Oxfendazole - 4.5 mg/kg
- ☒ Levamisole @ 7.5 mg/Kg effective against immature worm, second dose repeated 2 weeks later
- ☒ Diethylcarbmazine – 22 mg/kg intramuscularly for three days or a single dose of 50 mg/kg is primarily effective against the immature parasites

Control

- ☒ Animal must be removed from infected ground, placed on dry pastures and supplied with clean drinking water
- ☒ Moist pastures must be avoided for grazing
- ☒ Older animals should not be grazed together with the young stock
- ☒ Vaccine - irradiated attenuated infective larvae used as vaccine

Verminous bronchitis in pig

- ☒ **Etiology:** *Metastrongylus apri* (*M.elongatus*), *M. Salmi* and *M.Pudendotectus*
- ☒ **Transmission :** Transmitted by ingestion of earthworm intermediate host
- ☒ **Age :** Pigs at 4-6 months of age are highly susceptible

- ☒ Verminous bronchitis and pneumonia are possibly due to secondary bacterial infection
- ☒ Larvae of lung worm carry the swine influenza and swine fever virus
- ☒ Larvae may transmit viruses of Teschen disease

Life cycle

The eggs are excreted through faeces, hatching of egg in soil and release the L₁ Larvae, which is ingested by earth worm and L₂ and L₃ developmental stage, occur in earth worm. Pig infected by ingestion of infected earth worm. Further development inside the pig is similar to *D.viviparous*

Clinical signs and lesions

- ☒ Marked reduction in growth rate
- ☒ **Barking cough and pneumonia**
- ☒ Presence of greyish nodule near the ventral border of diaphragmatic lobe

Diagnosis

- ☒ Based on the demonstration of embryonated eggs in fresh faeces

Treatment

- ☒ Ivermectin, moxidectin and doramectin are recommended for lung worm infection in pigs
- ☒ Levamisole 15 mg/ kg b.wt
- ☒ Tetramisole 15 mg/ kg b.wt
- ☒ Fenbendazole 20-30 mg/kg b.wt

Control

- ☒ Infected pigs should be kept on dry ground or in sites with concrete floor and their faeces should be disposed off in such a way that they do not spread the infection
- ☒ Young pigs should run on clean fields
- ☒ Infected paddocks and fields may remain infected for a considerable time, since the intermediate stage can be in the earthworm for an unknown period
- ☒ A 3 % solution of carbathion applied in soil to kills earthworm

Trypanosmiasis

The group of disease caused by protozoa genus trypanasoma affect all domestic animals.

S.No	Trypanoso--m e sp	Distribution	Host affected	Main vector	Transmission	Predilection site	Disease

1.	<i>T.evansi</i>	Africa, Asia, South and cenral America	Horse, camel, dog, cattle and buffaloe	Tabanus sp	Mechanical	Blood	Surra
2.	<i>T. equiperdum</i>	Africa, Asia, Europe, America	Horse and donkey	-	Venereal transmission	Genital organ, subcutaneous tissue and nervous system	Dourine
4.	<i>T.theileri</i>	Worldwide	Cattle	Tabanus sp	Mechanical	Blood	Non pathogeni c
5.	<i>T.brucei brucei</i>	Tropical Africa	All domestic and wild animal	Glossinia sp	Cyclical transmission	Blood, myocardium, CNS and reproductive tract	Nagana
6.	<i>T. congolense</i>	Tropical Africa	All domestic and some wild animal	Glossinia sp (tsetse fly)	Cyclical transmission	Blood	Nagana Gambia fever
7.	<i>T.vivax</i>	Tropical Africa	Cattle, sheep, goat, camel and horse	Glossinia spp (tsetse fly)	Salivarian transmission	Blood	Nagana
8.	<i>T. cruzi</i>	South and central America and united states	Dog, cat and pigs	Rhodinius sp	-	Myocardium	Chagas disease
9.	<i>T.simiae</i>	Central Africa	Sheep, goat, pig		-	Blood	Non

			and camel				pathogenic
10.	<i>T. Melophagium</i>	Worldwide	Sheep and goat	Melophagus ovinus	-	Blood	Non pathogenic
11.	<i>T. brucei rhodesiense</i> and <i>T. brucei gambiense</i>	Africa	Human	Glossinia sp (tsetse fly)	-		Sleeping sickness

Developmental stage

Trypomastigote stage

- ☒ A blade like form with kinetoplast posterior to the nucleus and usually near the posterior extremity. An undulating membrane is well developed and free flagellum is often present
- ☒ This stage usually found in vertebrate host and also in arthropods.

Epimastigote stage

- ☒ Kinetoplast and axoneme lie anterior to the nucleus and undulating membrane is short
- ☒ In few species, this stage found in vertebrate host, but principally in arthropods.

Promastigote stage

- ☒ Kinetoplast and axoneme are at the anterior tip of body with no undulating membrane
- ☒ It is found in arthropods

Amastigote stage

- ☒ Rounded body with kinetoplast
- ☒ Absence of flagellum (or with short fibrils)
- ☒ Found in arthropods

Transmission of trypanosomes

Cyclical transmission

1. **Anterior station development/ Salivarian transmission**- highly pathogenic, the trypanosome multiply in digestive tract and proboscis, so new infection is transmitted when feeding. e.g *T.congolense*, *T.vivax* and *T.brucei*

2. **Posterior station development/stercorarian transmission**- non pathogenic the multiplication and transformation of trypanosomes occur in gut and the infective forms migrate to the rectum and passed with the faeces
e.g. *T.melophagium* , *T.theileri* and *T.cruzi*

Non cyclical/Mechanical transmission

- ☒ The trypanosomes are transmitted from one mammalian host to another by interrupted feeding of biting insects e.g. *Tabanus* sp and *stomoxys* sp.
- ☒ The trypanosomes transmitted by bite of vampire bat also can occur e.g. *T.evansi*
- ☒ Mechanical transmission can also occur in some salivarian group, transmitted cyclically in tse-tse flies. Eg. *T. vivax* and *T. brucei*
- ☒ Dog, cat and wild carnivores infected by ingestion of tissue from infected fresh carcass, died of trypanosomiasis

Surra (Mal De caderas)

It is caused by *trypanosoma evansi*, which is the first pathogenic trypanosome to be identified in 1880 in India.

Occurrence

The disease in South America is called as Maldecaderas or Murrina, Tibersa in camel. The incidence of Surra increases significantly in rainy season when there is large number of biting fly populations. The incidence of disease is high in dogs living around slaughter houses, which may be of peroral infection.

Host affected: All domestic animals are susceptible but the disease is only fatal in horse, camels and dogs

Pathogenesis

Inoculation of metacyclic trypanosome intradermally by blood sucking flies causes local skin reaction (chancre) within the chancre metacyclic trypanosome transform into trypomastigote form, which enter the blood stream through lymphatics produce **intermittent parasitemia**, which causes extravascular hemolysis through erythrophagocytosis in mononuclear phagocytic system of spleen, liver and lungs results in **anemia**, decreased haemoglobin synthesis, leukopenia and thrombocytopenia is due to immune mediated phagocytosis. The **lymphoid enlargement and splenomegally develops associated with plasma cell hyperplasia and hypergammaglobinemia** which is primarily due to an increase in IgM antibody. Concurrently there is a variable degree of suppression to immune response to other microbial antigen and vaccine results in **immunosuppression**. In chronic condition lymphoid organ and spleen become shrunken

due to exhaustion of their cellular element. Tissue damage as a result of localization and production of inflammatory reaction in skin, **skeletal muscle, central nervous system and myocardium and eye.**

Effect of trypanosomes on host metabolism

1. Depletion of essential nutrients
2. Production of toxic metabolites

1. Depletion of essential nutrients

The motility of trypanosomes in blood depends upon adequate supplies of glucose in the host. It was suggested that so much glucose was consumed by the trypanosomes results in **hosts' carbohydrate reserves became exhausted resulting in a breakdown of liver function** and the onset of **lethal hypoglycemia**. The Blood sugar levels return to normal, even in fasting animals, after administration of trypanocidal drugs. Feeding glucose to infected animals may prolong their lives for short periods but does not prevent the onset of terminal hypoglycemia.

2. Production of toxic metabolites

The major end-products of glucose breakdown by trypanosomes are **pyruvate and glycerol**, which are readily metabolized by the host to produce a considerable amount of energy. Although pyruvate is readily used by host tissues, the accumulation of pyruvate in the blood is directly proportional to the numbers of parasites present. High concentrations of pyruvate could lead to **depletion of alkali reserves, acidosis, and a lowered affinity of hemoglobin for oxygen**. It drew attention to the dark purple colour of blood in late stages of *T. brucei* infections. It is proposed that deficient oxygenation of hemoglobin coupled with mechanical blockage of the circulation leads to death by asphyxiation.

Clinical manifestation

Horse

Intermittent fever, progressive anemia, edema on dependent parts of the body (ventral abdomen and genitalia) and petechial haemorrhage on the serous membrane. Dullness, listlessness, loss of body condition despite good appetite, extreme emaciation, nasal and ocular discharge, the nervous signs includes incoordination, circling, staggering gait, head pressing, paraplegia, paralysis, delirium and convulsion.

Camel

The chronic form of disease in camel is called as **tibersa**. The chronic signs exhibited for upto 3 years, reduction in milk yield and working capacity and abortion in pregnant animals.

Dog

Intermittent fever (39° C– 41° C), oedema of the head, including larynx (**to be differentiated from rabies**), oedema of the abdominal wall and legs, anaemia, weakness, lack of appetite leading to emaciation, sexual excitement and paresis of the hindquarters. Ocular signs are most often observed in dogs, with conjunctivitis, lacrimation, keratitis, corneal opacity, and/or haemorrhagic signs, which can lead to fibrin deposits in the anterior chamber of the eye, sometime parasites have been observed in ocular aqueous fluid and myocarditis has been described and can be fatal

Cattle and buffaloe

The clinical signs occur as peracute, acute, subacute and chronic form

Peracute form

- ☒ Convulsion and death within 2-3 hrs

Acute form

- ☒ Dull, sleepy, staggering gait, circling, beating against wall, stamping of feet , bellowing, twitching of muscle and shivering of body, coma and death within 6-12 hrs.

Subacute and chronic form

- ☒ Emaciation, oedema, lacrimation and diarrhoea

Mild infection occur in endemic area, signs may include irregular estrus, high rate of abortion and still birth occur in pregnant animals and poor semen quality in bulls.

Pig

Infection is often chronic with not only intermittent fever, anaemia, loss of weight, abortion, and cutaneous rash, but also cause late nervous signs with hind leg paralysis.

Necropsy findings

- ☒ Liver, spleen and lymph node enlarged are enlarged in acute infection
- ☒ Non suppurative myocarditis and sometimes with hydropericarditis
- ☒ Atrophy of lymphoid organ

Diagnosis

1. Identification of the agent

- ☒ Parasite detection techniques are highly specific, but their sensitivity is relatively low (i.e. the proportion of false negative)
- ☒ Early acute phase of infection- Sensitivity is high as parasites are actively multiplying in the blood in the absence of immunological control
- ☒ Late chronic phase- during the chronic phase the sensitivity is low as, due to the immune response of the host, parasites are scanty and rarely seen in the blood.

I. Parasite detection technique

- a) **Wet films examination (Blood or lymph from lymph node biopsy)**

These are made by placing a droplet of blood or lymph (about 2 µl) on a clean microscope slide and covering with a cover-slip (22 × 22 mm). The blood is examined microscopically. The Trypanosomes can be recognised by their movement among the red blood cells (RBCs).

b) Thick films (prepared from blood or lymph taken from lymphnode biopsy)

These are made by placing a drop of blood (5– 10 µl) on a clean microscope slide and spreading it over an area of approximately 2 cm in diameter, the film is air dried and dehaemoglobinised by immersion in distilled water for a few seconds and dried before staining. The dry smear should be stained with Giemsa stain for 30 minutes and examined microscopically for general morphology of trypanosomes.

c) Thin smear films (prepared from blood or lymph taken from lymphnode biopsy)

Thin blood smears are made by placing a small drop of blood (about 5 µl). The thin blood smear is prepared in the shape of bullet, air dry the smear and stained with giemsa stain, examined under microscopically to identify the species of trypanosomes.

d) Microhaematocrit centrifugation technique (Woo method)

Fresh, usually ear vein blood (about 70 µl) is collected into heparinised capillary tube and sealed it and centrifuged at 9000 rpm for 5minutes. The plasma/white blood cell interface (buffy coat) is examined by slowly rotating the tube. Trypanosome movement can first be detected using the × 10 objective lens with reduced condenser or with phase contrast microscope or compound microscope. This is sensitive technique than wet flim examination.

e) Examination of body fluids like Cerebrospinal fluid and aqueous humour for trypanosomes

II. Chemical tests

- ☒ To detect the changes in the chemical composition of the blood produced by alteration of protein (albumin and globulin in serum)
- ☒ Non-specific and less reliable
- ☒ **Mercuric chloride test, Formal gel test, thymol turbidity test are commonly used for camels.**

a) Mercury chloride test

- ☒ Take 1 ml of 1:30,000 solution of mercuric chloride in test tube
- ☒ Add drop of serum
- ☒ Development of white precipitate within few seconds- positive

b) Formol gel test

- ☒ Take 1 ml of serum in test tube
- ☒ Add 2-3 drops of formalin
- ☒ Shake well and allow to stand

- ☒ Formation of gel within 2-3 hrs- positive

c) Thymol turbidity test

- ☒ Take 1 ml of aqueous thymol (1:100-1:2000 in test tube
- ☒ Add drop of inactivated serum (56° C for 30 minutes)
- ☒ White turbidity within few minutes-positive

d) Stilbamide test- Reliable test for diagnosis of surra in bovines in India

- ☒ Take about 0.5-2.5 ml of 10% solution of stilbamide
- ☒ Add drop of suspected inactivated serum (56° C for 30 minutes)
- ☒ Development of coagulant and settle down in 1 or 2 minutes and dissolve in 5-10 minutes-Positive

3. Isolation of trypanosomes in NNN Medium (Novy MacNeal Nicolle Medium)

4. Mice inoculation tests

About 0.5 ml of suspected blood or lymph aspirate inoculated intraperitoneally in the mice, heavy parasitemia found in 2-3 days of post infection

5. Immunodiagnostic tests-IFAT, CFT, ELISA & IHA are most reliable tests.

6. Molecular diagnosis by PCR in blood

7. Allergic tests – doubtful

About 1.0 ml antigen is injected at the side of the neck with a 2nd dose 48 hours after first dose. Infected cases show no painful edematous, uninfected animal possess circumscribed hard nodular swelling. Accuracy lacking in this test.

Differential diagnosis

- ☒ Theileriosis
- ☒ Anaplasmosis
- ☒ Rabies

Treatment

1. **Antrycide prosalt (Triquin)-** Combination of Quinapyramine sulphate (Methyl sulphate) and Quinapyramine chloride (Methyl chloride) @ 7.4 mg/ kg b.wt or 3.5 g in 15 ml distilled water given subcutaneously for **chemoprophylaxis**. Protection against trypanosomes occur 2-3 months only.
2. **Suramins -Sulphonated naphthalamines (Nagolol)-curative**
 - ☒ Effective against *T. evansi*, *T. brucei*, *T. equiperdum*.
 - ☒ Cattle: 0.5 g/ 45 kg followed by half dose after 2 weeks, I/V
 - ☒ Horse: 4g/ 45 kg as single dose, I/ V
 - ☒ Camel: 1-2 g/100 kg b.wt as single dose I/ V
 - ☒ Dog: 3.5 ml- 10% solution followed by 2nd dose at 3-4 weeks interval, I/V
3. **Diamidines: Diminazene aceturate (Berenil) – Curative @ 3.5 mg/ kg b.wt, S/C or I/M**

- ☒ Very effective against *T. congolense* and less effective against *T. evansi*, *T. brucei*, in bovines, ovines and caprines
- ☒ Dog and camels react to dogs; resistance may occur to the drug

4. Phenanthridine compounds

- ☒ Homidium bromide and Homidium chloride- 1-2% sol @ 1gm/kg b. wt, I/M route against *T. congolense* and *T. vivax* but not against *T. evansi*
- ☒ Pyrrithidium bromide - 4 % sol @ 2mg/ kg b. wt, S/C or I/M against *T. congolense* and *T. vivax*
- ☒ Isometamidium chloride - 0.5 mg / kg b. wt, deep I/M route against *T. simiae* in pigs

Control

Control is difficult and the factors responsible for the wide spread existence are

- ☒ Latent cases of cattle and buffalo
- ☒ wild animals act as reservoir
- ☒ Drug resistant strains
- ☒ Wide spread prevalence of vectors

The Control program consists of

- ☒ Treatment of affected animals
- ☒ Chemoprophylaxis
- ☒ Disposal of manures properly
- ☒ Proper drainage
- ☒ Regular spraying of insecticides: Sprinkling of kerosene oil over water in ponds, ditches, streams or insecticides to kill adults and aquatic larvae
- ☒ Removal of moist beddings, hay, dung from stable
- ☒ Keeping animals clean by regular grooming
- ☒ Segregation of sick animals from healthy during outbreak in fly-proof shed
- ☒ Identification of endemic areas by regular examination of blood for trypanosome sp for implementation of preventive measures
- ☒ Rearing of trypanotolerant animals e.g N-Dama cattle

Nagana (African trypanosomiasis)

It is caused by group of trypanosome species *Trypanosoma vivax*, *T. brucei brucei*, *T. Congolense* and *T. Simiae*

Host affected

- ☒ Horse, cattle, sheep, goat and pigs
- ☒ N-Dama cattles is genetically resistant to African trypanosomiasis

Pathogenesis – similar to *T. evansi*

Clinical manifestation

Cattle

Intermittent fever, loss of appetite occur intermittently during parasitemic peaks, anemia, generalized enlargement of superficial lymph glands, lethargy, progressive loss of body condition (emaciation), ocular and nasal discharge, icterus, edema of throat and underline. Typically, the disease is chronic, extending over several months, growth of young animals is stunted, while adult shows decreased fertility, abortion in pregnant animals or birth of weak neonates and paralysis. In chronic condition, size of lymph node is reduced. The death is associated with congestive heart failure is due to anemia and myocarditis.

Sheep

Fever, anorexia, loss of body weight, generalized lymphoid hyperplasia moderate anemia, subcutaneous edema and neurological disorder

Dog and cat

Fever, emaciation, edema of the eyelids and thorax, ocular and nervous manifestation may occur.

Necropsy findings

- ☒ Liver, spleen and lymph nodes are enlarged
- ☒ Petechiae on lymph node, pericardium and intestinal mucosa
- ☒ Non suppurative myocarditis, sometimes associated with hydropericarditis
- ☒ Non suppurative meningioencephalomyelitis
- ☒ Keratitis
- ☒ Atrophy of bone marrow
- ☒ Chronic stage- atrophy of lymphoid organ

Diagnosis, treatment and control -similar to *T. evansi*

Dourine

- ☒ It is caused by *T. Equiperdum*
- ☒ Horse and donkeys are highly susceptible
- ☒ Natural transmission occurs through coitus (Veneral transmission)
- ☒ Predilection site: genital organs, subcutaneous tissue, peripheral and central nervous system

Pathogenesis

Trypanosomes are deposited during coitus, penetrate the genital mucosa, and multiply locally in the extracellular space, and produce an edematous swelling. Subsequent

systemic invasion in subcutaneous tissue and nervous system causes vascular injury, leads to subcutaneous edema, incoordination.

Clinical manifestation

- ☒ Incubation period: 1-4 weeks

First stage

Stallion -Swelling and edema of penis, scrotum, prepuce and surrounding skin extending as forward upto chest, swollen inguinal lymph node, paraphymosis and mucopurulent urethral discharge

Mare- edematous swelling in the vulva, perineum, udder and abdominal floor, profuse vaginal discharge, hyperaemia and ulceration of vaginal mucosae

Second stage

Urticaria like plaques (2-5 cm) develop on the neck and body is called as silver dollar spots.

Tertiary stage

It is characterized by progressive anemia, emaciation, stiffness and weakness of limbs, ataxia, paralysis and atrophy of the hindquarters.

Diagnosis

- ☒ Demonstration of trypanosomes, edema fluid, cutaneous plaque and vaginal or urethra washing and blood
- ☒ Serological tests like CFT, IFAT, ELISA and capillary agglutination test

Differential diagnosis

- ☒ Coital exanthema
- ☒ Equine infectious anemia
- ☒ Purulent endometritis

Treatment

- ☒ Diminacine acetate (Berenil) @ 7.0 mg/kg b.wt I/M, second injection after 24 hours with half of the dose
- ☒ Suramin @10 mg/kg b.wt I/V for two to three treatment
- ☒ Quinapyramine sulphate @3-5 mg/ kg b.wt S/C
- ☒ For CNS involvement-Melarsoprol is to be used

Control

- ☒ Control of breeding
- ☒ Restriction of animal movement
- ☒ Testing and disposal of infected animals

Babesiosis

Babesiosis is a first vector borne disease of cattle, sheep, pigs, horses and dogs and is characterized by fever, intravascular haemolysis causing a syndrome of anaemia, haemoglobinaemia and hemoglobinuria. It is caused by *Babesia spp* of family *Babesidae* and transmitted by ixodid ticks.

Morphology

The *Babesia* sp is pleomorphic in nature and occurs in RBC as ovoid, round, elongate, pyriform and pear shaped form lieing in pairs with an acute narrow angle. Large forms lie with their narrow ends at an acute angle and small forms at obtuse angle.

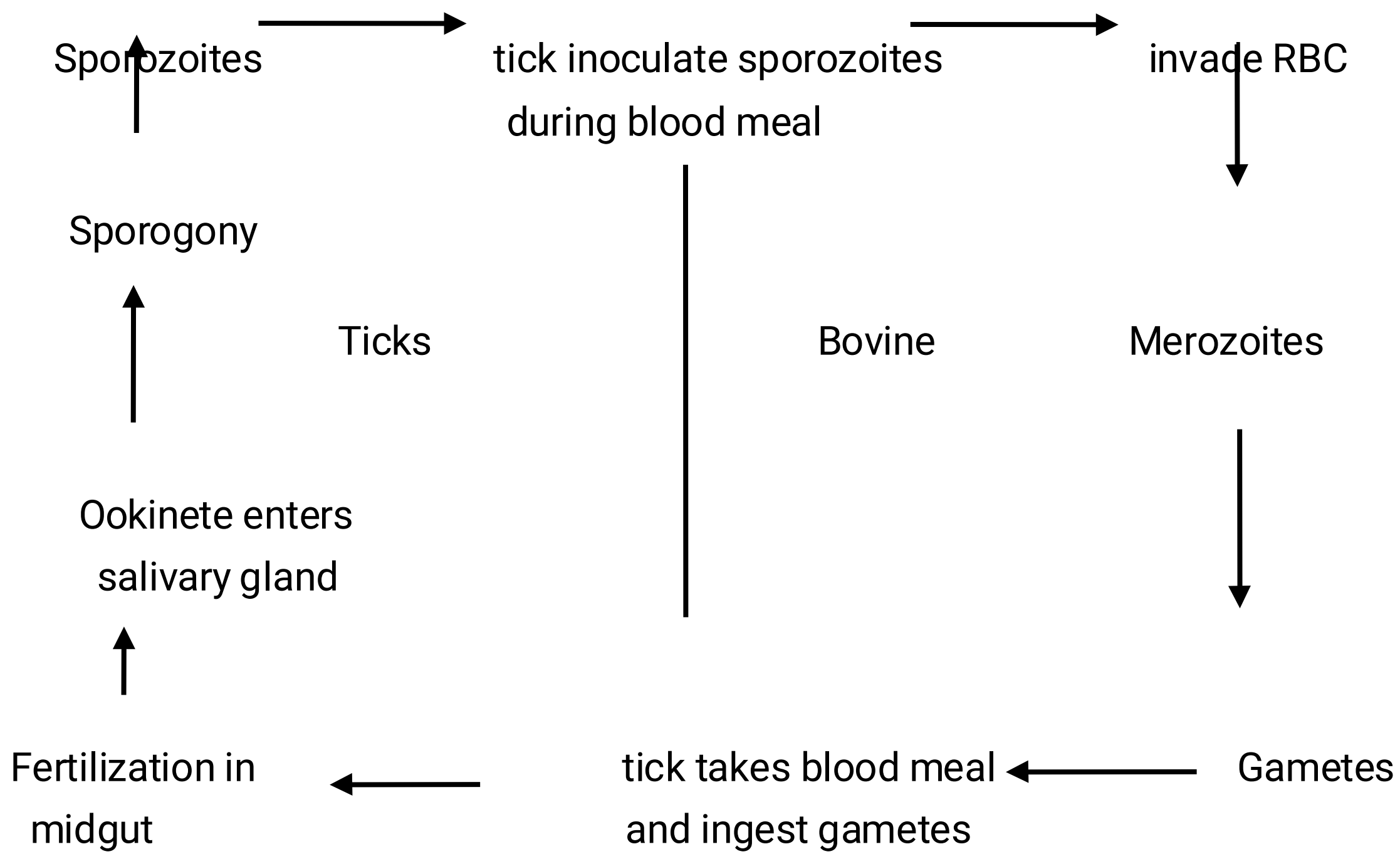
Babesia sp of domestic animals

S. N o.	<i>Babesia</i> spp	Vectors	Transmission	Disease
Cattle, Buffalo and Zebu				
1	<i>B. bigemina</i> (India)	<i>Boophilus annulatus</i> <i>B. microplus</i> <i>R. appendiculatus</i> <i>R. bursa</i> <i>Haemophysalis punctata</i>	Transovarian and Transtadial, also Intra uterine	Cattle Tick Fever Red Water Disease Bovine Piroplasmosis Texas Fever Triteza
2	<i>B. bovis</i> (India)	<i>B. microplus</i> <i>B. calcaratus</i> <i>R. bursa</i> <i>Ixodes ricinus</i> <i>I. Persulcatus</i>	Transovarian and Transtadial, also Intra uterine	Bovine Babesiosis Piroplasmosis Red Water Disease
3	<i>B. major</i>	<i>Ixodes ricinus</i>	Transovarian and Transtadial	Less severe than <i>B. bigemina</i>
4	<i>B. divergens</i>	<i>B. calcaratus</i> <i>Haemophysalis punctata</i>	Transovarian	Less severe than B. Bovis
Sheep and Goats				

1	<i>B. motasi</i>	<i>Dermacenter silvarian,</i> <i>Haemophysalis punctata</i> <i>R. Bursa</i>	Transovarian and Transtadial	Pathogenic
2	<i>B. ovis</i>	<i>R. bursa</i>	Transovarian and Transtadial Transplacental	Less severe than <i>B. Motasi</i>
Sheep				
1	<i>B. foliata</i> (India)	Not known	Not known	Not known
Goat				
1	<i>B. taylori</i> (India)	Not known	Not known	Benign type disease
Dogs				
1	<i>B. canis</i>	<i>R. sanguineus</i>	Transtadial	High pathogenic Biliary Fever Malignant jaundice Canine Piroplasmosis Canine babesiosis
2	<i>B. gibsoni</i> (India)	<i>R. sanguineus,</i> <i>Haemophysalis bispinosa</i>	Transtadial and Transovarian	Slightly Pathogenic Causes chronic disease with progressive anemia not readily treated with normal Babesiacidal drugs
Cats				
1	<i>B. felis</i> (India)	Not known	Not known	Less pathogenic than dog
2	<i>B. cati</i>	Not known	Not known	Less pathogenic

	(India)			than dog
Equines				
1	<i>B. equi</i> (India)	<i>Dermacenter spp</i> <i>R. sanguineus</i> <i>R. bursa</i> <i>Hyalomma anatolicum</i>	Transovarian and Transtadial, also Intra uterine	More pathogenic
2	<i>B. cabballi</i>	<i>Dermacenter spp</i> <i>R. sanguineus</i> <i>R. bursa</i> <i>Hyalomma anatolicum</i>	Transovarian and Transtadial, also Intra uterine	Less pathogenic
Pigs				
1	<i>B. trautamanni</i>	<i>Boophilus spp</i> <i>R. sanguineus</i> <i>Dermacenter spp</i>	Transovarian	Mild/ Fatal
2	<i>B. peroncitto</i>	Not known	Not known	Mild/ Fatal

Life cycle



Epidemiology

1. Virulence of parasite

Babesia bigemina and *Babesia bovis* are highly pathogenic, *B. divergens* and *B. canis* – relatively pathogenic; *B. major* and *B. ovis* – mildly pathogenic producing transient anaemia

2. Breed susceptibility - All breeds are equally susceptible

3. Inverse age resistance; young animals are less susceptible than older animals

4. Immune status of host

In endemic area the young animals acquire immunity through colostrums of the dam as a result of transient infection with mild clinical signs. The active immunity is depends on persistence of carrier status, and the phenomenon was termed as **premunity**.

5. Carrier status – infected animal act as carrier for over 2 years, if infected constantly, they act as carrier for life

6. Level of tick challenge.

In endemic area host immunity is maintained at high, when more number of ticks present in that area. Where there is few ticks or limited ticks, the population immunity is low. In these circumstances, the number of ticks suddenly increases due to favourable environmental condition or reduced dipping frequency. The incidence of clinical cases increases sharply. This situation is known as enzootic instability.

7. Risk factors

In endemic area occasional outbreak of disease often associated with stress such as parturition or presence of other diseases.

Transmission

- ☒ Transovarian transmission occur in one host tick & Transtadial transmission occur in two or three host tick
- ☒ Intra uterine transmission- *B. bigemina* and *B. bovis*
- ☒ Mechanically by contaminated needles and surgical instruments. eg. *B. bigemina* and *B. equi*. also *B. bovis*.

Pathogenesis

- ☒ The multiplication of parasite in red blood cells causes rapid destruction of erythrocytes with accompanying haemoglobinaemia, hemoglobinuria and fever
- ☒ The *Babesia bovis* is most pathogenic than *Babesia bigemina*
- ☒ *Babesia bovis* infection is associated with sludging of RBCs in the small capillaries. In the cerebrum this causes blockage of the vessels by clumps of infected RBC

leading to anoxia and tissue damage. The activation of certain plasma components (kinin, kallikrein) results in vasodilatation and accompanied by increased permeability leading to circulatory stasis, shock and intravascular coagulation

- ☒ Infection with *Babesia caballi* in horses causes Clogging of the capillaries of various organ by parasitized cells and free parasites, the stasis from this sludging causes degeneration of the endothelial cells of small blood vessels causes anoxia and accumulation of toxic metabolic products, capillary fragility, eventual escape of erythrocytes and macroscopic haemorrhage
- ☒ *Babesia canis* and *babesia gibsoni* causes multiorgan dysfunction by generation of cytokines, Nitic oxide and free radicals by host

Clinical manifestation

Cattle

- ☒ High fever (rectal temp. Reaching 41.5⁰c)
- ☒ Anorexia and ruminal atony
- ☒ Cattle stand with arched back, rough hair coat, dyspnoea and tachycardia
- ☒ Reddened mucous membrane initially later become pale and icteric
- ☒ Profound diarrhoea followed by constipation
- ☒ Anemia occur rapidly with 75% of erythrocyte destruction or more
- ☒ Weakness and loss of condition is due to anemia
- ☒ Profound diarrhoea followed by constipation
- ☒ Advanced stages: Severe anaemia, haemoglobinuria, jaundice, dark red to brown colour urine with a very stable froth and death after 24 hours
- ☒ Drop in milk production
- ☒ Abortion in pregnant animals
- ☒ Incoordination, convulsions, depressions and death occur in cerebral babesiosis apart from fever, anemia and hemoglobinuria

Sheep and goat: clinical signs similar to cattle

Dog and cat

Peracute form

- ☒ Rapid onset of collapse with hypotensive shock
- ☒ Pale mucous membrane (sometimes cyanotic)
- ☒ Tachycardia, weak pulse, weakness and depression
- ☒ Hemoglobinuria
- ☒ Multiple organ dysfunction with hypotension, hypoxaemia leading to coma and death

Acute form

- ☒ Fever, anemia, jaundice, marked thirst inapetence, prostration and death

- ☒ Petechial and ecchymotic haemorrhage may be observed on gums or ventral abdomen
- ☒ Involvement of circulatory system may produce edema, purpura, ascites and there may be gastritis and stomatitis
- ☒ Involvement of respiratory system may cause catarrh and dyspnoea
- ☒ CNS involvement causes locomotor disturbances, paresis or epileptic forms
- ☒ Cerebral babesiosis may be confused with rabies

Chronic form

- ☒ Fever is not high, mild jaundice and severe anemia, affected animals become weak and emaciated

Horse

- ☒ Pale mucous membrane, anemia and icterus
- ☒ Hemoglobinuria is rare
- ☒ Gastroenteritis
- ☒ Posterior paralysis

Necropsy findings

- ☒ Subepicardial and endocardial petechial haemorrhage
- ☒ Liver is yellowish brown colour
- ☒ Lungs are congested and edematous
- ☒ Pericardial sac may contain serosanguineous fluid
- ☒ Splenomegally with dark pulpy consistency
- ☒ Kidneys are pale often edematous
- ☒ Bladder is distended with dark reddish brown colour urine

Sample collection

- ☒ Live animals-peripheral blood smear and paired sera sample
- ☒ Dead animals- smear from heart muscle, kidney, liver, lung, brain and from a blood vessels in an extremity

Diagnosis

- ☒ Based on clinical signs
- ☒ Examination of piroplasms in RBC of peripheral blood smear stained with giemsa stain during acute phase of infection
- ☒ Presence of heavy infestation with ticks
- ☒ Haematology: TEC - less than 2 million/ cmm, Hb: less than 3.0g/dl and neutrophilia
- ☒ Biochemical changes: Reduced calcium, phosphorous, total serum protein; Increase in blood glucose, total serum bilirubin and SGOT in liver damage
- ☒ FAT- Acridine orange fixed with blood smear the organism appeared as red with yellow nucleus

- ☒ Necropsy: Splenomegaly, jaundice, haemoglobinuria, myocardial ecchymoses – highly suggestive
- ☒ Molecular diagnosis by PCR in chronic infection
- ☒ Immunodiagnosis in subclinical infections
- ☒ IHA- 80% effective in field infections
- ☒ IFA, ELISA, FAT, CFT, IHA and Capillary agglutination tests are used to identify the antibodies in serum
- ☒ Others: Latex agglutination tests, Card agglutination test, ELISA, Microplate enzyme immuno assay. IFA differentiates antibodies due to vaccination and natural infection

Differential diagnosis

- ☒ Eperythrozoonoses
- ☒ Theileriosis
- ☒ Leptospirosis
- ☒ Anaplasmosis
- ☒ Trypanosomosis
- ☒ Bacillary haemoglobinuria
- ☒ Phosphorous deficiency
- ☒ Copper toxicity
- ☒ Rabies

Treatment

- ☒ The most commonly used drugs are **Diaminacine aceturate @ 3.5 mg/ kg b.wt I/M**, Imidacarb @ 1.2 mg/Kg S/C, Amicarbalide @ 5-10mg/kg and trypan blue @ 10 mg/kg b.wt (1% solution)
- ☒ For cerebral babesiosis added dose is required. Diaminacine aceturate should be given on 2nd day of treatment
- ☒ Imidacarb @ 3.0 mg/Kg b.wt provide protection for 4 weeks and completely eliminate the *B.bigemina* and *B.bovis* from carrier animals
- ☒ The trypan blue is not effective against *B. bovis*
- ☒ Imidocarb has been successfully used as a chemoprophylactic that will prevent clinical infection for 2 months
- ☒ Injections of long acting oxytetracycline @ 20 mg/kg
- ☒ Sheep- **Diaminazene aceturate @ 3.5mg/Kg** on two successive days or 12mg/Kg as single dose
- ☒ Horse- Combination of **imidocarb and buparvaquone** appears to be effective and capable of eliminating *B.equi* and *B.caballi* from horse. (For *B.caballi* -Imidocarb @

2mg/Kg two doses at 24 hrs interval and Buparvaquone @ 4-6mg/Kg to control acute babesiosis due to *B.equi*)

- ☒ Dog- For *Babesia canis*- Imidocarb dipropionate @5.0 to 6.6 mg/Kg b.wt I/M
- ☒ For *Babesia gibsoni*- Tab. Atovaquone 13.4 mg/Kg TID and Azithromycin @10 mg/Kg once in day for 10 days with fatty meal. PCR testing was repeated at 30 and 60 days after the end of treatment
- ☒ Clindamicin can be effective against *babesia gibsoni* @ 25 mg/kg b.wt bid PO for 14 days
- ☒ For *Babesia felis* -Primaquine phosphate @ 0.5 mg/Kg twice at 24 hrs interval

Control

- ☒ Vaccination is commonly practised by inoculating blood from donor animals, that has recently recovered from infection
- ☒ A regime of four injections of long acting oxytetracycline @ 20 mg/kg at weekly intervals during their first month of grazing on tick infested pastures
- ☒ Control of ticks by spraying or dipping of animals in acaricidal agent at regular intervals
- ☒ Segregation and treatment of animals harbouring infection
- ☒ Killed vaccine- irradiated *Babesia* sp with freund' s incomplete adjuvant
- ☒ Chemoimmunization- administration of imidocarb and vaccines simultaneously
- ☒ Blood transfusion in severe anemic animals

Theileriosis

Theileriosis is a vector borne disease of cattle, sheep and goat characterized by fever and lymphoproliferative disorders, which may be associated with leukopenia and anaemia. It is caused by *Theileria* sp and transmitted by ixodid ticks.

Theileria sp of domestic animals

S. No.	Theile ria	Vectors	Disease	Pathogenic ity
Cattle and Buffalo				
1	<i>T.annulata</i> (India)	Hyalomma marginatum Hyalomma anatolicum (Transtadial	Tropical Theileriosis, Mediterranean coast fever, Tropical Piroplasmosis, Egyptian Fever	Highly pathogenic

		transmission)		
2	<i>T. parva</i>	Rhiphicephalus appendiculatus Hyalomma anatolicum (Transtadial transmission)	Bovine Theileriosis, East Coast fever, Turning Sickness, African Coast Fever, Corridor disease, Rhodesian tick Fever, Rhodesian Red Water disease	Highly pathogenic
3	<i>T. mutans</i> (India)	Rhiphicephalus appendiculatus Haemophysalis bispinosa Haemophysalis punctata (Transtadial transmission)	Benign Bovine Theileriosis, Turning Sickness African Coast Fever	Non-fatal/ Slightly pathogenic
Sheep and Goat				
1	<i>T. hirci</i> (India)	Rhiphicephalus bursa Hyalomma anatolicum anatolicum (Transtadial transmission)	Malignant ovine theileriosis	Highly pathogenic
2	<i>T. ovis</i>	Rhiphicephalus bursa Haemophysalis spp (Transtadial transmission)	Benign theileriosis	Mild pathogenic

Morphology

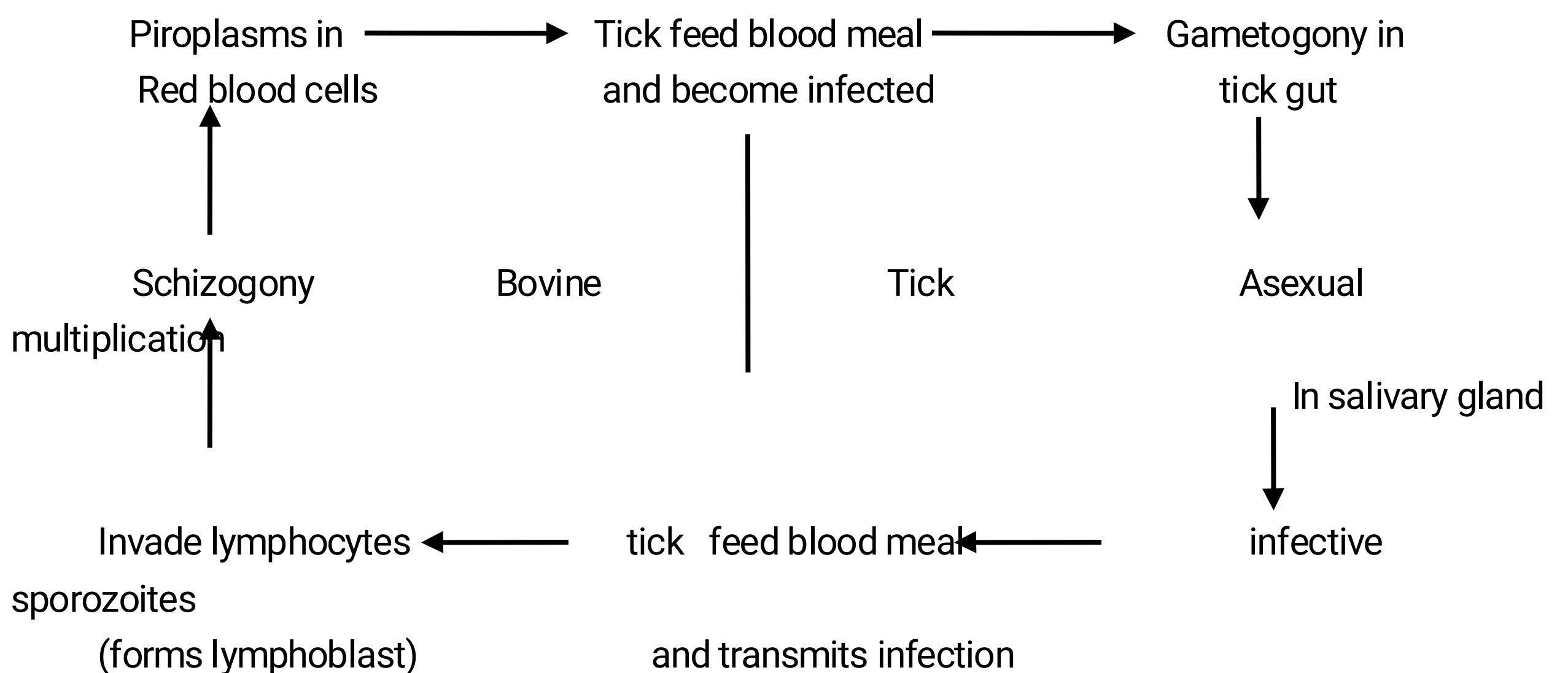
Piroplasm forms in RBC are highly pleomorphic and most commonly occur as rod, round, oval (signet ring), annular and comma shape. Macro and microschizonts (Koch blue bodies) in lymphocytes and endothelial cells of the spleen or lymph node. The macroschizonts divide into microschizonts and produce merozoites. The merozoites are released from lymphocyte and invade the RBC.

Tropical bovine theileriosis

- ☒ It is caused by *Theileria annulata*

Life cycle

- ☒ Asexual reproduction (Schizogony) occur in Vertebrate host
- ☒ Sexual reproduction (gamogony and sporogony) occur in vectors
- ☒ Transtadial transmission occurs in ticks. Transmission of Infection from tick larvae to nymphs or nymphs to adults.

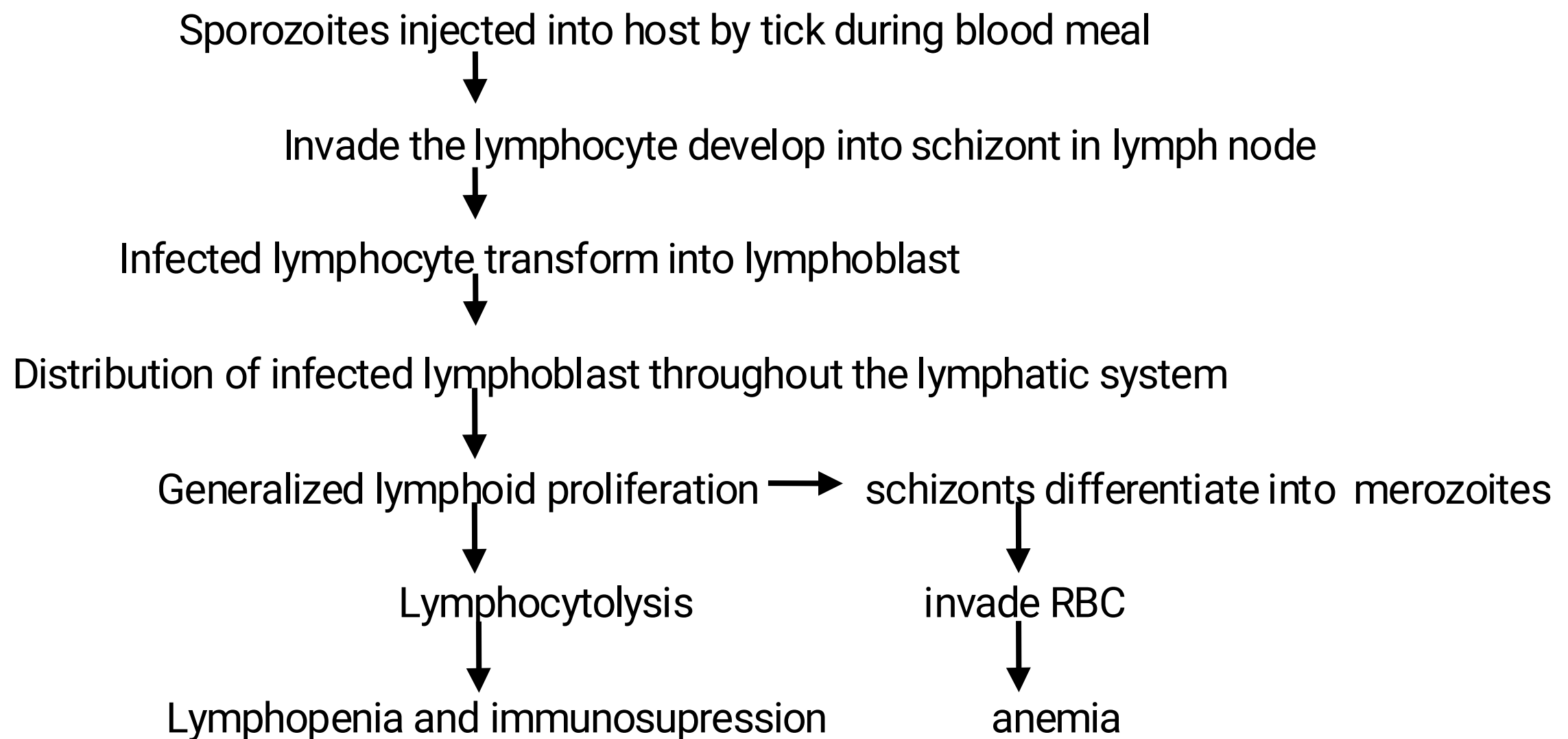


Epidemiology

- ☒ In India high incidence of disease mostly occur, during summer and rainy seasons (May - Oct) which is attributable to high incidence of tick vectors
- ☒ In tropical areas, ticks are active throughout the year
- ☒ Exotic and crossbred cattle are highly susceptible than zebu cattle
- ☒ Both young calves (even 2-3 days old) and adults are affected
- ☒ Mortality rate in non endemic area is 10-90%
- ☒ In endemic areas, all animals are virtually infected but the case- fatality is 10 -20% and is confined to calves
- ☒ **Transtadial transmission (stage to stage) by *Hyalomma spp* ticks** (Infection from tick larvae to nymph or nymph to adult occurs)
- ☒ Congenital transmission occasional in calves
- ☒ Indigenous cattle without clinical signs, buffalo, wild animals and clinically recovered animals are act as a carrier
- ☒ Recovery from infection leads to the development of premunity

- ☒ No cross immunity between theileria species
- ☒ Concurrent infection with *Anaplasma* and *Babesia* sp is common

Pathogenesis



Clinical signs

Incubation period: 9 to 25 days

The disease occurs as peracute, acute, subacute, mild and chronic forms

Peracute form

- ☒ Animal may die within 3-4 days of infection with high fever

Acute form

- ☒ Most common and may last for 3-4 days and may be prolonged for about 20 days
- ☒ Fever (40.5 – 41.5° C) is continuous or intermittent and persists for 3-20 days
- ☒ Inappetence, lacrimation, cessation of rumination, rapid heartbeat, respiratory distress, weakness, reduced milk production
- ☒ Enlargement of superficial lymph nodes and edema of the eyelids
- ☒ Laboured breathing, serous nasal discharge, coughing, restlessness and starchy coat
- ☒ Marked anaemia, bilirubinuria and bilirubinaemia, conjunctiva is icteric with petechial haemorrhage, jaundice, prostration and death
- ☒ Haemoglobinuria occurs in some cases
- ☒ Diarrhoea with blood and mucus in faeces, the animal is greatly emaciated (RBCs < 1 million/ cmm) and death occurs after 15 days of infection

Subacute form

- ☒ Fever is irregularly intermittent for 15 days, and then the animal recovers
- ☒ Pregnant animal may abort

Mild form

- ☒ Mild fever, inappetence, listlessness, slight digestive disturbances, lacrimation for few days and moderate anaemia

Chronic form

- ☒ Intermittent fever, inappetence, marked emaciation
Some degree of anemia and icterus may persist for 4 weeks or longer
- ☒ In some cases, acute form may supervene depending upon stress factors and animal may die within 1-2 days

Cerebral theileriosis

- ☒ Nervous signs are predominate and brain impression smears reveal Koch's Blue bodies.

Cutaneous form

- ☒ Small subcutaneous nodules containing schizonts rarely occur

Lesions

- ☒ Enlargement of the superficial lymph nodes, spleen and liver
- ☒ Distension of the gall bladder with thick bile
- ☒ Congestion and petechial haemorrhages on the kidney and lungs
- ☒ In few cases, necrotic infarcts on brain
- ☒ **Punched necrotic ulcers in the abomasum are pathognomonic**
- ☒ In young calves, abomasum is full of undigested clots of milk

Clinical pathology

- ☒ Haematological changes
- ☒ Haemoglobin level (< 2 gm/dl)
- ☒ Total RBC count (1.5-2 million/ Cmm)
- ☒ Decreased PCV
- ☒ Initial leukocytosis is followed by leukopenia in later stage of infection

Diagnosis

- ☒ Based on clinical signs and post mortem lesions
- ☒ Demonstration of erythrocytic forms (piroplasms) in peripheral blood smears
- ☒ Demonstration of Koch's blue bodies (schizonts) in lymphocytes from lymph nodes biopsy
- ☒ For latent cases, serological tests can be employed CFT, Capillary tube agglutination test, IFA, IHA, AGID, Dot-ELISA, LMIT, Conglutination Complement Absorption test (CCA), Gel agglutination test, immunoelectrophoresis to identify antibodies
- ☒ Leukocyte migration inhibition test is commonly used to assess the cell mediated immunity

Differential diagnosis

- ☒ Anaplasmosis
- ☒ Babesiosis,
- ☒ Trypanosomosis
- ☒ Ehrlichiosis

Treatment

- ☒ Naphthaquinone drug - Inj. **Buparvaquone (Butalex) - drug of choice.**
- ☒ Inj. Buparvaquone (Butalex) @ 2.5 mg/kg b.wt, deep I/M, The second should be given within 48 hrs in severe cases
- ☒ Hematinic supportive therapy can be given to combat anemia along with inj. Buparvaquone
- ☒ Parvaquone - 10 mg/kg, deep I/M, effective against piroplasmic stage
- ☒ Berenil (Diminazene aceturate) – 2.5 to 5 mg /kg, deep I/ M; Doubtful against erythrocytic stage
- ☒ Oxytetracycline - 20 mg /kg for 7 days; I/M; effective against of schizogony
- ☒ Halofuginone (stenorol) – 1 to 2 mg/kg as 2 doses, as tablets, Potent against schizont stage but toxic
- ☒ Others- 3% Tryphan blue solution

Prevention and control

- ☒ **Chemoprophylaxis , Immunoprophylaxis and Tick control**

1. **Chemoprophylaxis**- Long acting tetracycline two doses should be given intramuscularly

2. Immunoprophylaxis

- ☒ Vaccination of exotic and crossbred cattle with tissue culture vaccine (Rakshavac-T*)

Rakshavac-T

It is a live schizont grown in lymphoblast cell culture attenuated by prolonged invitro passage

Age of vaccination: 2 months and above

Dose : 3.0 ml S/C

Revaccination: every three years

- ☒ Infection and treatment method

Inoculation of ground up tick supernates (GUTS) into the disease free animal together with inj. Oxytetracycline LA or inj. Buparvaquone

It suppresses the development of the parasite and allow the effective CMI response with minimal disease response

- ☒ Subunit vaccine and recombinant vaccine also available for vaccination

3. Vector control

- ☒ Wide spread and strict application of acaricides on the animal and farm premises

East coast fever

- ☒ It is caused by *Theileria parva*

Morphology

Piroplasms in RBC mostly rod shaped, however round, oval, comma and ring shaped form may also occur. Several parasites may occur in individual RBC, but there is no evidence for multiplication of parasites in erythrocytes. The actively multiplying form schizonts (Koch blue bodies) occur in cytoplasm of lymphocytes and occasionally in the endothelial cells of lymph glands and spleen. The schizonts are circular or irregularly shaped.

Epidemiology

- ☒ The disease is mostly occur in East, Central and Southern Africa
- ☒ ECF outbreaks occur mostly during rainy season (increased tick activity following onset of rainfall)
- ☒ ECF affect mainly cattle, but also buffaloe

Life cycle: Similar to *T. annulata*

Pathogenesis- similar to *T.annulata*

Note: Immunosuppression is more severe than *T. annulata*

Clinical manifestation

- ☒ Incubation period- 10-25 days

Acute form

- ☒ Most common
- ☒ Fever (40-41.7⁰ c) depression, anorexia, drop in milk yield
- ☒ Nasal and ocular discharge
- ☒ Swelling of superficial lymph node
- ☒ Swelling of eyelids and ears
- ☒ Rapid heart beat
- ☒ Diarrhoea, sometimes dysentery may occur
- ☒ Emaciation, weakness and recumbency
- ☒ Pulmonary edema responsible for immediate cause of death
- ☒ Death due to asphyxia
- ☒ Occasionally, involvement of brain causes turning sickness, characterized by circling movement and abduction of hind limbs
- ☒ Immunity is solid after recovery

Sub acute form

- ☒ Frequent in calves but symptoms are less pronounced

Necropsy finding: same as that of *T. annulata*

Diagnosis

- ☒ Demonstration of Schizonts in lymph node biopsy
- ☒ Demonstration of erythrocytic forms are difficult in blood smear
- ☒ Differentiation from the *Theileria spp* is difficult
- ☒ Serological tests: IHA, IFA, Capillary agglutination test are used to identify antibody

Treatment: same as that of *T.annulata*

Theileriosis of sheep and goat

- ☒ It is caused by *T. hirci* (*T.lestoquardi*) and *T.ovis*
- ☒ *T. hirci* (*T.lestoquardi*)-highly pathogenic causes Malignant ovine theileriosis
- ☒ *T.ovis*-mildly pathogenic causes benign ovine theileriosis
- ☒ Morphology: Oval and rod shaped.
- ☒ Life cycle: Similar to cattle
- ☒ Pathogenesis- Similar to Bovine Tropical theileriosis

Clinical manifestation

- ☒ Acute: High fever, listlessness, nasal discharge, anaemia, jaundice, atony of rumen, weakness, transitory haemoglobinuria, lung edema and enlargement of lymph nodes
- ☒ No cross immunity between *T. hirci* and *T. ovis*
- ☒ Necropsy: petechial haemorrhage on abomasal mucosa, edematous lungs, haemorrhage in submucous, subserous and subcutaneous tissue
- ☒ Diagnosis: Examination of stained smear of blood, lymph nodes or spleen.
- ☒ Treatment: Single dose of Buparvaquone @ 2.5 mg/ kg b.wt.

Trichomoniasis

It is a sexually transmitted disease of cattle caused by *Trichomonas foetus* and clinically characterized by infertility, early abortion, pyometra and inapparent infection in bulls.

1. Distribution

- Worldwide
- India - Reported in West Bengal, Bihar, Rajasthan, Jammu and Kashmir and Orissa.

2. Host affected

- o Cattle, zebu, pig, horse and deer are affected, but pathogenic effect is seen only in bovines

3. Location

- o Cows: Uterus and internally the vagina
- o Bulls: Preputial cavity

4. Morphology

- o Body is spindle to spear shaped (25um to 15 um)
- o It has 3 anterior flagella and one posterior flagella and no pelta

5. Strain of the parasite

- o Three serologically distinct strains are Belfast, Manley and Brisbane
- o Outbreaks due to Manley strains are few

6. Incidence

- o Highest in those countries where bulls are kept for natural service

7. Susceptible age group - All age groups are susceptible

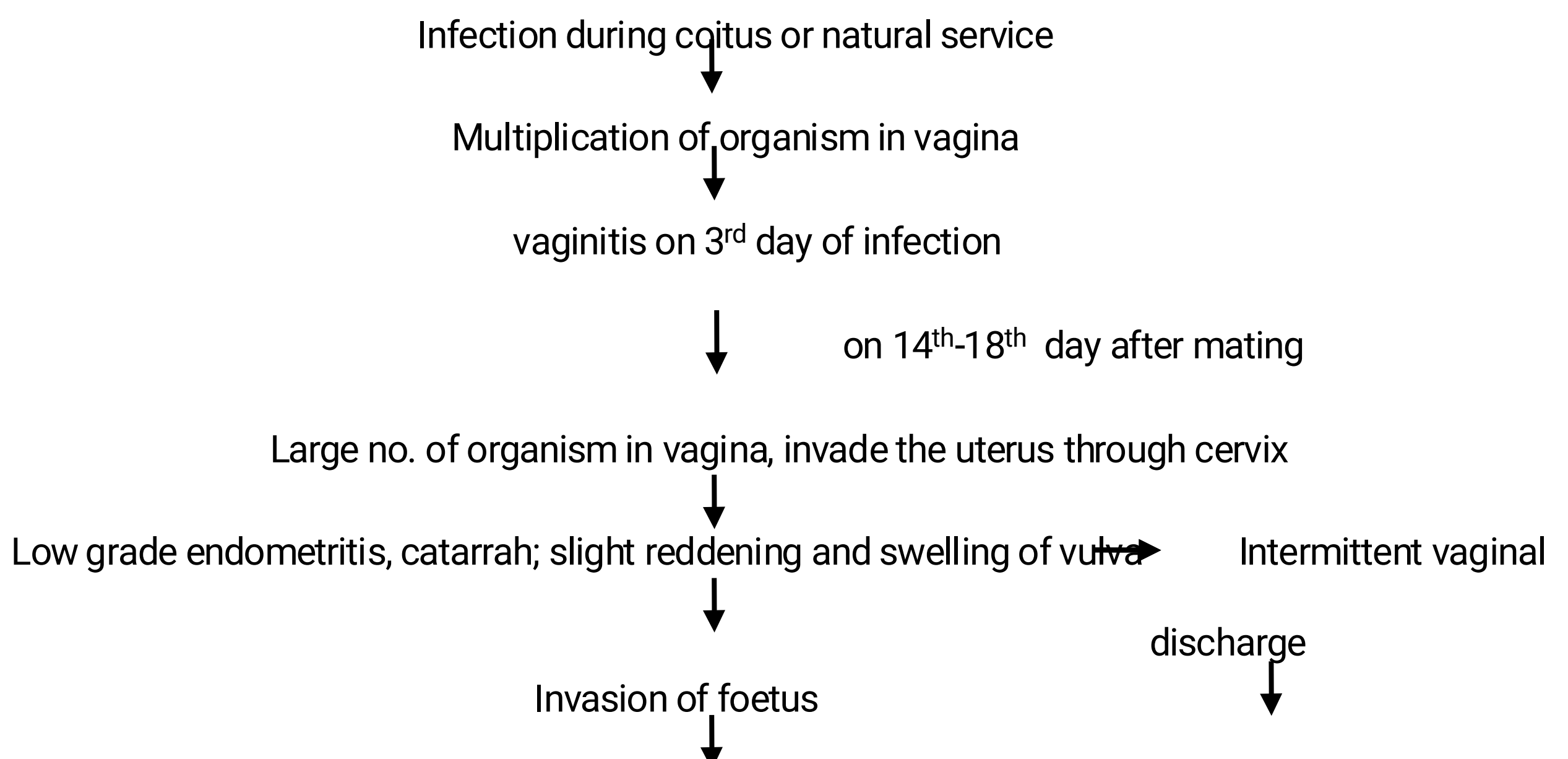
Transmission

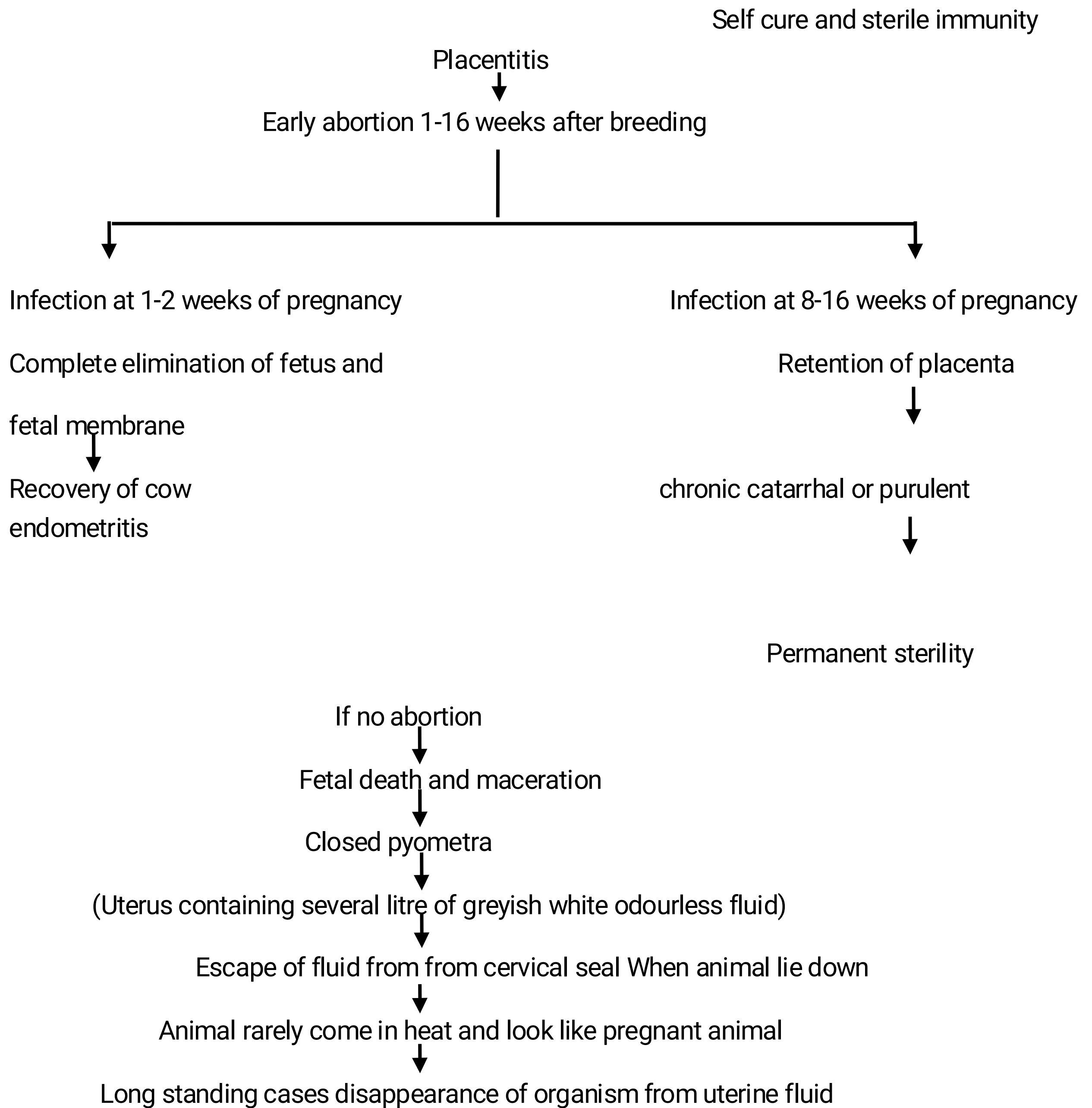
- ☒ By coitus under natural conditions
- ☒ By teaser bulls
- ☒ It may also be transmitted by artificial insemination of liquid semen and gynaecological examination of cow with help of speculum

Carrier status

- ☒ Bulls are carrier and permanent sources of infection
- ☒ In infected Cows, it is self-limiting, but untreated cows act as carriers for more than 1 year

Pathogenesis





Clinical manifestation in cow

- ☒ Mucopurulent discharge (Potato soup appearance) is due to vaginitis
- ☒ Low level of infection causes vaginal discharge and roughness of vaginal mucosa
- ☒ Maximum number of organisms are present in vagina, 14- 18 days after natural service
- ☒ Some cows infected with small no. of organism may conceive normally, carry full term and give birth
- ☒ Invasion of organism in placenta causes placentitis and abortion at 8-16 weeks after natural service
- ☒ Following abortion fetal membranes are completely eliminated

- ☒ In some animals fetus and fetal membranes are not completely eliminated and maceration occurs leading to chronic catarrhal sometimes purulent endometritis
- ☒ Persistent vaginal discharge noticeable when animal lies down
- ☒ Pyometra occurs when cervix are closed
- ☒ Anoestrus

Clinical manifestation in Bulls

- ☒ Organisms are most commonly found in preputial cavity, but also found in testes, epididymitis and seminal vesicles
- ☒ Pain on micturition, disinclination to serve cows, mucopurulent discharge
- ☒ Small red nodules in preputial mucosal surface
- ☒ Signs disappear 1-2 weeks after infection
- ☒ Spontaneous recovery is rare

Immunity

- ☒ Little evidence as bulls become immune to infection
- ☒ Infected non-pregnant animals recover without treatment, but a small proportion of cows act as carriers and permanent sterility may occur despite immunity
- ☒ Humoral antibodies are produced from uterus reaches the systemic circulation, which is readily detected by capillary agglutination test
- ☒ Vaginal antibodies develop locally
- ☒ Antibodies persist for 17 - 55 days, calves immunity acquired through colostrum

Diagnosis

- ☒ Clinical history, signs of early abortion, repeated return to service or irregular oestrus cycle
- ☒ History of recent purchase of mature bull for breeding
- ☒ Confirmation by demonstration of organism in placental fluid, stomach content of aborted foetus, uterine washing, pyometra or vaginal mucous and preputial washing
- ☒ Following abortion, organisms disappear within 48 hours
- ☒ The organisms will be observed as jerky movements and rotation of cells on their long axis, lashing flagella when examined in warm saline
- ☒ Examination of the above preparations by Giemsa staining, if the organisms are too low to allow diagnosis
- ☒ Culturing in CPLM (Cysteine-Peptone- Liver extract- Maltose serum) trichomonas medium or Diamond's media
- ☒ Serological tests - **Cervical mucus agglutination tests is the most satisfactory test.**
The mucus should be collected few days after oestrus from the anterior end of vagina

- ☒ Vaginal mucus should be collected and tested until 3 consecutive test should be negative to declare cow free of infection
- ☒ For Bulls 6 consecutive test be negative at weekly interval to declare free from infection
- ☒ Alternatively, breed the bull to two or more virgin heifers to assess freedom from infection

Treatment

- ☒ Sexual rest for 3 months in female
- ☒ Bulls: Difficult to treat, best to sell it for slaughter
- ☒ Pudental anaesthesia is used to relax retractor penis muscle
- ☒ The penis is washed with weak solution of detergent, dried and less than 1% acriflavin ointment or solution introduced into the preputial cavity and massaged in for 15-20 minutes
- ☒ Berenil (1% solution) 100-150 ml is injected into prepuce and retained for 15 minutes with massage daily for five consecutive days
- ☒ Oral administration of Dimetridazole @ 50 mg/kg b.wt daily for five days or intravenous injection of 10mg/ kg daily for five days for bulls/cows

Control

- ☒ Slaughter/ castration of infected bulls
- ☒ Avoid non tested bulls used for breeding purpose
- ☒ Aborted cows should be given sexual rests for 3 consecutive oestrus cycle
- ☒ Artificial insemination with proper care will reduce the incidence of disease
- ☒ Maintenance of proper records of service date, calving etc., at the farm helps in early diagnosis and control
- ☒ Examination of breeding bulls for *T. fetus* before purchase

Avian trichomonosis

Etiology: *Trichomonas gallinae*

Location: Oesophagus, crop and proventriculus

Epidemiology

- ☒ **Pigeons are most commonly affected**, turkey, chicken and raptors also affected fairly
- ☒ Young birds are mostly affected than adult birds
- ☒ **Pigeons transmit the infection to the squabs through pigeon milk**
- ☒ Chickens, turkeys and other birds acquire infection through drinking of water contaminated with discharge from mouth of infected bird, not by faeces

- ☒ Jones barn strain (JB strain) is most virulent and causes diseases
- ☒ Immunity: Varying degrees of immunity; relatively harmless strain produces immunity against virulent strains
- ☒ Morbidity rate: 80-90%

Pathogenesis

Invasion of organisms into pharyngeal glands, underlying tissues, reach liver where abscess formation leads to death.

Clinical manifestation

- ☒ Loss of body weight
- ☒ Stand huddled with ruffled feathers, and may fall over when forced to move
- ☒ Yellow necrotic lesions are present in oesophagus and crop
- ☒ Accumulation of greenish fluid or cheesy material in the mouth or crop may exude from the beak
- ☒ In turkey poults and chickens, drowsiness, **pendulus crop**, foul odour from mouth is usually observed
- ☒ Recovered birds are immune to re-infection but survive as carriers

Necropsy finding

- ☒ Whitish or yellowish caseous nodules in the crop. The circumscribed disc shaped lesion is called as yellow button
- ☒ The caseous nodules also found in liver and lungs

Sample collection

- ☒ Mouth or oesophageal content, Swab from crop, pharynx and oesophagus

Diagnosis

- ☒ Demonstration of organism in greenish fluid material or in the lesion
- ☒ Culture of organism in diamond's medium
(trypticase-yeast-extract-maltose-cysteine-serum medium)

Treatment

- ☒ Nitroimidazole compounds such as dimetrizole and metronidazole are effective against trichomoniasis
- ☒ Furazolidone-@ 25 – 30 mg/kg in gelatin capsules for 7 days

Control

- ☒ Control of trichomoniasis in chicken and turkey is achieved by preventing access of wild pigeon to drinking water by shielding the drinking water sources
- ☒ Elimination of carrier adult birds

Coccidiosis in poultry

Coccidiosis is a disease of universal importance in poultry production. It is caused by *Eimeria* sp. and clinically characterized by bloody diarrhoea and high mortality in young chicken.

Etiology : Multiple species of *Eimeria* causes disease in poultry as follows

S.No.	Species	Location	Pathogenicity
1	<i>E. tenella</i>	Caecam	Most pathogenic - Caecal Coccidiosis
2	<i>E. brunette</i>	Lower small intestine, Caeca, rectum, Cloaca	Most pathogenic- Rectal Coccidiosis
3	<i>E. necatrix</i>	Jejunum, mid-gut and other parts of large intestine	Most pathogenic- Intestinal coccidiosis
4	<i>E. maxima</i>	Mid-gut	Medium/ highly pathogenic intestinal Coccidiosis
5	<i>E. acervulina</i>	Duodenum	Mildly pathogenic
6	<i>E. mivati</i>	Duodenum and Rectum	More pathogenic
7	<i>E. mitis</i>	Duodenum and jejunum	Mild pathogenic
8	<i>E. praecox</i>	Duodenum	Mild/ Non- pathogenic
9	<i>E. hagoni</i> (uncommon)	Duodenum	Slightly pathogenic

These are intracellular parasites of intestinal epithelium. Both asexual (Schizogony) and sexual (gametogony) occur within the host and sporulation usually takes place outside the host (environment). The time taken for sporulation is 2-4 days. The sporulated oocyst outer wall enclosing four sporocysts each containing two sporozoites. This is referred to as **infective stage**. The oocyst commonly found as spherical/ovoid/ellipsoidal shape. It has refractile shell and some species with small pore at one end is called as micropyle covered with prominent polar cap.

Epidemiology

☒ Distribution: Worldwide

- ☒ The disease is more common in deep litter and intensive system of rearing with dirty environment
- ☒ Overcrowding predispose to infection
- ☒ Sporulation of oocyst can occur within two days of being passed in faeces
- ☒ Oocysts have a considerable longevity and can persist for several years

Host affected

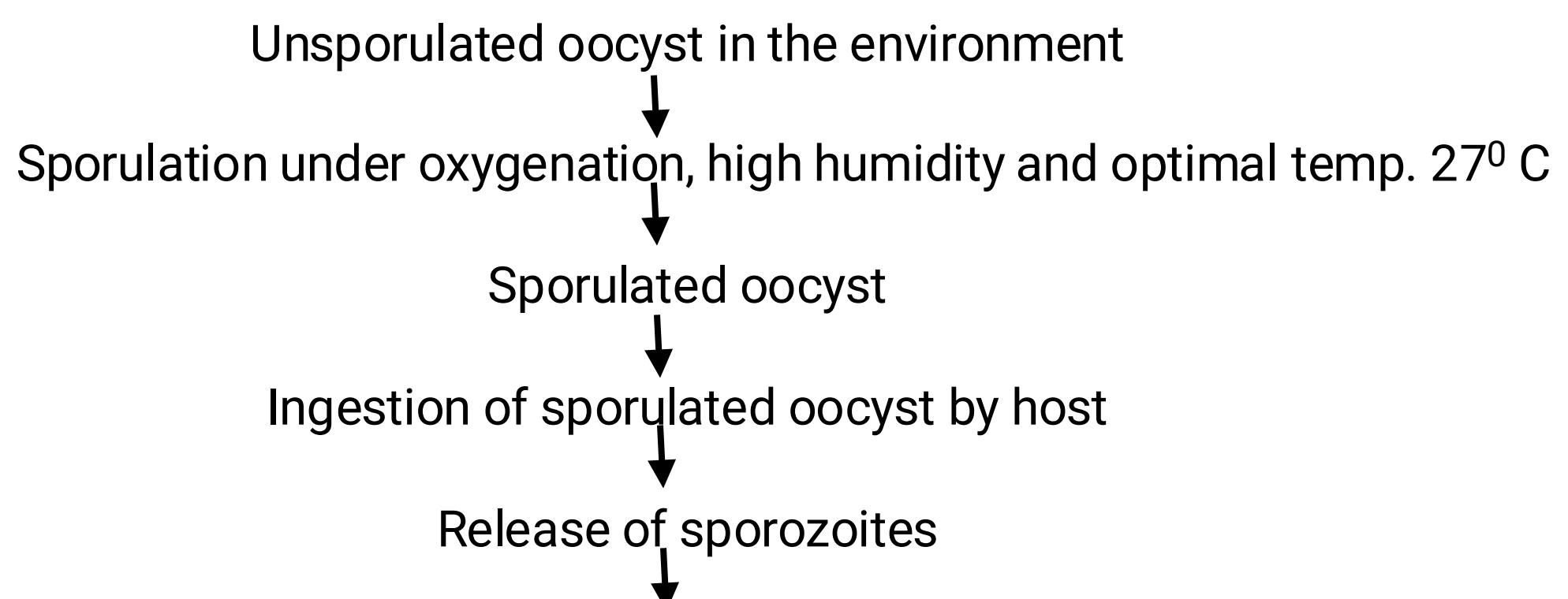
- ☒ Chicken and broilers are susceptible
- ☒ Breeder and layer pullets are at greatest risk, Barred Plymouth Rock, Jersey White Giant and Light Sussex are susceptible. White Leghorn, Rhode Island and Newhampshire are resistant
- ☒ **The disease is most common at the age of 4-6 weeks**
- ☒ Chicks > 6 weeks are resistant to infection, whereas susceptibility increases with age in *E.tenella*
- ☒ Transmission depends on housing, management and ingestion of sporulated oocyst in the environment
- ☒ Sparkling beetles, migratory birds and birds recovered from infection act as mechanical carrier

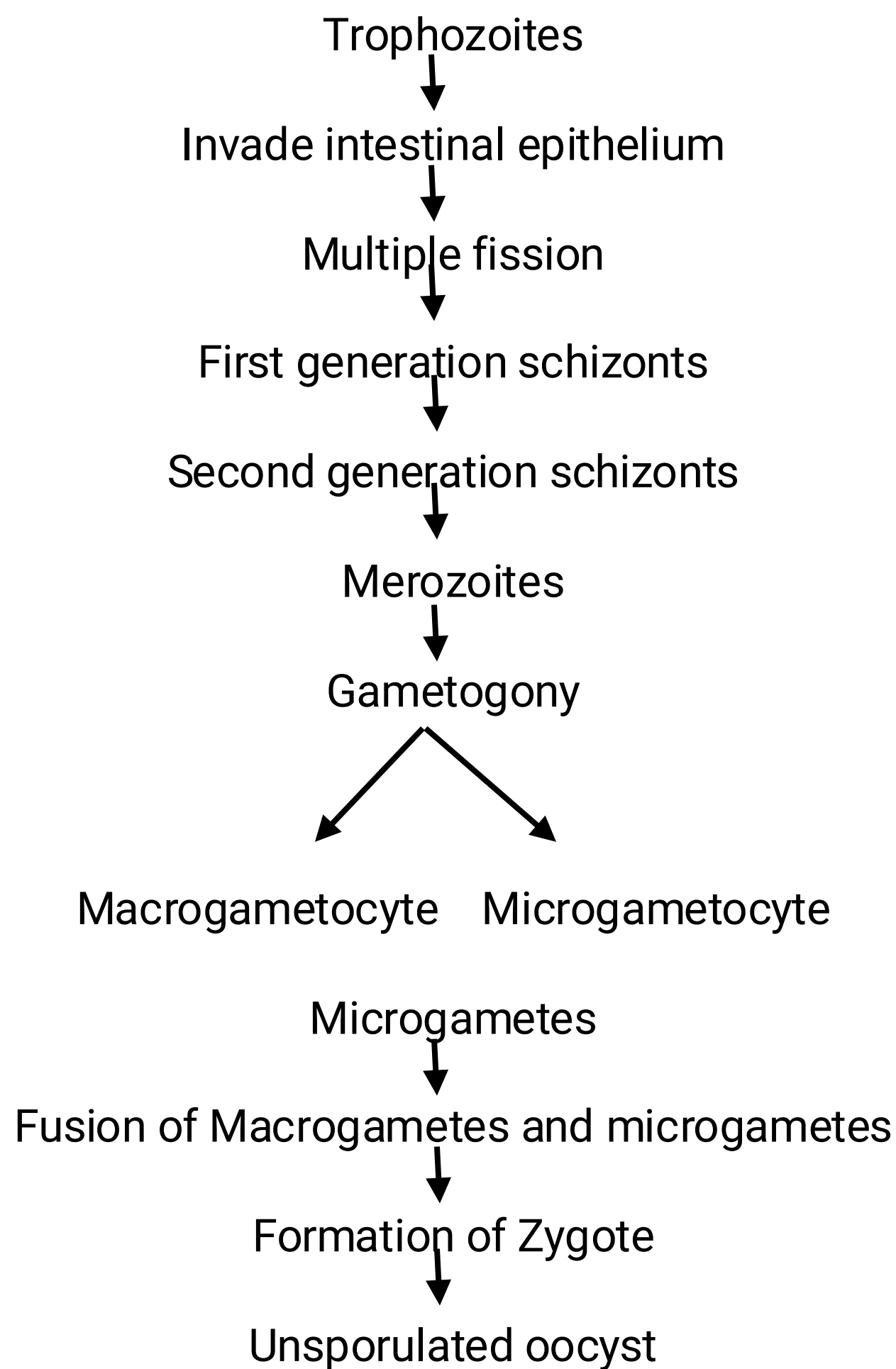
Immunity

- ☒ Previous exposure results in species - specific resistance. Some species induces a solid immunity; other species need several infections and is self – limiting in nature. Both CMI and humoral immunity are elicited
- ☒ Immunity is due to 2nd generation schizogony only
- ☒ Trickle infection: Continuous ingestion of low level of oocysts induces very strong immunity but cause damage the intestine
- ☒ Severity of disease increases with concurrent infections; Marek's disease increases the severity of coccidiosis

Life cycle

Prepatent period: 5 days





Pathogenesis

Caecal Coccidiosis

- ☒ *E. tenella* is primarily responsible for caecal coccidiosis although the gametogonous stages of *E. necatrix* also take place in the caecum and occasionally some stages of *E. Brunetti*
- ☒ The *E. tenella* infection primarily occurs at age of 3-7 weeks
- ☒ The severity of infection is related to parasitic density and parasites within the mucosa
- ☒ Heavy infection with developing schizonts are found deep in the mucosa or submucosa. The mature schizonts are released by destruction of intestinal epithelium leads to severe haemorrhage. Which occurs after 72 hrs of infection
- ☒ The prepatent period is 7 days

Clinical manifestation

- ☒ It occurs due to the ingestion of large number of oocysts over a short period
- ☒ Presence of soft faeces with blood, dullness, listlessness, drooping feathers, anaemia, paralysis and death

- ☒ The subclinical infection causes impaired absorption results in poor weight gain and poor feed conversion ratio

Intestinal coccidiosis

- ☒ It is mainly caused by *E.acervulina*, *E. maxima*, *E.necatrix*, *E.mitis* and *E.Praecox*
- ☒ The prepatent period vary from 4 to 7 days
- ☒ The subclinical infection is more common

Clinical signs

- ☒ Chronic watery diarrhoea with blood (mucous droppings in *E. maxima* infection)
- ☒ Stunted growth, reduced egg production, listlessness, anorexia, soiling of vent and feathers
- ☒ Recovered birds are unthrifty and emaciated for a long time

Rectal Coccidiosis

- ☒ It occur mainly due to infection with *E.brunetti*
- ☒ It causes coagulative necrosis with slight haemorrhage

Clinical signs

- ☒ White fluid droppings mixed with blood and mucus

Necropsy finding

Type	Species	Lesions
Intestinal coccidiosis	<i>E.acervulina</i>	White plaques arranged transversely in duodenum
	<i>E.maxima</i>	Thickened intestinal wall with blood tinged mucoid exudates
	<i>E.necatrix</i>	Ballooning of intestine with white spots, petechia, mucoid blood filled exudates (salt and pepper appearance)
Caecal coccidiosis	<i>E.tenella</i>	Haemorrhage coarse clotted blood in the lumen of caecum
Rectal coccidiosis	<i>E.brunetti</i>	Coagulation necrosis, mucoid bloody exudates in rectum

Diagnosis

- ☒ History of the flock
- ☒ Finding various types of lesions and their locations by P.M. examination
- ☒ Faecal examination for the presence of oocysts, but it is little significance in severe Outbreaks and clinical cases, as the mortality starts before the discharge of oocysts

- ☒ Examination of gut scrapings for the presence of schizonts and unsporulated oocysts
- ☒ Study of sporulated oocysts by allowing sporulation of oocysts after mixing faeces with several volumes of 2.5 % Potassium dichromate solution for 1 day to 2 weeks.

Treatment

- ☒ **Sulphonamide drugs are the most widely used** and it is recommended that these are given for two periods of three days in the drinking water, with an interval of two days between treatments.
- ☒ Sulphaquinoxaline sometimes potentiated with diaveridine, or sulphadimidinc are the drugs of choice. Where resistance has occurred to sulphonamides, mixtures of amprolium and ethopabate have given good results.
- ☒ **Sulphonamides have the great effect on II stage schizont** without inhibiting I stage schizogony and increases their resistance by acquired immunity
- ☒ Fully recommended dose should be given and withdrawn gradually by reducing half of the dose, otherwise it is lethal

S.No.	Drug	Dose	Coccidia	Stage of coccidian
Coccidiostats				
1.	Amprolium	125 ppm in feed / water for 6 days	<i>E. tenella</i> , <i>E. necatrix</i> and also <i>E. maxima</i>	I and II Schizonts
2.	Amprolium + Ethopabate	125 ppm + 8 ppm in feed/ water for 6 days	Extended spectrum	I and II Schizonts
3.	Sulphaquinoxaline	0.5 % in feed, 0.043 % in drinking water for 2 days with 3-5 days interval.	<i>E. tenella</i> , <i>E. necatrix</i> and <i>E. acervulina</i>	II Schizonts

4.	Sulphaquinoxaline + Pyrimethamine	Synergistic; 14 ppm + 45ppm in feed/ water	<i>E. tenella</i> , <i>E. necatrix</i> and <i>E. acervulina</i>	I and II Schizonts
5.	Nitrofurazone + Furazolidone	55 mg/kg + 55 mg/ kg	<i>E. tenella</i> and <i>E. necatrix</i>	II Schizonts
6.	Ionophore compounds			
	a) Monensin	0.121 % in feed	Superior to amprolium	
	b) Lasolacid	0.005 - 0.0075% in feed	High degree of activity	
	c) Salinomycin	0.01% in feed	Significant activity	
7.	Coccidiocidal			
	Robenidine	0.0066 % in feed	-	I Schizonts

Prevention

- ☒ Primary prophylaxis in vaccination

Coccivac - vaccine with live attenuated oocysts of *E. tenella*, *E. maxima*, *E. acervulina*, *E. mivati* of chicken isolates administered to day old healthy chicken through eye spray, cabinet spray or in feed

Coccivac-D (8 species) for breeders and layers

Other vaccines are Livacox - Q (attenuated quadrivalent vaccine) and *Immunocox*

Control

- ☒ **Trickle infection**

Initial infection of 100 -700 oocysts/ each species followed by repeated daily doses of 1-5 oocysts for 20 days produces immunity, which is reinforced by reinfection of oocysts in the litter

- ☒ **Shuttle programme**

Use of one product in the starter and another in the grower is called as shuttle programme

This programme is mainly intended to reduce the buildup of drug resistance

- ☒ **Step- down programme (Natural infection + anticoccidials)**

This is followed for control of coccidiosis in breeders and layers

In this approach the birds are naturally allowed for infection with moderate number of coccidial oocyst for certain period along with administration of broad spectrum anticoccidials drugs at the lowest approval level to provide protection for upto 6-12 wks. It helps to stimulate host immune mechanism in birds to develop protection against the severe outbreak of disease

- ☒ **Switch programme**-where drugs are switched within life-span of each batch
- ☒ Selective breeding of domestic fowl is to increase the genetic resistance of the host
- ☒ Nutritional supplement/ antagonism- Vitamin- A for recovery of the host; Vitamin-k to reduce the mortality due to haemorrhage and a high protein diet
- ☒ Isolation of sick birds
- ☒ Reduced drug rate to older birds allows limited exposure to developing coccidia so that it leads to acquired immunity
- ☒ Heaping the litter for 24 hours to reach a temperature of 50° C and forking for destruction of oocysts
- ☒ Avoidance of contamination of feeders and waterers with droppings
- ☒ Continuation of anticoccidials in water for 5 days
- ☒ Burning the infected materials and sprinkling the quick lime @2-3 kg / 10 sq.ft over the litter
- ☒ Avoidance of overcrowding in poultry houses
- ☒ Keep the waterers and feeders at higher level
- ☒ Feeding chicks with coccidiostat in chick mash.
- ☒ Thorough disinfection and fumigation (Ammonia + Potassium permanganate) in the premises before the entry of newly hatched chicks

Coccidiosis in turkey

Etiology

- ☒ *E. adenoids* is the primary cause of turkey coccidiosis
- ☒ *E.meleagrids*
- ☒ *E.meleagrimitis*

Epidemiology

- ☒ **Age:** Young birds are highly susceptible especially at 4-5 weeks
- ☒ **Risk Factors**
 - Intensive breeding (result in heavy loss)
 - Over crowding
 - Poor sanitation- dampness of house
- ☒ **Sporulation period-** one day

- ☒ Prepatent period- 5 days
- ☒ Pathogenesis by gametogony and schizogony

Clinical manifestation

- ☒ Watery diarrhoea with mucus and flecks of blood

Necropsy finding

Species	Lesions
<i>E.adenoids</i>	Intestines with catarrhal inflammation with or without haemorrhage
<i>E.meleagrids</i>	Cream coloured caeca, formation of caseous plug, petechial haemorrhage
<i>E.meleagrimitis</i>	Spotty congestion, petechial haemorrhage from duodenum to ileum with dilation of jejunum

Diagnosis : Similar to chicken

Treatment

- ☒ Sulphonamides are again the drugs of choice for treatment
- ☒ For prophylaxis, coccidiostats are normally incorporated in the feed for the first 12-16 weeks of life
- ☒ The drugs commonly used are sulphaquinoxaline, a mixture of amprolium, ethopabate and sulphaquinoxaline, or halofuginone at a low concentration.

Prevention and control: same as that of chicken

Coccidiosis in Geese

Etiology

- ☒ *E.truncata*- causes renal coccidiosis
- ☒ *E. anseris*-causes intestinal coccidiosis
- ☒ Mortality: 100%

Clinical manifestation

- ☒ Haemorrhagic enteritis
- ☒ Nephritis, marked emaciation and muscular incoordination

Lesion

- ☒ Enlargement of kidney
- ☒ Pin head sized grayish white foci or haemorrhagic petechia with accumulation of urates on kidney

Diagnosis

- ☒ History of the flock
- ☒ Finding various types of lesions and their locations by P.M. examination
- ☒ Faecal examination for the presence of oocysts is of little significance in severe outbreaks and clinical cases, as the mortality starts before the discharge of oocysts
- ☒ Examination of gut scrapings for the presence of schizonts and unsporulated oocysts
- ☒ Study of sporulated oocysts by allowing sporulation of oocysts after mixing faeces with several volumes of 2.5 % Potassium dichromate solution for 1 day to 2 weeks

Coccidiosis in cattle

Coccidiosis is a common disease of young cattle (1– 2 months to 1 yr) and usually is sporadic during the wet seasons of the year. “ Summer coccidiosis” and “ winter coccidiosis” in range cattle probably result from severe weather stress and crowding around a limited water source, which concentrates the hosts and parasites within a restricted area. Although particularly severe epidemics have been reported in feedlot cattle during extremely cold weather, cattle confined to feedlots are susceptible to coccidiosis throughout the year. Outbreaks usually occur within the first month of confinement. Cows may contribute to environmental contamination of *E bovis* oocysts through a periparturient rise of fecal oocyst counts.

Etiology

Twelve *Eimeria* spp have been identified in the feces of cattle worldwide, but only three (*E zuernii*, *E bovis*, and *E alabamensis*) are most often associated with clinical disease.

Morphology of oocyst

- ☒ Oval shape with either presence or absence of micropyle and residual cyst

Pathogenesis

- ☒ Time to onset of diarrhea after infection is 16– 23 days for *E bovis* and *E zuernii* and 3– 4 days for *E alabamensis*

Clinical manifestation

- ☒ The most typical syndrome of coccidiosis is chronic or subclinical disease in groups of growing animals. Calves may appear unthrifty and have faecal-stained perineal areas
- ☒ In light infections, cattle appear healthy and oocysts are present in normally formed faeces, but feed efficiency is reduced
- ☒ The most characteristic sign of clinical coccidiosis is watery faeces, with little or no blood, and animals show only slight discomfort for a few days. Severe infections are rare

- ☒ Severely affected cattle develop thin, **bloody diarrhoea that may continue for >1 wk**, or thin feces with streaks or clots of blood, shreds of epithelium, and mucus. They may develop a fever; become anorectic, depressed, dehydrated and loss body condition.
- ☒ Tenesmus is common because the most severe enteritis is confined to the large intestine
- ☒ Nervous signs (eg, muscular tremors, hyperesthesia, clonic-tonic convulsions with ventroflexion of the head and neck, nystagmus) and a high mortality rate (80%–90%) are seen in some calves with acute clinical coccidiosis
- ☒ Outbreaks of this “ nervous form” are seen most commonly during, or after, severely cold weather in midwinter in Canada and the northern USA; there are no reports of the “ nervous form” outside this geographic location
- ☒ Affected calves may die <24 hr after the onset of dysentery and nervous signs, or they may live for several days, commonly in a laterally recumbent position with a mild degree of opisthotonos

Diagnosis

- ☒ Based on clinical signs
- ☒ Finding of oocysts on faecal flotation or direct smear or by the McMaster technique
- ☒ Quantitative oocyst counts on individual rectal samples from at least five calves in a pen are helpful to confirm coccidiosis as a cause of clinical disease

Differential diagnosis

- ☒ Salmonellosis
- ☒ Bovine viral diarrhoea
- ☒ Malnutrition
- ☒ Internal parasitism
- ☒ Copper deficiency
- ☒ Cobalt deficiency
- ☒ Hypomagnesaemia

Treatment

- ☒ Coccidiosis is a self-limiting disease
- ☒ Spontaneous recovery without specific treatment is common when the multiplication stage of the coccidia has passed.
- ☒ Therapy for clinically affected animal
 - Sulphadimidine @100mg/Kg b.wt is given either orally or parenterally and repeated at half the initial dose level on another two days
 - Sulfaquinoxaline (6 mg/lb/day for 3– 5 days)
 - Amprolium (10 mg/kg/day for 5 days)

Sulfaquinoxaline is particularly useful for weaned calves that develop bloody diarrhea after arrival at a feedlot

Supportive oral or parenteral fluid therapy

Corticosteroids are contraindicated, because they increase shedding of oocysts and have induced clinical disease in subclinically infected calves.

Prevention

- ☒ Sulfonamides in the feed at 25– 35 mg/kg for ≥ 15 days
- ☒ Amprolium @ 5 mg/kg/day for 21 days
- ☒ Decoquate @ 22.7 mg/45 kg/day for 28 days
- ☒ lasalocid @1 mg/kg/day to a maximum of 360 mg/head/day
- ☒ Monensin @100– 360 mg/head/day can be used
- ☒ Toltrazuril administered at 15 mg/kg as a single oral dose, 14 days after animals are moved into group housing,

Control

- ☒ Isolation of sick animals
- ☒ All feed and water supplies should be high enough off the ground to avoid fecal contamination
- ☒ Mass medication of the feed and water supplies may be indicated in an attempt to prevent new cases and to minimize the effects of an epidemic.
- ☒ Overcrowding of animal should be avoided
- ☒ Calving grounds should be well drained and kept as dry as possible
- ☒ Inadequate housing and ventilation should be corrected

Coccidiosis in sheep and goat

Clinical coccidiosis of sheep and goats occurs mainly in young lambs and kids and there appears to be an increasing prevalence under conditions of intensive husbandry. Although the majority of sheep, particularly those under one year old, carry coccidia only two of the 11 species are known to be highly pathogenic

Etiology

- ☒ *E. arlongi*
- ☒ *E. crandallis*
- ☒ *E. ovinoidalis*

Epidemiology

- ☒ Prepatent period – 15 days
- ☒ Age: Upto 6 months of age in lambs and kids; usually 4-7 weeks of age group are affected

- ☒ *Transmission*: Newborn lambs / kids pick up the infection by suckling their soiled dams through soiled litter, drinking water and infected pastures
- ☒ Unhygienic condition
- ☒ Intensive grazing
- ☒ Flare-up of infection during onset of rains and winter.

Pathogenesis

- ☒ Diarrhoea with foul smelling and with or without streaks of blood
- ☒ Constipation, severe abdominal pain, tenesmus, anaemia, inappetence, unthriftiness, loss of weight, slight rise of body temperature, ailing kids lie down due to abdominal pain, soiling of hind quarters with faeces attracting blow flies

Diagnosis, Treatment and Control: Similar to cattle.

Coccidiosis in pig

- ☒ It is caused by *Isospora suis* and *E. deblickei*
- ☒ Young piglets at 2 weeks are highly affected
- ☒ Source of infection: oocyst produced from periparturient sow
- ☒ Transmission: **Coprophagia**
- ☒ Clinical signs: severe enteritis, diarrhoea is biphasic due to reinvasion from tissue
- ☒ Treatment: Amprolium in feed
- ☒ Prevention: administration of amprolium one week prior to farrowing to 3 weeks post farrowing

Coccidiosis in rabbit

Etiology

- ☒ *E. Stiedae*-Hepatic coccidiosis
- ☒ *E. Flavescens*- Intestinal coccidiosis
- ☒ *E. intestinalis*-intestinal coccidiosis

Epidemiology

- ☒ Age: Commonest around weaning and young rabbits are highly susceptible
- ☒ Risk Factors: Intensive breeding and rearing under poor sanitation
- ☒ Prepatent period for *E. Stiedae* 18 days and for others 5-7 days

Clinical manifestation

Hepatic coccidiosis

- ☒ Wasting, diarrhoea, ascites and polyuria
- ☒ Lesion- Enlargement of liver with studded white nodule

Intestinal Coccidiosis

- ☒ Diarrhoea, indigestion, bloating, inappetence, reduced weight gain, sudden death without symptoms or convulsion and paralysis
- ☒ Lesion: thickened intestinal wall with grayish white deposits, contents are pasty, diarrhoeic and with blood streaks.

Diagnosis

- ☒ Demonstration of oocysts in the faeces and postmortem examination.

Treatment

- ☒ Sulphonamides like sulphamezathine in drinking water @ 0.2 % is effective

Control

- ☒ Daily cleaning of cages/hutches/pens
- ☒ Provision of clean feeding trough
- ☒ Rearing on wire floor
- ☒ Incorporation of coccidiostat such as amprolium, clopidol or robenidine in the feed

Coccidiosis in Dogs and Cat

- ☒ *Isospora canis*-dog-mildly pathogenic
- ☒ *I. Rivolta*-cat-pathogenic
- ☒ *I. Felis*-cat-mildly pathogenic
- ☒ Young puppies and kittens are more prone to infection
- ☒ Risk factors - Overcrowding and poor sanitation
- ☒ Coccidiosis is usually associated with other infectious agent, immunosuppression and stress
- ☒ Clinical coccidiosis, although not common, has been reported in kittens and puppies
- ☒ In kittens clinical signs are seen during weaning stress
- ☒ Clinical signs in kittens- diarrhoea sometimes blood mixed, weight loss and dehydration
- ☒ **Treatment** - rarely practiced, because of self limiting in nature
 - In clinically affected cats, trimethoprim-sulfonamides @ 30-60 mg/Kg for 6 days
 - Dog- In severe cases sulfonamides such as sulphadimethoxine @ 50mg/Kg on first day and 25mg/Kg for 2-3 weeks along with supportive fluid therapy
- ☒ Control: Removal of faeces frequently, faecal contamination of feed and water should be prevented, disinfection of cages and utensils and insect control

Canine eperythrozoon infection

- ☒ Canine eperythrozoon infection is caused by *Eperythrozoon canis*. It comes under family Rickettsiaceae genus *Eperythrozoon*
- ☒ Based on phylogenetic similarities, several members of the genus *Eperythrozoon* have recently been transferred to the genus *Mycoplasma*
- ☒ Dog-*Mycoplasma haemocanis*
- ☒ Cat- *Mycoplasma haemofelis*
- ☒ *Eperythrozoon* sp. is transmitted mechanically by arthropods (*Rhipicephalus sanguineus*)
- ☒ Transmission may occur via surgical procedures through blood contamination of instruments

Clinical manifestation

- ☒ Eperythrozoonosis in dogs regarded as an innocuous disease and usually causes only mild anemia
- ☒ Animals with clinical disease have inappetence, wasting, anemia, malaise, and depression. body temperature may be elevated but is often normal
- ☒ Haematology shows a macrocytic hemolytic anemia, with anisocytosis, poikilocytosis, and a marked left shift in erythrocyte maturation. The leukocyte count is normal or slightly elevated

Lesion

- ☒ Subcutaneous haemorrhage
- ☒ Hepatomegally and splenomegally
- ☒ Ascites
- ☒ Bleeding in gastrointestinal mucosa

Diagnosis

- ☒ *Eperythrozoon* sp. is usually coccoid (0.5-1.0 µm diameter)
- ☒ In Giemsa-stained peripheral blood smears, they appear attached to the surface of erythrocytes
- ☒ Occasionally, rod-shaped forms (1-3 µm diameter) can be seen
- ☒ It is important to differentiate between acute disease, in which the organism is readily identified in peripheral blood smears, and chronic, subclinical disease, in which animals present with a secondary infection and the organism can be difficult to detect in blood smears

Differential diagnosis

- ☒ Canine Ehrlichiosis
- ☒ Infectious canine hepatitis

Treatment

- ☒ In severely affected animals, transfusion may be required before treatment can begin
- ☒ Tetracyclines, Oxytetracycline and Doxycycline are used at recommended dose rates for 2-3 weeks

Control

- ☒ The use of disposable needles and correct sterilization of surgical instruments will minimize accidental transmission
- ☒ Control of arthropods

Emerging exotic viral diseases of global importance

Emerging and exotic animal diseases present a growing threat to human and animal health and jeopardize food security. Increases in human and animal populations, with accompanying environmental degradation and globalized trade and travel, enhance opportunities for transfer of pathogens within and between species. The resulting diseases pose enormous challenges now and in the future.

Emerging and re-emerging animal diseases have in recent years been associated with outbreaks that have serious consequences for animal and human health. The World Organisation for Animal Health (OIE) defines an emerging disease as “ *a new infection or infestation resulting from the evolution or change of an existing pathogenic agent, a known infection or infestation spreading to a new geographic area or population, or a previously unrecognised pathogenic agent or disease diagnosed for the first time* and which has a significant impact on animal or public health.” A known or endemic disease is considered to be re-emerging if it shifts its geographical setting, expands its host range, or significantly increases its prevalence.

Factors responsible for emerging and re-emerging diseases includes

- ☒ Increasing human population
- ☒ Increasing numbers of food producing animals
- ☒ Human and domestic animal encroachment into wildlife habitat and resulting exposure to wild animals
- ☒ Environmental degradation
- ☒ Climate change
- ☒ Interspecies transfer of pathogens and
- ☒ Globalization of travel and trade

Emerging disease of global importance

1. Rinderpest

Rinderpest is an acute, highly contagious, viral disease of cattle, domesticated buffalo, and some species of wildlife. It is not a zoonotic disease, yet its introduction into Africa in the 19th century caused starvation and resulted in massive human fatalities. In 1889, **cattle shipped from India carried the rinderpest virus to Africa, causing an epidemic that established the virus on the continent.** Initially, approximately 90 percent of the cattle in sub-Saharan Africa died as well as many sheep and goats. Wild buffalo, giraffe, and wildebeest populations were decimated. This rinderpest epidemic is considered by some to have been the most catastrophic natural disaster ever to affect Africa. The Global Rinderpest Eradication Program is a large-scale international collaboration involving vaccination, local and international trade restrictions, and surveillance. This effort may be one of veterinary medicine's greatest achievements and rinderpest, the second disease (after smallpox) have been eradicated globally.

2. Foot and Mouth Disease

Foot and mouth disease (FMD) is perhaps the most important disease of livestock worldwide. It is a highly contagious viral disease that primarily affects cloven-hooved livestock and wildlife. FMD was once found worldwide, but has been eradicated from 70 countries. However, more than 100 countries are still considered either endemically or sporadically infected with the disease. Where the disease is endemic, it affects animal production and performance, including the capacity of draft animals to help farmers produce their crops. Countries which are infected are unable to access international markets. Outbreaks of FMD in countries that were previously free of the disease can be devastating to the economy due to loss of trade and the cost of eradication. An outbreak of FMD in the United Kingdom in 2001 is estimated to have cost \$10 billion. More than four million cattle and sheep were destroyed in order to eliminate the virus and re-establish trade. The World Organization for Animal Health (OIE) and the Food and Agriculture Organization of the United Nations (FAO) have recently launched a global campaign to eradicate FMD.

3. Blue tongue

Bluetongue is a non-zoonotic, non-contagious vector-borne disease, caused by a virus of the Reoviridae family with 25 known serotypes worldwide. It affects all domestic and wild ruminants, with sheep and cattle experiencing the highest rates of infection. The severity of the disease depends on the strain and morbidity can be very high in susceptible animals. Bluetongue outbreaks also cause direct economic losses through disease and mortality, loss of production, loss of milk yield, and declines in fertility. As bluetongue is not zoonotic, it poses no risks to human health and cannot be contracted or spread through food. Bluetongue is transmitted by biting midges that are infected with the virus after

ingesting blood from infected animals. **The spread of the disease depends mainly on those ecological and climatic factors that favour biting midge populations;** outbreaks are often seasonal, occurring at or shortly after the season of peak midge activity. Although bluetongue emerged in Africa and is endemic in many tropical areas, it has experienced dramatic geographical expansion in recent years, due to environmental factors and the growth of international trade links. *A series of bluetongue outbreaks in Europe starting in 1998 illustrate the role played by a warming climate and the increased risks of spreading disease vectors (insects or other living carriers that transmit an infectious agent) through trading routes.* As climate change becomes more of a concern in the future, the risks of bluetongue causing an epidemic will continue to increase. According to a leading expert on vector-borne diseases, “ through changes to climate, one of the most competent vector species of midge has spread around southern Europe and further north than it had before” while at the same time midge vectors have become “ better at being able to transmit the virus.” The bluetongue control measures include the surveillance of susceptible animals, quarantine, zoning, insect control and vaccination. Since controlling midge populations is not possible – they are too numerous – vaccination is the most effective practical measure to minimise losses related to the disease in endemic regions. A voluntary vaccination programme, combined with movement restrictions, successfully led to the control of the bluetongue virus-8 outbreak in the United Kingdom in 2008.

4. West Nile Virus

In 1999, the New York City area of the United States experienced an **outbreak of encephalitis in humans of unknown etiology.** At the same time, increased deaths were observed in a variety of bird species. Dr. Tracy McNamara, a veterinarian at the Bronx Zoo, hypothesized a **link between the disease in birds and humans, and submitted samples from dead zoo birds** to the USDA National Veterinary Services Laboratories (NVSL). NVSL, the U.S. Centers for Disease Control and Prevention (CDC), and the University of California, Irvine identified the virus as West Nile, which had never previously occurred in the Western Hemisphere. This virus progressed across North America in the next few years. It is now endemic in North, Central and South America and there is no feasible way to eradicate it. **West Nile (WNV) virus infects wild birds, which are the primary reservoir hosts.** It is mainly transmitted by mosquitoes, although it can spread between some birds and reptiles by direct contact. Although most humans are infected asymptotically or develop flu-like symptoms, WNV can cause encephalitis, meningitis, or a paralytic syndrome in less than one percent of infected people. **This virus can cause fatal encephalitis in horses and occasionally in other mammals.** It also affects alligators, causing severe outbreaks among farmed animals, and it can kill some species of birds. The veterinary vaccine industry, working in cooperation with the USDA Center for Veterinary Biologics, quickly developed an

effective vaccine to prevent the disease in horses. WNV vaccine is now considered one of the core equine vaccines in the U.S. and in WNV endemic areas of Canada. The vaccine has also been used off label to protect some endangered birds, such as California condors.

5. Rift valley fever

Rift Valley fever (RVF) is a zoonotic, mosquito– borne viral disease which affects ruminants. Sheep, cattle, and goats may be severely affected and serve as the primary host and amplifiers of the virus. Dogs are highly susceptible to infection and cats can be infected as well. In all affected species, a very high rate of abortion and death in neonates is observed. RVF is endemic in sub-Saharan Africa. Epidemics occur after heavy rainfalls when infected mosquito eggs hatch near large numbers of susceptible animals. RVF first appeared outside Africa in 2000, when outbreaks were reported in Saudi Arabia and Yemen.

The virus can also produce mild to severe disease in humans. Between November 2006 and February 2007, RVF caused more than 200 human deaths in Kenya. Control efforts included restricting the movement or slaughter of animals, vaccination of livestock, insect vector control, and public education. **RVF could spread to countries outside of Africa via an infected mosquito, animal, or human**, and could become established in a new region following a pattern similar to that of the spread of WNV throughout the Americas.

6. Influenza (H5 and H1)

Influenza viruses are highly prone to mutation and transfer between species. Expanding human and animal populations provide greater opportunities for influenza viruses to mutate and emerge in new species. In 1997, a highly virulent H5N1 influenza virus emerged in domestic waterfowl and poultry in Asia and has been sporadically transmitted to people. *Novel influenza viruses also have recently emerged to cause severe disease in dogs (H3N8) and cats (H5N1/H1N1) for apparently the first time.* In 2009, a novel H1N1 virus with genetic elements from swine, human, and avian influenza viruses emerged and caused a worldwide pandemic in humans.

7. Nipah Virus in Malaysia

In 1998, a never-before-observed virus emerged in the Malaysian **pig population, causing severe respiratory and neurologic signs.** The virus spread to swine caretakers resulting in more than 265 cases and over 105 deaths in Malaysia and Singapore. A novel paramyxovirus (Nipah virus) was isolated from a human patient. Both the U.S. CDC and the Australian Animal Health Laboratory sent teams of veterinarians and other specialists to Malaysia within days of isolation of the virus. Working closely with the Malaysian government and scientists, these teams developed diagnostic tests and control strategies (including the culling of more than one million pigs), that resulted in eradication of the virus from the swine population. Between 2001 and 2009, there were no new cases of Nipah virus infection in Malaysia. **Scientists have discovered that the virus is carried by healthy**

fruit bats. The virus is still present in fruit bats in Southeast Asia. Human cases of Nipah virus encephalitis have been reported repeatedly since 2000 in Bangladesh, in some cases due to infection from fruit or fruit juices contaminated by fruit bat saliva or urine. Subsequent human-to-human transmission has also occurred.

8. Ebola Virus in Africa

Ebola virus is a non-segmented RNA virus, which, together with Marburg virus, makes up the filovirus family. This now notorious group of viruses was discovered in 1967 when Marburg virus was identified as the etiologic agent of a **hemorrhagic fever outbreak** in research facilities in Europe, which handled tissues from African green monkeys imported from Uganda. Subsequently, Ebola viruses were shown to be the cause of simultaneously occurring hemorrhagic fever outbreaks in 1976 in the Democratic Republic of Congo (DRC, formerly Zaire) and Sudan. These outbreaks were shown to be caused by two different subtypes of Ebola virus, which became known as the Zaire and Sudan subtypes. Mortality rates of up to 80% were recorded in these and more recent outbreaks in DRC and Gabon in 1995– 1996. Epidemiologic data from recent outbreaks indicate that close contact is necessary for efficient transmission of Ebola virus from one individual to another, and little evidence can be found for aerosol transmission of the virus (5). Despite considerable efforts to identify the natural reservoir for Ebola and Marburg viruses, the host species remains an enigma. Although **non human primates have been implicated as the source of introduction of the virus into humans** during several of the identified outbreaks, they are not considered likely to represent reservoir species because of their susceptibility to high mortality hemorrhagic disease similar to that seen in humans. Little genetic difference has been detected between Ebola-Zaire viruses isolated 20 years apart and from locations over 1,000 km from one another, suggesting that ecological rather than genetic factors may play the dominant role in initiation of Ebola hemorrhagic fever outbreaks.

Control

Development of international cooperation; effective local, regional, and global networks for strong infectious disease surveillance and research collaboration to enable sharing of biological and study materials to enhance antimicrobial product development and vaccine trials; collaboration between animal and human health sciences to strengthen capacity for identification of microbial agents with epidemic potential so as to prevent their emergence.

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